

LIFE SCIENCES SUITE

Standard Operating Procedures Handbook

Procedures, demonstrations, good practices and diagrams for the 22 Life Sciences Suite modules and the related OCA modules — Odoo 19 Community

ENGLISH

About this handbook

This handbook gathers the Standard Operating Procedures (SOPs) that the Life Sciences Suite can support. Each SOP rests on a workflow, a control or a record that really exists in the modules. The workflows, buttons, menus, roles and controls quoted in this book were read from the module source code and from a database in which the 22 LS modules were installed on Odoo 19.0 Community Edition.

What this handbook is — and is not

- ✓ It is a **working basis** for your quality assurance: every sheet has blank fields (version, effective date, approval) to be completed under your own document control.
- ✓ The SOP codes (SOP-QA-001, etc.) are a **proposed numbering**, to be adapted to your document system.
- ✓ The **regulatory content** of each SOP (criteria, time limits, acceptance levels, local requirements) remains the responsibility of your QA and regulatory functions. The software helps to apply a procedure; it does not replace it and does not make an organisation compliant.
- ✓ References to regulations and standards are given at chapter or article level. Check the version in force before approval.

How the book is organised

- **Foundations** — GMP principles, document hierarchy, the quality cycle, how the modules hand over to each other, data integrity (ALCOA+), segregation of duties, and how to write an SOP in ls_qms.
- **Part I — Quality management system**: documents, training, deviations and CAPA, change control, complaints, audits, risk, suppliers.
- **Part II — Data integrity and compliance**: audit trail, electronic signatures, validation, calibration, environmental monitoring, recalls.
- **Part III — Industry modules**: pharmaceutical, cosmetics, medical devices, medical plastics, laboratory.
- **Part IV — Regulatory affairs**: import-export (phase 1).
- **Part V — Procedures supported by OCA modules**.
- **Part VI — Procedures needed but not covered by the suite**, with guidance on what they must contain.
- **Annexes** — SOP register, glossary, known limitations observed while preparing this book.

Screenshots, diagrams and demonstrations

Screenshots are captures of the running software (Odoo 19.0 Community, LS modules, demonstration data). The interface is in English: the modules ship no French translation, so the French handbook quotes the English button labels and gives their meaning. Lifecycle diagrams are drawn from the states actually defined by each module; explanatory diagrams (pyramids, V-model, charts) illustrate a concept and use illustrative values, which are labelled as such. Each SOP ends with a worked demonstration describing a realistic case step by step; the people and values in these demonstrations are fictitious.

Anatomy of a SOP sheet

Section	Content
Header	Code, title, module, menu, suggested owner, related SOPs; blank version, effective date and approval fields.
1 Objective · 2 Scope	Why the procedure exists and what it covers.
3 Responsibilities	Roles named after the security groups of the module, so that access rights and the SOP match.
4 Definitions	Terms used in the SOP.

Section	Content
5 Workflow in the software	Lifecycle diagram generated from the module states, and explanatory diagrams.
6 Procedure	Numbered steps. Buttons are shown as  ; menus as Menu ► Sub-menu .
7 Controls enforced by the software	Rules the system refuses to break (it blocks the action and shows a message). The SOP does not need to police them, but users must know them.
8 Good practices · 9 Pitfalls	Experience-based advice and frequent errors.
10 Records · 11 Indicators	Evidence produced and suggested KPIs.
12 References · 13 Demonstration	Regulatory basis and a worked example.

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Foundations

Why written procedures

Good Manufacturing Practice rests on a simple chain: **say what you do, do what you say, prove it**. A procedure says what must be done, by whom and when; the people trained on it do it; the records prove that it was done, and the quality system checks that the result is right. The suite supports each link: ls_qms and ls_document_management hold the procedures, ls_training proves the competence, each module keeps the records, and the audit, CAPA and risk modules close the loop.

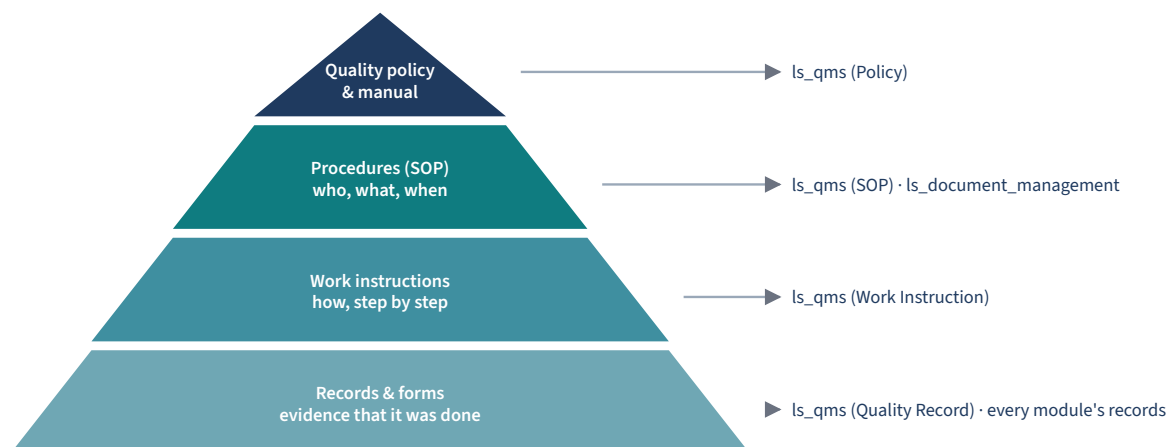


Figure 1 — Document hierarchy and where each level lives in the suite.

The quality cycle

ISO 9001, ISO 13485 and ICH Q10 describe a quality system as a continuous improvement cycle. The modules map onto the Plan–Do–Check–Act cycle as shown below. The cycle only works if the outputs of one stage reach the next one: an audit finding that never becomes a CAPA, or a CAPA whose action is never managed as a change, breaks the loop.

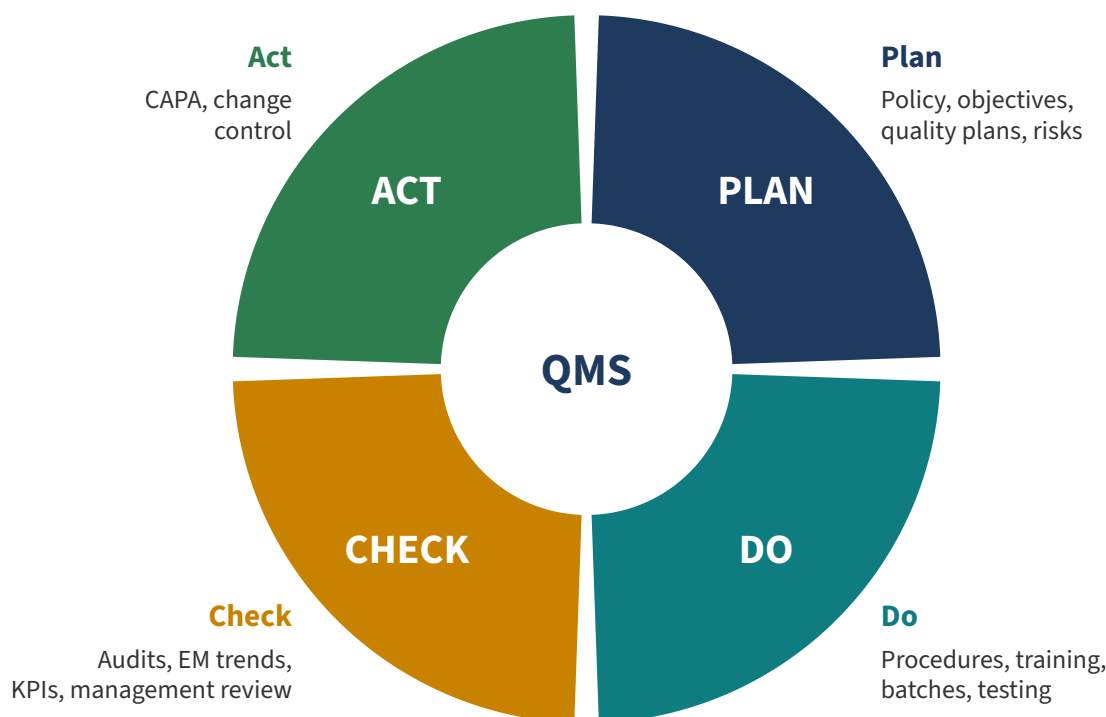
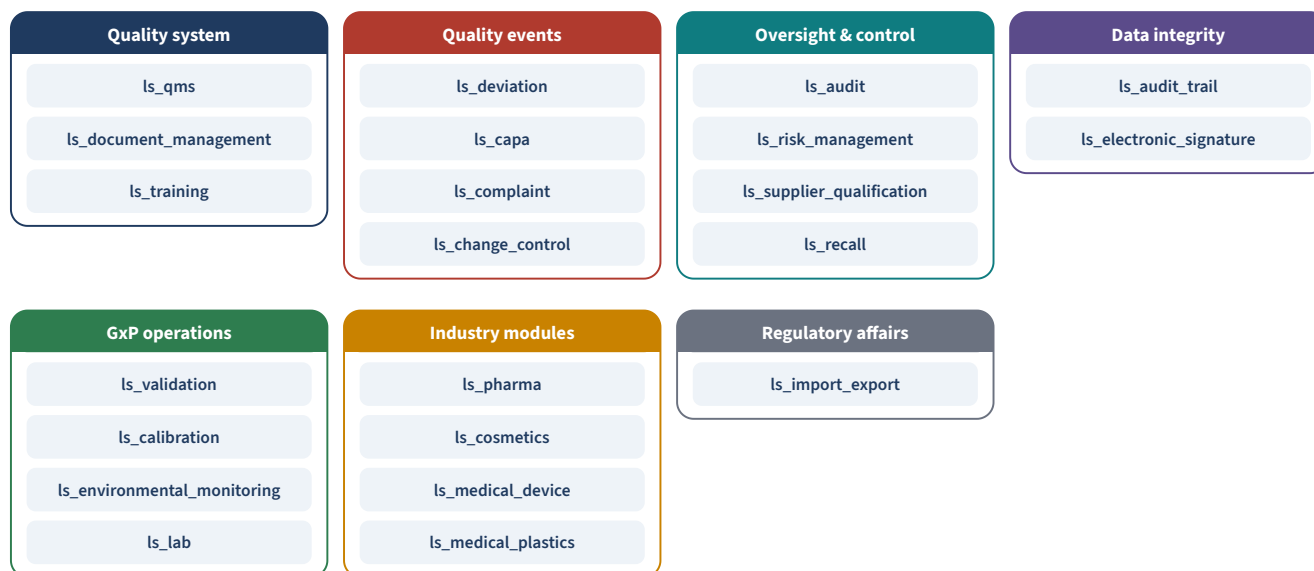


Figure 2 — Plan-Do-Check-Act cycle and the modules that support each stage.

The suite at a glance



No LS module holds a relational link to another LS module; hand-offs (deviation → CAPA, audit finding → CAPA, excursion → deviation...) are made through a reference field typed by the user. The procedures must therefore require these references to be entered and checked.

Figure 3 — The 22 Life Sciences Suite modules grouped by domain.

Each module can be installed on its own. The price of that independence is important for the procedures: **the modules do not create records in each other**. When a deviation needs a CAPA, the deviation reaches the state *CAPA Required* and records a CAPA reference, but the CAPA itself is opened by a person in *ls_capa*. The SOPs of this book therefore specify who opens the downstream record, within which time, and how the reference is written back.

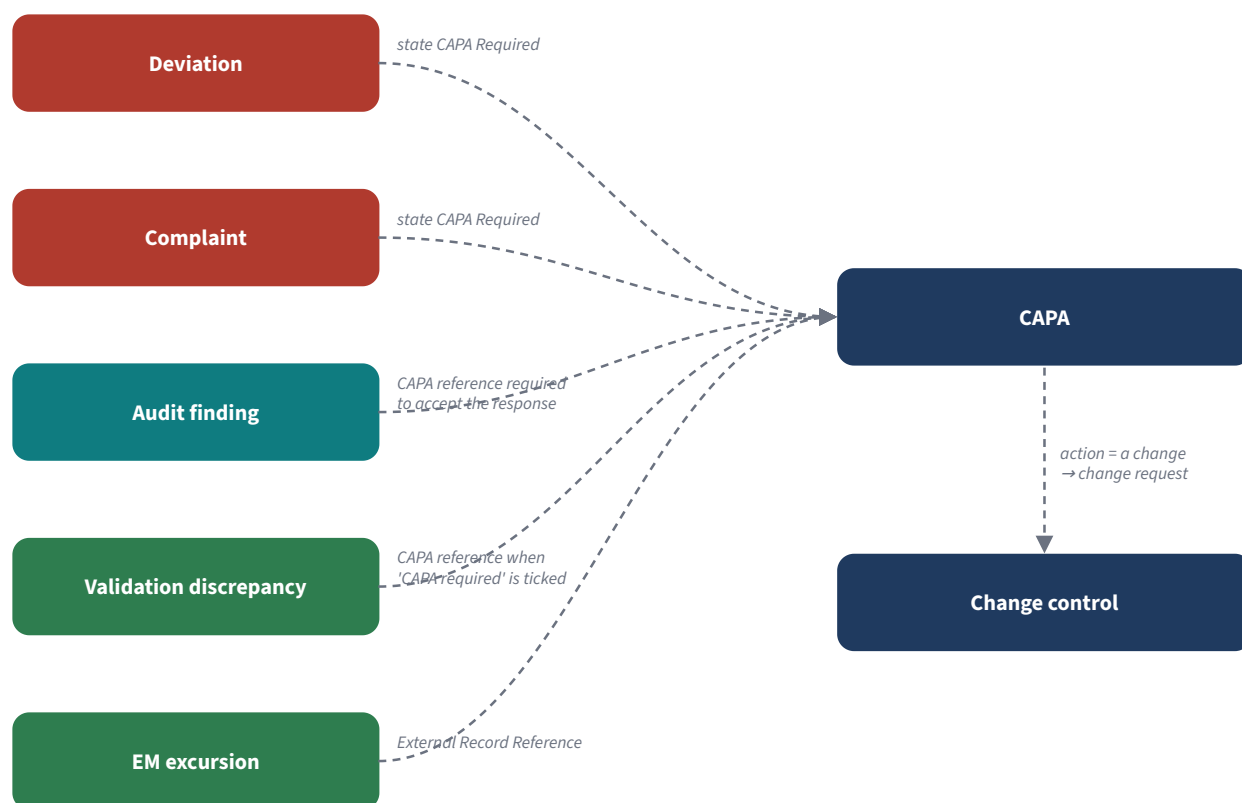


Figure 4 — Hand-offs between quality events, as implemented: by state and reference field, not by automatic link.

Data integrity — ALCOA+

Regulators expect GxP data to be Attributable, Legible, Contemporaneous, Original and Accurate, and also Complete, Consistent, Enduring and Available. The suite contributes technical controls — immutable records after approval, append-only logs, hash chains, mandatory reasons for change — but data integrity is first a behaviour: record at the time of the activity, never share an account, never record for someone else, and correct by a traced amendment rather than by deletion.



Figure 5 — ALCOA+ principles.

Segregation of duties (four eyes)

Most lifecycles in the suite separate the person who prepares or performs from the person who reviews and the person who approves. Where the rule is coded, the software refuses the forbidden combination and says why. Access rights must be designed so that every critical workflow has at least two eligible people, including during holidays.



The software refuses the same person in two roles on one record wherever the rule exists

(e.g. result review, SOP approval, batch release, excursion closure, protocol approval).

Figure 6 — Segregation of duties enforced by the software.

Roles and access rights

Each module defines its own security groups (typically Viewer, User or Operator, Reviewer or Approver, Manager). The responsibilities tables of the SOPs use these group names so that the procedure and the access matrix say the same thing. Grant the least privilege needed, review the grants periodically (see SOP-IT-001), and never give a production account a manager role "for convenience".

Writing a SOP in ls_qms

The *Standard Operating Procedure* model of ls_qms has the sections Purpose, Scope, Definitions and Abbreviations, Responsibilities, Procedure and Reference Documents. Purpose, scope and procedure are mandatory before review. The sheets of this handbook follow the same structure: copy each section into the corresponding field, adapt it, then submit it through the approval workflow described in SOP-QA-001.

Rules of good writing

- ★ One procedure, one process. Split when two teams or two triggers are involved.
- ★ Write imperative, numbered steps; one action per step; name the role, not the person.
- ★ State the time limits and the acceptance criteria as numbers, not as "promptly" or "as appropriate".
- ★ Quote the exact button and menu of the software, so that training and audit use the same words.
- ★ Say what to do when something goes wrong (the exception path), not only the nominal path.
- ★ Keep the "why" in the objective and the references; keep the steps short.

I

Quality management system

The procedures that organise the quality system itself: documents and records, training, quality events (deviations, CAPA, complaints, changes), audits, risk management and suppliers.

Document system — ls_qms,
ls_document_management

Training — ls_training

Deviations and CAPA — ls_deviation, ls_capa

Change control — ls_change_control

Complaints — ls_complaint

Audits — ls_audit

Risk management — ls_risk_management

Suppliers — ls_supplier_qualification

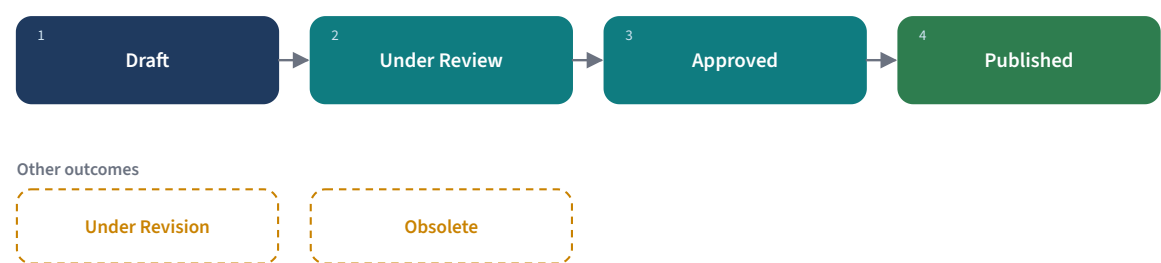
Document system

ls_qms · ls_document_management

Two modules carry controlled documentation. **ls_qms** holds the structured quality documents — quality policy, SOPs, work instructions and quality plans — whose content is typed in sections, plus quality objectives and quality records. **ls_document_management** holds controlled *files* (PDF, drawings, forms) organised in folders, with immutable versions protected by a SHA-256 checksum, an approval circuit per document and retention policies. Use ls_qms for documents you write in the system and ls_document_management for documents you manage as files.

Menu	Content
Quality Management ► Quality Policy / Procedures / Work Instructions / Quality Planning	Structured controlled documents (ls_qms)
Quality Management ► Quality Objectives, Quality Records	Objectives with measurements; retained quality records
Quality Management ► Reporting ► Procedures Due for Review	Documents whose periodic review is due
Documents ► Documents / Versions / My Approvals	Controlled files (ls_document_management)
Documents ► Configuration ► Folders / Retention Policies / Tags	Structure and retention rules

Group	Main rights
ls_qms: Viewer · User · Approver · Manager	User writes and submits; Approver approves, rejects and publishes; Manager sets obsolete, reopens, disposes of records.
ls_document_management: Viewer · Editor · Approver · Manager	Editor creates documents and versions; named approvers decide; only the Manager publishes and archives.



Model: ls.qms.sop
Figure 7 — Lifecycle of an ls_qms controlled document (policy, SOP, work instruction, quality plan).



Model: ls.document.document
Figure 8 — Lifecycle of a controlled file in ls_document_management.

Quality Management
Quality Policy
Quality Objectives
Procedures
Work Instructions
Quality Planning
Quality Records
Reporting
Configuration
7
14
My Company (San Francisco)

New
Standard Operating Procedures
[SOP-0001] Control of Documented Information...
Instructions 1
Attachments 0
1 / 1

New Revision
Set Obsolete
Draft
Under Review
Approved
Published
Send message
Log note
Activity

Title
Control of Documented Information
SOP-0001 - version 1

OWNERSHIP

Author ? Mitchell Admin
Owning Department Quality Assurance (demo)
Reviewers ?
Approved By Mitchell Admin
Training Required ? ☒
Company My Company (San Francisco)

LIFECYCLE

Submitted On
Approved On
Effective Date ? Jun 27
Review Period (Months) ? 24
Next Review Due ? Jun 27, 2028
Review Status ? Up To Date

Purpose
Scope
Definitions
Responsibilities
Procedure
References
Work Instructions
Revision

Define how quality documents are drafted, reviewed, approved, published, revised and withdrawn.

Today
OdooBot 10:40 AM
Standard Operating Procedure created

Figure 9 — A procedure in *ls_qms*: sections, status bar and approval buttons. (Screenshot of the running software (Odoo 19.0 Community, *LS* modules, demonstration data).)

Drafting, review, approval and publication of quality documents

Module	ls_qms · ls_document_management	Version	_____
Menu	Quality Management ► Procedures (and Quality Policy, Work Instructions, Quality Planning)	Effective date	_____
Owner (suggested)	Head of Quality Assurance	Approved by	_____
Related SOPs	SOP-QA-002, SOP-QA-003, SOP-QA-012, SOP-QA-030		

1 Objective

Ensure that every quality document (quality policy, SOP, work instruction, quality plan) is written to a common standard, reviewed and approved by competent people other than its author, and made effective in a controlled way, so that only approved, current instructions are used.

2 Scope

All quality documents managed in ls_qms, and controlled files (forms, drawings, external documents) managed in ls_document_management. Excludes records, which follow SOP-QA-004.

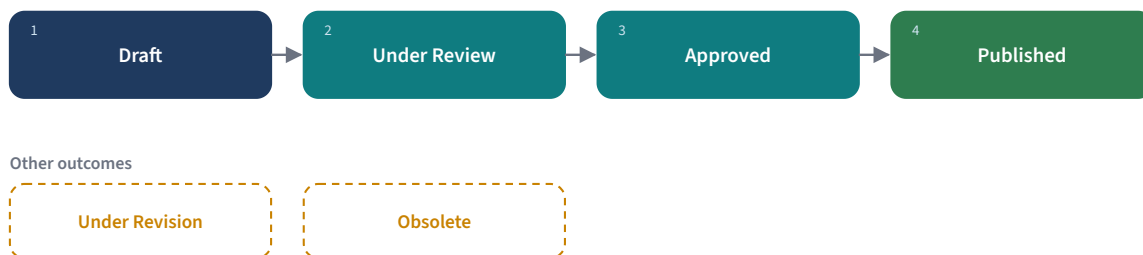
3 Responsibilities

Role (software group)	Responsibility
QMS User (author)	Drafts the document, completes all mandatory sections, proposes reviewers, submits for review.
Reviewers (named on the document)	Check technical content and consistency with other documents; return comments within the agreed time.
QMS Approver	Approves or rejects with a reason; publishes the approved revision on its effective date.
QMS Manager	Owens the document system, numbering and templates; resolves disputes; withdraws documents.
Document Manager (ls_document_management)	Publishes and archives controlled files once all named approvers have approved.

4 Definitions

Term	Definition
Revision	A numbered version of a document. Only one revision of a document may be published at a time.
Effective date	Date from which the published revision applies. It cannot precede the approval date.
Reason for change	Mandatory statement explaining why a new revision is created.

5 Workflow in the software



Model: ls.qms.sop

Figure 10 — Lifecycle of a quality document in ls_qms.

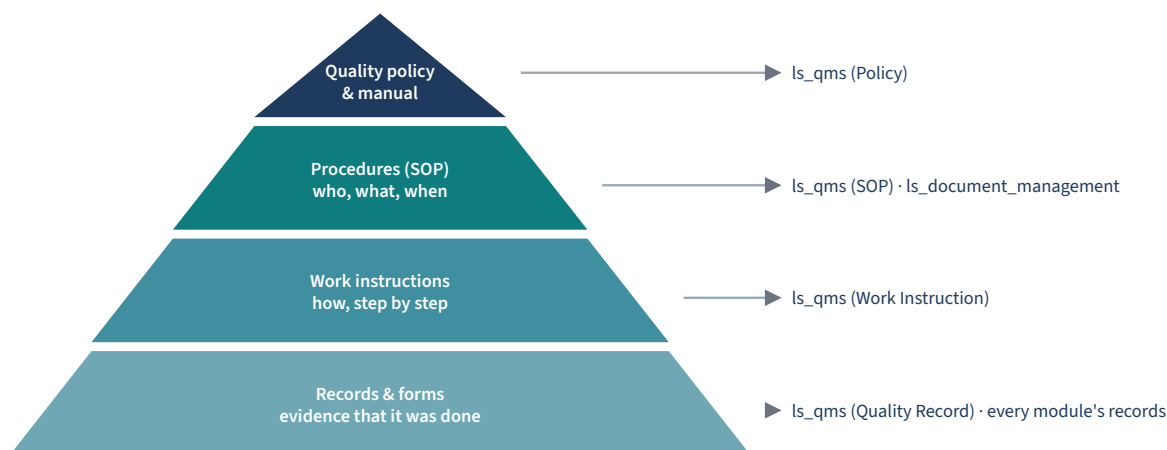


Figure 11 — Where each document type sits in the hierarchy.

6 Procedure

- 1 **Check the need.** Search the register ([Quality Management ► Procedures](#)) to avoid duplicates. For a change to an existing document, use SOP-QA-002.
- 2 **Create the draft.** Click **New**. Enter Title, Owning Department, Review Period (months; the default comes from the QMS parameter, 24 months at installation) and tick *Training Required* when users must be trained before applying the procedure. The reference and version are allocated by the system.
- 3 **Write the content.** Complete Purpose, Scope, Definitions and Abbreviations, Responsibilities, Procedure and Reference Documents. For a work instruction, select the *Parent Procedure*; for a quality plan, add at least one control line; for the policy, write the policy statement.
- 4 **Name the reviewers.** Add the reviewers who cover every affected function (production, QC, engineering, regulatory as relevant).
- 5 **Submit.** Click **Submit for Review**. The document moves to *Under Review* and its content is locked for the author.
- 6 **Review.** Reviewers record comments in the chatter. Blocking comments lead the approver to **Reject** with a written reason; the document returns to Draft.
- 7 **Approve.** A QMS Approver who is not the author clicks **Approve**. The approval date is recorded.
- 8 **Plan the training.** If *Training Required* is ticked, launch SOP-QA-012 before the effective date.
- 9 **Publish.** On the effective date the approver clicks **Publish**. The previous revision, if any, becomes *Obsolete* automatically.
- 10 **Distribute.** Inform the users (chatter, e-mail, briefing). Withdraw any paper copy of the previous revision.

- 11 Controlled files.** For a file in `ls_document_management`: create the document in its folder, upload the file with **New Version**, add at least one approver, **Submit for Review**; each approver decides in **Documents ▶ My Approvals**; the Document Manager then clicks **Publish**.

7. Controls enforced by the software

- ✓ Purpose, scope and procedure of an SOP must be filled before it can be submitted.
- ✓ The author of a document cannot approve it (segregation of duties).
- ✓ Only one revision of a document may be published at a time; publishing retires the previous one.
- ✓ The effective date cannot precede the approval date.
- ✓ Published or obsolete content cannot be modified — a new revision is required.
- ✓ Only draft documents can be deleted; others must be made obsolete.
- ✓ A work instruction cannot be published while its parent procedure is not published.
- ✓ `ls_document_management`: a document cannot be submitted without a version (file) and at least one approver; only a Document Manager can publish or archive; each version carries a SHA-256 checksum verified for integrity.

8. Good practices

- ★ Use one template per document type and keep the section headings of `ls_qms`.
- ★ Choose reviewers by function, not by availability; two reviewers are usually enough.
- ★ Set the effective date far enough ahead to complete training (SOP-QA-012).
- ★ Write the reason for change even for the first revision ("new procedure").
- ★ Quote the exact software buttons and menus in procedures that use the suite.

9. Pitfalls to avoid

- ✗ Publishing before the users are trained.
- ✗ Approving your own document through a second account — a data integrity breach.
- ✗ Keeping uncontrolled copies (desktop, printouts) after a new revision is published.
- ✗ Using the chatter to change content after approval: content is frozen, create a revision.

10 Records and evidence

- The document revision with its approval and publication dates
- Chatter history of review comments
- Rejection reasons
- Training records (SOP-QA-012)

11 Performance indicators

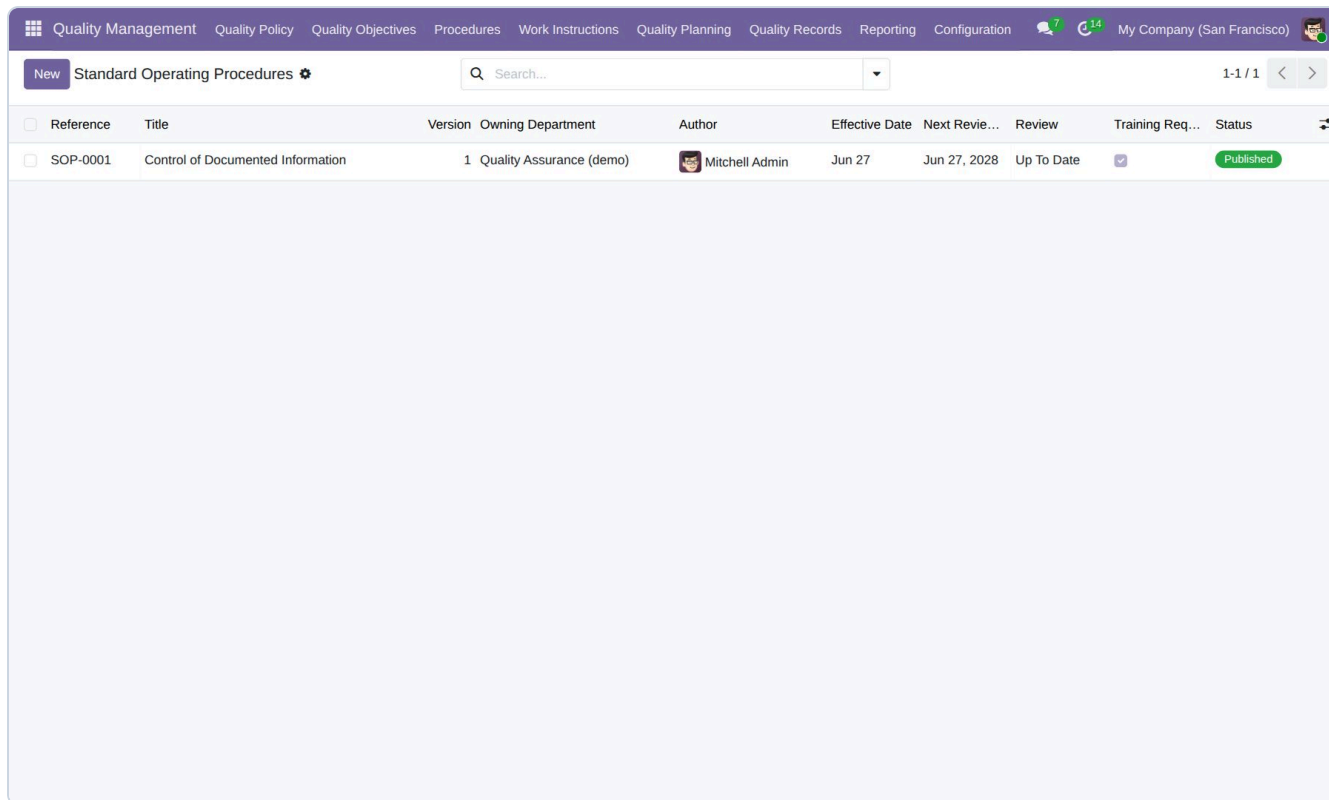
- Median time from submission to approval
- Number of rejections per document
- Documents published without training completed (target 0)

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 4 — Documentation
- ISO 9001:2015 — Quality management systems — Requirements
- ISO 13485:2016 — Medical devices — Quality management systems
- 21 CFR 211.100 — Written procedures; deviations
- EU GMP Annex 11 — Computerised Systems (check the revision in force)

13. Demonstration

Case. The warehouse needs a procedure for temperature mapping. The QA officer creates *SOP – Temperature mapping of storage areas*, sets a 24-month review period and ticks *Training Required*. She names the warehouse manager and the validation engineer as reviewers and clicks **Submit for Review**. The validation engineer asks for the acceptance criterion to be stated in °C; the approver clicks **Reject** with this reason. After correction the document is resubmitted and approved by the Head of QA (not the author). Training sessions are held (SOP-QA-012); on 1 March the approver clicks **Publish**: the SOP becomes effective and its next review date is computed automatically (1 March two years later).



Quality Management Quality Policy Quality Objectives Procedures Work Instructions Quality Planning Quality Records Reporting Configuration 7 14 My Company (San Francisco)										
New	Standard Operating Procedures		Search...		1-1 / 1					
Reference	Title	Version	Owning Department	Author	Effective Date	Next Review Date	Review	Training Required	Status	
SOP-0001	Control of Documented Information	1	Quality Assurance (demo)	Mitchell Admin	Jun 27	Jun 27, 2028	Up To Date	☒	Published	

Figure 12 — Procedure register with status and next review date. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Revision, withdrawal and archiving of documents

Module	ls_qms · ls_document_management	Version	_____
Menu	Quality Management ► Procedures ► New Revision / Set Obsolete	Effective date	_____
Owner (suggested)	QMS Manager	Approved by	_____
Related SOPs	SOP-QA-001, SOP-QA-003, SOP-QA-004, SOP-QA-030		

1 Objective

Control changes to effective documents so that every change is justified, reviewed and approved, the previous revision is withdrawn from use at the right moment and remains retrievable for the retention period.

2 Scope

Published documents of ls_qms and ls_document_management that must be changed, replaced or withdrawn.

3 Responsibilities

Role (software group)	Responsibility
Document owner / QMS User	Requests the revision, writes the reason for change and the new content.
QMS Approver	Approves and publishes the new revision.
QMS Manager	Withdraws (sets obsolete) documents that are no longer needed; checks references to them.
Document Manager	Starts revisions and archives files in ls_document_management.

4 Definitions

Term	Definition
Obsolete	A document that is no longer applicable. It remains readable for history but must not be used.
Archived (files)	End state of a published controlled file; its retention period may start at archiving.

5 Workflow in the software



Model: ls.document.document

Figure 13 — Lifecycle of a controlled file, including archiving.



Only one revision is published at a time; published content is frozen.

Figure 14 — A revision replaces the previous one, which becomes obsolete.

6 Procedure

- 1 **Decide the path.** Minor editorial change or substantive change? Substantive changes affecting product, process or validated state also require a change request (SOP-QA-030).
- 2 **Create the revision.** On the published document click **New Revision**. Enter the *Reason for Change* (mandatory). A new draft revision linked to the previous one is created; the current revision moves to *Under Revision* and remains the applicable revision until the new one is published.
- 3 **Edit and route.** Edit the draft and route it through review and approval as in SOP-QA-001.
- 4 **Publish.** Publishing the new revision retires the previous one to *Obsolete* automatically; its obsolete date is recorded.
- 5 **Withdraw without replacement.** If the document is no longer needed, the QMS Manager clicks **Set Obsolete** after checking that no other published document refers to it (for a SOP, check its work instructions).
- 6 **Controlled files.** In *Is_document_management*, click **Start Revision** on a published document, add a new version with **New Version**, then submit it for approval. To withdraw a file, the Document Manager clicks **Archive**.
- 7 **Communicate.** Announce the change and its effective date; collect paper copies of the old revision.

7. Controls enforced by the software

- ✓ A reason for change is required to create a new revision.
- ✓ Only one revision may be published; publishing retires the previous one.
- ✓ Obsolete and published content cannot be edited.
- ✓ Only draft documents can be deleted; published documents are made obsolete, never deleted.
- ✓ *Is_document_management*: only published documents can enter revision or be archived; a published document cannot be deleted; versions are immutable.

8. Good practices

- ★ Summarise what changed in the reason for change ("§5.3 temperature limit changed from 25 °C to 30 °C").
- ★ Check the documents that refer to the revised one; revise them together if necessary.
- ★ Keep the effective date and the training of the new revision aligned (SOP-QA-012).

9. Pitfalls to avoid

- ✗ Making a substantive change without a change request.
- ✗ Setting a SOP obsolete while its work instructions are still published.
- ✗ Leaving the old revision printed at the workstation.

10 Records and evidence

- Chain of revisions (Previous Revision / Following Revisions)
- Reason for change
- Obsolete date
- Archived file versions with checksums

11 Performance indicators

- Revisions per document per year
- Documents with overdue review
- Number of obsolete documents still referenced

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 4 — Documentation
- ISO 13485:2016 — Medical devices — Quality management systems
- 21 CFR 211.180 to 211.198 — Records and reports

13. Demonstration

Case. An audit shows that the cleaning SOP still names a detergent that is no longer purchased. The owner opens the published SOP, clicks **New Revision** and writes: "\$6.2 detergent X replaced by detergent Y (change CC-2026-014)". The new draft is reviewed and approved. On publication, revision 3 becomes *Published* and revision 2 becomes *Obsolete* with today's date, and remains available for inspection.

Periodic review of documents

Module	ls_qms	Version	_____
Menu	Quality Management ► Reporting ► Procedures Due for Review	Effective date	_____
Owner (suggested)	QMS Manager	Approved by	_____
Related SOPs	SOP-QA-001, SOP-QA-002, SOP-QA-004		

1 Objective

Confirm at defined intervals that every effective document is still accurate, applicable and consistent with practice, regulations and other documents.

2 Scope

All published ls_qms documents (policy, SOP, work instructions, quality plans) and controlled files whose owner defines a review.

3 Responsibilities

Role (software group)	Responsibility
Document owner	Performs the review and records its conclusion.
QMS Manager	Monitors the list of documents due, escalates overdue reviews, reports the indicator to management review.
QMS Approver	Approves the revision when the review concludes that a change is needed.

4 Definitions

Term	Definition
Review period	Number of months between two reviews (field <i>Review Period (Months)</i>).
Next review due	Computed by the system: effective date + review period.

5 Workflow in the software



Only one revision is published at a time; published content is frozen.

Figure 15 — Periodic review cycle computed from the effective date.

6 Procedure

- List the documents due.** Each month open [Quality Management ► Reporting ► Procedures Due for Review](#) and assign each document to its owner, with a due date.
- Review the content.** The owner checks the document against current practice, recent deviations, CAPA, changes, audit findings and regulatory updates.

- 3 **Conclude.** Either *no change needed* or *revision needed*. Record the conclusion as a quality record of type *Document Review Record* (SOP-QA-004) that refers to the document.
- 4 **Revise if needed.** Create a new revision (SOP-QA-002). The review date restarts from the effective date of the new revision.
- 5 **No change.** If no change is needed, the review period restarts only when a new revision becomes effective; to reset the next review date without content change, publish a revision whose reason for change states "*periodic review — no change*", according to your policy.
- 6 **Escalate.** Reviews overdue by more than one month are reported to the Head of QA and to management review.

7. Controls enforced by the software

- ✓ The next review date is computed from the effective date and the review period; the period cannot be negative.
- ✓ Published content cannot be edited in place — any change goes through a new revision.

8. Good practices

- ★ Plan reviews in batches by process so related documents are reviewed together.
- ★ Use the review to simplify: remove duplicated or unused documents.
- ★ Decide in your document policy how a "no change" review is evidenced (review record only, or a new revision).

9. Pitfalls to avoid

- ✗ Signing off a review without reading the document against current practice.
- ✗ Letting documents drift years past their review date.

10 Records and evidence

- Quality record *Document Review Record* per review
- New revision when applicable

11 Performance indicators

- % of documents reviewed on time
- Number of documents overdue by more than 3 months (target 0)

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 4 — Documentation
- ISO 9001:2015 — Quality management systems — Requirements
- ISO 13485:2016 — Medical devices — Quality management systems

13. Demonstration

Case. In January, the report lists 12 procedures due. The QMS Manager assigns them. For SOP-QA-040 the owner finds that the acknowledgement time was shortened from 5 to 2 days by the complaint category configuration; she creates a new revision to align the text. For SOP-MET-001 no change is needed; she records a *Document Review Record* stating "reviewed against CAPA-2025-031 and change CC-2025-044 — no change".

Management and retention of quality records

Module	ls_qms · ls_document_management	Version	_____
Menu	Quality Management ► Quality Records · Documents ► Configuration ► Retention Policies	Effective date	_____
Owner (suggested)	QMS Manager	Approved by	_____
Related SOPs	SOP-QA-001, SOP-DI-004		

1 Objective

Ensure that quality records are complete, protected against unauthorised change, retained for the required period and disposed of only after that period, under control.

2 Scope

Quality records of ls_qms (management review, meeting minutes, process verification, document review, objective review, customer feedback, internal communication, QMS change evaluation) and controlled files with a retention policy. Records produced by the other modules (batch records, deviations, calibrations...) are retained in their module and follow the same principles.

3 Responsibilities

Role (software group)	Responsibility
QMS User	Creates and completes the record; confirms it.
QMS Manager	Sets retention periods, corrects confirmed records when justified (logged), archives and disposes of records.
Document Manager	Maintains retention policies and legal holds of controlled files.

4 Definitions

Term	Definition
Retention period	Minimum time a record must be kept (months). <i>Retain Until</i> is computed from the record date.
Legal hold	Flag that suspends the end of retention of a controlled file (litigation, inspection).
Disposal	Controlled destruction of a record after its retention period.

5 Workflow in the software



Model: ls.qms.quality_record

Figure 16 — Lifecycle of a quality record.

6 Procedure

- 1 **Define retention rules.** The QMS Manager lists retention periods per record type from the regulations applicable to the products (for example batch-related records, medical device records) and configures them: *Retention Period (Months)* on quality records; in **Documents ► Configuration ► Retention Policies** the duration, unit, trigger (publication or archiving date), end-of-retention action (notify or archive) and advance notice.
- 2 **Create the record.** In **Quality Management ► Quality Records** click **New**, choose the *Record Type*, date, department, description, conclusion and the related policy, procedure, plan or objective. Attach supporting files.
- 3 **Confirm.** When complete, click **Confirm**. From then on only a QMS Manager can correct it, and the correction is logged.
- 4 **Archive.** When the record is no longer active, click **Archive**.
- 5 **Monitor retention.** Use **Quality Management ► Reporting ► Records Past Retention** and the retention notices of *Is_document_management*.
- 6 **Dispose.** After the *Retain Until* date and checking there is no legal hold, open inspection or litigation, the QMS Manager clicks **Dispose** and records the disposal decision.

7. Controls enforced by the software

- ✓ Only draft quality records can be deleted; others must be retained until the end of their retention period.
- ✓ Confirmed records can only be corrected by a QMS Manager, and the correction is logged.
- ✓ Only confirmed records can be archived; only archived records can be disposed of, and not before *Retain Until*.
- ✓ Only a QMS Manager can dispose of records.
- ✓ *Is_document_management*: retention duration limited to 200 years; legal hold available on documents.

8. Good practices

- ★ Base retention periods on the longest applicable requirement (product lifetime, market, contracts).
- ★ Record the disposal decision itself as a quality record.
- ★ Test periodically that archived records can still be read (see SOP-DI-004 for audit trail data).

9. Pitfalls to avoid

- ✗ Disposing of records needed for an ongoing investigation or litigation.
- ✗ Keeping records in personal drives outside the controlled system.

10 Records and evidence

- Quality records with state history
- Disposal decisions
- Retention policy configuration

11 Performance indicators

- Records past retention not yet processed
- Records confirmed more than 30 days after the activity

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 4 — Documentation
- 21 CFR 211.180 to 211.198 — Records and reports
- ISO 13485:2016 — Medical devices — Quality management systems
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)

13. Demonstration

Case. After the quarterly quality meeting, the secretary creates a quality record of type *Quality Meeting Minutes*, links it to the objective *Complaint closure time*, attaches the signed attendance list and clicks **Confirm**. The retention period is 120 months; *Retain Until* is computed automatically. Ten years later the record appears in *Records Past Retention*; the QMS Manager checks that no legal hold applies, clicks **Archive** then **Dispose**.

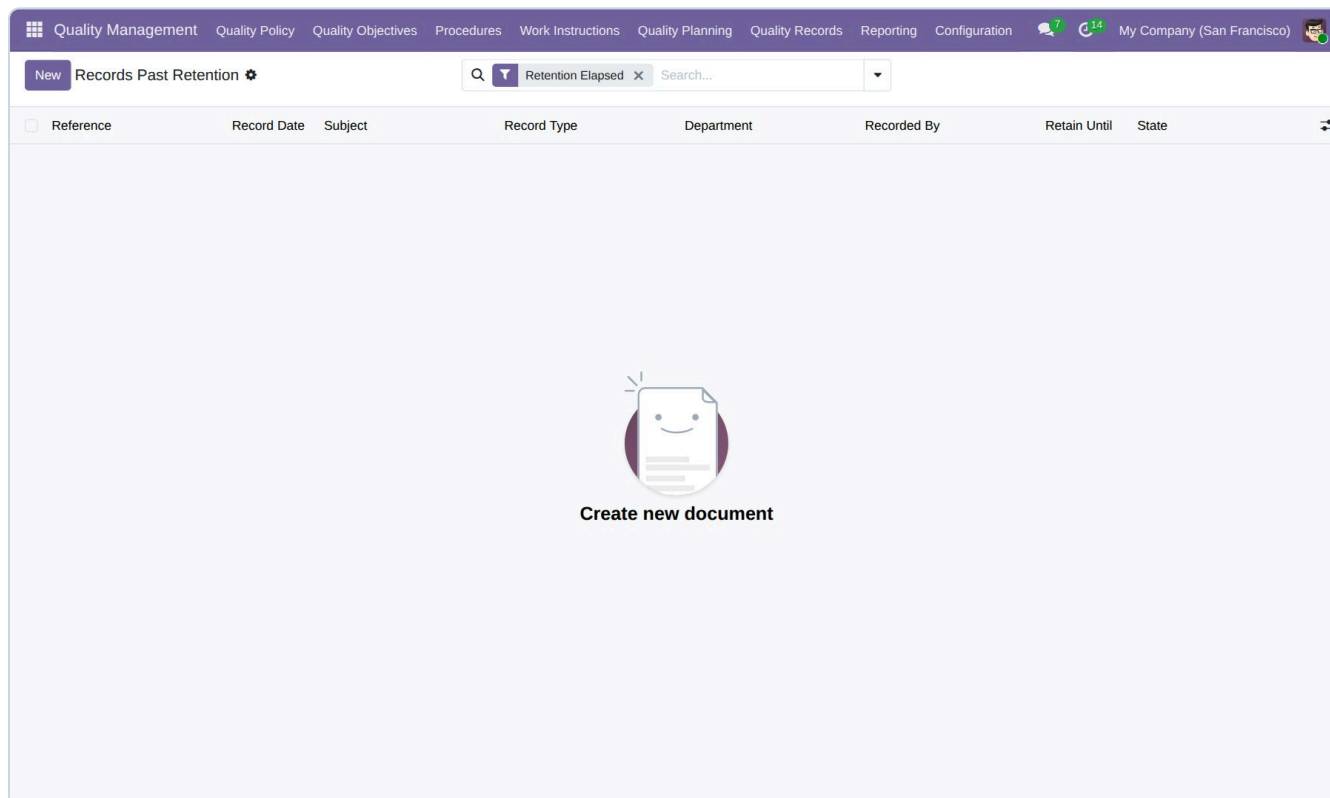


Figure 17 — Quality records with their retention date. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Documents Documents My Approvals Versions Linked Records Configuration 7 14 My Company (San Francisco)						
New Retention Policies Search... 1-1 / 1						
☐	Code	Policy Name	Retention...	Duration Unit	Retention Trigger	End of Retention Action
☐	GMP-5Y	GMP records - 5 years after publication		5 Years	Publication Date	Notify Document Managers
						90

Figure 18 — Retention policies of controlled files. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Quality objectives and measurement of their achievement

Module	ls_qms	Version	_____
Menu	Quality Management ► Quality Objectives	Effective date	_____
Owner (suggested)	Head of Quality / Management	Approved by	_____
Related SOPs	SOP-GEN-001, SOP-GEN-002, SOP-QA-004		

1 Objective

Translate the quality policy into measurable objectives, follow them with dated measurements and conclude on their achievement at the end of the period.

2 Scope

All quality objectives of the company and its departments recorded in ls_qms.

3 Responsibilities

Role (software group)	Responsibility
Top management	Sets the objectives consistent with the quality policy and allocates resources.
Objective owner (Responsible)	Records measurements, explains deviations from the trajectory, proposes actions.
QMS Manager	Maintains the list of objectives, prepares the review for management review.

4 Definitions

Term	Definition
Direction	<i>Increase, Decrease or Maintain</i> : the expected movement of the indicator.
Baseline / Target / Tolerance	Starting value, value to reach, and accepted margin.
Performance status	Computed: No Measurement, On Track, At Risk, Off Track, Target Reached.

5 Workflow in the software



Other outcomes



Model: ls.qms.objective

Figure 19 — Lifecycle of a quality objective.

6 Procedure

- 1 **Define.** In **Quality Management ► Quality Objectives** click **New**. Enter the objective, the related *Quality Policy*, department, responsible, start and target dates, direction, unit, baseline, target and tolerance.
- 2 **Start.** Click **Start** at the beginning of the period.
- 3 **Measure.** Add dated measurements at the planned frequency (monthly for example). The current value, achievement % and performance status are computed.
- 4 **React.** When the status becomes *At Risk* or *Off Track*, the owner records the analysis in the chatter and, if the cause is systemic, raises a CAPA (SOP-QA-022).
- 5 **Close.** At the target date click **Close as Achieved** or **Close as Not Achieved**. Record the conclusion as a quality record of type *Quality Objective Review*.
- 6 **Review.** Present all objectives in management review (SOP-GEN-001) and set the next period's objectives.

7. Controls enforced by the software

- ✓ For an *Increase* objective the target must be greater than the baseline; for *Decrease* it must be lower.
- ✓ A measurement cannot be dated before the start date of the objective.
- ✓ Only draft objectives can be started or deleted; others are cancelled.
- ✓ An objective must be in progress to be closed.

8. Good practices

- ★ Use SMART objectives: one indicator, one number, one date, one owner.
- ★ Limit the number of objectives so that each receives attention.
- ★ Record the data source of each measurement in its note.

9. Pitfalls to avoid

- ✗ Changing the target during the period to make it achievable — close and create a new objective instead.
- ✗ Objectives without a link to the policy or to a real process risk.

10 Records and evidence

- Objective with measurements and closure
- Quality Objective Review records
- Management review minutes

11 Performance indicators

- % of objectives achieved
- Objectives Off Track without action for more than one month

12 References

- ISO 9001:2015 — Quality management systems — Requirements
- ISO 13485:2016 — Medical devices — Quality management systems
- ICH Q10 — Pharmaceutical Quality System

13. Demonstration

Case. Objective *Reduce the median complaint closure time*: direction Decrease, unit days, baseline 48, target 30, tolerance 3, from January to December. Monthly measurements: 46, 44, 41... In June the value is 39 and the status turns *At Risk*; the owner notes that investigations wait for samples and raises CAPA-2026-022. In December the measurement is 29: the owner clicks **Close as Achieved**.

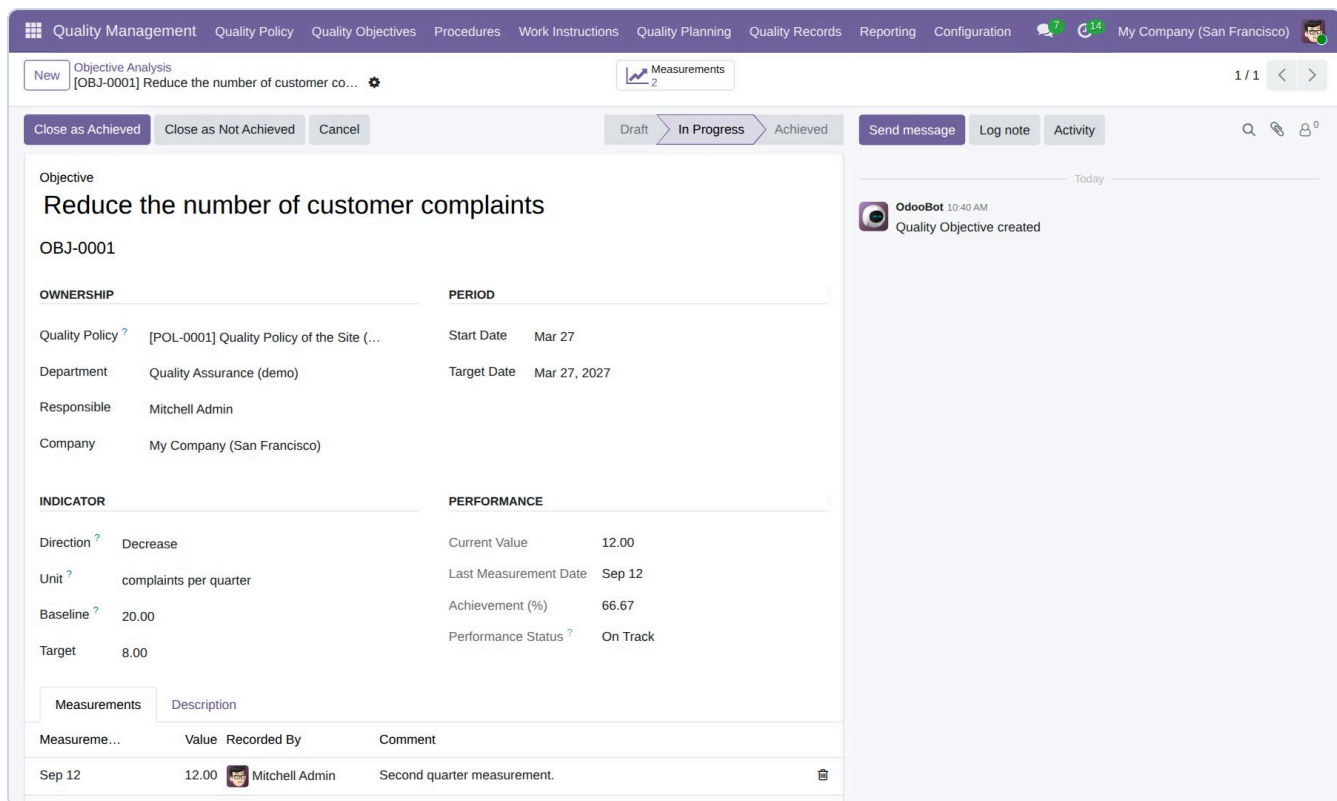


Figure 20 — Quality objective with baseline, target and measurements. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Training

ls_training

ls_training manages a controlled course catalogue (Draft → Under Review → Approved → Obsolete), sessions with attendance and assessment scores, certifications issued automatically when a session closes, training requirements by job position, department or employee, a training matrix, and competency assessments that are distinct from attendance. Certification status (Valid, Expiring Soon, Expired, Revoked) is refreshed nightly and reminders are sent before expiry.

Menu	Content
Training ► Configuration ► Courses / Competencies / Training Requirements	Catalogue and rules
Training ► Operations ► Sessions / Attendance / Certifications / Competency Assessments	Delivery and evidence
Training ► Operations ► Expiring and Expired	Certifications to renew
Training ► Reporting ► Training Matrix	Employee × course compliance matrix

Group	Main rights
Learner · Viewer	Read own records / all records
Trainer	Runs sessions, records attendance and scores
Manager	Approves courses, manages requirements, revokes certifications

Regulatory position stated by the module

The module README states that ls_training does not implement 21 CFR Part 11 (no electronic signature, no field-level audit trail of its own). If training records are GxP records in your organisation, cover them with ls_audit_trail rules (SOP-DI-001) and state the approach in your validation.

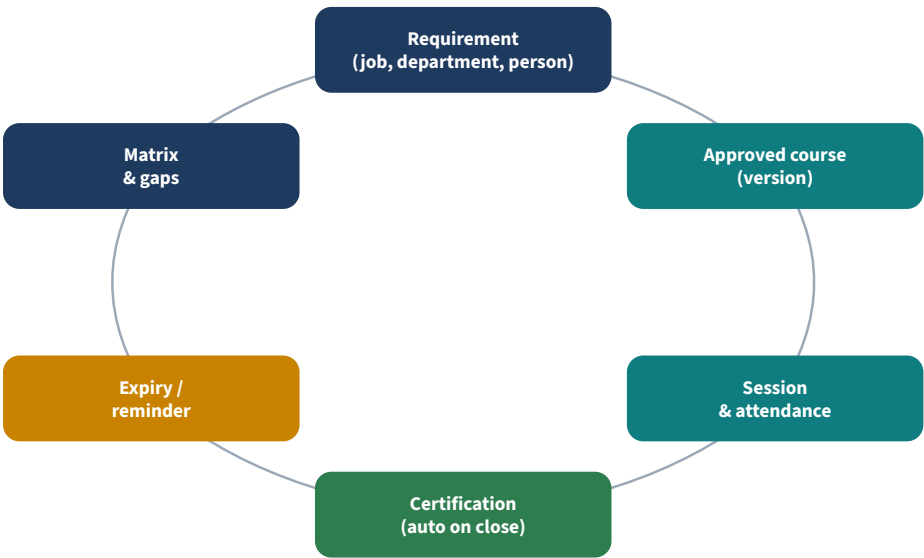
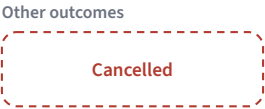


Figure 21 — The training loop.



Model: ls.training.session

Figure 22 — Lifecycle of a training session.

Training

Operations

Reporting

Configuration

7

14

My Company (San Francisco)

New

Training Courses

Search...

1-3 / 3

☐	Course Code	Course Title	Version	Course Type	Delivery Mode	Duration (H...	Certificat...	Session ...	Status	Company	
☐	TRN/CRS/DEMO1	GMP Fundamentals	2.0	GMP / GxP	Classroom	4.00	24	1	Approved	My Company (San Francisco)	
	TRN/CRS/DEMO2	Aseptic Gowning Qualification	1.0	Technical	On-the-Job	2.00	12	0	Approved	My Company (San Francisco)	
☐	TRN/CRS/DEMO3	Site Induction	1.0	Induction	Self-Study	1.00	0	0	Approved	My Company (San Francisco)	
						7.00					

Figure 23 — Course catalogue. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Creation and approval of training courses

Module	ls_training	Version	_____
Menu	Training ► Configuration ► Courses	Effective date	_____
Owner (suggested)	Training Manager	Approved by	_____
Related SOPs	SOP-QA-011, SOP-QA-012, SOP-QA-013		

1 Objective

Ensure that every course used as evidence of qualification has approved content, learning objectives, a defined assessment rule and a validity period.

2 Scope

All courses of the catalogue: induction, GMP/GxP, procedure/SOP, technical, health and safety, quality system and regulatory courses, whatever the delivery mode.

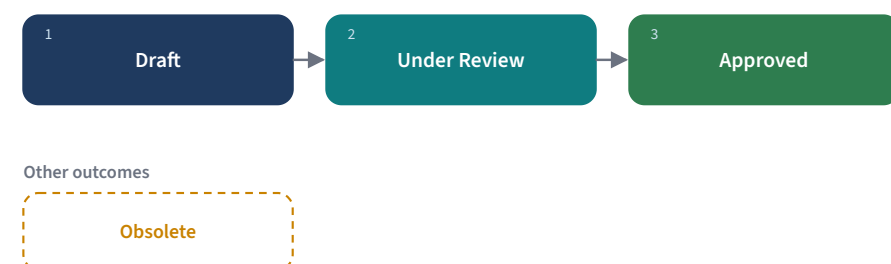
3 Responsibilities

Role (software group)	Responsibility
Course author (Trainer)	Writes the content and objectives, proposes the assessment and validity.
Subject-matter reviewer	Checks technical accuracy.
Training Manager	Approves the course, makes obsolete courses that are no longer valid.

4 Definitions

Term	Definition
Delivery mode	Classroom, On-the-Job, Self-Study or External e-Learning (an e-Learning URL is then mandatory).
Certification validity	Months during which the certification stays valid; 0 means no expiry.
Pass score	Score (%) to reach when the course requires an assessment.

5 Workflow in the software



Model: ls.training.course

Figure 24 — Lifecycle of a course.

6 Procedure

- 1 **Create.** In **Training ► Configuration ► Courses** click **New**. Enter name, code, version, course type, delivery mode, duration (hours), learning objectives and content.
- 2 **Define assessment.** Tick *Requires Assessment* and set the pass score if knowledge must be tested (for example a written quiz for a critical SOP).
- 3 **Set validity.** Enter the certification validity in months according to the risk (for example 12 months for aseptic behaviour, 0 for a one-off induction module).
- 4 **Link competencies.** Link the competencies the course contributes to.
- 5 **Submit and approve.** Click **Submit for Review**; the Training Manager checks and clicks **Approve**. Only approved courses can be scheduled.
- 6 **Retire.** When the course is replaced, close or cancel its open sessions, then click **Make Obsolete**.

7. Controls enforced by the software

- ✓ Duration must be greater than zero; the pass score must lie between 0 and 100; validity cannot be negative.
- ✓ An External e-Learning course requires an e-Learning URL.
- ✓ Only a draft course can be submitted; only a course under review can be approved; only an approved course can be made obsolete.
- ✓ A course with open sessions cannot be made obsolete.
- ✓ The course version in force is frozen onto every certification issued.

8. Good practices

- ★ For a SOP course, use the SOP reference as course code and its revision as course version.
- ★ Write measurable learning objectives ("The operator can perform a line clearance").
- ★ Prefer on-the-job training with competency assessment for practical skills.

9. Pitfalls to avoid

- ✗ Using the same course for several SOP revisions without changing its version.
- ✗ A pass score set so low that the assessment proves nothing.

10 Records and evidence

- Approved course with version
- Course content

11 Performance indicators

- Courses under review for more than 30 days
- Courses without assessment for critical tasks

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 2 — Personnel
- 21 CFR 211.25 — Personnel qualifications
- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 22716:2007 — Cosmetics — Good Manufacturing Practices

13. Demonstration

Case. For SOP-PRD-002 *Line clearance*, the trainer creates course code *SOP-PRD-002*, version 3, type *Procedure / SOP*, delivery *On-the-Job*, duration 1 h, with an assessment (pass score 100 %: the trainee must perform the clearance without error) and a 24-month validity. The Training Manager approves it.

The screenshot shows the Odoo 19.0 Community interface for the 'Training Courses' module. The header bar includes 'Training', 'Operations', 'Reporting', and 'Configuration'. The user is logged in as 'My Company (San Francisco)'. The breadcrumb trail is 'New > Training Courses > [TRN/CRS/DEMO1] GMP Fundamentals (v2.0)'. The form is titled 'GMP Fundamentals' and has tabs for 'Sessions' (1) and 'Certifications' (5). The form is in the 'Draft' state, with buttons for 'Reset to Draft', 'Make Obsolete', 'Draft', 'Under Review', and 'Approved'. There are also buttons for 'Send message', 'Log note', and 'Activity'. The form is divided into three main sections: 'IDENTIFICATION', 'DELIVERY', and 'ASSESSMENT'. The 'IDENTIFICATION' section includes fields for 'Course Code' (TRN/CRS/DEMO1), 'Version' (2.0), 'Course Type' (GMP / GxP), and 'Company' (My Company (San Francisco)). The 'DELIVERY' section includes fields for 'Delivery Mode' (Classroom), 'Duration (Hours)' (04:00), and 'Certification Validity (Months)' (24). The 'ASSESSMENT' section includes a checkbox for 'Requires Assessment' (checked) and a 'Pass Score (%)' field (80.00). Below these sections are tabs for 'Content', 'Competencies', and 'Training Requirements'. The 'Content' tab is active, showing the text 'Understand the principles of Good Manufacturing Practice and their application on the shop floor.' and a 'Course content' field.

Course Title
GMP Fundamentals

IDENTIFICATION

Course Code ?	TRN/CRS/DEMO1
Version ?	2.0
Course Type	GMP / GxP
Company	My Company (San Francisco)

DELIVERY

Delivery Mode	Classroom
Duration (Hours)	04:00
Certification Validity (Months) ?	24

ASSESSMENT

Requires Assessment ? ☒

Pass Score (%) 80.00

Content Competencies Training Requirements

Understand the principles of Good Manufacturing Practice and their application on the shop floor.

Course content

Today

OdooBot 10:40 AM
Training Course created

Figure 25 — Course form: type, delivery mode, validity, pass score. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Session planning and issue of certifications

Module	ls_training	Version	_____
Menu	Training ► Operations ► Sessions	Effective date	_____
Owner (suggested)	Training Manager	Approved by	_____
Related SOPs	SOP-QA-010, SOP-QA-013		

1 Objective

Plan and deliver training sessions so that attendance and assessment results are recorded as evidence and certifications are issued only to people who met the course rule.

2 Scope

All sessions of approved courses, internal or external trainer. Externally delivered training may also be recorded as a manual certification.

3 Responsibilities

Role (software group)	Responsibility
Training coordinator (Trainer)	Creates sessions, registers employees, records attendance and scores, closes the session.
Trainer	Delivers the session and assesses attendees.
Training Manager	Supervises, revokes certifications when needed.

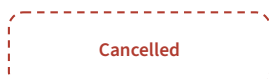
4 Definitions

Term	Definition
Outcome	Computed from recorded evidence: presence and, when required, score $\geq$ pass mark. Absence yields <i>Pending</i> , not <i>Failed</i> .
Certification	Append-only evidence issued at session closure; renewal creates a new record.

5 Workflow in the software

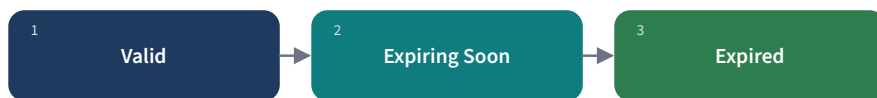


Other outcomes



Model: ls.training.session

Figure 26 — Lifecycle of a session.



Other outcomes



Model: *ls.training.certification*

Figure 27 — *Statuses of a certification.*

6 Procedure

- 1 **Create the session.** In **Training ► Operations ► Sessions** click **New**, select the approved course, start and end, trainer (internal or external), location and maximum attendees.
- 2 **Register.** Click **Register Employees**: select employees, a department, a job position or a training requirement; optionally exclude employees who already hold a valid certification.
- 3 **Confirm.** Click **Confirm** once the list is final (at least one attendee).
- 4 **Deliver.** Click **Start** at the beginning; in the *Attendees* tab tick *Attended* for each person present and enter the assessment *Score*.
- 5 **Close.** Click **Close and Issue Certifications**. Each attendee who meets the rule receives a certification carrying the course version; validity is computed from the course.
- 6 **Handle absentees.** Absent employees remain *Pending*; register them in a later session.
- 7 **Revoke if needed.** If a certification was issued wrongly, a Manager opens it and clicks **Revoke** with a documented reason. Certifications are never deleted.

7. Controls enforced by the software

- ✓ A session must end after it starts and cannot exceed its maximum attendees.
- ✓ A session names either an internal or an external trainer, not both.
- ✓ Only a draft session can be confirmed and it needs at least one attendee; only a confirmed session can start; only a session in progress can be closed.
- ✓ A session cannot be closed while an attendee has no recorded outcome.
- ✓ Attendance scores lie between 0 and 100; an employee of another company cannot attend.
- ✓ Once closed, attendance evidence and the course, dates and trainer are frozen; the session can no longer be cancelled.
- ✓ Certifications cannot be deleted; revocation needs a reason.

8. Good practices

- ★ Keep the signed attendance sheet of classroom sessions as an attachment.
- ★ Record the score as soon as the assessment is corrected, not days later.
- ★ For external training, attach the provider's certificate to the manual certification.

9. Pitfalls to avoid

- ✗ Marking an employee attended because "he read the SOP" without assessment when the course requires one.
- ✗ Closing the session before scores are entered.

10 Records and evidence

- Session with attendance and scores
- Certifications (PDF Training Certificate)
- Individual Training Record (PDF per employee)

11 Performance indicators

- Sessions closed more than 5 days after the end
- Pass rate per course
- Certifications revoked

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 2 — Personnel
- 21 CFR 211.25 — Personnel qualifications
- ISO 13485:2016 — Medical devices — Quality management systems

13. Demonstration

Case. Session *SOP-PRD-002 v3 — 12 May* for the packaging team: 8 employees registered through the job position *Packaging operator*, 2 excluded because already certified. After the practical assessment, 7 score 100 %; one scores 80 % and fails. On **Close and Issue Certifications**, 7 certifications valid 24 months are issued; the failed employee is registered in the next session.

The screenshot displays the Odoo 19.0 Community Training module interface. The top navigation bar includes 'Training', 'Operations', 'Reporting', and 'Configuration'. The user is logged in as 'My Company (San Francisco)'. The main view shows a training session titled 'TRN/SES/2026/00001 - GMP Fundamentals'. The session status is 'Closed'. The course is '[TRN/CRS/DEMO1] GMP Fundamentals (v2.0)' and the company is 'My Company (San Francisco)'. The schedule is for September 20, 8:00 AM to 12:00 PM, in Training Room 1, with a maximum of 20 attendees. The trainer is an internal trainer. The results show 5 registered, 5 passed, and 0 failed. The attendees table lists four employees: Abigail Peterson (100.00, Passed), Anita Oliver (100.00, Passed), Audrey Peterson (80.00, Passed), and Barty McBly (80.00, Passed). The right sidebar shows a chat log with messages from OdooBot.

Employee	Attended	Assessment...	Result
Abigail Peterson	✓	100.00	Passed
Anita Oliver	✓	100.00	Passed
Audrey Peterson	✓	80.00	Passed
Barty McBly	✓	80.00	Passed

Figure 28 — Session with attendees, outcomes and issued certifications. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Training Operations Reporting Configuration				My Company (San Francisco)	
Certifications				1-5 / 5	
Reference	Employee	Course	Granted On	Expires On	Status
TRN/CER/2026/00005	Barty McBly	[TRN/CRS/DEMO1] GMP Fundamentals (v2.0)	Sep 20	Sep 20, 2028	Valid
TRN/CER/2026/00004	Audrey Peterson	[TRN/CRS/DEMO1] GMP Fundamentals (v2.0)	Sep 20	Sep 20, 2028	Valid
TRN/CER/2026/00003	Anita Oliver	[TRN/CRS/DEMO1] GMP Fundamentals (v2.0)	Sep 20	Sep 20, 2028	Valid
TRN/CER/2026/00002	Abigail Peterson	[TRN/CRS/DEMO1] GMP Fundamentals (v2.0)	Sep 20	Sep 20, 2028	Valid
TRN/CER/2026/00001	Mitchell Admin	[TRN/CRS/DEMO1] GMP Fundamentals (v2.0)	Sep 20	Sep 20, 2028	Valid

Figure 29 — Certifications issued automatically when the session was closed. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Training linked to the publication of a new document

Module	ls_qms · ls_training	Version	_____
Menu	Quality Management ► Procedures · Training ► Configuration ► Training Requirements	Effective date	_____
Owner (suggested)	QMS Manager with the Training Manager	Approved by	_____
Related SOPs	SOP-QA-001, SOP-QA-010, SOP-QA-011, SOP-QA-013		

1 Objective

Ensure that nobody applies a new or revised procedure before being trained on it, and that the training evidence refers to the exact revision.

2 Scope

Every ls_qms document with *Training Required* ticked, and every revision of such a document.

3 Responsibilities

Role (software group)	Responsibility
Document author	Ticks <i>Training Required</i> , identifies the target population and the training method.
Training Manager	Creates or revises the course, the requirement and the sessions.
QMS Approver	Publishes only when the training completion criterion is met.
Line managers	Release their staff for training; do not assign untrained staff to the task.

4 Definitions

Term	Definition
Target population	Job positions, departments or named employees who apply the document.
Completion criterion	Proportion of the target population to be trained before publication (for example 100 % of the people on duty).

5 Procedure

- 1 Flag the need.** In the draft document, tick *Training Required* and describe in the reason for change what is new.
- 2 Create or version the course.** On approval, the Training Manager creates the course (or a new version of it) with code = document reference, version = document revision (SOP-QA-010).
- 3 Define the requirement.** In **Training ► Configuration ► Training Requirements** create the requirement: required course, job position / department / employee, *Mandatory*, and a grace period in days for newcomers.
- 4 Train.** Run sessions (SOP-QA-011). Use *Register Employees* by training requirement so nobody is forgotten.
- 5 Check completion.** Generate the matrix (**Training ► Reporting ► Training Matrix**) filtered on the course and check the gaps.
- 6 Publish.** The approver publishes the document once the completion criterion is met (SOP-QA-001) and records the training status in the chatter.

7 Catch up. Employees absent at publication are trained before performing the task; line managers enforce it.

6. Controls enforced by the software

- ✓ There is no automatic link between a published document and a training course: the SOP flag *Training Required* is informative, and the course, requirement and sessions are created by people. This SOP is the control.
- ✓ A training requirement must target an employee, a job position or a department, and cannot be duplicated.
- ✓ Only approved courses can be scheduled; the course version is frozen on each certification.

7. Good practices

- ★ Use "read and understand" self-study with a short quiz only for minor revisions; use classroom or on-the-job training for substantive changes.
- ★ Plan the effective date after the training capacity is known.
- ★ Keep one course per SOP, versioned, rather than a new course per revision.

8. Pitfalls to avoid

- ✗ Publishing on the approval date by default, before anyone is trained.
- ✗ Forgetting night shifts and temporary staff in the target population.

9 Records and evidence

- Course version
- Training requirement
- Certifications
- Training matrix extract attached to the publication note

10 Performance indicators

- % of target population trained at publication
- Tasks performed by untrained staff (from audits; target 0)

11 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 2 — Personnel
- EudraLex Vol. 4, EU GMP Part I, Chapter 4 — Documentation
- 21 CFR 211.25 — Personnel qualifications

12. Demonstration

Case. SOP-QA-020 revision 5 changes the deviation classification. The author ticks *Training Required*. After approval the Training Manager creates version 5 of course SOP-QA-020 and a requirement for the job positions *QA officer* and *Production supervisor*. Two sessions train 22 of 24 people. The matrix shows two people on leave; the approver publishes with the note "22/24 trained, 2 on leave, trained on return before any deviation assessment".

Follow-up of qualifications and requalification

Module	ls_training	Version	_____
Menu	Training ► Reporting ► Training Matrix · Operations ► Competency Assessments	Effective date	_____
Owner (suggested)	Training Manager	Approved by	_____
Related SOPs	SOP-QA-011, SOP-QA-012, SOP-DI-011		

1 Objective

Know at any time who is qualified for which task, renew qualifications before they expire, and assess competence where attendance alone is not enough.

2 Scope

All mandatory training requirements, certifications with an expiry, and competencies requiring periodic assessment.

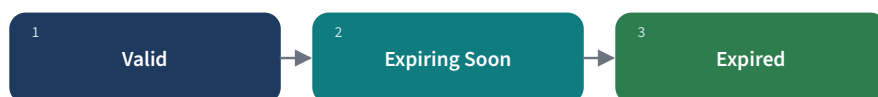
3 Responsibilities

Role (software group)	Responsibility
Training Manager	Runs the matrix monthly, plans renewals, reports gaps.
Line manager / assessor	Performs competency assessments of their staff (not of themselves).
Employee	Follows reminders and attends renewal sessions.

4 Definitions

Term	Definition
Matrix cell status	Valid, Expiring Soon, Expired, Revoked or Not Trained for each employee × course.
Competency assessment	Evidence that a person can do the job: named assessor, demonstrated level (Not Demonstrated, Developing, Proficient, Expert), evidence, confirmation lock.

5 Workflow in the software



Other outcomes



Model: ls.training.certification

Figure 30 — Certification statuses refreshed nightly.

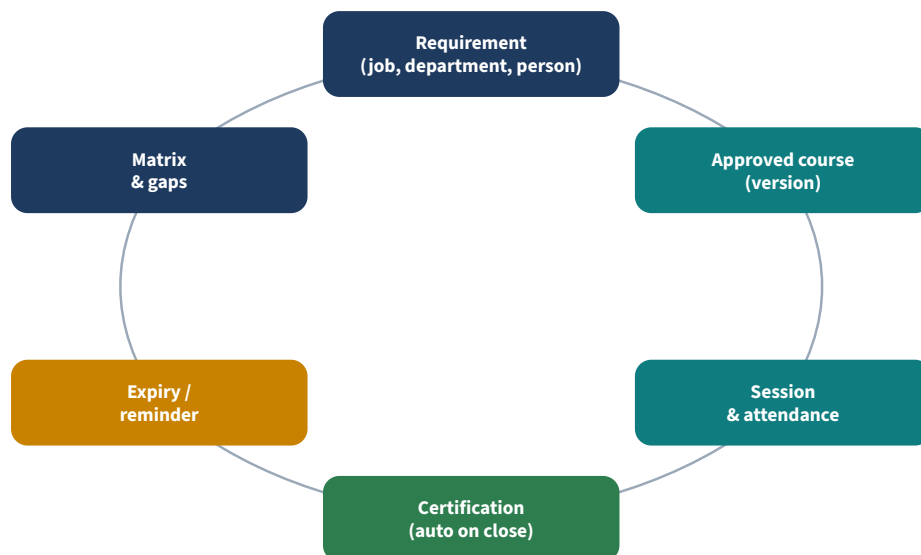


Figure 31 — Requirement, certification, expiry and matrix.

6 Procedure

- 1 **Generate the matrix.** Monthly, open **Training ► Reporting ► Training Matrix**, click **Generate Matrix** and apply the *Gaps* filter.
- 2 **Treat expiring certifications.** Use **Training ► Operations ► Expiring and Expired**; schedule renewal sessions before the expiry date. The system sends daily reminders within the warning window.
- 3 **Act on expired or missing training.** The line manager removes the employee from the task until retrained; record the decision.
- 4 **Assess competence.** In **Training ► Operations ► Competency Assessments** create the assessment: employee, competency, assessor, date, demonstrated level, evidence; set the next assessment date when the competency requires reassessment; click **Confirm**.
- 5 **Report.** Report the compliance rate per department to management review.

7. Controls enforced by the software

- ✓ Certification status is computed from the expiry date and refreshed nightly; certifications are never deleted.
- ✓ An employee cannot assess their own competency.
- ✓ A competency assessment cannot be confirmed without evidence and cannot be modified or deleted once confirmed.
- ✓ The next assessment date cannot precede the assessment date.

8. Good practices

- ★ Link critical access rights (e.g. batch release) to a valid qualification and review them together (SOP-IT-001).
- ★ Use competency assessments for aseptic operators, analysts and auditors.
- ★ Plan renewals in the annual training plan rather than waiting for reminders.

9. Pitfalls to avoid

- ✗ Treating an expired certification as "administrative" and letting the employee keep working.
- ✗ Line managers assessing competencies they have not observed.

10 Records and evidence

- Training matrix extracts
- Certifications and renewals
- Competency assessments

11 Performance indicators

- Compliance rate (mandatory requirements met)
- Expired certifications of active staff (target 0)
- Competency reassessments overdue

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 2 — Personnel
- 21 CFR 211.25 — Personnel qualifications
- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 22716:2007 — Cosmetics — Good Manufacturing Practices

13. Demonstration

Case. The March matrix shows 3 *Expired* cells for the course *Aseptic gowning* in the filling team. The Training Manager informs the supervisor, who removes the three operators from Grade A/B work. A renewal session is held 4 days later; the new certifications are valid 12 months. In parallel, the supervisor confirms a competency assessment *Aseptic behaviour — Proficient* for each operator, based on an observed media fill.

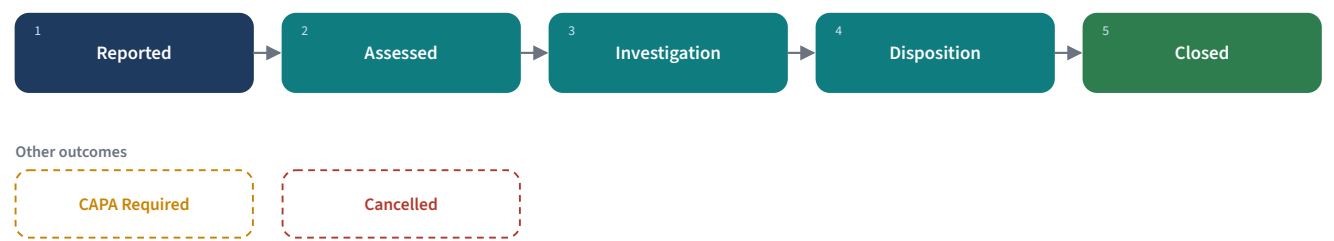
Deviations and CAPA

ls_deviation · ls_capa

ls_deviation records departures from approved instructions or specifications: classification (type, category, severity), impact assessment, immediate actions, investigations with a root cause analysis method, product dispositions and closure. Every state change is written in an append-only *Transition Log*. **ls_capa** manages corrective and preventive actions from any source (audit, deviation, complaint, inspection, management review, risk assessment...): impact assessment, root cause analyses (Five Whys, Ishikawa, FMEA), actions (optionally turned into project tasks), effectiveness checks and closure.

Menu	Content
Quality Management ► Deviations ► All Deviations / My Deviations	Deviation records
Quality Management ► Deviations ► Immediate Actions / Investigations / Product Dispositions	Sub-records
Quality Management ► Reporting ► Deviation Analysis / Transition Log	Analysis and audit trail of transitions
Quality ► CAPA ► CAPA Records / Root Cause Analyses / Actions / Effectiveness Verification	CAPA records
Settings: closure target (days) per severity	Configured per company (default 30 days for minor, major and critical)

Group	Main rights
Deviation: Viewer · Reporter · Investigator · Manager	Reporter declares; Investigator assesses and investigates; Manager approves dispositions, closes, and alone may edit closed records.
CAPA: Viewer · Investigator · Coordinator · Manager	Investigator performs analyses and actions; Coordinator drives the record; Manager closes.



Model: ls_deviation

Figure 32 — Lifecycle of a deviation.



Model: ls_capa.issue

Figure 33 — Lifecycle of a CAPA record.

Quality Management
Deviations
Reporting
Configuration

7
14
My Company (San Francisco)

New
Deviations
DEV/2026/00002
1 / 1

Complete Assessment
Extend Target Date
Cancel
Reported
Assessed
Investigation
Disposition
Closed

Send message
Log note
Activity

DEV/2026/00002

Storage temperature excursion in the finished goods quarantine are:

CLASSIFICATION

Deviation Type	Environmental Excursion
Category	Warehouse and Storage
Severity	Major
Planned Deviation	☐
Tags	
Company	My Company (San Francisco)

CHRONOLOGY AND OWNERSHIP

Occurrence Date	Sep 24, 10:41 AM
Detection Date	Sep 24, 10:41 AM
Report Date	Sep 27, 10:41 AM
Target Closure Date	Sep 26
Is Overdue	☒
Days Open	3
Reported By	OdooBot
Assigned Investigator	
QA Reviewer	
Department	

SUBJECT OF THE DEVIATION

Product	Equipment
Quantity Affected	0.00
Manufacturing Order	Affected Lots/Serials

Today

QA Reviewer (demo)
12:12 PM
Critical → Major (Severity)

OdooBot
10:45 AM

Subject: Overdue deviation DEV/2026/00002

Deviation DEV/2026/00002 has passed its target closure date.

- Title: Storage temperature excursion in the finished goods quarantine area
- Severity: Critical
- Current state: Reported
- Target closure date: 2026-09-26
- Days open: 3

Either progress the deviation to closure or record a justified extension of the target closure date.

OdooBot
10:41 AM

GxP Deviation created

Figure 34 — Deviation form: classification, chronology, subject, status bar. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Reporting, assessment and classification of deviations

Module	ls_deviation	Version	_____
Menu	Quality Management ► Deviations ► All Deviations	Effective date	_____
Owner (suggested)	Head of Quality Assurance	Approved by	_____
Related SOPs	SOP-QA-021, SOP-QA-022, SOP-QA-030		

1 Objective

Make sure that every departure from an approved procedure, specification or condition is reported without delay, contained, and classified by QA on the basis of a documented impact assessment.

2 Scope

Unplanned deviations and planned deviations (temporary departures approved in advance) in production, quality control, warehouse, engineering and utilities.

3 Responsibilities

Role (software group)	Responsibility
Any employee (Deviation Reporter)	Reports the event immediately and takes safe immediate actions.
Supervisor	Checks the report on the same shift, completes immediate actions.
Deviation Investigator / QA	Documents the impact assessment, sets the severity, assigns the investigator.
Deviation Manager (QA)	Confirms the classification; decides on planned deviations.

4 Definitions

Term	Definition
Deviation	Departure from an approved instruction, specification or established standard.
Planned deviation	Departure approved before execution; it requires a justification recorded beforehand.
Severity	Minor, Major or Critical, according to the potential impact on patient safety, product quality, regulatory compliance and validated state.

5 Workflow in the software



Model: ls_deviation

Figure 35 — Deviation lifecycle; this SOP covers Reported → Assessed.

6 Procedure

- 1 **Contain.** Stop or secure the activity if needed; segregate the material concerned.
- 2 **Report.** In [Quality Management ► Deviations](#) click [New](#). Enter title, description, deviation type, category, occurrence and detection dates, product, lots, manufacturing order, equipment, location and quantity affected. For a planned deviation, tick *Planned Deviation* and write the justification before execution.
- 3 **Record immediate actions.** Add each immediate action (containment, segregation, notification) with its owner; mark it done with completion evidence.
- 4 **Assess the impact.** QA completes the impact questions (patient safety, product quality, regulatory, validation, other batches) and the impact assessment text.
- 5 **Classify.** Set the severity and assign the investigator and QA reviewer. The target closure date is proposed from the company setting for that severity.
- 6 **Complete the assessment.** Click [Complete Assessment](#). The state becomes *Assessed* and the transition is logged.
- 7 **Send back if incomplete.** QA uses [Send Back](#) when the report is not usable, with an explanation in the chatter.
- 8 **Extend if necessary.** A target date can only be extended with [Extend Target Date](#) and a rationale.

7. Controls enforced by the software

- ✓ Occurrence date ≤ detection date ≤ recording date.
- ✓ A planned deviation requires a justification recorded before execution.
- ✓ An affected quantity requires a product.
- ✓ Assessment requires a severity, a documented impact assessment and an assigned investigator.
- ✓ An immediate action cannot be marked done without completion evidence.
- ✓ A deviation that progressed beyond *Reported* cannot be deleted — only cancelled with a reason.
- ✓ Every transition is recorded in the append-only Transition Log (cannot be edited or deleted).

8. Good practices

- ★ Report within the shift; a late deviation report is itself a finding.
- ★ Describe facts (what, where, when, how much), not opinions or blame.
- ★ Classify on the potential impact, not on the likely outcome of the investigation.
- ★ Check the list of open deviations at the daily production meeting.

9. Pitfalls to avoid

- ✗ Correcting the situation and "forgetting" the report.
- ✗ Grouping unrelated events in one deviation.
- ✗ Downgrading severity to meet the closure target.

10 Records and evidence

- Deviation record with transition log
- Immediate actions with evidence
- Impact assessment

11 Performance indicators

- Time from occurrence to report
- Deviations by type/category (trend)
- % classified within 2 working days

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 1 — Pharmaceutical Quality System
- EudraLex Vol. 4, EU GMP Part I, Chapter 5 — Production
- 21 CFR 211.100 — Written procedures; deviations
- 21 CFR 211.192 — Production record review
- ICH Q10 — Pharmaceutical Quality System
- ICH Q9(R1) — Quality Risk Management

13. Demonstration

Case. During filling, an operator sees that the room differential pressure alarm had been active for 40 minutes. The supervisor stops filling, quarantines the 1 200 units filled in that period and reports deviation DEV/2026/00017: type *Environmental Excursion*, category *Production*, product and lot filled. Immediate actions: quarantine (done, evidence: quarantine label photo) and notification of QA (done). QA assesses a potential impact on product quality and on other batches, sets severity *Major*, assigns the investigator and clicks **Complete Assessment**.

Quality Management Deviations Reporting Configuration										
New Deviations										
Open Search...										
1-3 / 3										
☐	Reference	Title	Deviation Type	Category	Severity	Detection Date	Target Clos...	Days Open	Assigned Inves...	State
☐	DEV/2026...	Storage temperature excursion in the finished goods quar...	Environmental...	Warehouse a...	Major	Sep 24, 10:41 AM	Sep 26	3		Reported
☐	DEV/2026...	Theoretical yield below the lower limit on batch compression	Yield Discrepa...	Production	Major	Sep 22, 10:41 AM	Oct 22	5		Reported
☐	DEV/2026...	Missing second-person verification signature on a dispens...	Documentatio...	Documentation	Minor	Sep 19, 10:41 AM	Oct 17	8		Reported

Figure 36 — Deviation register with severity and overdue flag. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Investigation, extension to other batches and product disposition

Module	ls_deviation	Version	_____
Menu	Quality Management ► Deviations ► Investigations / Product Dispositions	Effective date	_____
Owner (suggested)	Head of Quality Assurance	Approved by	_____
Related SOPs	SOP-QA-020, SOP-QA-022, SOP-QA-090		

1 Objective

Find the root cause of each deviation, determine whether other batches or products are affected, decide on the fate of the affected product, and close with a written conclusion and follow-up.

2 Scope

All assessed deviations. Complaint and laboratory OOS investigations follow SOP-QA-040 and SOP-QC-013.

3 Responsibilities

Role (software group)	Responsibility
Assigned investigator	Leads the investigation, records findings, contributing factors and root cause.
QA reviewer	Challenges the investigation, checks the extension to other batches.
Deviation Manager	Approves or rejects product dispositions; closes the deviation.

4 Definitions

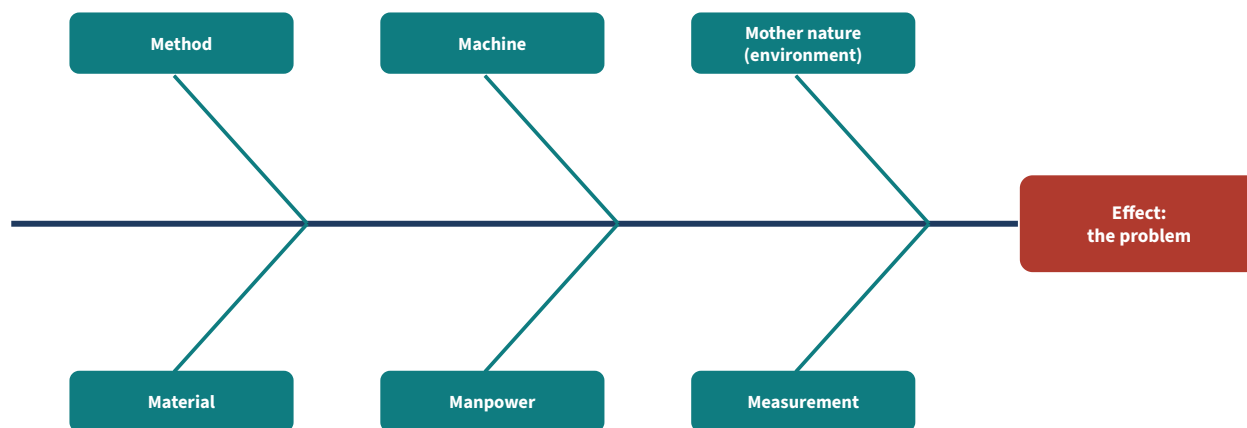
Term	Definition
Extension rationale	Documented reasoning on whether other batches, products or equipment are affected (required by 21 CFR 211.192).
Disposition	Decision on the affected quantity: Use As Is, Rework, Reprocess, Quarantine, Reject, Destroy, Return to Supplier.

5 Workflow in the software



Model: ls_deviation

Figure 37 — This SOP covers Assessed → Closed.



Ishikawa (6M) diagram — one of the root-cause methods offered in Is_capa (with Five Whys and FMEA).

Figure 38 — Ishikawa diagram used during investigation.

6 Procedure

- 1 **Open the investigation.** Add an investigation record: subject, RCA method (from [Configuration ► Root Cause Analysis Methods](#)), investigator, start date. Click **Start**.
- 2 **Start the phase.** On the deviation click **Start Investigation** (at least one investigation record must exist).
- 3 **Investigate.** Collect facts (batch record, logbooks, EM data, interviews), apply the method (5 Whys, Ishikawa...), record findings, contributing factors and root cause; tick *Root Cause Determined* when confirmed.
- 4 **Complete.** Click **Complete** on the investigation (findings and root cause required).
- 5 **Extend.** Write the *Other Batches Evaluated* / extension rationale: which other lots, products or equipment could be affected and why they are or are not.
- 6 **Propose dispositions.** For each affected lot add a disposition: product, lot, quantity, decision and justification. Click **Proceed to Disposition**.
- 7 **Approve dispositions.** The Deviation Manager clicks **Approve** or **Reject** on each disposition.
- 8 **Decide on CAPA.** If a systemic cause exists click **Require CAPA** and open the CAPA in Is_capa (SOP-QA-022); otherwise record the rationale for no CAPA.
- 9 **Close.** Click **Close Deviation**: enter the conclusion, the follow-up, the CAPA decision and CAPA reference, and the QA reviewer.

7. Controls enforced by the software

- ✓ The investigation phase needs at least one investigation record.
- ✓ An investigation cannot be completed without findings and a recorded root cause; completion date $\geq$ start date.
- ✓ Disposition requires all investigations completed and an extension rationale; if an impact was assessed, at least one disposition is required.
- ✓ Only a Deviation Manager can approve or reject a disposition; the lot must belong to the product.
- ✓ Closure is refused while an immediate action is open or a disposition is still draft; a CAPA reference is mandatory when CAPA is required.
- ✓ A closed or cancelled deviation can only be modified by a Deviation Manager.

8. Good practices

- ★ Distinguish the direct cause from the root cause; ask "why" until you reach a process or system cause.
- ★ Human error is rarely a root cause: look for what made the error possible.
- ★ Base the extension on data (same equipment, same material lot, same period).
- ★ Close within the target; if not possible, extend formally with a reason.

9. Pitfalls to avoid

- ✗ "Operator retrained" as the only action for a systemic cause.
- ✗ Releasing product before the disposition is approved.
- ✗ Closing with "no impact" without evidence.

10 Records and evidence

- Investigation records
- Dispositions with approval
- Conclusion, follow-up, CAPA reference
- Transition log

11 Performance indicators

- Median time to closure by severity
- % closed within target
- Recurrence of the same root cause

12 References

- 21 CFR 211.192 — Production record review
- EudraLex Vol. 4, EU GMP Part I, Chapter 1 — Pharmaceutical Quality System
- EudraLex Vol. 4, EU GMP Part I, Chapter 5 — Production
- ICH Q9(R1) — Quality Risk Management
- ICH Q10 — Pharmaceutical Quality System

13. Demonstration

Case (continued from SOP-QA-020). The investigator uses Ishikawa: the alarm was not escalated because the BMS pager had been muted after a maintenance intervention (method + machine). Root cause: no procedure to restore alarm routing after maintenance. Extension: two earlier batches filled in the same room since the maintenance are reviewed; their EM data are within limits — no impact. Disposition for the 1 200 units: *Reject*, approved by the Deviation Manager. CAPA required (restore-after-maintenance checklist); CAPA-2026-031 opened and referenced at closure.

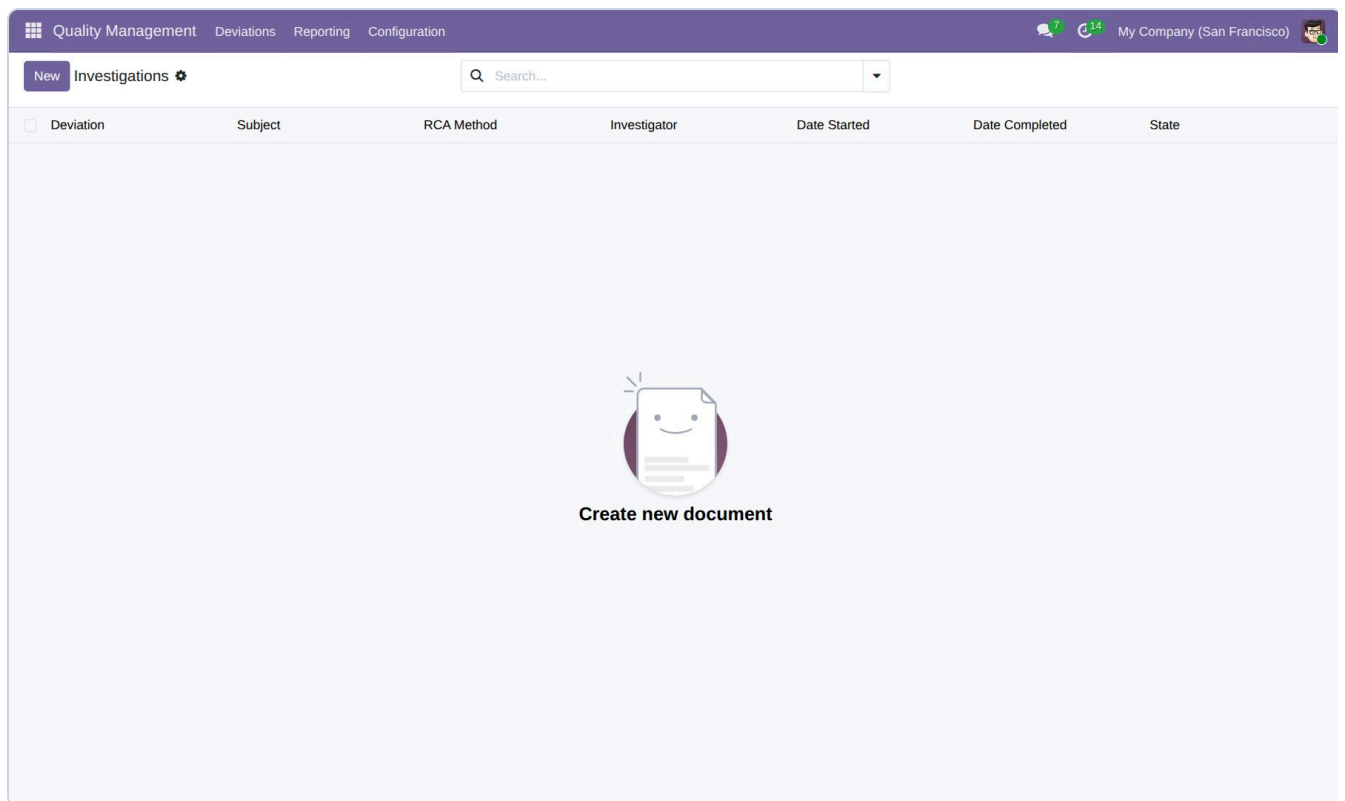


Figure 39 — Deviation investigations. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

CAPA management from identification to closure

Module	ls_capa	Version	_____
Menu	Quality ► CAPA ► CAPA Records	Effective date	_____
Owner (suggested)	CAPA Manager (QA)	Approved by	_____
Related SOPs	SOP-QA-021, SOP-QA-023, SOP-QA-030, SOP-QA-040, SOP-QA-053		

1 Objective

Eliminate the causes of detected or potential nonconformities by a structured process: identify, assess, analyse root causes, plan and implement actions, verify effectiveness, close.

2 Scope

Corrective and preventive actions from internal and external audits, inspections, deviations, complaints, nonconformities, management review, risk assessments, environmental monitoring, suppliers and change control.

3 Responsibilities

Role (software group)	Responsibility
Initiator (any user)	Identifies the need and creates the record with its source reference.
CAPA Owner / Coordinator	Drives the CAPA through its stages and meets the due date.
CAPA Investigator	Performs root cause analyses and actions.
CAPA Manager	Checks each gate, closes the CAPA.

4 Definitions

Term	Definition
Correction	Immediate fix of the nonconformity (recorded as <i>Immediate Action / Correction</i>).
Corrective action	Action on the cause to prevent recurrence.
Preventive action	Action on the cause of a potential nonconformity to prevent occurrence.

5 Workflow in the software



Model: ls_capa.issue

Figure 40 — CAPA lifecycle with its gates.

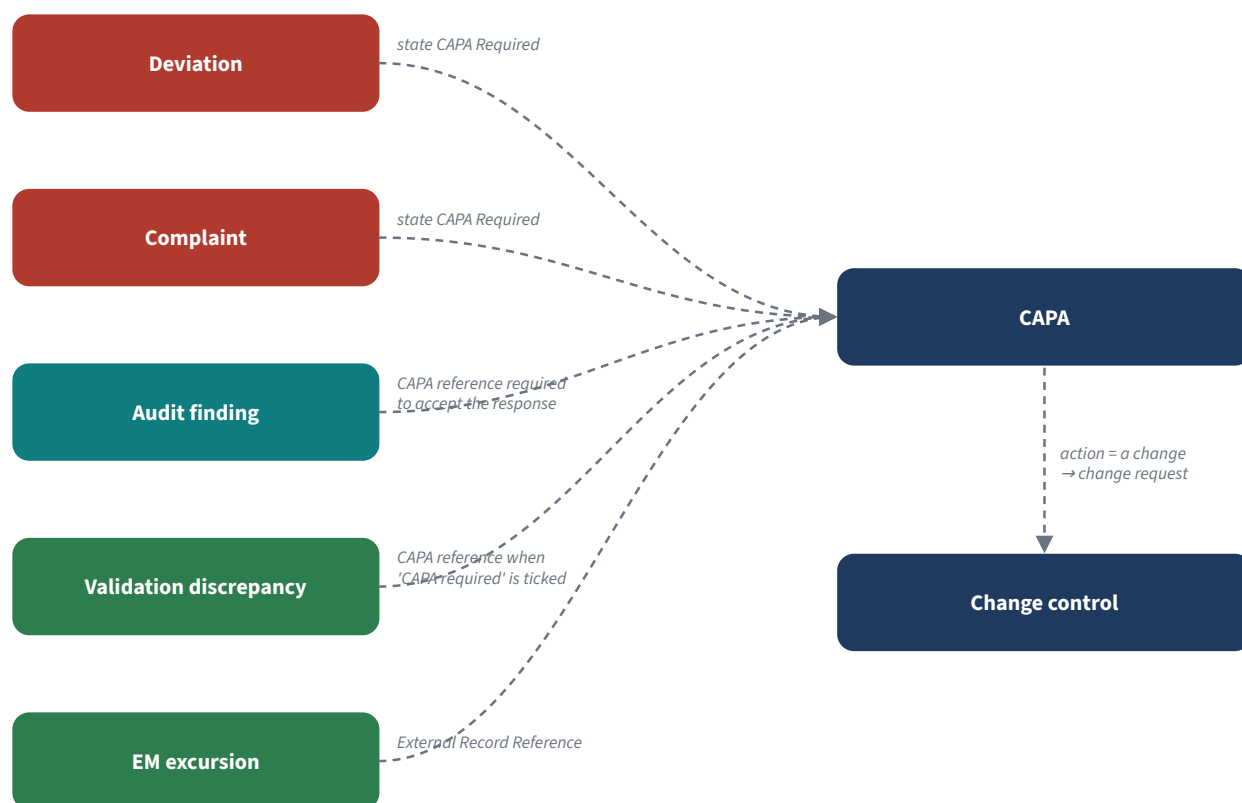


Figure 41 — Sources feeding the CAPA system.

6 Procedure

- 1 **Identify.** In **Quality ► CAPA ► CAPA Records** click **New** : title, description, source (e.g. *Deviation*) and *Source Reference* (e.g. DEV/2026/00017), type (corrective, preventive or both), severity, priority, category, owner and due date.
- 2 **Assess.** Document the impact assessment (product quality, regulatory, patient/user safety) and the correction already taken. Click **Assess** .
- 3 **Investigate.** Click **Start Investigation** . Add one or more root cause analyses (Five Whys, Ishikawa, FMEA or other method), each with a root cause statement; mark the primary root cause; click **Confirm** on each.
- 4 **Plan actions.** Click **Plan Actions** . Add actions: type, description, root cause addressed, responsible, planned completion date. Use **Create Task** to follow an action in a project.
- 5 **Execute.** Click **Start Execution** . Each responsible clicks **Start** then **Mark Done** with completion evidence. Cancelled actions need a reason.
- 6 **Complete.** When all actions are done or cancelled, click **Complete** .
- 7 **Verify effectiveness.** Plan and perform effectiveness checks (SOP-QA-023); when at least one concludes *Effective* and all are done, click **Verify Effectiveness** .
- 8 **Close.** Click **Close CAPA** , write the closure summary. Write the CAPA reference back into the source record (deviation, complaint, audit finding).

7. Controls enforced by the software

- ✓ Each transition is only allowed from the expected status.
- ✓ Assessment requires a documented impact assessment.
- ✓ Action planning requires at least one confirmed root cause analysis.
- ✓ Execution requires at least one planned action; completion requires all actions done or cancelled.
- ✓ An action cannot be completed without evidence nor cancelled without a reason; a completed action cannot be cancelled.
- ✓ Verification requires every effectiveness check completed and at least one *Effective*.
- ✓ Closure requires a closure summary; due date $\geq$ identification date.
- ✓ A CAPA beyond *Identified* cannot be deleted — it is archived.
- ✓ Five Whys needs at least the first Why; Ishikawa needs a category; FMEA ratings are 1–10; a confirmed root cause cannot be reset once execution started.

8. Good practices

- ★ Size the CAPA to the risk: not every deviation needs a CAPA, but every recurring cause does.
- ★ One action, one owner, one date, one piece of evidence.
- ★ Check that the action addresses the confirmed root cause (link *Root Cause Addressed*).
- ★ Review overdue CAPAs weekly in the quality meeting.

9. Pitfalls to avoid

- ✗ Closing on completion of actions without effectiveness verification.
- ✗ Extending due dates repeatedly without escalation.
- ✗ Opening a CAPA without writing its reference back into the source record — the link is manual.

10 Records and evidence

- CAPA record with root causes, actions, evidence, effectiveness checks and closure summary

11 Performance indicators

- Overdue CAPAs
- Median time to closure
- % of CAPAs verified effective at first check
- Recurring root causes

12 References

- ICH Q10 — Pharmaceutical Quality System
- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 9001:2015 — Quality management systems — Requirements
- EudraLex Vol. 4, EU GMP Part I, Chapter 1 — Pharmaceutical Quality System
- 21 CFR Part 820 — Quality Management System Regulation (incorporates ISO 13485:2016 by reference; applicable since 2 February 2026)

13. Demonstration

Case. CAPA-2026-031, source *Deviation*, reference DEV/2026/00017, type corrective, severity major, due in 60 days. Impact assessment documented; Five Whys analysis confirmed: *no restore-after-maintenance step for alarm routing*. Actions: (1) add a restore checklist to the maintenance work order template — engineering, 30 days; (2) train maintenance technicians — 45 days; (3) weekly automatic alarm routing test — IT, 45 days. Each action is closed with evidence (new template, training certificates, test report). Effectiveness check planned at +3 months: *no alarm routing failure after 12 maintenance interventions*.

CAPA/2026/00002

Gowning procedure not followed in Grade C corridor

CLASSIFICATION		OWNERSHIP AND DATES	
Source ?	Internal Audit	Identified By	OdooBot
Source Reference ?	IA-2026-004	CAPA Owner ?	OdooBot
Capa Type ?	Corrective	Identification Date ?	Sep 21
Severity ?	Minor	Due Date ?	Oct 21
Priority	★☆☆	Days To Due Date ?	24
Category ?	Production	Action Progress (%) ?	33 %
		Company	My Company (San Francisco)

Issue | Impact Assessment | Root Cause Analysis | Actions | Effectiveness

An internal audit observed two operators entering the Grade C corridor without the second pair of gloves required by SOP-CR-004.

IMMEDIATE ACTION / CORRECTION

Alarm routing restored; affected units quarantined.

Activity Log:

- QA Reviewer (demo)** 12:12 PM: No → Yes (Product Quality Impact). Identified → In Progress (Status).
- QA Reviewer (demo)** 12:12 PM: Status changed to In Progress.
- QA Reviewer (demo)** 12:12 PM: Status changed to Action Planning.
- QA Reviewer (demo)** 12:12 PM: Status changed to Investigation.
- QA Reviewer (demo)** 12:12 PM: Status changed to Assessed.
- OdooBot** 10:40 AM: CAPA Record created.

Figure 42 — CAPA record form. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Quality
CAPA
Configuration

7
14
My Company (San Francisco)

New
Root Cause Analyses
RCA-1
1 / 1

Reset to Draft
Draft
Confirmed
Send message
Log note

RCA-1

CAPA	CAPA/2026/00002	Analyst	QA Reviewer (demo)
Analysis Method	Five Whys	Analysis Date	Sep 27
Primary Root Cause ?	☒		

ROOT CAUSE STATEMENT

No step to restore alarm routing after maintenance interventions.

FIVE WHYS

Why 1 Why was the alarm not seen? Routing to the pager was disabled.

Why 2 Why disabled? Muted during maintenance.

Why 3 Why not restored? No restore step in the work order.

Why 4 Why no step? Template written before BMS alarms were routed.

Why 5 Why not updated? No change control link between BMS and maintenance templates.

Actions

Reference	Action Type	Action Description	Responsible	Planned Co...	Status

QA Reviewer (demo) 12:12 PM

CAPA Root Cause Analysis created

Figure 43 — Five Whys root cause analysis, confirmed. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Quality
CAPA
Configuration

7
14
My Company (San Francisco)

New
CAPA Actions
Search...
1-4 / 4

Reference	CAPA	Action Type	Action Description	Responsible	Planned Co...	Actual Com...	L...	Status
☐ ACT-2	CAPA/202...	Corrective	Train maintenance technicians on the new checklist.	QA Mana...	Nov 11			Draft
☐ ACT-3	CAPA/202...	Preventive	Weekly automatic test of alarm routing.	QA Mana...	Nov 11			Draft
☐ CAPA-AC...	CAPA/202...	Corrective	Recalibrate dispensing balance BAL-014 and verify performance against certified reference weights before returning it to service.	OdooBot	Sep 22			Draft
☐ CAPA-AC...	CAPA/202...	Preventive	Configure the calibration schedule so that instruments past their due date are blocked from dispensing operations, and extend the control to all weighing equipment.	OdooBot	Oct 22			Draft

Figure 44 — CAPA actions with owners, dates and states. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Verification of CAPA effectiveness

Module	ls_capa	Version	_____
Menu	Quality ► CAPA ► Effectiveness Verification	Effective date	_____
Owner (suggested)	CAPA Manager (QA)	Approved by	_____
Related SOPs	SOP-QA-022, SOP-QA-031		

1 Objective

Prove, with objective data collected after implementation, that the CAPA actions eliminated or reduced the cause, and trigger a new CAPA when they did not.

2 Scope

Every CAPA before its verification and closure.

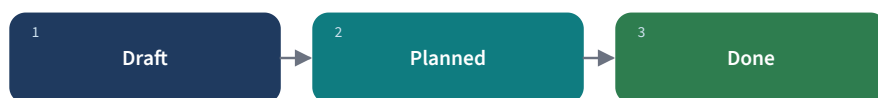
3 Responsibilities

Role (software group)	Responsibility
CAPA Owner	Defines the method and acceptance criteria when planning actions.
Verifier	Collects the data at the planned date and concludes; ideally independent of the action owner.
CAPA Manager	Accepts the conclusion; raises a follow-up CAPA when not effective.

4 Definitions

Term	Definition
Verification method	Data Review, Follow-up Audit, Process Monitoring, Retesting, Training Verification or Other Documented Method.
Acceptance criteria	Measurable condition that proves effectiveness (e.g. "zero recurrence in 3 months").

5 Workflow in the software



Model: ls_capa.effectiveness

Figure 45 — Lifecycle of an effectiveness check.

6 Procedure

- 1 Define.** In the CAPA, add an effectiveness check: method, acceptance criteria, planned verification date (not before the identification date), verifier.
- 2 Plan.** Click **Plan** once the actions are implemented; the check appears in **Quality ► CAPA ► Effectiveness Verification**.
- 3 Collect data.** At the planned date, collect the data defined by the method over the whole observation period.

4 **Conclude.** Record the conclusion and click **Conclude Effective** or **Conclude Not Effective**.

5 **Not effective.** Click **Raise Follow-up CAPA**: a new CAPA is created and linked; the root cause analysis must be revisited.

6 **Verify the CAPA.** Return to the CAPA and click **Verify Effectiveness**, then close (SOP-QA-022).

7. Controls enforced by the software

- ✓ Only draft checks can be planned; a check cannot be planned before the CAPA identification date.
- ✓ A check cannot be completed while its result is *Pending* or without a conclusion.
- ✓ A follow-up CAPA can only be raised from a check concluded *Not Effective*, and only once.
- ✓ The CAPA cannot be verified until every check is completed and at least one is *Effective*.

8. Good practices

- ★ Set the criteria before implementation, so that they are not tailored to the results.
- ★ Choose an observation period long enough to see recurrence (number of batches, audits, interventions).
- ★ Training alone is verified by observing the task, not by the attendance list.

9. Pitfalls to avoid

- ✗ "Actions completed" used as effectiveness evidence.
- ✗ Verification by the person who implemented the action, without a second look.

10 Records and evidence

- Effectiveness checks with data, conclusion and date
- Follow-up CAPA when applicable

11 Performance indicators

- % effective at first verification
- Follow-up CAPAs raised
- Checks overdue

12 References

- ICH Q10 — Pharmaceutical Quality System
- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 9001:2015 — Quality management systems — Requirements

13. Demonstration

Case. For CAPA-2026-031 the check is: method *Process Monitoring*, criterion "*alarm routing verified operational after 100 % of maintenance interventions over 3 months, minimum 10 interventions*". At +3 months the verifier reviews 12 work orders: all carry the completed restore checklist and the weekly routing tests passed. She clicks **Conclude Effective**; the CAPA owner clicks **Verify Effectiveness** and closes the CAPA.

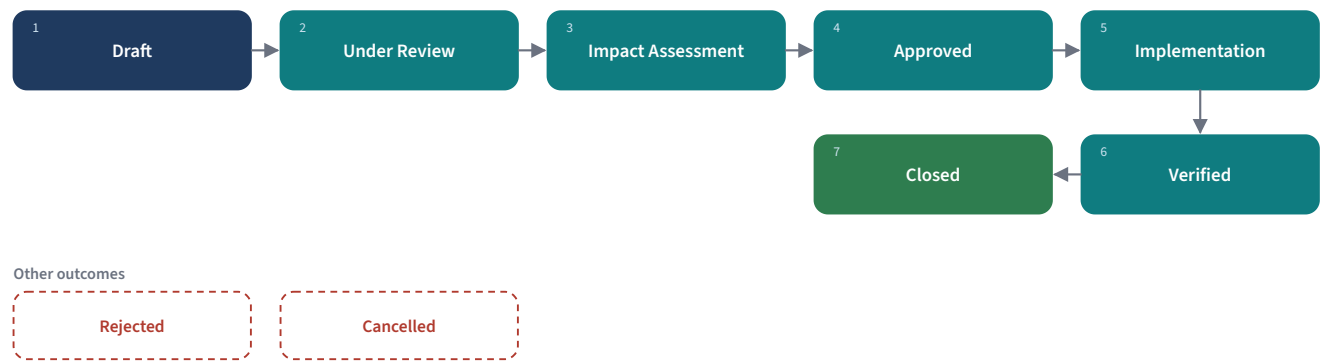
Change control

ls_change_control

ls_change_control manages change requests from draft to closure: classification (minor, major, critical), permanent or temporary change, impact areas with an impact assessment per area, approvals generated from the category's approval templates, implementation actions with evidence, and effectiveness verifications. The request is approved automatically once every assessment is completed and every approval is favourable; any rejection rejects the request.

Menu	Content
Change Control ► Change Requests ► All / My Change Requests / Impact Assessments	Requests and assessments
Change Control ► Approvals ► My Approvals	Decisions waiting for the connected user
Change Control ► Execution ► Implementation Actions / Effectiveness Verifications	Execution
Change Control ► Configuration ► Change Categories / Impact Areas / Settings	Categories, approval templates, delays

Group	Main rights
Viewer · Requester	Read; create and submit requests
Approver	Record approval decisions assigned to them
Change Control Manager	Drives the workflow, starts assessment and implementation, closes, rejects, cancels



Model: ls.change_control.request

Figure 46 — Lifecycle of a change request.

Change Control

Change Requests

Approvals

Execution

Reporting

Configuration

7

14

My Company (San Francisco)

New

Change Requests

CC/2026/00003 - Install a new tablet press in th...

Assessed1

Approved0

Actions Closed0

1 / 1

Reject

Cancel Request

...

Under Review

Impact Assessment

Approved

Implementation

Verified

Closed

Send message

Log note

Activity

6

Change Request ?

CC/2026/00003

Install a new tablet press in the solid dosage form area

Outstanding items before the next step:

Assessment of area 'Utilities' is not completed.

Assessment of area 'Equipment' is not completed.

Approval 'Quality Assurance (QA Manager (demo))' is still pending.

Approval 'Engineering and Maintenance (QA Manager (demo))' is still pending.

Approval 'Validation (QA Manager (demo))' is still pending.

CLASSIFICATION

Change Category ?

Equipment Change

Change Type ?

Permanent

Classification ?

Critical

Priority ?

☆

RESPONSIBILITIES

Requester ?

Mitchell Admin

Requesting Department ?

Change Control Manager ?

QA Manager (demo)

Company ?

My Company (San Francisco)

DATES

Request Date ?

Sep 27, 10:40 AM

Requested Target Date ?

EXECUTION

Actual Implementation Date ?

Planned Verification Date ?

Planned Activities

Today: "Change control approval requested" for QA Manager (demo)

Your decision is expected on change request CC/2026/00003 for the role Quality Assurance (QA Manager (demo)).

Mark Done Edit Cancel

Today: "Change control approval requested" for QA Manager (demo)

Your decision is expected on change request CC/2026/00003 for the role Engineering and Maintenance (QA Manager (demo)).

Mark Done Edit Cancel

Today: "Change control approval requested" for QA Manager (demo)

Your decision is expected on change request CC/2026/00003 for the role Validation (QA Manager (demo)).

Mark Done Edit Cancel

Today

QA Manager (demo) 12:13 PM

Under Review → Impact Assessment (Status)

QA Manager (demo) 12:13 PM

Impact assessment completed for area 'Facilities and Premises' with impact level 'Medium'.

QA Manager (demo) 12:13 PM

None → QA Manager (demo) (Change Control Manager)

Draft → Under Review (Status)

QC Analyst (demo) 12:13 PM

Change request submitted for review.

OdooBot 10:40 AM

Change Control Request created

Figure 47 — Change request form. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Change request and impact assessment

Module	ls_change_control	Version	_____
Menu	Change Control ► Change Requests	Effective date	_____
Owner (suggested)	Change Control Manager	Approved by	_____
Related SOPs	SOP-QA-031, SOP-QA-002, SOP-QA-060, SOP-VAL-005		

1 Objective

Ensure that any change that may affect product quality, the validated state, regulatory compliance or safety is proposed, justified, assessed by every affected function and approved before implementation.

2 Scope

Changes to products, processes, equipment, facilities, utilities, materials and suppliers, specifications, test methods, computerised systems and quality documents with a substantive effect. Temporary changes are included.

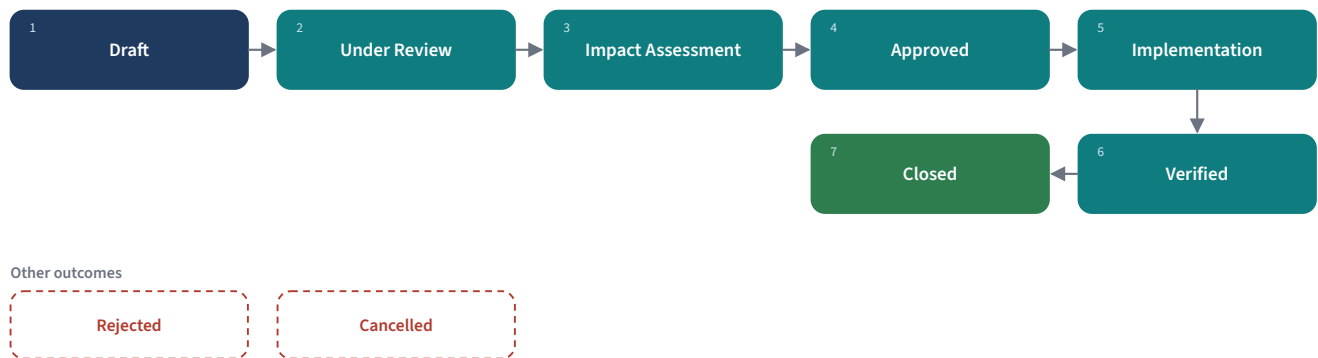
3 Responsibilities

Role (software group)	Responsibility
Requester	Describes the current situation, the proposed change and its justification; submits.
Change Control Manager	Screens the request, completes classification and impact areas, starts the assessment, arbitrates.
Assessors (per impact area)	Assess the impact of their area and describe the required actions.
Approvers	Approve or reject, with a comment when rejecting.

4 Definitions

Term	Definition
Classification	Minor, Major or Critical.
Temporary change	Change limited in time; it must declare its end date.
Regulatory impact	No Regulatory Action, Notification to the Authority, Variation of the Marketing Authorisation, Prior Approval Required.

5 Workflow in the software



Model: `ls.change_control.request`

Figure 48 — This SOP covers Draft → Approved.

6 Procedure

- 1 **Create.** In **Change Control ► Change Requests** click **New** : title, category, permanent/temporary (end date if temporary), classification, priority, current situation, proposed change, justification, requested target date.
- 2 **Declare impacts.** Tick the impact flags (GMP, product quality, regulatory, revalidation, training, document update, customer notification) and select the impact areas; reference the risk assessment if one exists (SOP-QA-060).
- 3 **Submit.** Click **Submit for Review** . The content is frozen for the requester.
- 4 **Screen.** The Change Control Manager checks completeness; if incomplete, **Return to Draft** with a comment.
- 5 **Start the assessment.** Click **Start Impact Assessment** : the system generates one assessment per impact area and the approvals defined by the category, and notifies the approvers.
- 6 **Assess.** Each assessor opens **Change Requests ► Impact Assessments**, sets the impact level (No Impact, Low, Medium, High), writes the assessment and, where there is an impact, the required actions; clicks **Complete Assessment** .
- 7 **Decide.** Each approver opens **Approvals ► My Approvals** and clicks **Approve** or **Reject** (a comment is required to reject).
- 8 **Approval.** When all assessments are completed and all approvals favourable, the request becomes *Approved* automatically and a planned implementation date is proposed from the category delay. A single rejection rejects the request.

7. Controls enforced by the software

- ✓ Once submitted, the request content is frozen; only a Change Control Manager may modify it — a change of the change itself requires a new request.
- ✓ Workflow fields (dates, states, deciders) cannot be written directly.
- ✓ A temporary change must declare an end date; a permanent change must not.
- ✓ Assessments and decisions can only be recorded while the request is in *Impact Assessment*; only the assigned assessor/approver (or the manager for assessments) may record them.
- ✓ A written assessment is required; if an impact is declared the required actions must be described.
- ✓ A rejection requires a comment; decided approvals cannot be modified or deleted.
- ✓ A submitted request cannot be deleted; requester and manager must be internal users.

8. Good practices

- ★ Describe the change so that a person who was not in the meeting understands it.
- ★ Involve regulatory affairs early: a variation can take months.
- ★ Use a risk assessment for major and critical changes.
- ★ Link documents to revise, validations to perform and training to deliver as implementation actions.

9. Pitfalls to avoid

- ✗ Implementing before approval ("it was urgent").
- ✗ Approving without reading the assessments.
- ✗ Using a temporary change that becomes permanent without a new request.

10 Records and evidence

- Change request with assessments, approvals and dates
- Referenced risk assessment

11 Performance indicators

- Time from submission to approval
- Changes implemented before approval (target 0)
- Temporary changes past their end date

12 References

- ICH Q10 — Pharmaceutical Quality System
- EU GMP Annex 15 (2015) — Qualification and Validation
- ISO 13485:2016 — Medical devices — Quality management systems
- EudraLex Vol. 4, EU GMP Part I, Chapter 1 — Pharmaceutical Quality System
- EU GMP Annex 11 — Computerised Systems (check the revision in force)

13. Demonstration

Case. Engineering proposes to replace the tablet press punches with a new supplier's tooling. Request CC-2026-021: category *Equipment*, permanent, classification *Major*, regulatory impact *No Regulatory Action* (equivalent design), requires revalidation and document update. Impact areas: Production, Quality Control, Validation, Purchasing. The manager clicks **Start Impact Assessment**; four assessments and three approvals are generated. Validation assesses *High* impact (OQ of the press with new tooling). All approve; the request becomes *Approved*.

Change Control									
Change Requests									
Approvals									
Execution									
Reporting									
Configuration									
My Company (San Francisco)									
New Change Requests									
1-3 / 3									
Reference	Title	Change Category	Classificat...	Requester	Change Control ...	Planned Im...	Status	Activities	
CC/2026/...	Install a new tablet press in the solid dosage form area	Equipment Change	Critical	Mitchell Admir	QA Manager (d		Impact As...	Change control approval r...	
CC/2026/...	Update the blister artwork of Product B after a leaflet...	Packaging and Labelling...	Minor	Mitchell Admir			Draft		
CC/2026/...	Replace the HPLC column of the assay method for P...	Analytical Method Change	Major	Mitchell Admir			Draft		

Figure 49 — Change request register. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Change Control																							
Change Requests																							
Approvals																							
Execution																							
Reporting																							
Configuration																							
My Company (San Francisco)																							
New Change Requests																							
1 / 1																							
Assessed 1 Approved 0 Actions Closed 0																							
Reject Cancel Request ... Under Review Impact Assessment Approved Implementation Verified Closed																							
Send message Log note Activity																							
Change Request ?																							
CC/2026/00003																							
Install a new tablet press in the solid dosage form area																							
Outstanding items before the next step:																							
Assessment of area 'Utilities' is not completed.																							
Assessment of area 'Equipment' is not completed.																							
Approval 'Quality Assurance (QA Manager (demo))' is still pending.																							
Approval 'Engineering and Maintenance (QA Manager (demo))' is still pending.																							
Approval 'Validation (QA Manager (demo))' is still pending.																							
CLASSIFICATION					RESPONSIBILITIES																		
Change Category ?	Equipment Change	Requester ?	Mitchell Admin			Requesting Department ?																	
Change Type ?	Permanent	Change Control Manager ?	QA Manager (demo)			Company ?	My Company (San Francisco)																
Classification ?	Critical																						
Priority ?	☆																						
DATES					EXECUTION																		
Request Date ?	Sep 27, 10:40 AM	Actual Implementation Date ?				Planned Verification Date ?																	
Requested Target Date ?																							
Planned Activities																							
Today: "Change control approval requested" for QA Manager (demo)																							
Your decision is expected on change request CC/2026/00003 for the role Quality Assurance (QA Manager (demo)).																							
✓ Mark Done ✎ Edit ✕ Cancel																							
Today: "Change control approval requested" for QA Manager (demo)																							
Your decision is expected on change request CC/2026/00003 for the role Engineering and Maintenance (QA Manager (demo)).																							
✓ Mark Done ✎ Edit ✕ Cancel																							
Today: "Change control approval requested" for QA Manager (demo)																							
Your decision is expected on change request CC/2026/00003 for the role Validation (QA Manager (demo)).																							
✓ Mark Done ✎ Edit ✕ Cancel																							
Today																							
QA Manager (demo) 12:13 PM	Under Review → Impact Assessment (Status)																						
QA Manager (demo) 12:13 PM	Impact assessment completed for area 'Facilities and Premises' with impact level 'Medium'.																						
QA Manager (demo) 12:13 PM	None → QA Manager (demo) (Change Control Manager)																						
	Draft → Under Review (Status)																						
QC Analyst (demo) 12:13 PM	Change request submitted for review.																						
OdooBot 10:40 AM	Change Control Request created																						

Figure 50 — Change request in Impact Assessment, with generated assessments and approvals. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Change Control

Change Requests

Approvals

Execution

Reporting

Configuration

7

14

My Company (San Francisco)

New

Impact Assessments

To Complete

Search...

1-2 / 2

☐ Change Request	Impact Area	Assessor	Impact Level	Status	
☐ CC/2026/00003 - Install a new tablet press in the solid dosage form area	Utilities	QA Manager (demo)	No Impact	Draft	
☐ CC/2026/00003 - Install a new tablet press in the solid dosage form area	Equipment	QA Manager (demo)	No Impact	Draft	

Figure 51 — Impact assessments per area. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Approval, implementation and verification of changes

Module	ls_change_control	Version	_____
Menu	Change Control ► Execution	Effective date	_____
Owner (suggested)	Change Control Manager	Approved by	_____
Related SOPs	SOP-QA-030, SOP-QA-023, SOP-VAL-005, SOP-QA-012		

1 Objective

Implement approved changes as planned, with evidence for each action, verify that the change achieved its purpose without adverse effect, and close the request formally.

2 Scope

All approved change requests.

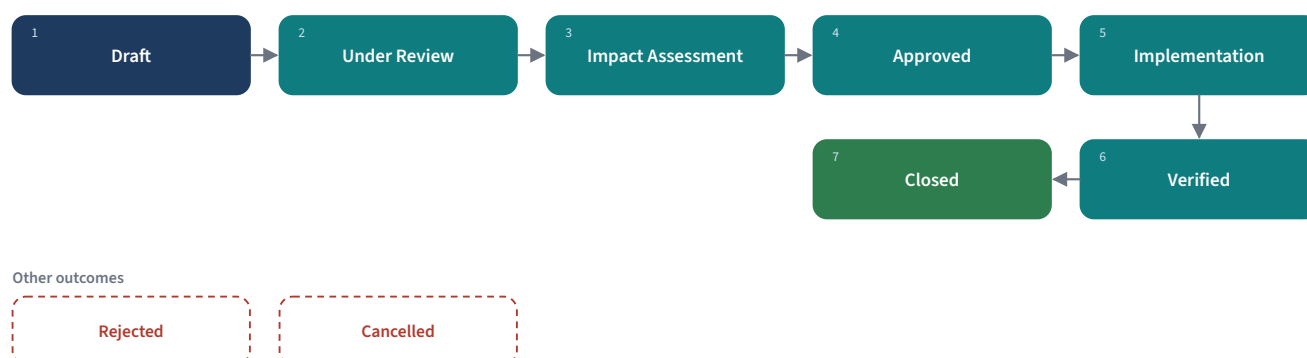
3 Responsibilities

Role (software group)	Responsibility
Action owners	Perform the implementation actions and record evidence.
Change Control Manager	Starts implementation, declares the change verified, closes; cancels actions when justified.
Verifier	Performs the effectiveness verification against the acceptance criteria.

4 Definitions

Term	Definition
Implementation action	Task with an owner and an evidence reference (e.g. document revision, validation report, training record).
Effectiveness verification	Check planned on the request with a method and acceptance criteria; required when the category demands it.

5 Workflow in the software



Model: ls.change_control.request

Figure 52 — This SOP covers Approved → Closed.

6 Procedure

- 1 **Plan the actions.** On the approved request add implementation actions (owner, description, due date) derived from the assessments: document revisions, validation, training, supplier notification, regulatory filing.
- 2 **Start implementation.** The manager clicks **Start Implementation** (at least one action is required).
- 3 **Execute.** Each owner clicks **Start** then **Mark as Done** with the evidence reference. Only the manager can cancel an action (**Cancel Action**), with a reason.
- 4 **Verify effectiveness.** At the planned date the verifier records the verification result and conclusion and clicks **Conclude: Effective** or **Conclude: Not Effective** (the latter requires a follow-up reference, e.g. a new change request or CAPA).
- 5 **Declare verified.** When all actions are closed and verifications complete, the manager clicks **Declare Verified**.
- 6 **Close.** Click **Close** and write the closure statement (what was changed, effective date, evidence).
- 7 **Temporary change.** At the end date, verify the return to the original state and record it before closing.

7. Controls enforced by the software

- ✓ Implementation actions can only be executed once the change is approved; only the owner or the manager may update an action.
- ✓ An action cannot be closed without an evidence reference; cancellation requires a reason and the manager role; a started action cannot be deleted.
- ✓ Verifications can only be completed during *Implementation*; a conclusion is required; *Not Effective* requires a follow-up.
- ✓ Verification of the request is refused while blocking items remain; closure requires a closure statement.
- ✓ A category requiring effectiveness verification must define a verification delay (on the category or the company).

8. Good practices

- ★ Do not declare the change effective on the day of implementation: allow a representative period.
- ★ Keep the evidence reference precise (document number and revision, report number).
- ★ Close the loop with training (SOP-QA-012) and revalidation (SOP-VAL-005).

9. Pitfalls to avoid

- ✗ Actions closed with "done" and no evidence.
- ✗ Leaving requests in *Implementation* for months after the change is in use.

10 Records and evidence

- Implementation actions with evidence
- Effectiveness verifications
- Closure statement

11 Performance indicators

- Actions overdue
- Requests closed within planned date
- % of changes needing a follow-up

12 References

- ICH Q10 — Pharmaceutical Quality System
- EU GMP Annex 15 (2015) — Qualification and Validation
- ISO 13485:2016 — Medical devices — Quality management systems

13. Demonstration

Case (continued). CC-2026-021 actions: (1) OQ of the press with new tooling — report VAL-OQ-2026-008; (2) revise SOP-PRD-014 rev 4; (3) train operators. After 10 batches the verifier checks tablet weight, hardness and friability trends: all within control limits → **Conclude: Effective**. The manager clicks **Declare Verified** then **Close**.

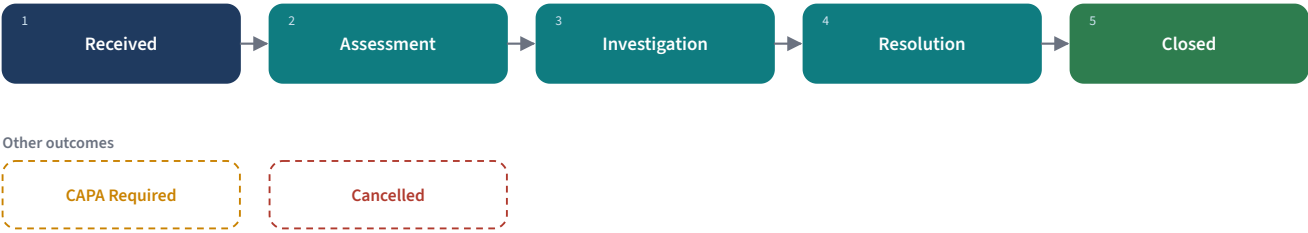
Complaints

ls_complaint

ls_complaint records product quality complaints from any channel, with product, lot and sample, assessment, investigations (several methodologies, approval by a different user), resolutions, adverse events with reportability assessment, and closure. Complaint categories carry target times for acknowledgement, investigation and closure, and the reporting deadline for adverse events.

Menu	Content
Complaints ► Complaint Handling ► All / My Complaints / Investigations / Resolutions	Complaint handling
Complaints ► Vigilance ► Adverse Events	Adverse events and authority reports
Complaints ► Configuration ► Complaint Categories	Target days and reporting deadline
Complaints ► Reporting ► Complaint Analysis	Trending

Group	Main rights
Complaint Viewer · Investigator · Reviewer · Manager	Investigator investigates; Reviewer approves investigations; Manager closes and cancels.



Model: ls.complaint

Figure 53 — Lifecycle of a complaint.

Complaints
Complaint Handling
Vigilance
Reporting
Configuration
7
14
My Company (San Francisco)

New
Complaints
CMP/2026/00003 - Skin reaction reported after ...
Investigations 2
Resolutions 0
Adverse Events 1
1 / 1

CAPA Required
Start Resolution
Cancel
Received
Assessment
Investigation
Resolution
Closed
Send message
Log note
Activity

CMP/2026/00003

Skin reaction reported after administration

RECEPTION		HANDLING	
Receipt Date ?	Sep 27, 10:40 AM	Responsible ?	QA Reviewer (demo)
Occurrence Date ?		Reviewer ?	QA Manager (demo)
Channel	Email	Priority	☆☆
Received By	OdooBot	Category	Product Quality Defect
Company	My Company (San Francisco)	Severity	Minor
		Complaint Type	Safety or Adverse Event

Complaint
Product
Assessment
Investigations
Adverse Events
CAPA
Resolutions

COMPLAINANT		TARGETS	
Complainant		Acknowledgement Due Date	Sep 29
Complainant Type	Healthcare Professional	Investigation Due Date	Oct 17
Contact Name		Closure Due Date	Oct 27
Contact Email			

Today

OdooBot 12:13 PM
None → **QA Reviewer (demo)** (Responsible)
None → **QA Manager (demo)** (Reviewer)
Critical → **Minor** (Severity)
Received → **Investigation** (Status)

QA Reviewer (demo) 12:13 PM
Investigation started.

QA Reviewer (demo) 12:13 PM
Assessment started.

OdooBot 10:40 AM
Product Quality Complaint created

Figure 54 — Complaint form. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Reception, assessment and investigation of complaints

Module	ls_complaint	Version	_____
Menu	Complaints ► Complaint Handling	Effective date	_____
Owner (suggested)	Complaint Manager (QA)	Approved by	_____
Related SOPs	SOP-QA-041, SOP-QA-022, SOP-QA-081		

1 Objective

Register every complaint, acknowledge it on time, assess its seriousness, investigate the cause when justified, respond to the complainant and feed the CAPA and recall processes.

2 Scope

Product quality complaints received from patients, consumers, healthcare professionals, customers, distributors, pharmacies, hospitals, authorities or internal sources, for all products of the company.

3 Responsibilities

Role (software group)	Responsibility
Receiver (any employee)	Forwards the complaint to QA the same day; creates the record if trained.
Complaint Investigator	Assesses and investigates, requests the sample, writes conclusions.
Complaint Reviewer	Approves or rejects investigations (not their own).
Complaint Manager	Decides on waiver of investigation and CAPA, closes the complaint.

4 Definitions

Term	Definition
Complaint	Any written, electronic or oral communication alleging a deficiency of a product after release.
Waiver of investigation	Documented decision not to investigate (e.g. identical complaint already investigated).

5 Workflow in the software



Model: ls.complaint

Figure 55 — Complaint lifecycle.

6 Procedure

- 1 Register.** In **Complaints ► Complaint Handling** click **New**: receipt date, channel, received by, complainant and type, subject and description, product, lot, quantity, market, sample availability.

- 2 **Acknowledge.** Contact the complainant within the category's acknowledgement target; record *Customer Notified* and the date.
- 3 **Assess.** Click **Start Assessment** ; set category, complaint type, severity, potential safety and regulatory impact, assessment summary. If an adverse event is involved, create it (SOP-QA-041).
- 4 **Investigate or waive.** Click **Start Investigation** and add an investigation (investigator, methodology, plan). Or click **Waive Investigation** with a justification.
- 5 **Investigate.** Examine the sample, the retained samples, batch record, other complaints on the same lot; record the summary, root cause category and description, conclusion (Confirmed, Not Confirmed, Inconclusive) and the batch impact. Click **Complete** ; a reviewer clicks **Approve** .
- 6 **CAPA.** If a CAPA is needed click **CAPA Required** , open the CAPA (SOP-QA-022) and record its reference.
- 7 **Resolve.** Click **Start Resolution** ; add resolutions (reply letter, replacement, credit) and mark them done with evidence.
- 8 **Close.** Click **Close** with a closure summary. Consider a recall (SOP-QA-081) if the investigation shows a risk for other units on the market.

7. Controls enforced by the software

- ✓ Receipt date not in the future; occurrence date $\leq$ receipt date; expiry date $\geq$ manufacturing date; the lot must belong to the product.
- ✓ A product must be identified before leaving the *Received* state.
- ✓ A description is mandatory when the channel or type is *Other*.
- ✓ Investigation: completion date $\geq$ start date; mandatory fields checked at completion; root cause description required unless *Not Determined*; the approver must differ from the investigator; a rejection needs a reason.
- ✓ Resolution requires completion evidence; cancellation requires a reason.
- ✓ Closure requires a closure summary; cancellation requires a reason; a complaint that left *Received* cannot be deleted.

8. Good practices

- ★ Record the complainant's words verbatim in the description.
- ★ Ask for the sample and the lot number at first contact.
- ★ Search for complaints on the same lot before concluding.
- ★ Treat suspected falsified products as urgent and inform the authority according to local rules.

9. Pitfalls to avoid

- ✗ Closing "not confirmed" without examining a sample or retained sample.
- ✗ Missing the reportability assessment when health is involved.

10 Records and evidence

- Complaint record, investigations with approval, resolutions, closure summary
- Correspondence with the complainant

11 Performance indicators

- Acknowledgement within target
- Median closure time
- Complaints per million units per product
- % confirmed

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 8 — Complaints and Product Recall
- 21 CFR 211.198 — Complaint files
- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 22716:2007 — Cosmetics — Good Manufacturing Practices
- Regulation (EC) No 1223/2009, Article 23 — Serious undesirable effects

13. Demonstration

Case. A pharmacy reports by phone that a blister of lot 24B117 contains one broken tablet. The receiver registers the complaint (channel *Phone*, complainant type *Pharmacy*, type *Product Quality Defect*). QA acknowledges within 2 days, assesses severity *Minor*, starts an investigation (methodology *Batch Record Review*): no event recorded, IPC hardness within limits, 2 similar complaints in 12 months across 3 lots → not systemic. Conclusion *Complaint Confirmed*, root cause *Transport or Storage* (crushed box visible on the photo). Resolution: replacement and letter. Closed in 18 days.

Complaints										
Complaints										
Priority	Reference	Receipt Date	Subject	Complainant	Product	Category	Severity	Responsible	Closure Du...	Status
☐	CMP/202...	Sep 27, 10:40 AM	Skin reaction reported aft...		[DEMO-TAB-10] Demo Film-Coat...	Product Q...	Minor	QA Reviewer	Oct 27	Investiga...
☐	CMP/202...	Sep 27, 10:40 AM	Leaflet missing from carton	Demo Ph...	[DEMO-TAB-10] Demo Film-Coat...	Packaging...	Minor		Oct 27	Received
☐	CMP/202...	Sep 27, 10:40 AM	Chipped tablets found in b...	Demo Ph...	[DEMO-TAB-10] Demo Film-Coat...	Product Q...	Major		Oct 27	Received

Figure 56 — Complaint register. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Complaints
Complaint Handling
Vigilance
Reporting
Configuration
7
14
My Company (San Francisco)

New
Investigations
CMP/2026/00003 / CMP/INV/2026/00001
1 / 1

Start
Draft
In Progress
Completed
Approved
Send message
Log note
Activity
1

CMP/INV/2026/00001

CONTEXT

ComplaintCMP/2026/00003 - Skin reaction reported ...
InvestigatorQA Reviewer (demo)
Sequence10
CompanyMy Company (San Francisco)

TIMING

Date Started
Date Completed
Approved By
Approval Date

Analysis
Batch Impact

Methodology
Root Cause Category
Conclusion
Investigation Plan
Investigation Summary ?
Root Cause Description

Today

QA Reviewer (demo) 12:13 PM
Complaint Investigation created

Figure 57 — Complaint investigation (batch record review). (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Adverse events: reportability assessment and reporting deadlines

Module	ls_complaint	Version	_____
Menu	Complaints ► Vigilance ► Adverse Events	Effective date	_____
Owner (suggested)	Vigilance officer / Complaint Manager	Approved by	_____
Related SOPs	SOP-QA-040, SOP-MD-007		

1 Objective

Identify, among complaints, the events that must be reported to a competent authority, assess reportability and causality in a documented way, and submit within the legal deadline.

2 Scope

Adverse events and incidents linked to complaints for medicinal products, medical devices and cosmetics. Pharmacovigilance systems for medicinal products may be managed in a dedicated tool; this SOP then covers the hand-over.

3 Responsibilities

Role (software group)	Responsibility
Complaint Investigator	Creates the adverse event from the complaint and collects facts.
Vigilance officer	Assesses seriousness, causality and reportability; submits the report.
Complaint Manager	Configures the reporting deadlines per category; supervises timeliness.

4 Definitions

Term	Definition
Awareness date	Date when the company first became aware of the event; the reporting clock starts from it.
Report due date	Computed: awareness date + reporting deadline (days) configured on the complaint category.
Causality	Certain, Probable, Possible, Unlikely, Unrelated, Not Assessable.

5 Workflow in the software



Model: ls.complaint.adverse_event

Figure 58 — Lifecycle of an adverse event.

6 Procedure

- 1 **Configure deadlines.** For each complaint category set *Adverse Event Reporting Deadline (days)* according to the applicable rule. A single deadline per category: create separate categories when deadlines differ by seriousness (for example, for medical devices under the MDR: 2, 10 or 15 days depending on the case; for US medical device reporting: 30 calendar days, or 5 work days in defined cases; for EU pharmacovigilance: 15 days for serious and 90 days for non-serious reactions).
- 2 **Create.** From the complaint, add an adverse event: awareness date, event date, description, seriousness and criteria, outcome, subject pseudonym, age range, sex.
- 3 **Assess.** Record causality and its rationale, *Reportable to Authority*, the reportability rationale and the competent authority. Click **Assess**.
- 4 **Submit.** Prepare and send the report through the authority's channel before the due date; record the submission date, the authority acknowledgement reference and who submitted; click **Record Submission**.
- 5 **Follow up.** Record follow-up information; tick *Follow Up Required* when a follow-up report is due.
- 6 **Close.** Click **Close** with closure notes.

7. Controls enforced by the software

- ✓ Awareness date not in the future; event date $\leq$ awareness date.
- ✓ Reportability rationale mandatory once assessed; causality assessment mandatory.
- ✓ A reportable event requires the competent authority; a non-reportable event cannot record a submission.
- ✓ The authority acknowledgement reference is mandatory at submission.
- ✓ Closed events cannot be modified; non-draft events cannot be deleted.

8. Good practices

- ★ Assess reportability on the day of awareness; when in doubt, report.
- ★ Use pseudonyms — never store the patient's name in the complaint system.
- ★ Monitor the overdue filter daily.

9. Pitfalls to avoid

- ✗ Starting the clock at the date QA received the complaint instead of the first awareness in the company.
- ✗ One category used for all seriousness levels with a single, inappropriate deadline.

10 Records and evidence

- Adverse event with assessment, submission and acknowledgement

11 Performance indicators

- % of reportable events submitted on time (target 100 %)
- Median time from awareness to assessment

12 References

- MDR Articles 87 to 92 — Vigilance
- 21 CFR Part 803 — Medical Device Reporting
- EMA Good Pharmacovigilance Practices, Module VI — Management and reporting of adverse reactions
- Regulation (EC) No 1223/2009, Article 23 — Serious undesirable effects
- ISO 13485:2016 — Medical devices — Quality management systems

13. Demonstration

Case. A hospital reports that a catheter tip detached during use, without injury. Awareness 3 March. The vigilance officer assesses seriousness *Serious* (could have led to serious deterioration), causality *Probable*, reportable to the national competent authority. The category *Medical device — serious incident* has a 15-day deadline: the report is due 18 March. It is submitted on 12 March; the authority's reference is recorded; a follow-up report is planned after the investigation.

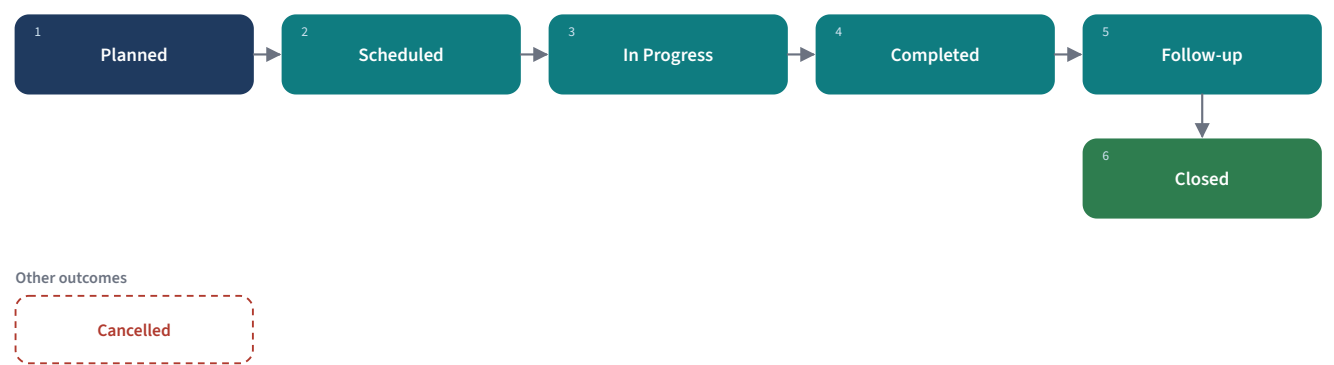
Audits

ls_audit

ls_audit supports the internal audit process: auditable areas, audit types, versioned checklists, auditor qualifications (with lead-auditor status, qualified scope and expiry), audit programmes, individual audits with impartiality checks, checklist responses with objective evidence, findings with response, CAPA reference and verification, and audit reports reviewed and approved by people other than their author.

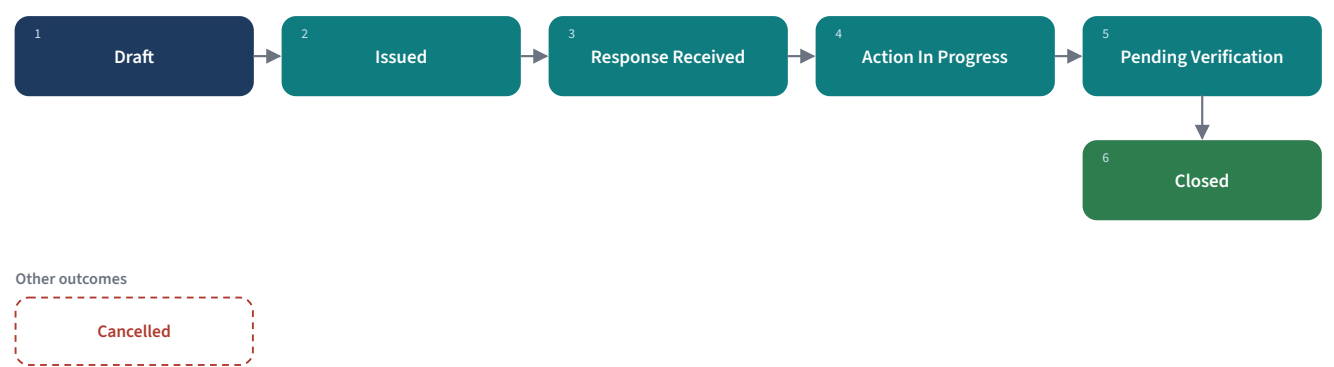
Menu	Content
Quality ► Audits ► Audit Programmes / Audits / Audit Calendar / Checklists	Planning and execution
Quality ► Findings ► All Findings / My Findings	Findings
Quality ► Reports ► Audit Reports / Checklist Responses	Reporting
Quality ► Configuration ► Audit Types / Auditable Areas / Auditor Qualifications / Finding Categories	Reference data

Group	Main rights
Auditee	Answers findings assigned to them
Auditor · Lead Auditor	Execute audits, raise findings, prepare reports; the lead auditor leads the audit
Audit Manager	Approves programmes and checklists, manages qualifications, approves reports



Model: ls.audit.schedule

Figure 59 — Lifecycle of an audit.



Model: ls.audit.finding

Figure 60 — Lifecycle of an audit finding.

Quality
Audits
Findings
Reports
Configuration

7
14
My Company (San Francisco)

New
Audits
[AUD/2026/0002] Documentation Control - Raw...

Findings
0
Reports
0

1 / 1

Load Checklist
Start Fieldwork
Cancel
Planned
Scheduled
In Progress
Completed
Follow-up
Closed

Send message
Log note
Activity

AUD/2026/0002

Title
Documentation Control - Raw Material Warehou:

PLANNING

AUDIT TEAM

Audit Programme
[APG/2026/0001] Internal Audit Pr...

Lead Auditor
Amina Lead Auditor

Audit Type
Internal System Audit

Audit Team

Planned Date
Nov 11

Auditees
Sofiane Production Manager

Actual Start

Audited Organisation

Actual End

Checklist Used

Duration (hours)
00:00

Company
My Company (San Francisco)

Audited Areas
Algiers Manufacturing Site / Raw M...

Objective, Scope and Criteria
Checklist
Findings
Reports

OBJECTIVE

Determine whether goods receipt and storage records conform to the internal

Today

QA Manager (demo)
12:13 PM
Planned → Scheduled (Status)

OdooBot
10:40 AM
Audit created

Figure 61 — Audit form. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Internal audit programme

Module	ls_audit	Version	_____
Menu	Quality ► Audits ► Audit Programmes	Effective date	_____
Owner (suggested)	Audit Manager	Approved by	_____
Related SOPs	SOP-QA-051, SOP-QA-052, SOP-QA-053, SOP-QA-071		

1 Objective

Plan internal audits (self-inspections) so that every process and area of the quality system is audited at a frequency based on risk and past results, and follow the programme to completion.

2 Scope

All auditable areas of the company over the programme period (usually one year).

3 Responsibilities

Role (software group)	Responsibility
Audit Manager	Builds the programme, gets it approved, monitors execution, closes it.
Head of Quality / management	Approves the programme and provides resources.
Lead auditors	Plan and conduct the audits assigned.

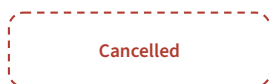
4 Definitions

Term	Definition
Audit programme	Set of audits planned for a period with a common objective.
Auditable area	Process, department or site that can be audited; areas can be nested.

5 Workflow in the software



Other outcomes



Model: ls.audit.program

Figure 62 — Lifecycle of an audit programme.

6 Procedure

- 1 Prepare reference data.** Maintain auditable areas, audit types and finding categories (with response deadlines) in [Quality ► Configuration](#).
- 2 Assess risk.** Rank areas by risk: GMP criticality, results of previous audits, deviations, complaints, changes, inspection findings.

- 3 **Create the programme.** In **Audit Programmes** click **New**: title, start and end dates, owner, objective. Add the audits (title, type, areas, planned date inside the period).
- 4 **Approve.** Click **Approve** (the programme must contain at least one audit).
- 5 **Start.** Click **Start** at the beginning of the period; audits can then be scheduled (SOP-QA-052).
- 6 **Monitor.** Review the *Audit Calendar* monthly; reschedule with justification; add for-cause audits when events require it.
- 7 **Close.** At the end of the period, when every audit is closed or cancelled, click **Close** and report the programme outcome to management review.

7. Controls enforced by the software

- ✓ The end date cannot precede the start date; audits must be planned within the programme period.
- ✓ A programme can only be approved from Draft and must contain at least one audit; it must be approved before it starts and in progress before it closes.
- ✓ A programme cannot close while audits are neither closed nor cancelled.
- ✓ An audit cannot be scheduled while its programme is not approved.

8. Good practices

- ★ Audit every GMP area at least once per defined cycle and more often where risk is high.
- ★ Include the quality system itself (document control, CAPA, training) and the data integrity controls.
- ★ Keep capacity for for-cause audits.

9. Pitfalls to avoid

- ✗ Programmes copied from year to year without risk review.
- ✗ Postponing the same audit repeatedly.

10 Records and evidence

- Approved programme
- Audit calendar
- Programme closure summary

11 Performance indicators

- % of programme audits completed on time
- For-cause audits added

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 9 — Self Inspection
- ISO 19011:2018 — Guidelines for auditing management systems
- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 9001:2015 — Quality management systems — Requirements

13. Demonstration

Case. Programme *Internal audits 2027*: 14 audits covering 11 areas, with warehouse and sterile filling audited twice because of 2026 deviations. The Head of QA approves it in December; the Audit Manager clicks **Start** on 2 January.

Quality Audits Findings Reports Configuration

7 14 My Company (San Francisco)

New

Audit Programmes
[APG/2026/0001] Internal Audit Programme

Audits
2

1 / 1

Close Cancel

Draft Approved In Progress Closed

Send message Log note Activity

1 1

APG/2026/0001

Title ?
Internal Audit Programme

PERIOD

Start Date ? Jan 1
End Date ? Dec 31

GOVERNANCE

Programme Owner ? Nadia Quality Manager
Approved By ? QA Manager (demo)
Approval Date ? Sep 27, 12:13 PM
Completion Rate (%) ? 0%
Company ? My Company (San Francisco)

Objective and Rationale Audits Notes

Cover every GMP-critical process at least once during the year. Frequency is set from process criticality, regulatory impact and the results of the previous programme.

QA Manager (demo) 12:13 PM
None → 09/27/2026 12:13:06 PM (Approval Date)
None → QA Manager (demo) (Approved By)
Draft → In Progress (Status)

OdooBot 10:40 AM
Audit Programme created

Figure 63 — Audit programme with its audits. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Auditor qualification and impartiality

Module	ls_audit	Version	_____
Menu	Quality ► Configuration ► Auditor Qualifications	Effective date	_____
Owner (suggested)	Audit Manager	Approved by	_____
Related SOPs	SOP-QA-050, SOP-QA-052, SOP-QA-013		

1 Objective

Ensure that audits are conducted by qualified auditors, led by qualified lead auditors, within their qualified scope, and that no auditor audits their own work.

2 Scope

All internal auditors and lead auditors, and every audit of ls_audit.

3 Responsibilities

Role (software group)	Responsibility
Audit Manager	Qualifies auditors, maintains qualification records and expiry, assigns audit teams.
Auditors	Maintain their competence, declare conflicts of interest.

4 Definitions

Term	Definition
Qualified scope	Auditable areas an auditor is qualified to audit.
Impartiality	An auditor must not audit an area they own or work they performed.

5 Procedure

- 1 Define criteria.** Write the qualification criteria (training in ISO 19011 principles, GMP knowledge, number of audits as observer, supervised audits) in this SOP.
- 2 Record the qualification.** In [Quality ► Configuration ► Auditor Qualifications](#) click **New**: auditor (user), employee, *Lead Auditor* flag, qualification date, expiry date, certificate reference, qualified scope (areas), notes.
- 3 Maintain.** Monitor *Qualification Status* and *Days To Expiry*; requalify before expiry (performance review, refresher training).
- 4 Assign teams.** When planning an audit, choose a lead auditor qualified as lead and team members qualified for the areas; list the auditees.
- 5 Check impartiality.** The system refuses auditors who are also auditees or who own an audited area; record other conflicts (recent job in the area) in the notes and replace the auditor.

6. Controls enforced by the software

- ✓ Expiry date cannot precede the qualification date; qualified scope must stay within the company.
- ✓ Only a user registered as qualified lead auditor can lead an audit.
- ✓ An audit is refused if a person appears both in the audit team and among the auditees, or if a team member owns an audited area.

7. Good practices

- ★ Rotate auditors across areas to keep a fresh view.
- ★ Pair new auditors with experienced lead auditors.
- ★ Link auditor training to ls_training certifications (SOP-QA-013).

8. Pitfalls to avoid

- ✗ Qualification records not updated after an auditor changes department (new conflict of interest).
- ✗ Lead auditors whose qualification expired still leading audits.

9 Records and evidence

- Auditor qualification records
- Training and competency evidence

10 Performance indicators

- Qualified lead auditors available
- Qualifications expiring within 60 days

11 References

- ISO 19011:2018 — Guidelines for auditing management systems
- EudraLex Vol. 4, EU GMP Part I, Chapter 9 — Self Inspection
- ISO 13485:2016 — Medical devices — Quality management systems

12. Demonstration

Case. A former production supervisor moves to QA and is qualified as auditor on 1 June, scope *Warehouse, QC laboratory*, expiry after 3 years. When the Audit Manager tries to add her to the audit of *Production — packaging*, which she owned until May, the impartiality check refuses her: she is replaced.

Quality Audits Findings Reports Configuration 7 14 My Company (San Francisco)						
New Auditor Qualifications Search... 1-2 / 2						
☐ Auditor	Employee	Lead Auditor	Qualificatio...	Expiry Date	Qualification Status	Company
☐ Amina Lead Auditor		☒	Sep 27, 2025	Sep 27, 2028	Valid	My Company (San Francisco)
☐ Karim Auditor		☐	Mar 27	Sep 27, 2027	Valid	My Company (San Francisco)

Figure 64 — Auditor qualifications with status and days to expiry. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Conducting an audit: preparation, fieldwork, findings

Module	ls_audit	Version	_____
Menu	Quality ► Audits ► Audits	Effective date	_____
Owner (suggested)	Lead auditor	Approved by	_____
Related SOPs	SOP-QA-050, SOP-QA-051, SOP-QA-053		

1 Objective

Conduct each audit in a planned, evidence-based way, and record nonconformities as findings with objective evidence and a clear requirement reference.

2 Scope

All internal audits of the approved programme and for-cause audits.

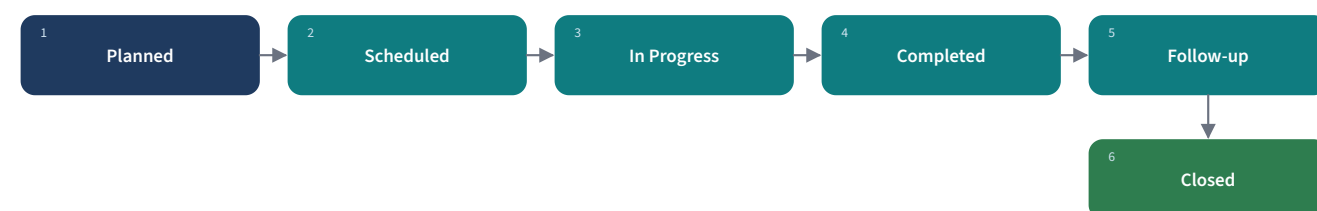
3 Responsibilities

Role (software group)	Responsibility
Lead auditor	Prepares, leads the opening and closing meetings, issues findings.
Audit team	Performs the fieldwork, records responses with evidence.
Auditees	Provide access and information; nominate the person responsible for each finding response.

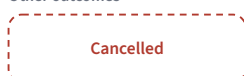
4 Definitions

Term	Definition
Checklist response	Result per question: Conform, Non-conform, Observation, Not Applicable (Pending until answered).
Finding	Nonconformity or observation, with statement, objective evidence, requirement reference, category and severity.

5 Workflow in the software



Other outcomes



Model: ls.audit.schedule

Figure 65 — Lifecycle of an audit.

6 Procedure

- 1 **Plan.** In the audit record (from the programme): objective, scope, criteria, lead auditor, team, auditees, planned date. Click [Schedule](#).
- 2 **Load the checklist.** Click [Load Checklist](#) and choose an approved checklist for the audit type; questions become responses to fill.
- 3 **Prepare.** Review previous findings, deviations and CAPAs of the area; send the agenda.
- 4 **Open fieldwork.** Click [Start Fieldwork](#) at the opening meeting.
- 5 **Collect evidence.** Answer each question with the result and the objective evidence (documents seen, records sampled, observations).
- 6 **Raise findings.** From a Non-conform or Observation response click [Raise Finding](#); complete the statement, evidence, requirement reference, category and severity, area and responsible auditee.
- 7 **Close fieldwork.** Hold the closing meeting; click [Complete Fieldwork](#).
- 8 **Issue findings.** Click [Issue to Auditee](#) on each finding; the response due date follows the finding category (SOP-QA-053).
- 9 **Checklist maintenance.** Checklists are approved by the Audit Manager; to change an approved checklist use [New Version](#).

7. Controls enforced by the software

- ✓ Objective evidence is required for any Non-conform or Observation response; a finding can only be raised from such a response, once per question.
- ✓ Responses can be edited only while the audit is scheduled or in progress; they can be removed only while it is scheduled.
- ✓ A finding can only be issued while the audit is in progress; it cannot be assigned to the person who raised it; its area must be in the audit scope.
- ✓ An approved checklist cannot be edited; a used checklist cannot return to draft — create a new version. A checklist without questions cannot be approved.
- ✓ Audits that left *Planned* cannot be deleted, only cancelled.

8. Good practices

- ★ Sample records randomly and write the identifiers in the evidence.
- ★ State the requirement breached (SOP section, GMP chapter) for each nonconformity.
- ★ Separate facts from opinions; share findings at the closing meeting — no surprises in the report.

9. Pitfalls to avoid

- ✗ Findings without evidence ("procedure not followed").
- ✗ Auditing only documents, never the shop floor.

10 Records and evidence

- Audit with checklist responses and evidence
- Findings
- Attendance of opening/closing meetings

11 Performance indicators

- Conformity rate per audit
- Findings per audit by severity

12 References

- ISO 19011:2018 — Guidelines for auditing management systems
- EudraLex Vol. 4, EU GMP Part I, Chapter 9 — Self Inspection
- ISO 13485:2016 — Medical devices — Quality management systems

13. Demonstration

Case. Audit AUD-2027-004 *Warehouse*: the lead auditor loads checklist *WH-01 v3* (42 questions). Question 17 "Are temperature excursions investigated?": Non-conform — evidence: logger report of 14 July shows 27.8 °C for 3 h, no deviation found. She clicks **Raise Finding** : category *Major*, requirement *SOP-WH-005 §6.2*, responsible auditee: warehouse manager. After the closing meeting, **Complete Fieldwork** and **Issue to Auditee** .

The screenshot displays the Odoo 19.0 Community interface for a versioned audit checklist. The top navigation bar includes 'Quality', 'Audits', 'Findings', 'Reports', and 'Configuration'. The user is logged in as 'My Company (San Francisco)'. The main header shows 'New Checklists [CHK-DOC v1] Documentation Control' with a '1 / 1' indicator. Below the header, there are tabs for 'Reset to Draft', 'New Version', and 'Set Obsolete'. The checklist is currently in the 'Approved' state, with 'Draft' and 'Obsolete' also visible. On the right, there are buttons for 'Send message', 'Log note', and 'Activity'. The checklist details include: Name 'Documentation Control', CHK-DOC 'version 1', Audit Type 'Internal System Audit', Approved By 'Nadia Quality Manager', Approval Date 'Jul 27, 10:40 AM', Question Count '4', and Company 'My Company (San Francisco)'. A table lists the questions with their references and mandatory status:

Reference	Question	Mandatory
SOP-QA-001 §4.2	Are only current, approved versions of procedures available at the point of use?	☒
SOP-QA-001 §4.6	Are obsolete documents removed from the point of use or clearly identified as obsolete?	☒
SOP-PR-014 §6.1	Are batch records completed contemporaneously and signed by the operator performing the step?	☒
SOP-QA-007 §3.3	Are corrections to records made so that the original entry remains readable?	☐

On the right side, there is a chat window with a message from 'OdooBot' at 10:40 AM stating 'Audit Checklist Template created'.

Figure 66 — Versioned audit checklist. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Quality

Audits

Findings

Reports

Configuration

7

14

My Company (San Francisco)

New

Audits

Open

Search...

1-2 / 2

☐	Reference	Title	Audit Type	Audit Programme	Planned Date	Lead Auditor	Findings	Open Fi...	Status	Company	
☐	AUD/2026...	Documentation Control - Raw Material War...	Internal Syste...	[APG/2026/0001] Internal Audit Prog...	Nov 11	Amina Lead ...	0	0	Scheduled	My Company (San Fra...	
☐	AUD/2026...	Documentation Control - Sterile Filling	Internal Syste...	[APG/2026/0001] Internal Audit Prog...	Oct 11	Amina Lead ...	0	0	Planned	My Company (San Fra...	

Figure 67 — Audits of the programme. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Audit report and follow-up of findings

Module	ls_audit	Version	_____
Menu	Quality ► Reports ► Audit Reports · Findings	Effective date	_____
Owner (suggested)	Audit Manager	Approved by	_____
Related SOPs	SOP-QA-052, SOP-QA-022		

1 Objective

Report the audit results to the responsible management through an independently reviewed and approved report, and follow each finding until its correction is verified.

2 Scope

All audits after fieldwork, and all issued findings.

3 Responsibilities

Role (software group)	Responsibility
Lead auditor	Prepares the report; verifies finding actions.
Reviewer	Reviews the report independently.
Audit Manager	Approves and issues the report; accepts or rejects finding responses.
Responsible auditee	Responds to findings: root cause, correction, corrective action, CAPA reference.

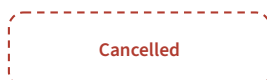
4 Definitions

Term	Definition
Distribution list	Managers who must receive the report; it cannot be empty.
Response due	Deadline to respond, derived from the finding category.

5 Workflow in the software

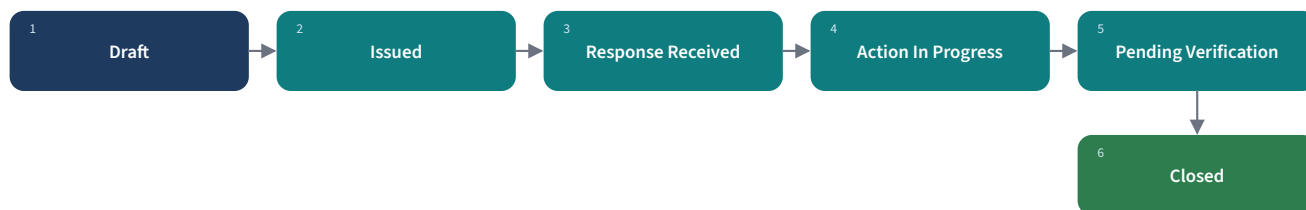


Other outcomes

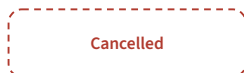


Model: ls.audit.report

Figure 68 — Lifecycle of an audit report.



Other outcomes



Model: ls.audit.finding

Figure 69 — Lifecycle of a finding.

6 Procedure

- 1 **Draft the report.** In **Audit Reports** click **New** : audit, date, executive summary, conclusion, positive observations, distribution list.
- 2 **Review.** Click **Submit for Review** ; a reviewer other than the preparer clicks **Record Review** .
- 3 **Approve and issue.** A user other than the preparer clicks **Approve** , then **Issue** .
- 4 **Respond.** The auditee opens **Findings ► My Findings** and clicks **Submit Response** : root cause, immediate correction, corrective action, CAPA reference when a CAPA is opened (SOP-QA-022), action due date.
- 5 **Accept or reject.** The Audit Manager clicks **Accept Response** or **Reject Response** (with reason).
- 6 **Implement.** When the actions are done, the auditee clicks **Action Completed** .
- 7 **Verify and close.** The lead auditor verifies on site or on records, documents the verification method and result, and clicks **Verify and Close** .
- 8 **Close the audit.** When all findings are closed, click **Move to Follow-up** then **Close Audit** on the audit.

7. Controls enforced by the software

- ✓ A report can only be submitted when the fieldwork is complete.
- ✓ The reviewer and the approver must differ from the preparer; approval requires a prior review; issue requires a distribution list.
- ✓ A response can only be accepted with a documented root cause and a CAPA reference when required.
- ✓ A finding cannot be closed without the verification method and result.
- ✓ Issued findings and reports cannot be deleted, only cancelled; final records cannot be modified.

8. Good practices

- ★ Issue the report within the time set in this SOP (for example 10 working days after closing meeting).
- ★ Reject responses that only correct the example found and do not address the cause.
- ★ Verify effectiveness of major findings at the next audit of the area.

9. Pitfalls to avoid

- ✗ Accepting "will be done" responses without dates or owners.
- ✗ Closing findings on the auditee's word without verification.

10 Records and evidence

- Approved and issued report
- Findings with responses and verification

11 Performance indicators

- Report issue time
- Findings overdue
- Repeat findings

12 References

- ISO 19011:2018 — Guidelines for auditing management systems
- EudraLex Vol. 4, EU GMP Part I, Chapter 9 — Self Inspection
- ISO 13485:2016 — Medical devices — Quality management systems

13. Demonstration

Case (continued). The report of AUD-2027-004 is prepared by the lead auditor, reviewed by a second auditor and approved by the Audit Manager, then issued to the site director and warehouse manager. The warehouse manager responds: root cause — logger alarms e-mailed to a former employee; correction — deviation opened; corrective action — alarm distribution list managed by role; CAPA-2027-012. Accepted; after implementation the auditor tests an alarm and clicks **Verify and Close** .

Risk management

ls_risk_management

ls_risk_management follows ICH Q9(R1) and ISO 14971 concepts: configurable risk matrices (severity and probability levels, cells mapped to risk levels and acceptability), a risk register with hazard, hazardous situation and harm, successive risk assessments (initial, residual, periodic review, production/post-production information), risk control measures with the ISO 14971 option priority, residual risk acceptance, and FMEA worksheets whose failure modes can raise register entries.

Menu	Content
Risk Management ► Risk Management ► Risk Register / Risk Assessments / Risk Control Measures	Register and assessments
Risk Management ► FMEA ► FMEA Worksheets / Failure Modes	FMEA
Risk Management ► Reporting ► Reviews Due	Periodic review
Risk Management ► Configuration ► Risk Matrices / Risk Categories	Matrices

Group	Main rights
Risk Viewer · Risk Analyst	Read; perform assessments and FMEA
Risk Manager	Approves matrices and assessments, accepts residual risk, closes risks

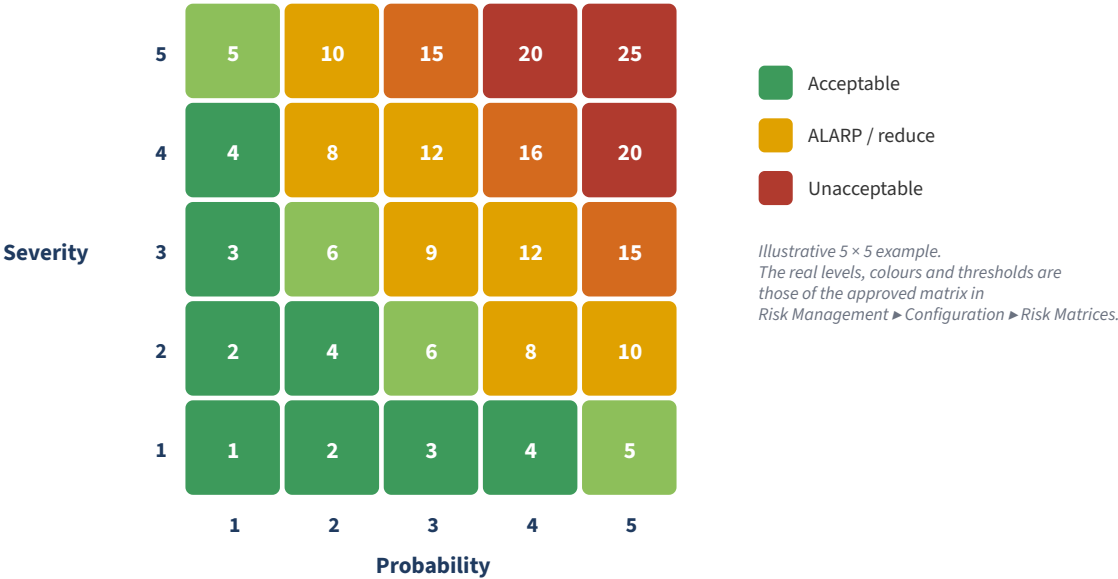
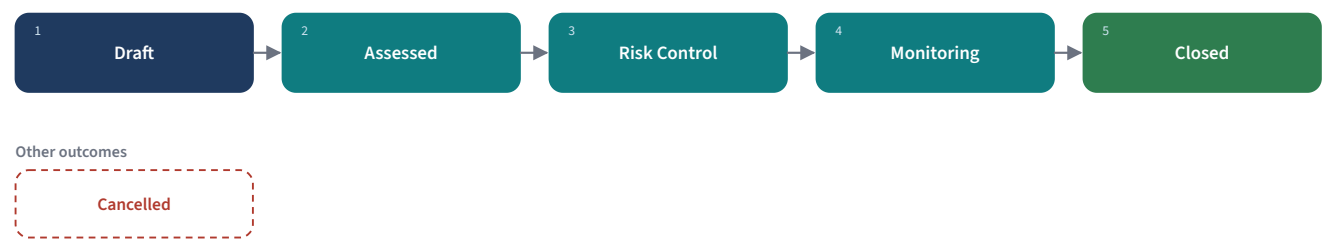


Figure 70 — Illustrative risk matrix.



Model: ls.risk.register

Figure 71 — Lifecycle of a risk register entry.

Quality risk management: matrix, assessment, control and acceptance of residual risk

Module	ls_risk_management	Version	_____
Menu	Risk Management ► Risk Register · Configuration ► Risk Matrices	Effective date	_____
Owner (suggested)	Risk Manager	Approved by	_____
Related SOPs	SOP-QA-061, SOP-QA-062, SOP-QA-030, SOP-MD-003		

1 Objective

Identify, analyse, evaluate and control quality risks with a common, approved method, and accept residual risks formally and knowingly.

2 Scope

Product, process, equipment, facility, utility, material, supplier, computerised system, laboratory, packaging, regulatory and occupational risks recorded in the register. Medical device risk files follow SOP-MD-003 in addition.

3 Responsibilities

Role (software group)	Responsibility
Risk Manager	Approves the matrix and assessments, confirms control completeness, accepts residual risk, closes.
Risk owner	Maintains the risk entry and its measures.
Risk Analyst	Performs assessments (not the approval of their own work).
Measure owners	Implement control measures and provide evidence.

4 Definitions

Term	Definition
Risk level / acceptability	Given by the matrix cell for a severity × probability pair: Negligible to Very High; Acceptable, Acceptable Only With Risk Control, Not Acceptable.
Control option priority	Inherent safety by design, then protective measures, then information for safety (ISO 14971).

5 Workflow in the software



Model: ls.risk.register

Figure 72 — Risk register entry lifecycle.



Model: *ls.risk.mitigation*

Figure 73 — Risk control measure lifecycle.

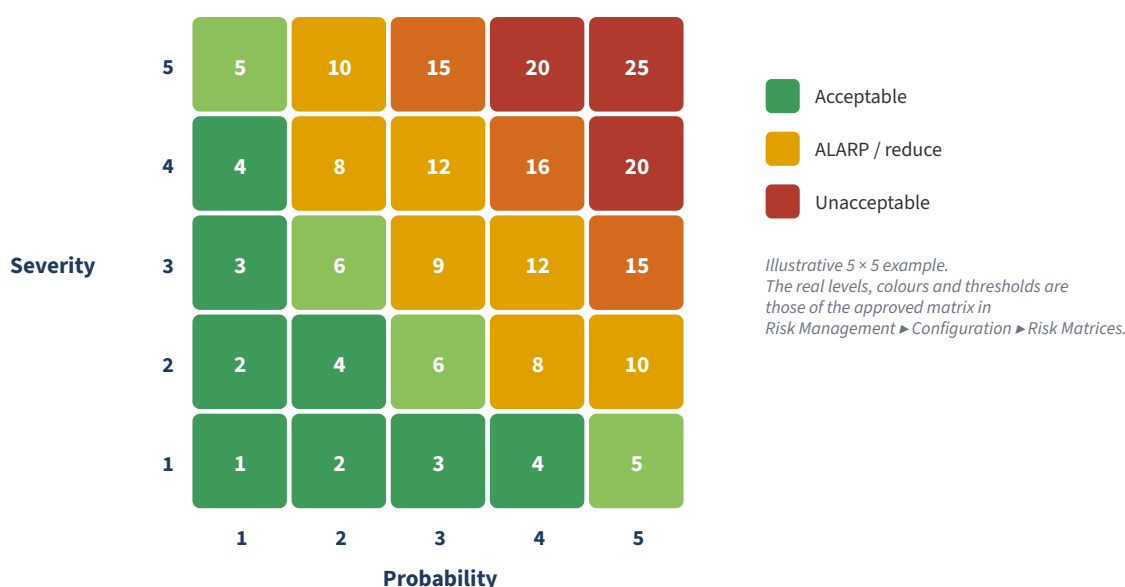


Figure 74 — Illustrative matrix: the real one is configured and approved.

6 Procedure

- Set up the matrix.** In **Configuration ► Risk Matrices** define severity and probability levels, click **Generate Missing Cells**, set the level and acceptability of each cell, reference the governing document, and click **Approve**. Mark one approved matrix as the company default.
- Identify.** In the *Risk Register* click **New**: title, type, category, owner, identification date, hazard, hazardous situation, sequence of events, harm, matrix, review interval.
- Assess.** Click **Assess** to create the initial assessment: severity, probability, estimation rationale. The analyst clicks **Confirm**; a Risk Manager other than the assessor clicks **Approve**. Then click **Mark Assessed**.
- Control.** Click **Start Risk Control**. Add control measures (description, option, responsible, due date); each is approved, started, marked implemented with evidence, then its effectiveness verified with evidence by a different person.
- Residual assessment.** Record a *Residual* assessment after the measures.
- New risks.** If a measure introduces a new hazard, tick *Introduces New Risk*, describe it and click **Create Risk Record for New Risk**.
- Complete and accept.** Click **Confirm Control Completeness** once every measure is verified or cancelled; if the residual risk is not plainly acceptable, click **Accept Residual Risk** with the justification and benefit-risk analysis.
- Monitor.** Click **Move to Monitoring**; periodic reviews follow SOP-QA-062; close with a reason when the risk no longer exists.

7. Controls enforced by the software

- ✓ The matrix must be complete (one cell per combination) and approved before assessments are approved; only one active default matrix per company.
- ✓ An assessment requires a rationale before confirmation; the assessor cannot approve it; approved assessments are frozen (cancel and record a new one).
- ✓ A risk has one initial assessment; later estimations use the other types.
- ✓ A risk cannot leave Draft without an approved initial assessment.
- ✓ Control completeness requires all measures verified or cancelled; monitoring requires completeness and, when the acceptability demands it, a residual risk acceptance decision.
- ✓ Measures: implementation evidence and effectiveness evidence are mandatory; the person who verified the implementation cannot verify the effectiveness; verified measures are frozen.
- ✓ Only a Risk Manager may perform the protected operations; assessed risks cannot be deleted.

8. Good practices

- ★ Write the rationale with data (failure history, detection capability), not "expert judgement" alone.
- ★ Prefer design measures over procedural ones; training is the weakest control.
- ★ Link the risk to the record it concerns (*Linked Model / Linked Record ID*) — e.g. a change request.

9. Pitfalls to avoid

- ✗ Lowering the probability in the residual assessment without any implemented measure.
- ✗ Accepting residual risk by default without a benefit-risk justification.

10 Records and evidence

- Risk register entry with assessments, measures, evidence, acceptance

11 Performance indicators

- Risks Not Acceptable without an active measure (target 0)
- Measures overdue
- Reviews overdue

12 References

- ICH Q9(R1) — Quality Risk Management
- ISO 14971:2019 — Medical devices — Application of risk management
- ISO 31000:2018 — Risk management — Guidelines
- EudraLex Vol. 4, EU GMP Part I, Chapter 1 — Pharmaceutical Quality System

13. Demonstration

Case. Risk *Cross-contamination on the shared granulator*: severity 4, probability 3 → level High, *Acceptable Only With Risk Control*. Measures: dedicated product-contact parts (design), cleaning validation with visual and swab limits (protective), label on the equipment status board (information). After verification, residual assessment: severity 4, probability 1 → Medium, acceptable with control. Completeness confirmed; the Risk Manager accepts the residual risk with justification; the risk moves to *Monitoring* with a 12-month review.

Risk Management

Risk Management

FMEA

Reporting

Configuration

7

14

My Company (San Francisco)

New

Risk Register

RISK/2026/00003 - Audit trail disabled on a chr...

Assessments

1

Controls

0

1 / 1

<

>

Assess

Mark Assessed

Cancel

Draft

Assessed

Risk Control

Monitoring

Closed

Send message

Log note

Activity

Q

0

RISK/2026/00003

Audit trail disabled on a chromatography system

CLASSIFICATION

OWNERSHIP

Risk Type

Computerised System

Risk Owner

Mitchell Admin

Category

Data Integrity

Identified By

Mitchell Admin

Risk Matrix ?

[DEMO-3X3] Demonstration 3x3 Matrix

Identification Date

Sep 27

Company

My Company (San Francisco)

CURRENT EVALUATION

Initial Risk Level

Current Acceptability

Current Risk Level

Current Assessment ?

Risk Analysis

Assessments

Risk Control

Residual Risk

Monitoring

Traceability

Description

The audit trail of a computerised laboratory system may be switched off.

ANALYSIS INPUTS

Intended Use and Reasonably

Today

OdooBot

10:39 AM

Risk Register Entry created

Figure 75 — Risk register entry. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

FMEA (failure modes and effects analysis)

Module	ls_risk_management	Version	_____
Menu	Risk Management ► FMEA ► FMEA Worksheets	Effective date	_____
Owner (suggested)	Risk Manager	Approved by	_____
Related SOPs	SOP-QA-060, SOP-QA-062		

1 Objective

Analyse systematically how a design, process, system or use can fail, rate severity, occurrence and detection, and drive actions where the risk priority exceeds the thresholds.

2 Scope

Design, process, system and use/application FMEA.

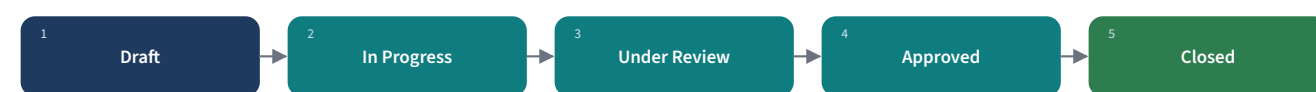
3 Responsibilities

Role (software group)	Responsibility
Facilitator	Leads the team, records the worksheet, submits it.
Team members	Contribute knowledge of the item analysed.
Reviewer and approver	Independent of the facilitator; review and approve the worksheet.

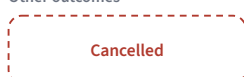
4 Definitions

Term	Definition
RPN	Risk Priority Number = $S \times O \times D$, each rated 1–10.
Thresholds	RPN action threshold and severity action threshold set on the worksheet: above them an action is required.

5 Workflow in the software



Other outcomes



Model: ls.risk.fmea

Figure 76 — Lifecycle of an FMEA worksheet.

6 Procedure

- Create.** In **FMEA Worksheets** click **New**: title, FMEA type, scope, assumptions and boundaries, revision, facilitator, team, start date, RPN and severity thresholds. Click **Start**.
- Analyse.** For each item and function add failure modes: effect, severity, cause, occurrence, current controls, detection. The RPN is computed and lines above a threshold are flagged.

- 3 **Plan actions.** For flagged lines write the recommended action, responsible and target date.
- 4 **Review and approve.** Click **Submit for Review**; a reviewer other than the facilitator clicks **Record Review**, then an approver (also not the facilitator) clicks **Approve**.
- 5 **Implement and re-rate.** Record actions taken and the revised S, O and D together; the revised RPN and the reduction are computed.
- 6 **Raise risks.** Use **Create Risk** on a failure mode to open a register entry (SOP-QA-060) when the risk must be managed over time.
- 7 **Close or revise.** Click **Close** when no action remains open; use **Create Revision** for a later update.

7. Controls enforced by the software

- ✓ Only a draft worksheet can be started; only an in-progress worksheet with failure modes can be submitted.
- ✓ The facilitator cannot review or approve; approval requires prior review; lines above threshold must have a recommended action.
- ✓ Revised S, O and D must be given together.
- ✓ Closure requires no open action; only approved or closed worksheets can be revised.
- ✓ Raising a risk from a failure mode requires an approved default matrix; one register entry per failure mode.

8. Good practices

- ★ Rate severity on the effect, occurrence on the cause, detection on the current controls — never mix them.
- ★ Treat high severity even when the RPN is low (severity threshold).
- ★ Keep a cross-functional team: operators know the real failure modes.

9. Pitfalls to avoid

- ✗ Reducing detection scores by adding inspections instead of removing causes.
- ✗ Worksheets never revised after process changes.

10 Records and evidence

- Approved FMEA worksheet with revisions
- Linked register entries

11 Performance indicators

- Lines above threshold without action
- Average RPN reduction

12 References

- IEC 60812:2018 — Failure modes and effects analysis (FMEA and FMECA)
- ICH Q9(R1) — Quality Risk Management
- ISO 14971:2019 — Medical devices — Application of risk management

13. Demonstration

Case. Process FMEA of the blister packaging line: failure mode *missing tablet in blister*, effect *patient under-dosed* S=8, cause *feeder blockage* O=4, control *camera inspection* D=3 → RPN 96, above the RPN threshold of 80. Action: add feeder vibration sensor. After implementation O=2 → RPN 48. The worksheet is reviewed, approved, then closed when all actions are complete.

Risk Management

Risk Management

FMEA

Reporting

Configuration

7

14

My Company (San Francisco)

New

FMEA

FMEA/2026/0001 - Granulation process FMEA (rev. 1)

1 / 1

Start

Cancel

Draft

In Progress

Under Review

Approved

Closed

Send message

Log note

Activity

Q

0

The 1-10 severity, occurrence and detection scales and the Risk Priority Number are an industry convention, not a requirement of ISO 14971:2019. The action thresholds below are defined by this organisation.

FMEA/2026/0001

Granulation process FMEA

Fmea Type	Process FMEA	RPN Action Threshold ?	100
Revision	1	Severity Action Threshold ?	9
Facilitator	Mitchell Admin	Failure Mode Count	3
Start Date	Sep 27	Highest RPN	120
Company	My Company (San Francisco)	Lines Requiring Action	2
		Open Actions	2

Failure Modes

Scope

Team

Review and Approval

Item	Function	Failure M...	Effect of F...	Severity ...	Potential ...	Occurre...	Current C...	Detectio...	
Granulator	Mix powd...	Inadequat...	Content uniformity outside specification	9	Mixing time below the validated range	3	In-process content uniformity testing	4	108
Binder ad...	Deliver bi...	Binder ov...	Oversized granules and clow...	6	Pump calibration drift	4	Quarterly pump calibration	5	120

OdooBot

10:39 AM

FMEA Worksheet created

Today

Figure 77 — FMEA worksheet. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Periodic review of risks

Module	ls_risk_management	Version	_____
Menu	Risk Management ► Reporting ► Reviews Due	Effective date	_____
Owner (suggested)	Risk Manager	Approved by	_____
Related SOPs	SOP-QA-060, SOP-QA-061, SOP-GEN-001		

1 Objective

Keep risk assessments current by reviewing them at defined intervals and whenever new information (deviations, complaints, changes, post-market data) appears.

2 Scope

All risk register entries in *Monitoring* and approved FMEA worksheets.

3 Responsibilities

Role (software group)	Responsibility
Risk owner	Performs the review with the relevant data.
Risk Manager	Approves the periodic assessment, decides on further control or closure.

4 Definitions

Term	Definition
Review interval	Months between two reviews, set on each risk; <i>Next Review Date</i> is computed.
Periodic Review assessment	Assessment type used to record the re-estimation.

5 Procedure

- 1 **List.** Monthly open [Risk Management ► Reporting ► Reviews Due](#).
- 2 **Collect data.** Gather deviations, complaints, CAPA, changes, trends and post-market information related to the risk since the last review.
- 3 **Re-estimate.** Create an assessment of type *Periodic Review* (or *Production / Post-Production Information* when triggered by such data), confirm and have it approved.
- 4 **Decide.** If the level increased, return to risk control (new measures); if the hazard disappeared, close the risk with a reason.
- 5 **Event-driven reviews.** Any major deviation, complaint trend or change affecting a risk triggers an immediate review, not waiting for the interval.
- 6 **Report.** Summarise the risk profile in management review.

6. Controls enforced by the software

- ✓ Approved assessments cannot be modified; a new assessment is recorded for each review.
- ✓ The assessor cannot approve their own assessment.

7. Good practices

- ★ Align review intervals with risk level (shorter for High).
- ★ Record "no change" reviews as assessments too — the absence of change is information.

8. Pitfalls to avoid

- ✗ Reviews done on paper and never recorded.
- ✗ Ignoring post-market data in the review of product risks.

9 Records and evidence

- Periodic review assessments
- Decisions (new measures, closure)

10 Performance indicators

- Reviews overdue
- Risks whose level increased at review

11 References

- ICH Q9(R1) — Quality Risk Management
- ISO 14971:2019 — Medical devices — Application of risk management
- ICH Q10 — Pharmaceutical Quality System

12. Demonstration

Case. The risk *Temperature excursion during transport* is due for review. Since last year, 3 complaints and 1 deviation relate to it. The owner records a *Production / Post-Production Information* assessment: probability raised from 2 to 3, level Medium → High. The Risk Manager approves it and the risk returns to *Risk Control* with a new measure (validated shipper for summer lanes).

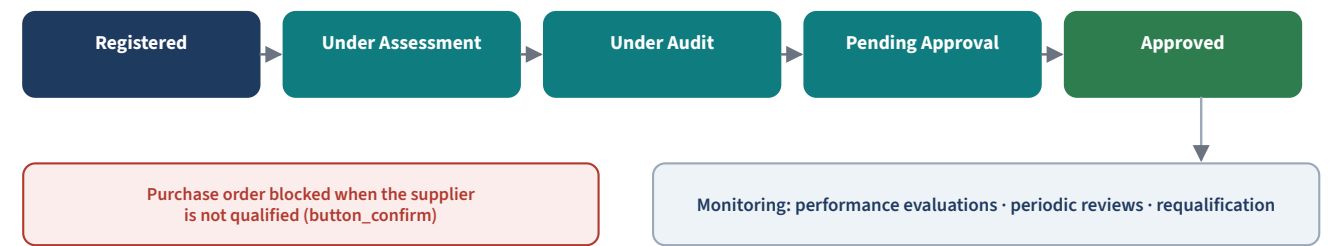
Suppliers

ls_supplier_qualification

ls_supplier_qualification manages one qualification dossier per supplier and company: scored assessments from templates, supplier audits with graded findings, the qualified scope (materials and services), performance evaluations (on-time delivery, quality acceptance, documentation, responsiveness, rated A to D), periodic reviews, and an append-only, SHA-256-chained signature log of qualification decisions. A company setting controls purchase order confirmation for suppliers that are not approved: *none*, *warn* (default) or *block*.

Menu	Content
Supplier Qualification ► Qualification ► Qualification Dossiers / Qualified Scope / Expiring Approvals	Dossiers
Supplier Qualification ► Evaluation ► Assessments / Audits / Audit Findings	Evaluation
Supplier Qualification ► Monitoring ► Performance Evaluations / Periodic Reviews / Signature Log	Monitoring
Supplier Qualification ► Configuration ► Supplier Categories / Assessment Templates / Criteria / Reference Standards	Configuration

Group	Main rights
Supplier Viewer · Supplier Assessor	Read; perform assessments, audits and reviews
Supplier Manager	Approves dossiers, changes status, configures



Other states: Conditionally Approved, Suspended, Expired, Disqualified.

Figure 78 — Supplier qualification lifecycle and purchase control.

Supplier Qualification

Qualification

Evaluation

Monitoring

Configuration

7

14

My Company (San Francisco)

New

Qualification Dossiers

SQ/2026/00001

Scope

1

Assessments

1

Audits

0

Performance

0

Reviews

0

Signatures

0

1 / 1

Start Audit

Submit for Approval

Reset to Draft

Registered

Under Assessment

Under Audit

...

Send message

Log note

Activity

0

SQ/2026/00001

Not ready for approval yet:

No completed assessment with a Pass or Conditional result is attached to the dossier.

No closed audit with an acceptable outcome is attached to the dossier.

No qualified material or service defines the approval scope.

SUPPLIER

Supplier

Demo Pharma Ingredients SARL

Supplier Category

Active Substance Manufacturer

Criticality

Critical

Responsible

OdooBot

Company

My Company (San Francisco)

VALIDITY

Registration Date

Jun 29

Approval Date

Approved By

Requalification Interval

Months

36

Approval Valid Until

Next Audit Date

Next Review Date

INDICATORS

Risk Level

Medium

Latest Assessment

Latest Assessment

Score (%)

0%

MONITORING

Latest Performance

Score

0.00

Latest Performance

Rating

OdooBot

10:41 AM

Life Sciences Supplier Qualification Dossier created

Figure 79 — Qualification dossier. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Qualification and approval of suppliers

Module	ls_supplier_qualification	Version	_____
Menu	Supplier Qualification ► Qualification ► Qualification Dossiers	Effective date	_____
Owner (suggested)	Supplier Manager (QA)	Approved by	_____
Related SOPs	SOP-QA-071, SOP-QA-072, SOP-QA-073, SOP-QC-001		

1 Objective

Approve suppliers of GMP-relevant materials and services only after an assessment proportionate to their criticality, and document the approval, its scope, its conditions and its validity.

2 Scope

Suppliers of active substances, excipients, packaging, components, contract services, laboratories, logistics and other GMP-relevant goods and services.

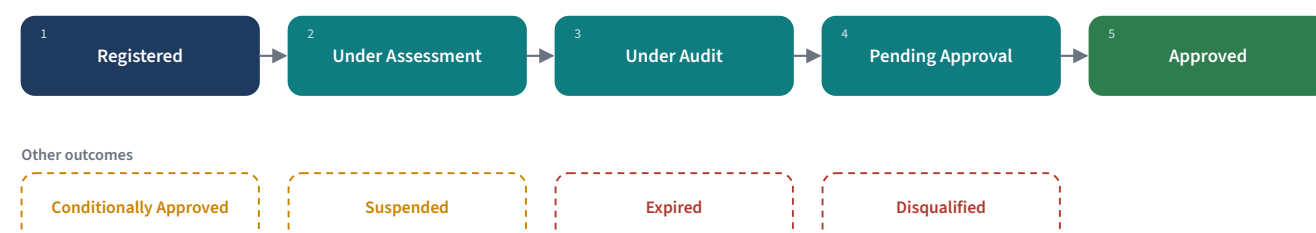
3 Responsibilities

Role (software group)	Responsibility
Supplier Assessor	Performs assessments and audits; cannot approve dossiers they produced evidence for (when segregation is enforced).
Supplier Manager	Configures categories and templates, submits and approves dossiers, sets conditions.
Purchasing	Requests new suppliers and does not order before approval.

4 Definitions

Term	Definition
Supplier category	Criticality, whether assessment, initial audit and periodic audit are required, audit interval, requalification and review intervals.
Qualified scope	Materials or services for which the supplier is qualified, with validity dates.
Conditional approval	Approval with documented conditions (e.g. increased incoming testing).

5 Workflow in the software



Model: ls.supplier_qualification

Figure 80 — Lifecycle of a qualification dossier.

6 Procedure

- 1 **Configure.** Define supplier categories and assessment templates (criteria, weights, maximum score, minimum score for mandatory criteria, pass and conditional thresholds). In the company settings choose *Enforce Segregation of Duties* and the purchase order control level (see SOP-QA-073).
- 2 **Register.** In **Qualification Dossiers** click **New** : supplier, category, criticality, responsible, registration date, requalification interval. One active dossier per supplier and company.
- 3 **Assess.** Click **Start Assessment** ; create an assessment, choose the template, click **Load Criteria from Template** , **Start** , score each criterion (comment every criterion below the minimum), click **Complete** ; a different assessor clicks **Review** .
- 4 **Audit if required.** Click **Start Audit** when the category requires an initial audit (SOP-QA-071).
- 5 **Define the scope.** Add the qualified scope lines (product or service, qualification date, validity end) and click **Qualify** on each.
- 6 **Submit.** Click **Submit for Approval** ; the system lists any missing prerequisite.
- 7 **Approve.** The Supplier Manager clicks **Approve** and confirms approval date, validity and, for a conditional approval, the conditions. The decision is written in the signature log.
- 8 **Communicate.** Inform purchasing and the warehouse; the supplier appears as approved on purchase orders.

7. Controls enforced by the software

- ✓ One active dossier per supplier and company — requalify the existing one instead of creating another.
- ✓ Transitions follow the workflow: only a registered dossier moves to assessment; submission lists missing prerequisites; only a pending dossier can be approved.
- ✓ Segregation of duties (company setting, enabled by default): a person who produced assessment or audit evidence cannot approve the dossier; an assessor cannot review their own assessment.
- ✓ Assessment scoring: scores within the maximum; conditional threshold $\leq$ pass threshold; low scores require comments; criteria are loaded only in draft.
- ✓ An approved dossier needs its mandatory information; a conditional approval needs its conditions; expiry after approval date.
- ✓ The signature log is append-only and chained; its chain can be verified (**Verify Chain**).

8. Good practices

- ★ Base the depth of qualification on criticality: questionnaire for low risk, audit for high risk.
- ★ Qualify the manufacturer, not only the distributor, for active substances.
- ★ Keep quality agreements as attachments of the dossier.

9. Pitfalls to avoid

- ✗ Approving under commercial pressure with incomplete evidence.
- ✗ Qualifying a supplier generically when only one material was assessed — use the qualified scope.

10 Records and evidence

- Dossier with assessments, audits, scope, approval and signature log

11 Performance indicators

- Suppliers approved per category
- Time to qualify
- Conditional approvals open

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 5 — Production
- EudraLex Vol. 4, EU GMP Part I, Chapter 7 — Outsourced Activities
- ICH Q7 — Good Manufacturing Practice for Active Pharmaceutical Ingredients
- ISO 13485:2016 — Medical devices — Quality management systems
- 21 CFR 211.84 — Testing and approval or rejection of components
- ICH Q10 — Pharmaceutical Quality System

13. Demonstration

Case. New excipient supplier, category *Excipient — major*: assessment with template *Excipient GMP questionnaire* scores 82 % (pass threshold 80 %); desk review of certificates of analysis for 3 lots; no audit required for this category. Scope: *Microcrystalline cellulose PH-102*, validity 3 years. A Supplier Manager who did not perform the assessment approves the dossier; the signature log records *Approved*.

SA/2026/00001

SUBJECT

Qualification	SQ/2026/00001
Supplier	Demo Pharma Ingredients SARL
Template	Full Supplier Assessment
Assessment Type	Initial Qualification

EXECUTION

Date	Jul 29
Assessor	OdooBot
Reviewer	
Review Date	

SCORING RULES (FROZEN)

Max Score Per Criterion	5
Mandatory Min Score	3
Pass Threshold	80.00
Conditional Threshold	60.00

RESULT

Total Weighted Score	0.00
Total Max Weighted Score	0.00
Score (%)	0%
Mandatory Failed Count	0
Finding Count	0
Result	

Scored Criteria

Criterion	Domain	Weight	Mandatory	Score	Max Sco...	Evidence	Comment	Raise Fin...
Add a line								

Activity

Today

OdooBot 10:41 AM
Life Sciences Supplier Assessment created

Figure 81 — Scored supplier assessment. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Supplier audits

Module	ls_supplier_qualification	Version	_____
Menu	Supplier Qualification ► Evaluation ► Audits	Effective date	_____
Owner (suggested)	Supplier Manager (QA)	Approved by	_____
Related SOPs	SOP-QA-070, SOP-QA-072, SOP-QA-052		

1 Objective

Verify on site, remotely or on documents that a supplier's quality system and practices meet the requirements, and obtain corrective actions for the findings.

2 Scope

Initial, surveillance, for-cause and requalification audits of suppliers, whatever the format (on-site, remote, desktop, documentation review).

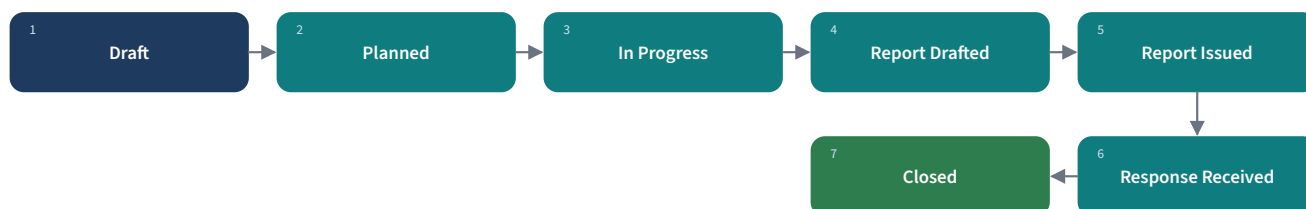
3 Responsibilities

Role (software group)	Responsibility
Lead auditor	Plans, conducts, reports, verifies findings.
Audit team	Supports the lead auditor.
Supplier contact	Answers findings with corrective actions.
Supplier Manager	Takes the outcome into the qualification decision.

4 Definitions

Term	Definition
Outcome	Acceptable, Acceptable with Actions, Not Acceptable.
Finding severity	Critical, Major, Minor, Observation.

5 Workflow in the software

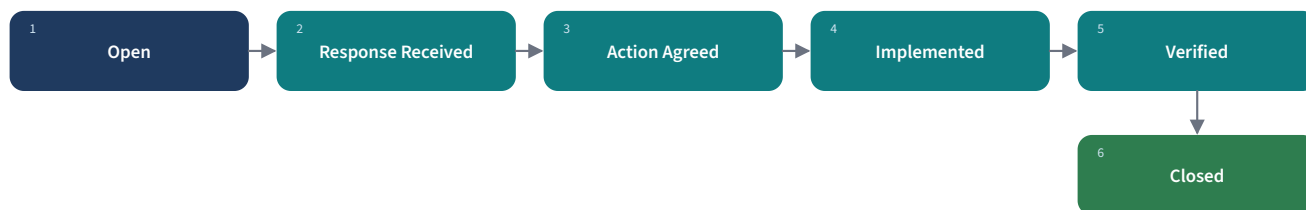


Other outcomes

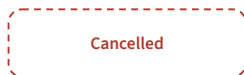
Cancelled

Model: ls.supplier.audit

Figure 82 — Lifecycle of a supplier audit.



Other outcomes



Model: ls.supplier.audit.finding

Figure 83 — Lifecycle of a supplier audit finding.

6 Procedure

- 1 **Plan.** From the dossier click **Start Audit** or create an audit: type, scope, basis (reference standards), planned date, lead auditor, team, supplier contact. Click **Plan**.
- 2 **Conduct.** Click **Start** on the first day; record dates; add findings with severity and description.
- 3 **Report.** Click **Draft Report**; set the outcome and write the conclusion; click **Issue Report**. The response due date defaults from the company *Audit Response Lead Time* (30 days by default).
- 4 **Collect responses.** For each finding click **Register Response** with the supplier's answer; then **Agree Action** with the corrective action and due date.
- 5 **Follow up.** When the supplier reports completion click **Mark Implemented**; verify (evidence, follow-up audit) and click **Verify** with the verification description, then **Close**.
- 6 **Close the audit.** Register the overall response (**Register Response** on the audit) and click **Close Audit**; update the dossier (SOP-QA-070 / 072).

7. Controls enforced by the software

- ✓ Dates: end $\geq$ start; response due $\geq$ report date; response not before the report is issued.
- ✓ An audit with a critical finding cannot conclude *Acceptable*.
- ✓ The report cannot be issued without outcome and written conclusion; states follow Plan $\rightarrow$ Start $\rightarrow$ Draft Report $\rightarrow$ Issue Report $\rightarrow$ Response $\rightarrow$ Close.
- ✓ Findings: response before agreeing an action; the agreed action and its due date are required; verification requires a description; a finding cannot close before the report is issued; critical/major findings need a documented corrective action.
- ✓ A planned audit cannot be deleted, only cancelled.

8. Good practices

- ★ Send the agenda and the document request list before the audit.
- ★ Grade findings consistently with your internal audit categories.
- ★ For critical findings, involve the Supplier Manager immediately: suspension may be required.

9. Pitfalls to avoid

- ✗ Accepting supplier answers without evidence of implementation.
- ✗ Audits whose report is issued months later.

10 Records and evidence

- Audit, report, findings with responses and verification

11 Performance indicators

- Open findings per supplier
- Report issue time
- Audits due vs done

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 7 — Outsourced Activities
- ICH Q7 — Good Manufacturing Practice for Active Pharmaceutical Ingredients
- ISO 19011:2018 — Guidelines for auditing management systems
- ISO 13485:2016 — Medical devices — Quality management systems

13. Demonstration

Case. On-site audit of a packaging printer: 1 major finding (no line clearance record between two customers' artworks), 2 minor. Outcome *Acceptable with Actions*, report issued 5 days after the audit; response due 30 days later. The supplier introduces a line clearance form; the auditor verifies 3 completed forms sent as evidence and closes the findings; the dossier stays approved.

Supplier performance evaluation and requalification

Module	ls_supplier_qualification	Version	_____
Menu	Supplier Qualification ► Monitoring	Effective date	_____
Owner (suggested)	Supplier Manager (QA)	Approved by	_____
Related SOPs	SOP-QA-070, SOP-QA-071, SOP-QA-073		

1 Objective

Monitor approved suppliers with objective performance data and periodic reviews, and requalify, restrict, suspend or disqualify them when evidence requires it.

2 Scope

All approved and conditionally approved suppliers.

3 Responsibilities

Role (software group)	Responsibility
Supplier Assessor / evaluator	Performs performance evaluations and reviews.
Supplier Manager	Takes status decisions and requalifies.

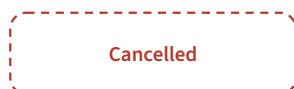
4 Definitions

Term	Definition
Performance score	Weighted score of on-time delivery, quality acceptance, documentation compliance and responsiveness (weights set on the company, default 30/40/15/15), rated A – Preferred, B – Acceptable, C – Improvement Required, D – Unacceptable.
Review decision	Maintain Approval, Maintain with Conditions, Requalification Required, Suspend, Disqualify.

5 Workflow in the software



Other outcomes



Model: ls.supplier.performance

Figure 84 — Lifecycle of a performance evaluation.

6 Procedure

- 1 **Evaluate performance.** At the defined frequency (for example quarterly) create a performance evaluation for the period; click **Recompute Delivery Counters** when the purchase-stock link is installed, or enter counters; enter lots inspected/rejected, nonconformities, complaints, documentation and responsiveness scores; click **Confirm**.
- 2 **Act on ratings.** Rating C or D: open a supplier finding or CAPA, inform the supplier, consider a for-cause audit.
- 3 **Periodic review.** When due (review interval of the category), create a periodic review: period, reviewer, click **Collect Evidence** (assessments, audits, performance evaluations of the period), write the summary and the decision with justification; click **Complete Review**.
- 4 **Change status.** For suspension or disqualification the Supplier Manager clicks **Change Status** with the reason; the decision is logged.
- 5 **Requalify.** Before approval expiry (reminders are raised by the company *Expiry Reminder* lead time, 60 days by default), click **Requalify** and repeat the relevant steps of SOP-QA-070.

7. Controls enforced by the software

- ✓ Period end after start; counters non-negative; late deliveries $\leq$ deliveries; rejected lots $\leq$ inspected lots; percentages 0–100.
- ✓ Two confirmed evaluations of the same dossier cannot overlap; confirmed evaluations cannot be deleted.
- ✓ Automatic delivery counters require the purchase-stock link (module *purchase_stock*).
- ✓ Reviews: an adverse decision must be justified; conditions must be documented when maintaining with conditions; completed reviews cannot be deleted or cancelled.
- ✓ Rating thresholds must decrease from A to C; at least one weight must be positive.

8. Good practices

- ★ Discuss results with key suppliers; share the scorecard.
- ★ Event-driven reviews after a recall, a critical complaint or a regulatory action at the supplier.

9. Pitfalls to avoid

- ✗ Suppliers left approved past their validity because nobody requalified them.
- ✗ Performance data entered by memory rather than from records.

10 Records and evidence

- Performance evaluations
- Periodic reviews with decisions
- Signature log entries

11 Performance indicators

- Suppliers rated C/D
- Approvals expired
- Reviews overdue

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 7 — Outsourced Activities
- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 9001:2015 — Quality management systems — Requirements
- ICH Q10 — Pharmaceutical Quality System

13. Demonstration

Case. Q2 evaluation of an ampoule supplier: 20 deliveries, 5 late (OTD 75 %), 12 lots inspected, 2 rejected (quality 83 %); score gives rating C. The Supplier Manager raises a supplier CAPA and schedules a for-cause audit. At the annual review the reviewer collects the audit, the two evaluations and decides *Maintain with Conditions*: tightened incoming inspection until two consecutive quarters rated B.

Control of purchases from non-approved suppliers

Module	ls_supplier_qualification · purchase	Version	_____
Menu	Company settings ► Purchase Order Control · Purchase ► Orders	Effective date	_____
Owner (suggested)	Supplier Manager with the Purchasing Manager	Approved by	_____
Related SOPs	SOP-QA-070, SOP-QA-072		

1 Objective

Prevent the purchase of GMP-relevant materials and services from suppliers that are not approved, whose approval expired, or for products outside their qualified scope.

2 Scope

Purchase orders in Odoo Purchase for suppliers with a qualification dossier and for GMP-relevant products.

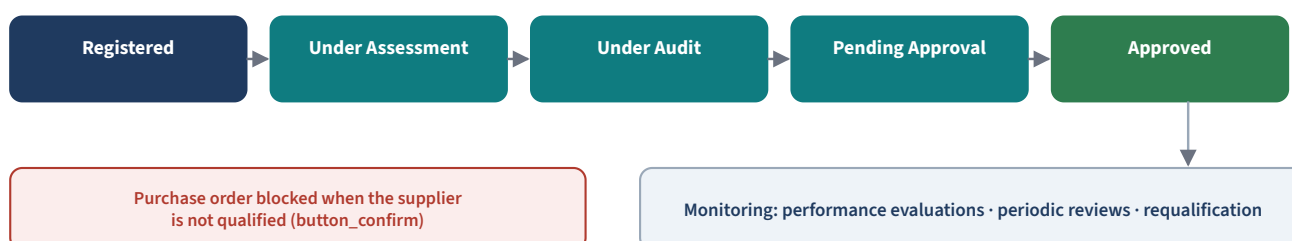
3 Responsibilities

Role (software group)	Responsibility
Supplier Manager	Sets the control level and keeps dossiers current.
Buyers	Check the qualification warning before confirming; never bypass a block.
QA	Handles exceptional purchases through a deviation or change request.

4 Definitions

Term	Definition
Control level	Company setting <i>Purchase Order Control</i> : <i>none</i> (no check), <i>warn</i> (issue posted in the order chatter — default at installation), <i>block</i> (confirmation refused).
Scope check	Optional setting <i>Check Qualified Scope on Purchase Orders</i> : each ordered product must be in the supplier's qualified scope, qualified and not expired.

5 Workflow in the software



Other states: Conditionally Approved, Suspended, Expired, Disqualified.

Figure 85 — The purchase order control closes the loop.

6 Procedure

- 1 Configure.** In the company settings of Supplier Qualification set *Purchase Order Control* to **block** for GMP operations and tick *Check Qualified Scope on Purchase Orders*. Record this configuration in the validation file.

- 2 **Order.** The buyer creates the order; the *Qualification Warning* field shows any issue before confirmation.
- 3 **Confirm.** On **Confirm Order** the system checks the supplier status (approved states), the approval expiry and, if enabled, the scope. When blocked, the confirmation is refused with the list of issues.
- 4 **Resolve.** Complete the qualification (SOP-QA-070), requalify an expired supplier (SOP-QA-072), or add the product to the qualified scope.
- 5 **Exceptional purchase.** If a purchase must be made before qualification (e.g. development batch), open a planned deviation or change request, approved by QA, and quarantine the goods until the decision.
- 6 **Verify periodically.** Quarterly, list confirmed orders to suppliers not approved at the order date (report from Purchase) to check that the control works.

7. Controls enforced by the software

- ✓ With level *block*, confirmation is refused for a supplier that is not approved or whose approval expired, and — with the scope check — for products outside the qualified scope.
- ✓ With level *warn*, the issue is only posted in the order chatter: the order can still be confirmed.
- ✓ The status is read in the company of the order, even when the buyer has no access to qualification dossiers.

8. Good practices

- ★ Use *block* in production companies; *warn* is acceptable only for non-GMP entities.
- ★ Keep the product list of the qualified scope aligned with the product catalogue.

9. Pitfalls to avoid

- ✗ Leaving the default *warn* level in a GMP company.
- ✗ Creating a new supplier record to bypass the block — the dossier is linked to the partner.

10 Records and evidence

- Company setting (with its change history)
- Blocked confirmation messages
- Planned deviations for exceptional purchases

11 Performance indicators

- Orders confirmed to non-approved suppliers (target 0)
- Blocked confirmations per month

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 5 — Production
- ICH Q7 — Good Manufacturing Practice for Active Pharmaceutical Ingredients
- ISO 13485:2016 — Medical devices — Quality management systems

13. Demonstration

Case. A buyer orders gelatin capsules from a supplier whose approval expired 10 days ago. On **Confirm Order** the system refuses: *"The approval of supplier X expired on ..."*. The Supplier Manager starts the requalification; meanwhile QA approves a planned deviation for a single delivery held in quarantine until requalification is complete.

II

Data integrity and compliance

The procedures that protect the reliability of GxP data and demonstrate that systems, instruments and environments are fit for use: audit trail, electronic signatures, validation, calibration, environmental monitoring and recalls.

Audit trail — ls_audit_trail	Calibration — ls_calibration
Electronic signatures — ls_electronic_signature	Environmental monitoring — ls_environmental_monitoring
Validation — ls_validation	Recalls — ls_recall

Audit trail

ls_audit_trail

ls_audit_trail records who changed what, when and from where, for the models and fields chosen in *audit rules*. Each change becomes an immutable entry of a per-company SHA-256 hash chain, so that altering, inserting or removing an entry is detectable. Integrity is verified on demand and daily by a scheduled action; *evidence packs* export a sealed ZIP (JSON, CSV and manifest) with its digest for inspectors; a controlled *retention run* removes aged entries while anchoring the chain. **Nothing is audited until a rule exists.** One-to-many and many-to-many relations and non-stored computed fields are never captured directly.

Menu	Content
Audit Trail ▶ Audit Log / Field Changes / Audit Dashboard	Entries and field changes
Audit Trail ▶ Integrity ▶ Verifications / Verify Now	Integrity checks
Audit Trail ▶ Evidence Packs	Sealed exports
Audit Trail ▶ Configuration ▶ Audit Rules / Retention Run	Scope and retention

Group	Main rights
Audit Trail / Viewer	Read the trail of their companies
Audit Trail / Auditor	Also verify integrity and produce evidence packs
Audit Trail / Administrator	Also configure rules and run approved retention

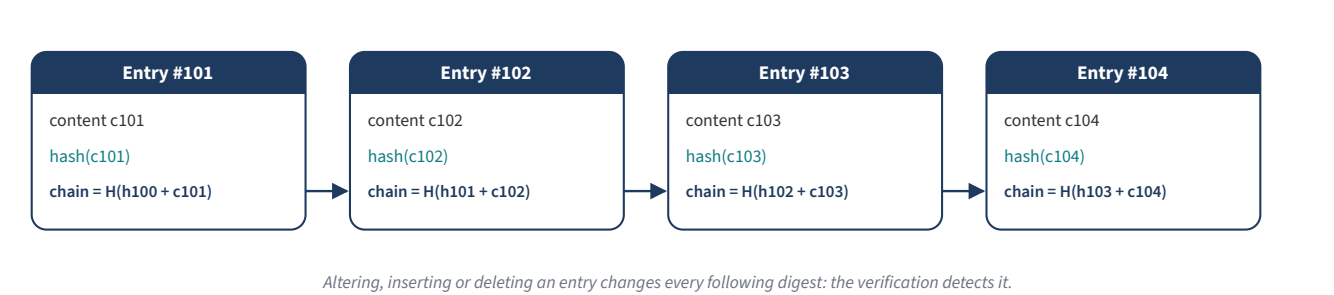


Figure 86 — How the hash chain makes tampering detectable.

Definition of the audit trail scope (rules per model)

Module	ls_audit_trail	Version	_____
Menu	Audit Trail ► Configuration ► Audit Rules	Effective date	_____
Owner (suggested)	Data Integrity lead / System owner	Approved by	_____
Related SOPs	SOP-DI-002, SOP-VAL-006, SOP-IT-002		

1 Objective

Define, justify and maintain the list of GxP records and fields whose creation, modification and deletion must be captured by the audit trail, based on a documented risk assessment.

2 Scope

All models of the installed modules that hold GxP data. The scope is part of the validated configuration of the system.

3 Responsibilities

Role (software group)	Responsibility
System owner / QA	Performs the data risk assessment and approves the scope.
Audit Trail Administrator	Creates and maintains audit rules; cannot hide changes from auditors.
Validation	Verifies the rules in the computerised system validation (SOP-VAL-006).

4 Definitions

Term	Definition
Audit rule	Model, operations audited (creation, modification, deletion), audited fields (empty = all stored fields except technical ones), excluded fields, justification.
GxP record	Data used to take a GMP decision or to demonstrate compliance.

5 Workflow in the software



Figure 87 — The audit trail supports ALCOA+ — attributable, contemporaneous, original.

6 Procedure

- 1 **Inventory.** List the models holding GxP data (e.g. deviations, CAPA, batch records, lab results, calibration records, stock lots) and their critical fields.
- 2 **Assess.** For each model decide which operations and fields need auditing; record the risk-based rationale.
- 3 **Create rules.** In [Audit Trail](#) ► [Configuration](#) ► [Audit Rules](#) click [New](#) : name, audited model, audit creations / modifications / deletions, audited fields or excluded fields, justification.
- 4 **Relations.** For data held in one-to-many or many-to-many relations (e.g. result lines), create a rule on the model on the other side of the relation.
- 5 **Verify.** Make a test change and check the entry in [Audit Log](#) and [Field Changes](#) before declaring the rule effective; include this test in the validation.
- 6 **Change control.** Any rule addition, change or removal goes through change control (SOP-QA-030) and is re-verified.

7. Controls enforced by the software

- ✓ Nothing is audited without a rule.
- ✓ Rules cannot target models of the audit engine itself, abstract models, transient models, or fields that do not belong to the model; a rule must audit at least one operation.
- ✓ Entries and field changes are immutable and cannot be deleted except through the retention run.
- ✓ Segregation of duties is enforced in the ORM: auditors cannot change the audited scope.

8. Good practices

- ★ Audit modifications and deletions of every GxP record type; creations where the creator/time matters.
- ★ Prefer an explicit field list for critical models, so that the scope is stable and reviewable.
- ★ Keep the rule list printed in the validation file and reconcile it at each periodic review.

9. Pitfalls to avoid

- ✗ Assuming that Odoo's chatter tracking is an audit trail — it is not tamper-evident and covers only tracked fields.
- ✗ Excluding fields to "reduce noise" without justification.

10 Records and evidence

- Audit rules with justification
- Data risk assessment
- Verification test evidence

11 Performance indicators

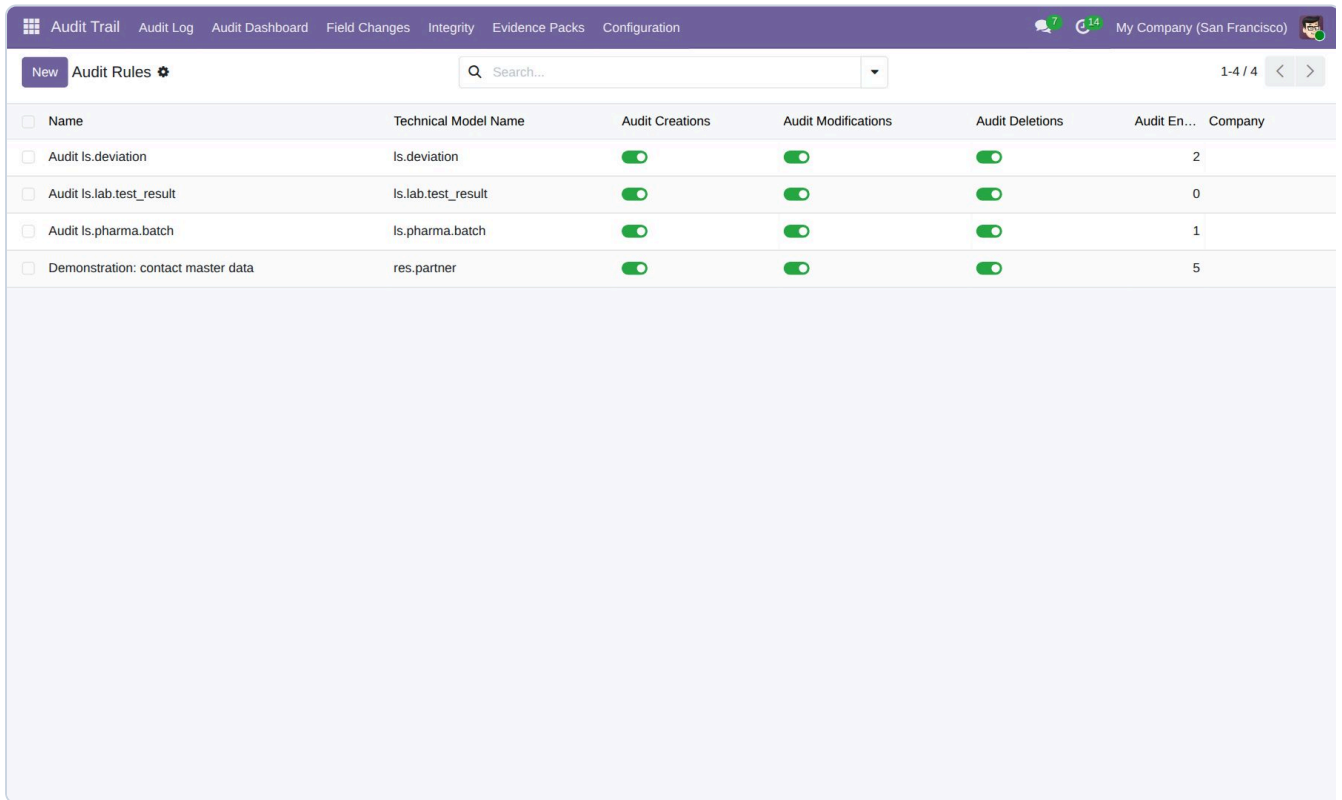
- GxP models without rule (target 0)
- Rule changes per year

12 References

- 21 CFR Part 11 — Electronic Records; Electronic Signatures
- EU GMP Annex 11 — Computerised Systems (check the revision in force)
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)
- MHRA — GxP Data Integrity Guidance and Definitions (2018)
- FDA Guidance — Data Integrity and Compliance With Drug CGMP, Questions and Answers (2018)

13. Demonstration

Case. For *ls.lab.test_result* the rule audits modifications and deletions of all stored fields; for *stock.lot* only the fields *expiration date* and *name*. A test edit of a result value creates an entry showing old value, new value, user, date and IP address — evidence kept in the OQ.



☐ Name	Technical Model Name	Audit Creations	Audit Modifications	Audit Deletions	Audit En...	Company
☐ Audit ls.deviation	ls.deviation	☒	☒	☒	2	
☐ Audit ls.lab.test_result	ls.lab.test_result	☒	☒	☒	0	
☐ Audit ls.pharma.batch	ls.pharma.batch	☒	☒	☒	1	
☐ Demonstration: contact master data	res.partner	☒	☒	☒	5	

Figure 88 — Audit rules defining the audited scope. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Audit trail review and integrity verification

Module	ls_audit_trail	Version	_____
Menu	Audit Trail ► Audit Log · Integrity ► Verify Now	Effective date	_____
Owner (suggested)	QA / Data Integrity lead	Approved by	_____
Related SOPs	SOP-DI-001, SOP-DI-003, SOP-QA-090, SOP-QC-012		

1 Objective

Review audit trail data at a frequency based on risk, as part of record review, and confirm regularly that the audit trail has not been altered.

2 Scope

Audit trail entries of all audited models; integrity of the hash chain of each company.

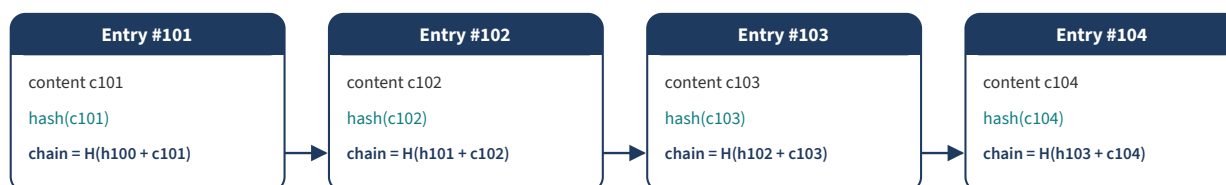
3 Responsibilities

Role (software group)	Responsibility
Record reviewers (QA, lab reviewers)	Review the audit trail of the records they approve (batch, results).
Audit Trail Auditor	Runs integrity verifications, investigates failures.
QA	Defines review frequencies and handles findings as deviations.

4 Definitions

Term	Definition
Review by exception	Focused review of changes that matter: modifications of results after entry, deletions, changes outside working hours, changes by unusual users.
Verification scope	Recent window, explicit range or complete chain.

5 Workflow in the software



Altering, inserting or deleting an entry changes every following digest: the verification detects it.

Figure 89 — Integrity verification walks the chain.

6 Procedure

- 1 Define frequency.** Set in this SOP which audit trails are reviewed with each record (e.g. results before approval, batch record before release) and which periodically (e.g. monthly for master data).
- 2 Review.** In **Audit Trail ► Audit Log** filter by model, record, user and period; open **Field Changes** to see old and new values; use **Open Audited Record** to see the context.

- 3 **Document.** Record the review (e.g. in the batch review or as a quality record) with its scope and outcome.
- 4 **Verify integrity.** The scheduled action *Audit Trail: Verify Integrity* runs daily; in addition an Auditor runs **Integrity ► Verify Now** monthly on the complete chain and before any inspection. Results are stored in **Integrity ► Verifications**.
- 5 **React.** A failed verification or a suspicious change opens a deviation (SOP-QA-020) with data integrity classification; preserve evidence; inform QA management.

7. Controls enforced by the software

- ✓ Verification records are immutable and cannot be deleted.
- ✓ Entries cannot be modified; a sealed entry cannot be modified; deletion only through the retention run.

8. Good practices

- ★ Train reviewers on what a suspicious pattern looks like (repeated edits of a result, deletions before approval).
- ★ Review the audit trail of the system configuration too (rules, users, access rights).

9. Pitfalls to avoid

- ✗ Reviewing thousands of entries without focus — the review becomes a signature without content.
- ✗ Ignoring failed scheduled verifications because "it is IT".

10 Records and evidence

- Review records
- Verification results
- Deviations on findings

11 Performance indicators

- Verifications failed (target 0)
- Records approved without audit trail review (target 0)

12 References

- EU GMP Annex 11 — Computerised Systems (check the revision in force)
- 21 CFR Part 11 — Electronic Records; Electronic Signatures
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)
- MHRA — GxP Data Integrity Guidance and Definitions (2018)

13. Demonstration

Case. Before approving sample LS-2026-0412, the reviewer filters the audit log on its results: one assay value was changed from 97.8 % to 98.4 % by the analyst 10 minutes after entry, with the comment "*transcription error, see printout p. 2*". The printout confirms 98.4 %. The reviewer documents the check in the review comment and approves.

Audit Trail

Audit Log

Audit Dashboard

Field Changes

Integrity

Evidence Packs

Configuration

7

14

My Company (San Francisco)

Audit Log

Q

Last 30 Days

Search...

1-8 / 8

☐	Chain P...	Event Date	Login	Model	Record ID	Record	Operation	Change...	Entry Type	Company	
☐		8 Sep 27, 12:12 PM	demo.reviewer	ls.pharma.batch	2	26T021 - Tablets 10 mg	Modification		1 Data Change	My Company (San Francisco)	
☐		7 Sep 27, 12:12 PM	demo.reviewer	ls.deviation	2	DEV/2026/00002	Modification		1 Data Change	My Company (San Francisco)	
☐		6 Sep 27, 12:12 PM	demo.reviewer	ls.deviation	2	DEV/2026/00002	Modification		1 Data Change	My Company (San Francisco)	
☐		5 Sep 27, 12:08 PM	__system__	res.partner	83	Production Operator 2 (demo)	Creation		2 Data Change	My Company (San Francisco)	
☐		4 Sep 27, 12:08 PM	__system__	res.partner	82	Production Operator (demo)	Creation		2 Data Change	My Company (San Francisco)	
☐		3 Sep 27, 12:07 PM	__system__	res.partner	81	QA Manager (demo)	Creation		2 Data Change	My Company (San Francisco)	
☐		2 Sep 27, 12:07 PM	__system__	res.partner	80	QA Reviewer (demo)	Creation		2 Data Change	My Company (San Francisco)	
☐		1 Sep 27, 12:07 PM	__system__	res.partner	79	QC Analyst (demo)	Creation		2 Data Change	My Company (San Francisco)	

Figure 90 — Audit trail entries created by the audit rules. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Audit Trail Audit Log Audit Dashboard Field Changes Integrity Evidence Packs Configuration 7 14 My Company (San Francisco)									
Integrity Verifications Q Search... 1-2 / 2									
☐	Reference	Executed On	Executed By	Source	Range Start	Range End	Entries V...	Result	Company
☐	VER/1/20260927 121201	Sep 27, 12:12 PM	QA Manager (demo)	Manual	Sep 26, 12:12 PM	Sep 27, 12:17 PM	8	Passed	My Company (San Francisco)
☐	VER/1/20260927 121201	Sep 27, 12:12 PM	QA Manager (demo)	Manual			8	Passed	My Company (San Francisco)

Figure 91 — Integrity verification: Passed. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Evidence packs for inspections

Module	ls_audit_trail	Version	_____
Menu	Audit Trail ► Evidence Packs	Effective date	_____
Owner (suggested)	Audit Trail Auditor (QA)	Approved by	_____
Related SOPs	SOP-DI-002, SOP-DI-004		

1 Objective

Provide inspectors and auditors with complete, sealed and verifiable extracts of the audit trail on request, without giving them uncontrolled access to the system.

2 Scope

Requests from regulatory inspectors, customer auditors, notified bodies and internal investigations.

3 Responsibilities

Role (software group)	Responsibility
Audit Trail Auditor	Prepares, generates and verifies the pack.
QA inspection coordinator	Receives the request, checks the scope, hands over the pack and records it.

4 Definitions

Term	Definition
Evidence pack	ZIP archive containing audit_entries.json, audit_entries.csv and manifest.json, with the SHA-256 digest of the archive stored on the record.
Chain positions	First and last chain positions exported, allowing the inspector to see the continuity.

5 Workflow in the software



Model: ls.audit_trail.evidence_pack

Figure 92 — An evidence pack is sealed at generation.

6 Procedure

- Record the request.** Note who asked, for what purpose, which records and period.
- Create the pack.** In **Audit Trail ► Evidence Packs** click **New**: purpose, from and to dates, model filter and user filter as needed.
- Generate.** Click **Generate**. The pack becomes immutable; the number of entries and field changes, chain positions and archive digest are recorded; an integrity verification is linked.
- Check.** Click **Verify Archive Digest** to confirm the stored archive matches its digest.

5 **Hand over.** Click **Download Archive** ; transmit it with the digest value; print the cover sheet if paper signatures are required.

6 **Keep.** The pack stays in the system; archive it if needed, never delete it.

7. Controls enforced by the software

- ✓ Only a draft pack can be generated; the criteria must match at least one entry; start ≤ end.
- ✓ Generated packs are immutable and cannot be deleted (archive instead).
- ✓ A mismatch between the stored archive and its digest is reported explicitly.

8. Good practices

- ★ Generate packs from the production system, not from a copy, and state it in the purpose.
- ★ Give the inspector the digest separately so they can check the file themselves.

9. Pitfalls to avoid

- ✗ Sending screenshots or spreadsheets edited by hand instead of the sealed pack.

10 Records and evidence

- Evidence pack record, archive and digest
- Request log

11 Performance indicators

- Time to answer an inspection request

12 References

- EU GMP Annex 11 — Computerised Systems (check the revision in force)
- 21 CFR Part 11 — Electronic Records; Electronic Signatures
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)

13. Demonstration

Case. An inspector asks for all changes made to the specifications of product P-100 in 2026. The Auditor creates a pack: purpose *"Inspection 2027-03, request #7"*, 01/01/2026–31/12/2026, model filter *ls.lab.specification*. After **Generate** and **Verify Archive Digest** , the ZIP and its digest are handed over within 30 minutes.

Audit Trail

Audit Log

Audit Dashboard

Field Changes

Integrity

Evidence Packs

Configuration

7

14

My Company (San Francisco)

New

Evidence Packs

EP-2026-001

1 / 1

Download Archive

Verify Archive Digest

Print Cover Sheet

Draft

Generated

Send message

Log note

1

1

Reference

EP-2026-001

SELECTION

Company ?

My Company (San Francisco)

From ?

Sep 26, 12:12 PM

To ?

Sep 27, 12:17 PM

Model Filter ?

User Filter ?

RESULT

Exported Entries

8

Exported Field Changes

13

First Chain Position

1

Last Chain Position

8

Archive Digest ?

bda6f4537c3213642f12211c56f6ec020134095862e51444362ff66e51e57dfb

Integrity Verification ?

VER/1/20260927 121201 (Passed)

Archive

EP-2026-001.zip

Generated By

QA Manager (demo)

Generated On

Sep 27, 12:12 PM

Active ?

PURPOSE

Inspection 2026-10,

request #7: changes to

deviations and batches.

QA Manager (demo)

12:12 PM

Audit Trail Evidence Pack created

Figure 93 — Generated evidence pack with its SHA-256 digest. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Retention and controlled purge of audit data

Module	ls_audit_trail	Version	_____
Menu	Audit Trail ► Configuration ► Retention Run	Effective date	_____
Owner (suggested)	QA with the Audit Trail Administrator	Approved by	_____
Related SOPs	SOP-QA-004, SOP-DI-003		

1 Objective

Keep audit trail data as long as the records it relates to, and remove aged entries only under an approved procedure that preserves the verifiability of the chain.

2 Scope

Audit trail entries of all companies.

3 Responsibilities

Role (software group)	Responsibility
QA	Sets the retention period consistent with record retention (SOP-QA-004) and authorises each run.
Audit Trail Administrator	Executes the approved run.

4 Definitions

Term	Definition
Retention anchor	Special entry written by the retention run that preserves the integrity of the remaining chain.
Retention period	Company setting <i>Audit Trail Retention (days)</i> ; runs also require <i>Allow Audit Trail Retention Runs</i> and a <i>Retention Procedure Reference</i> .

5 Procedure

- 1 Set retention.** Set the retention period in days to at least the longest retention of the audited records. Keep *Allow Audit Trail Retention Runs* unticked until a run is authorised.
- 2 Prepare.** Before a run, generate an evidence pack of the entries to be removed (SOP-DI-003) and archive it according to SOP-QA-004.
- 3 Authorise.** QA authorises the run with a change request or quality record; set the retention procedure reference (this SOP number) and tick *Allow Audit Trail Retention Runs*.
- 4 Execute.** The Administrator opens **Configuration ► Retention Run**, confirms the number of entries, enters the justification and clicks **Execute Retention Run**.
- 5 Verify.** Run a complete-chain verification (SOP-DI-002); untick *Allow* again.

6. Controls enforced by the software

- ✓ Entries cannot be deleted except by the retention run.
- ✓ The retention period cannot be negative; runs require a retention period and a retention procedure reference on the company.
- ✓ The run writes a retention anchor so that the remaining chain stays verifiable.

7. Good practices

- ★ Plan runs rarely (e.g. yearly) and under change control.
- ★ Test the readability of archived evidence packs before deleting source entries.

8. Pitfalls to avoid

- ✗ Purging to improve performance without regard to record retention.

9 Records and evidence

- Authorisation, evidence pack of removed entries, run justification, verification result

10 Performance indicators

- Runs executed without authorisation (target 0)

11 References

- EU GMP Annex 11 — Computerised Systems (check the revision in force)
- EudraLex Vol. 4, EU GMP Part I, Chapter 4 — Documentation
- 21 CFR 211.180 to 211.198 — Records and reports
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)

12. Demonstration

Case. Records must be kept 11 years. Retention is set to 4 018 days. In 2038, QA authorises a run under CC-2038-004; the pack of entries older than 11 years is generated and archived; the Administrator executes the run; the complete-chain verification passes.

Electronic signatures

ls_electronic_signature

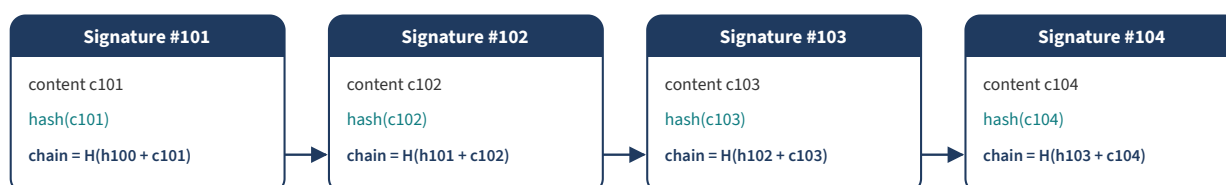
ls_electronic_signature provides electronic signatures designed to support 21 CFR Part 11 subparts B and C: printed name, date/time and meaning with every signature; identification code **and** password (or password only inside a continuous session); a binding statement; append-only signature log with a SHA-256 chain protected by a database trigger; logging of every attempt; lockout after repeated failures and immediate alert to the *Security Unit*. Signature *policies* say which model needs which signature meaning, how many signers, and whether a field transition is blocked until signed.

Important — current coverage

- ✓ A model can carry these signatures only if it inherits *ls.signature.mixin*. In the code examined, **no production model of the 22 LS modules inherits it** (only the test fixture module does). Signature policies can therefore not be applied to LS records until a model is extended by development and validated.
- ✓ Other modules have their own mechanisms: **ls_validation** re-authenticates the signer (login and password) and keeps immutable validation signatures; **ls_lab** records a *signature intent* (user, time, meaning) and states it is **not** a Part 11 signature; **ls_supplier_qualification** keeps an append-only, chained decision log **without** re-authentication.
- ✓ Your electronic signature SOPs must state, per record type, which mechanism applies — and use handwritten signatures on printed records where none is adequate.

Menu	Content
Electronic Signatures ► Signatures ► Signature Log / My Signatures / Signature Requests	Signatures and requests
Electronic Signatures ► Monitoring ► Signature Attempts / Signing Sessions / Integrity Checks / Verify Chain Now	Security monitoring
Electronic Signatures ► Configuration ► Signature Meanings / Signature Policies / Settings	Configuration

Group	Main rights
Signer · Viewer	Sign; read signatures
Auditor	Verify integrity, read attempts
Manager	Configure meanings and policies
Security Unit	Receives alerts on refused signatures and lockouts



Altering, inserting or deleting an entry changes every following digest: the verification detects it.

Figure 94 — Each signature is bound to the previous one.

Electronic signature policy: meanings, policies, signer commitment

Module	ls_electronic_signature	Version	_____
Menu	Electronic Signatures ► Configuration	Effective date	_____
Owner (suggested)	QA with the System Owner	Approved by	_____
Related SOPs	SOP-DI-011, SOP-DI-012, SOP-VAL-002		

1 Objective

Define where electronic signatures are used, what each signature means, who may sign, and obtain each signer's commitment that the electronic signature is the legally binding equivalent of their handwritten signature.

2 Scope

All electronic signatures executed in the system: ls_electronic_signature, validation signatures, lab signature intents and supplier decision log, as applicable.

3 Responsibilities

Role (software group)	Responsibility
QA	Owens the policy; approves meanings and policies.
Signature Manager	Configures meanings and policies after approval.
Each signer	Signs the commitment before first use; never shares credentials.
HR / QA	Keeps the signed commitments.

4 Definitions

Term	Definition
Meaning	Stated purpose of a signature (e.g. Authored, Reviewed, Approved, Released); code of 2–32 upper-case characters; optionally reason mandatory and restricted to authorised groups.
Policy	Rule on a model: meaning, trigger (manual or field transition from/to value), number of signatures, distinct signers, authorised signers, applicability domain, full credentials required.
Binding statement	Text shown to the signer at each signature (Settings).

5 Workflow in the software



Model: *ls.signature.request*

Figure 95 — Lifecycle of a signature request.

6 Procedure

- 1 **Map signature points.** List, per record type, the signature required by your procedures and the mechanism available (see the coverage box of this chapter). Where no adequate mechanism exists, keep a handwritten signature on the printed record.
- 2 **Define meanings.** In **Configuration ► Signature Meanings** create each meaning with a code, description, *Reason Mandatory* when needed and the authorised groups.
- 3 **Define policies.** For models inheriting the signature capability, create policies in **Configuration ► Signature Policies**: model, meaning, trigger, signatures required, distinct signers, authorised signers, domain, require full credentials.
- 4 **Set the binding statement.** In **Configuration ► Settings** enter the binding statement approved by legal/QA.
- 5 **Collect commitments.** Each signer signs a paper or electronic certification of their commitment before receiving the Signer role; for the US market, the organisation's certification to the FDA required by 21 CFR 11.100(c) is handled by Regulatory Affairs.
- 6 **Request signatures.** When needed, a user creates a signature request (record, meaning, expected signers, deadline) and clicks **Send Request**; signers use **Sign** or **Decline** with a reason.
- 7 **Review.** Review meanings and policies at each periodic system review (SOP-VAL-005).

7. Controls enforced by the software

- ✓ Meaning codes must follow the pattern (2–32 characters, upper case, digits, underscores).
- ✓ A policy can only target a model present in the registry that inherits the signature capability; a field-transition policy needs both the field and the target value; the domain must be a valid list.
- ✓ Signature records cannot be modified or deleted; a correction is a new signature with a reason.
- ✓ An operation governed by a policy is refused until the required number of valid signatures with the right meaning is present.
- ✓ Requests: only a draft request can be sent; only expected signers may sign; only pending requests can be declined; overdue requests expire by scheduled action.

8. Good practices

- ★ Few, clear meanings; avoid synonyms ("Checked" vs "Verified").
- ★ Require distinct signers when a policy needs two signatures.
- ★ Show the signature manifestation (name, date/time, meaning) on printed reports.

9. Pitfalls to avoid

- ✗ Assuming that clicking an *Approve* button is an electronic signature — it is not unless the mechanism re-authenticates and records meaning.
- ✗ Declaring the LS modules Part 11 compliant without validating the actual signature points.

10 Records and evidence

- Approved meanings and policies (configuration)
- Signer commitments
- Signature log

11 Performance indicators

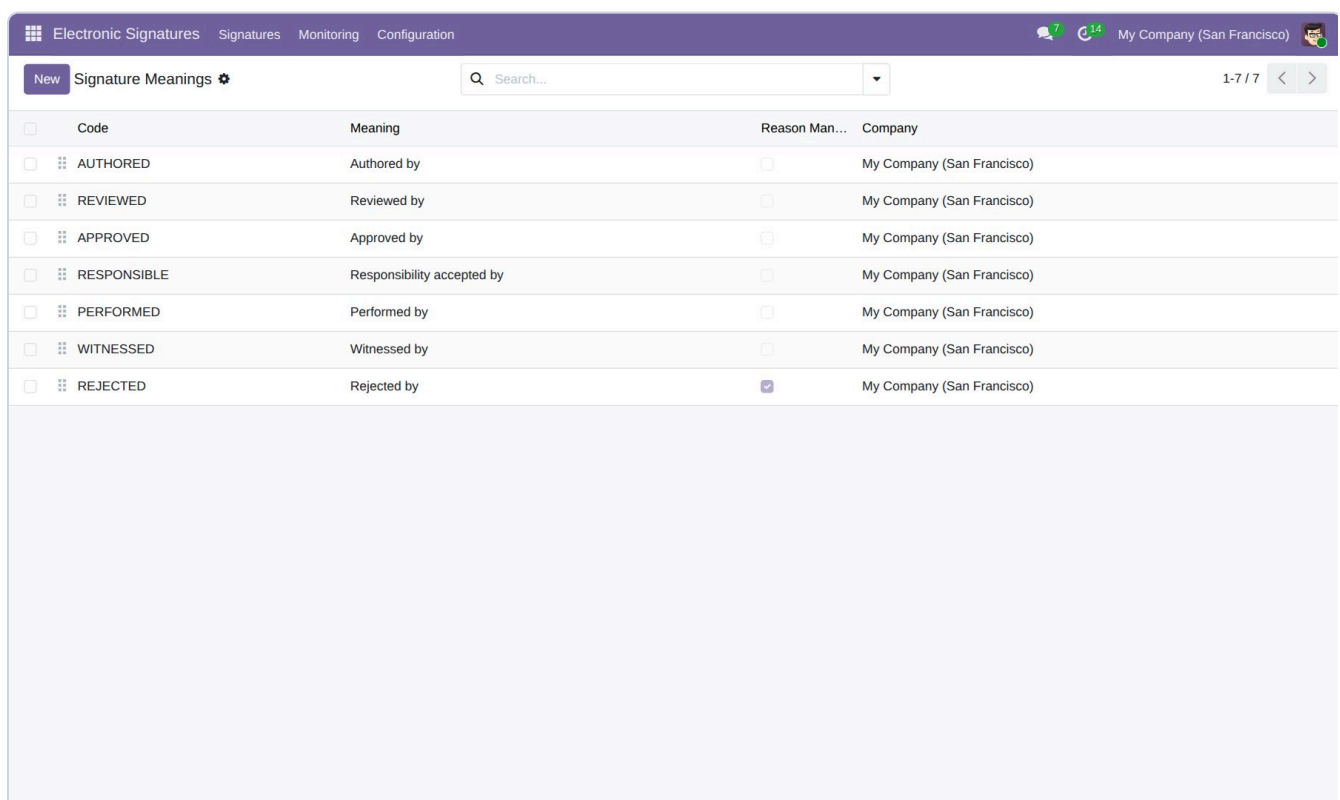
- Signers without commitment on file (target 0)
- Signature requests expired

12 References

- 21 CFR Part 11 — Electronic Records; Electronic Signatures
- EU GMP Annex 11 — Computerised Systems (check the revision in force)
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)

13. Demonstration

Case. QA maps signature points: validation protocols and reports → ls_validation signatures (re-authentication); lab result approval → lab signature intent, complemented by the printed CoA signed by hand until a Part 11 mechanism is validated; supplier approval → decision log plus signed approval form. Meanings AUTHORED, REVIEWED, APPROVED, RELEASED are created in ls_electronic_signature for future use.



The screenshot shows the 'Electronic Signatures' module configuration page in Odoo 19.0 Community. The page title is 'Signature Meanings' with a star icon. A search bar is present. The table below lists the configured signature meanings, each with a checkbox, a code, a meaning, a reason management checkbox, and a company name.

☐	Code	Meaning	Reason Man...	Company
☐	AUTHORED	Authored by	☐	My Company (San Francisco)
☐	REVIEWED	Reviewed by	☐	My Company (San Francisco)
☐	APPROVED	Approved by	☐	My Company (San Francisco)
☐	RESPONSIBLE	Responsibility accepted by	☐	My Company (San Francisco)
☐	PERFORMED	Performed by	☐	My Company (San Francisco)
☐	WITNESSED	Witnessed by	☐	My Company (San Francisco)
☐	REJECTED	Rejected by	☒	My Company (San Francisco)

Figure 96 — Signature meanings. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Management of credentials, account lockout and security alerts

Module	ls_electronic_signature	Version	_____
Menu	Electronic Signatures ► Configuration ► Settings · Monitoring ► Signature Attempts	Effective date	_____
Owner (suggested)	IT security with QA	Approved by	_____
Related SOPs	SOP-DI-010, SOP-IT-001, SOP-QA-013		

1 Objective

Ensure that each user identity is unique, credentials remain under the sole control of their owner, repeated failed signature attempts lock the identity, and security events are investigated.

2 Scope

User accounts of the system and the signature controls of ls_electronic_signature (and ls_validation re-authentication).

3 Responsibilities

Role (software group)	Responsibility
IT administrator	Creates, disables and never reassigns accounts; resets passwords only after identity check.
Security Unit (group)	Receives alerts, investigates refused signatures and lockouts.
Users	Keep passwords secret; report suspected compromise at once.

4 Definitions

Term	Definition
Continuous session	Period during which a signer may sign with password only; closed after the idle timeout.
Lockout	Refusal of signatures for an identification code after the configured number of failures, for the configured duration.

5 Procedure

- 1 Configure.** In [Electronic Signatures ► Configuration ► Settings](#) set *Continuous Session Timeout (minutes)* (default 15), *Failures Before Lockout* (default 3) and *Lockout Duration (minutes)* (default 15); check the *Database Immutability Trigger* state and the password verification API mode.
- 2 Assign the Security Unit.** Add the security officers to the group *Security Unit*; they receive an e-mail on each refused signature.
- 3 Manage accounts.** One account per person; never reuse or rename an account for another person; disable accounts of leavers the same day.
- 4 Monitor.** The Security Unit reviews [Monitoring ► Signature Attempts](#) weekly (results: incorrect password, identification code mismatch, not authorised, blocked by lockout, reason missing, precondition not met, system error).

- 5 **Investigate.** Each lockout or identification mismatch is checked with the user; suspected misuse becomes a deviation (data integrity).
- 6 **Close idle sessions.** The scheduled action closes idle signing sessions every 15 minutes; administrators can close a session manually.

6. Controls enforced by the software

- ✓ A signature requires identification code and password, or password only within a continuous session; both are required again after the session lapses.
- ✓ A signature under another person's identification code is refused.
- ✓ Every attempt is logged in an immutable security log; refusals trigger an alert to the Security Unit.
- ✓ An identification code is locked out after repeated failures.
- ✓ `ls_validation`: the login entered must match the connected user and the password must be verified, otherwise the signature is not recorded.

7. Good practices

- ★ Align password rules with IT policy (length, rotation where required, no reuse).
- ★ Never lower the failure threshold to zero or disable alerts "temporarily".

8. Pitfalls to avoid

- ✗ Shared "team" accounts in the laboratory or on the shop floor.
- ✗ Administrators resetting a password over the phone without identity verification.

9 Records and evidence

- Settings configuration
- Signature attempt log
- Account creation/disable records
- Investigation records

10 Performance indicators

- Lockouts per month
- Accounts of leavers not disabled within 24 h (target 0)

11 References

- 21 CFR Part 11 — Electronic Records; Electronic Signatures
- EU GMP Annex 11 — Computerised Systems (check the revision in force)
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)

12. Demonstration

Case. On Monday the Security Unit receives three alerts "*Identification code does not match the session*" from the same workstation in the packaging area. Investigation: an operator tried to sign with a colleague's login left open on the terminal. A deviation is opened; the session timeout is confirmed at 15 minutes; the team is retrained on account sharing.

Electronic Signatures

Signatures

Monitoring

Configuration

7

14

My Company (San Francisco)

Menu

Signature Attempts

Q

Failures Only

x

Search...

Attempted At

Login Attempted

Session User

Result

Ip Address

</

Figure 97 — Log of signature attempts. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Verification of the signature chain

Module	ls_electronic_signature	Version	_____
Menu	Electronic Signatures ► Monitoring ► Integrity Checks / Verify Chain Now	Effective date	_____
Owner (suggested)	Signature Auditor (QA)	Approved by	_____
Related SOPs	SOP-DI-010, SOP-DI-002		

1 Objective

Confirm regularly that no electronic signature has been altered, inserted or removed, and that each signature remains bound to the content that was signed.

2 Scope

The signature log of ls_electronic_signature, the validation signature records, and the supplier decision log chain.

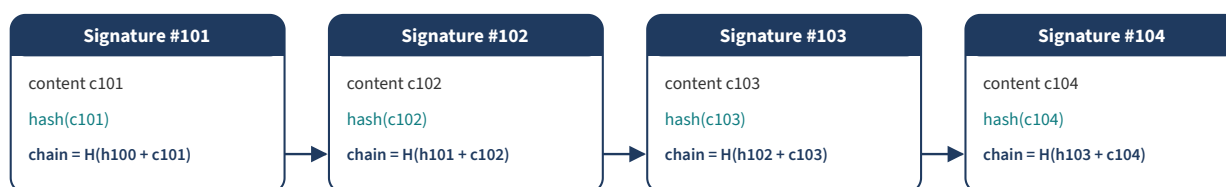
3 Responsibilities

Role (software group)	Responsibility
Signature Auditor	Runs and reviews verifications, investigates failures.
QA	Handles failures as critical data integrity deviations.

4 Definitions

Term	Definition
Record hash / chain hash	SHA-256 of the signed payload, and digest binding it to the previous signature of the same company.
Integrity check	Result Passed or Failed of a verification, manual or scheduled.

5 Workflow in the software



Altering, inserting or deleting an entry changes every following digest: the verification detects it.

Figure 98 — Signature chain verification.

6 Procedure

- Scheduled verification.** The scheduled action *Electronic Signatures: verify hash chain* runs daily; results appear in **Monitoring ► Integrity Checks**.
- Manual verification.** Monthly and before inspections, run **Monitoring ► Verify Chain Now** and record the result.

- 3 **Single signature.** To check one signature, open it in the signature log and click **Verify Integrity** ; **Open Signed Record** shows the record.
- 4 **Supplier decisions.** In **Supplier Qualification ► Monitoring ► Signature Log** use **Verify Chain** .
- 5 **Failure.** Stop signing on the affected scope, open a critical deviation, preserve database backups, involve IT and QA management.

7. Controls enforced by the software

- ✓ Signature records are append-only (ORM and database trigger refuse update and delete).
- ✓ A signature whose record no longer exists or whose payload no longer matches is reported by the verification.

8. Good practices

- ★ Include the chain verification in the periodic review of the system (SOP-VAL-005).
- ★ Keep the verification results with the validation evidence.

9. Pitfalls to avoid

- ✗ Restoring a database backup without re-verifying the chain afterwards.

10 Records and evidence

- Integrity check results
- Deviations on failures

11 Performance indicators

- Failed checks (target 0)
- Days since last successful verification

12 References

- 21 CFR Part 11 — Electronic Records; Electronic Signatures
- EU GMP Annex 11 — Computerised Systems (check the revision in force)

13. Demonstration

Case. The monthly manual verification of March passes for 1 482 signatures. In April, after a database restore on the test server, a verification on production is run again as a precaution: passed; the result is attached to the change record of the restore.

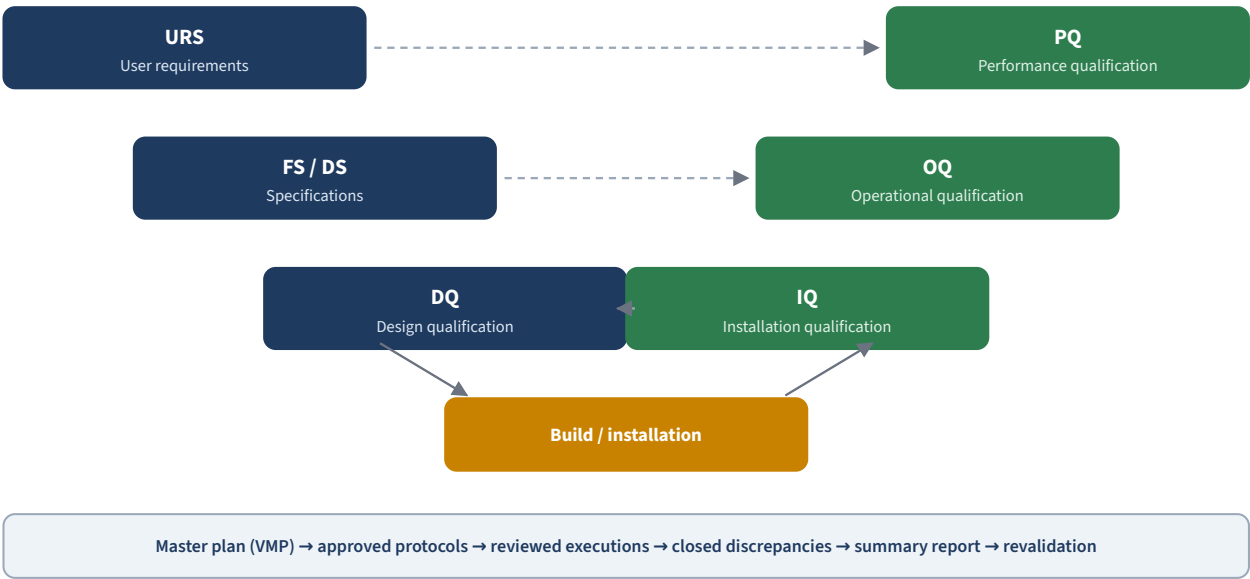
Validation

ls_validation

ls_validation manages validation master plans, validation items (equipment, utilities, facilities, processes, cleaning procedures, computerised systems, analytical methods, shipping lanes) with their validation status and validity, protocols of eight types (DQ, IQ, OQ, PQ, PV, CV, CSV, MV) with test cases and acceptance criteria, execution records with per-test verdicts, discrepancies, and summary reports. Key steps are signed with re-authentication (login and password) and the signatures are immutable.

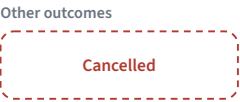
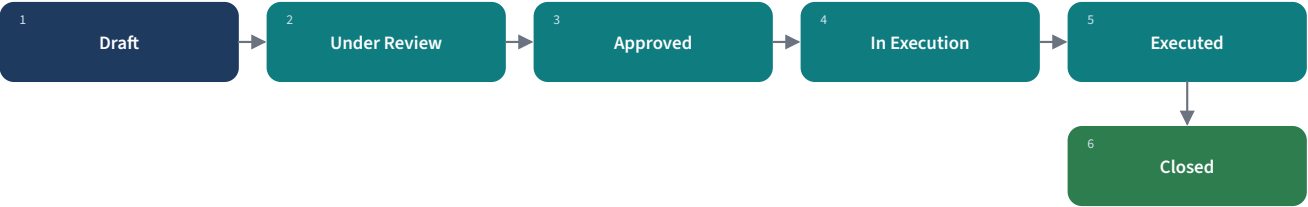
Menu	Content
Validation ► Planning ► Master Plans / Validation Items / Expiring or Expired	Planning
Validation ► Execution ► Protocols / Execution Records / Discrepancies	Execution
Validation ► Reporting ► Summary Reports / Electronic Signatures	Reports and signatures

Group	Main rights
Validation: Viewer · Engineer	Read; author protocols, execute tests
Validation: Approver	Review and approve with signature
Validation: Manager	Manage plans, items, revalidation



Dashed arrows link each specification level to the test that verifies it.

Figure 99 — The V-model of qualification.



Model: ls.validation.protocol

Figure 100 — Lifecycle of a protocol.

Validation master plan (VMP)

Module	ls_validation	Version	_____
Menu	Validation ► Planning ► Master Plans	Effective date	_____
Owner (suggested)	Validation Manager	Approved by	_____
Related SOPs	SOP-VAL-002, SOP-VAL-005, SOP-VAL-006		

1 Objective

Describe the validation policy, strategy, responsibilities and the list of items to validate, and keep it approved, active and current.

2 Scope

All GxP equipment, utilities, facilities, processes, cleaning procedures, computerised systems, analytical methods and transport lanes of the site.

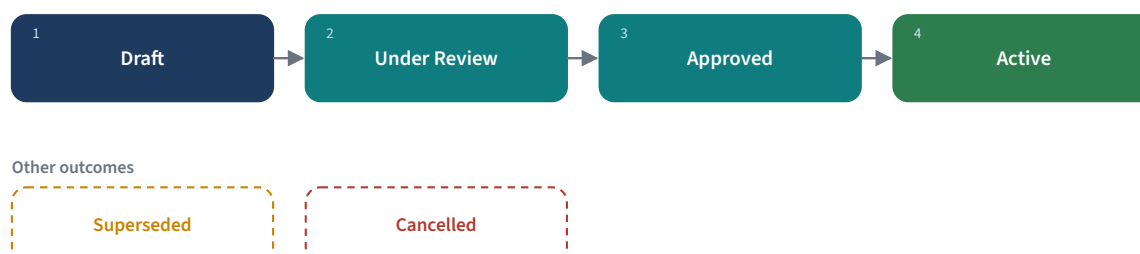
3 Responsibilities

Role (software group)	Responsibility
Validation Manager	Writes and maintains the VMP and the item list.
Reviewer	Reviews the VMP (QA, engineering).
Approver	Approves with electronic signature.

4 Definitions

Term	Definition
Validation item	Object to validate, with type, GxP impact (none, indirect, direct), criticality (low, medium, high), owner and periodic review interval.
Validation status	Not Validated, In Validation, Validated, Expiring Soon, Expired, Retired — derived from the approved reports.

5 Workflow in the software



Model: ls.validation.master.plan

Figure 101 — Lifecycle of a validation master plan.

6 Procedure

- 1 Register items.** In **Validation Items** create each item: code, designation, type, GxP impact, criticality, manufacturer, model, serial number, location, owner, periodic review interval.

- 2 **Write the VMP.** In **Master Plans** click **New** : title, scope, validation strategy (risk-based approach, V-model, deliverables per item type), responsibilities, review interval; add the items in scope.
- 3 **Review.** Click **Submit for Review** ; the reviewer comments; **Back to Draft** if changes are needed.
- 4 **Approve.** The approver clicks **Approve and Sign** (login and password), then **Activate** on the effective date.
- 5 **Maintain.** When scope or strategy changes, click **New Version** on the approved or active plan; the new version supersedes the old one when activated.
- 6 **Review periodically.** Review the VMP at its *Next Review Date* and report progress in management review.

7. Controls enforced by the software

- ✓ The scope must be documented before review; only a draft plan can be submitted; only a plan under review can be approved or returned to draft.
- ✓ Only an approved plan can be activated; a new version only from an approved or active plan; approved/active content is frozen.
- ✓ An active or superseded plan cannot be cancelled; plans that are not draft or cancelled cannot be deleted.
- ✓ Signatures require re-authentication and are immutable.

8. Good practices

- ★ Scale the deliverables to GxP impact and criticality (GAMP 5 categories for systems).
- ★ Include cleaning, transport and computerised systems — they are often forgotten.

9. Pitfalls to avoid

- ✗ A VMP that lists items but no strategy.
- ✗ Items validated long ago but never reviewed.

10 Records and evidence

- Approved VMP with signatures
- Validation item register

11 Performance indicators

- % of items validated
- Items expired
- VMP review on time

12 References

- EU GMP Annex 15 (2015) — Qualification and Validation
- ISPE GAMP 5 Guide, Second Edition (2022) — A Risk-Based Approach to Compliant GxP Computerized Systems
- FDA Guidance — Process Validation: General Principles and Practices (2011)
- ICH Q10 — Pharmaceutical Quality System

13. Demonstration

Case. VMP 2027 lists 64 items: 28 equipment, 6 utilities, 9 areas, 7 processes, 5 cleaning procedures, 6 computerised systems (including the Odoo LS suite), 3 transport lanes. The strategy states: direct-impact equipment → IQ/OQ/PQ; indirect → IQ/OQ; GAMP category 4 systems → CSV with risk-based testing. Approved and activated on 15 January.

Validation
Planning
Execution
Reporting

7
14
My Company (San Francisco)

New
Validation Master Plans
VMP/2026/0001 v1 - Site Validation Master Plan

Protocols
1
Signatures
1

1 / 1

New Version

Draft
Under Review
Approved
Active

Send message
Log note
Activity

VMP/2026/0001

Site Validation Master Plan

DOCUMENT CONTROL

Version
1

Author
OdooBot

Reviewer

Approved by
QA Manager (demo)

Approval Date
Sep 27, 12:10 PM

VALIDITY

Effective Date
Sep 27

Review Interval (months)
24

Next Review Date
Sep 27, 2028

Company
My Company (San Francisco)

Scope
Strategy
Responsibilities
Items in Scope
Protocols
Electronic Signatures

This demonstration plan covers the qualification of the production and quality control equipment of the site, and the validation of the cleaning procedures of production line A.

Today

QC Analyst (demo)
12:10 PM

None → 09/27/2026 12:10:59 PM (Approval Date)

None → QA Manager (demo) (Approved by)

None → 09/27/2026 (Effective Date)

Draft → Active (Status)

QA Manager (demo)
12:10 PM

Master plan is now active.

QA Manager (demo)
12:10 PM

Approved and signed electronically.

QC Analyst (demo)
12:10 PM

Submitted for review.

OdooBot
10:39 AM

Validation Master Plan created

Figure 102 — Validation master plan. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Drafting and approval of qualification protocols: DQ, IQ, OQ, PQ

Module	ls_validation	Version	_____
Menu	Validation ► Execution ► Protocols	Effective date	_____
Owner (suggested)	Validation Manager	Approved by	_____
Related SOPs	SOP-VAL-001, SOP-VAL-003, SOP-VAL-006		

1 Objective

Write qualification protocols whose test cases each have a clear procedure and a pre-defined acceptance criterion, and approve them before any execution.

2 Scope

Design, installation, operational and performance qualification protocols (DQ, IQ, OQ, PQ). Process, cleaning, method and computerised system validation protocols follow the same steps (SOP-VAL-006).

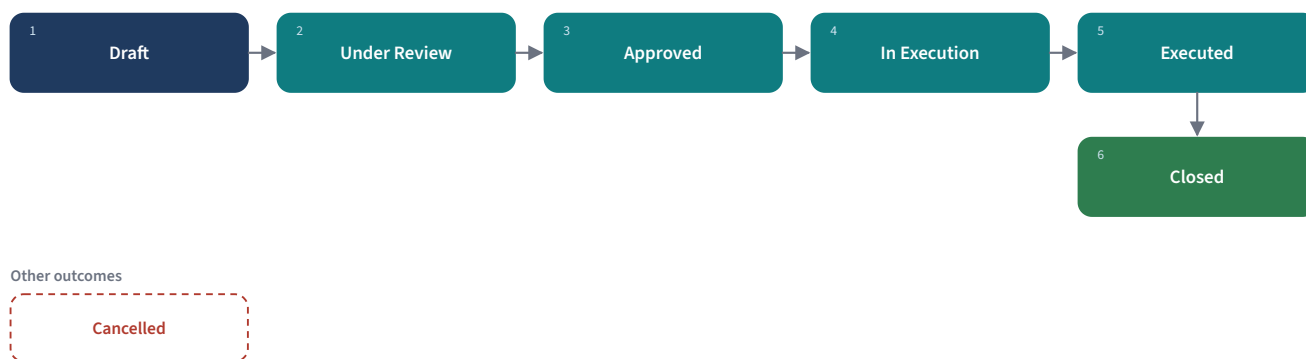
3 Responsibilities

Role (software group)	Responsibility
Validation Engineer (author)	Writes the protocol and test cases.
Reviewer	Checks traceability to requirements and adequacy of criteria.
Approver (QA)	Approves with signature before execution.

4 Definitions

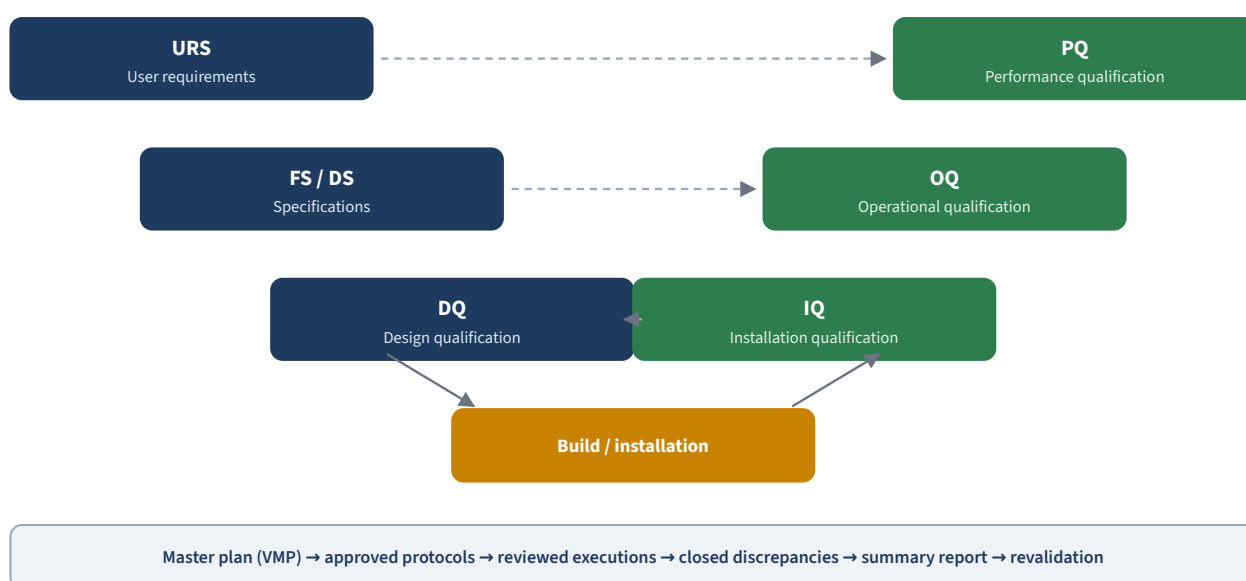
Term	Definition
DQ	Documented verification that the proposed design is suitable for the intended purpose.
IQ	Documented verification that the equipment is installed according to the approved design and manufacturer's recommendations.
OQ	Documented verification that it operates as intended throughout the anticipated operating ranges.
PQ	Documented verification that it performs effectively and reproducibly in routine conditions.
Critical test	Test case whose failure prevents the conclusion <i>Validated</i> .

5 Workflow in the software



Model: ls.validation.protocol

Figure 103 — Protocol lifecycle.



Dashed arrows link each specification level to the test that verifies it.

Figure 104 — Each qualification stage answers a specification level.

6 Procedure

- 1 **Create.** In **Protocols** click **New** : title, protocol type, validation item, master plan, objective, scope, prerequisites, reference documents, planned start and end.
- 2 **Write test cases.** Add test cases: number, title, objective, procedure, acceptance criterion (mandatory), expected evidence; tick *Critical Test* where applicable.
- 3 **Review.** Click **Submit for Review** ; use **Back to Draft** for corrections.
- 4 **Approve.** The approver clicks **Approve and Sign** and re-enters login and password.
- 5 **Change after approval.** If the protocol must change, click **New Version** ; never modify an approved protocol.

7. Controls enforced by the software

- ✓ Only a draft protocol can be submitted; a protocol without test cases cannot be approved; every test case needs an acceptance criterion.
- ✓ The item must be in the scope of the referenced master plan; planned end $\geq$ planned start.
- ✓ Test cases cannot be changed after approval — a new protocol version is required.
- ✓ Executions can only be created from an approved protocol.
- ✓ Protocols with execution records cannot be cancelled; approved or executed protocols are frozen.

8. Good practices

- ★ Trace each test to a user requirement or risk (reference in the objective).
- ★ Write criteria as measurable values with units and tolerances.
- ★ Mark as critical the tests linked to product quality or patient safety.

9. Pitfalls to avoid

- ✗ Criteria such as "works correctly".
- ✗ Starting tests before approval "to save time" — the results are not valid.

10 Records and evidence

- Approved protocol with signatures

11 Performance indicators

- Protocols approved before execution (target 100 %)
- Review cycles per protocol

12 References

- EU GMP Annex 15 (2015) — Qualification and Validation
- ISPE GAMP 5 Guide, Second Edition (2022) — A Risk-Based Approach to Compliant GxP Computerized Systems
- ISO 13485:2016 — Medical devices — Quality management systems

13. Demonstration

Case. OQ of autoclave AC-02: 18 test cases, e.g. *OQ-07 Chamber temperature uniformity, empty chamber, 121 °C/15 min: all probes 121.0–124.0 °C during the plateau* (critical). The protocol is reviewed by QA and approved with signature before the tests are scheduled.

Validation
Planning
Execution
Reporting

7
14
My Company (San Francisco)

New
Protocols
PROT/2026/0001 v1 - Operational Qualification ...

Executions
1
Signatures
1

1 / 1

New Execution
New Version
Cancel
Draft
Under Review
Approved
In Execution
Executed
Closed

Send message
Log note
Activity

PROT/2026/0001

Operational Qualification - Steam Sterilizer 500 L

SUBJECT

DOCUMENT CONTROL

Protocol Type
OQ - Operational Qualification

Validation Item
[EQ-0001] Steam Sterilizer 500 L

Master Plan
VMP/2026/0001 v1 - Site Validation Ma...

Company
My Company (San Francisco)

Version
1

Author
Odoobot

Reviewer

Approved by
QA Manager (demo)

Approval Date
Sep 27, 12:11 PM

Planned Start

Planned End

Test Cases
Objective
Scope
Prerequisites
Reference Documents
Executions
Electronic Signatures

Test Case	Title	Acceptance Criteria	Critical T...
TC-001	Chamber temperature con...	The chamber temperature remains within 121.0 degrees Celsius and 124.0 degrees Celsius during the whole exposure phase.	✓
TC-002	Chamber pressure control	The chamber pressure remains within the range declared in the operating manual during the whole exposure phase.	✓

Today

QC Analyst (demo)
12:11 PM

Approved → In Execution (Status)

QC Analyst (demo)
12:11 PM

None → 09/27/2026 12:11:00 PM (Approval Date)

None → QA Manager (demo) (Approved by)

Draft → Approved (Status)

QA Manager (demo)
12:11 PM

Approved and signed electronically.

QC Analyst (demo)
12:11 PM

Submitted for review.

Odoobot
10:39 AM

Validation Protocol created

Figure 105 — Protocol with test cases. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Protocol execution and management of discrepancies

Module	ls_validation	Version	_____
Menu	Validation ► Execution ► Execution Records / Discrepancies	Effective date	_____
Owner (suggested)	Validation Manager	Approved by	_____
Related SOPs	SOP-VAL-002, SOP-VAL-004, SOP-QA-022		

1 Objective

Execute approved protocols contemporaneously, record the observed result and verdict of each test, and handle every failure as a discrepancy investigated before approval.

2 Scope

All execution rounds of approved protocols.

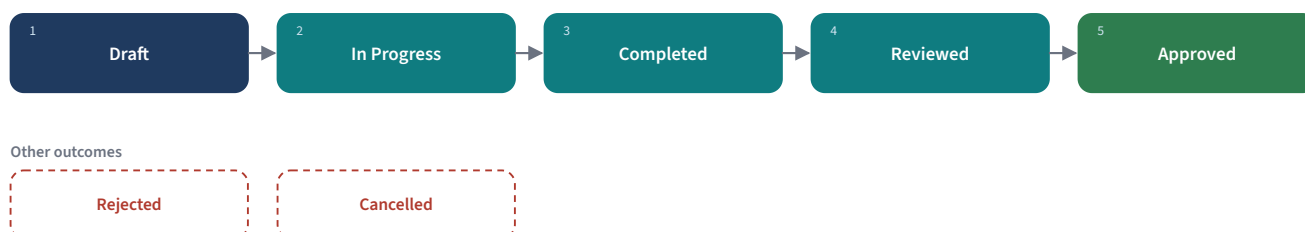
3 Responsibilities

Role (software group)	Responsibility
Executor (Validation Engineer)	Performs the tests and records results and evidence.
Reviewer	Reviews the execution; must be different from the executor.
Approver	Approves the execution with signature.
Discrepancy owner	Investigates and resolves discrepancies.

4 Definitions

Term	Definition
Execution round	Numbered execution of a protocol; a new round is created after a rejected execution.
Verdict	Pending, Pass, Fail or Not Applicable per test case.
Discrepancy	Failure or unexpected event, classified Minor, Major or Critical.

5 Workflow in the software

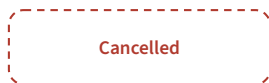


Model: ls_validation.execution

Figure 106 — Lifecycle of an execution record.



Other outcomes



Model: *ls.validation.discrepancy*

Figure 107 — Lifecycle of a validation discrepancy.

6 Procedure

- 1 **Create the execution.** From the approved protocol click **New Execution**; result lines are generated from the test cases (**Refresh Test Cases** while draft or in progress).
- 2 **Start.** Click **Start** when testing begins.
- 3 **Record results.** For each test record the observed result and verdict at the time of the test; attach raw data (printouts, logger files).
- 4 **Raise discrepancies.** For each failed test click the warning icon *Create a discrepancy* on the result line (or create one in **Discrepancies**): classification, description, immediate action.
- 5 **Investigate.** The owner clicks **Start Investigation**, documents root cause, corrective action and impact assessment, ticks *CAPA Required* and enters the CAPA reference if needed, clicks **Resolve**, then **Close and Sign**.
- 6 **Complete.** When all lines have a verdict and observed result, click **Complete**.
- 7 **Review and approve.** A reviewer (not the executor) clicks **Review and Sign**; the approver clicks **Approve and Sign**. If the execution is not acceptable, **Reject** with a reason and start a new round.
- 8 **Close the protocol.** When executions are approved, close the protocol (**Close**).

7. Controls enforced by the software

- ✓ Executions only from approved protocols; only a draft execution starts; only an execution in progress completes.
- ✓ Completion requires every line with a verdict and an observed result, and every failed test linked to a discrepancy.
- ✓ The reviewer must differ from the executor; approval only after review and with no open discrepancy.
- ✓ Discrepancies: investigation only from open; resolution requires root cause, corrective action and impact assessment; a CAPA reference is required when flagged; only resolved discrepancies close; closed ones are frozen and cannot be cancelled.
- ✓ Results of a completed execution cannot be modified — record a new round; an approved execution cannot be cancelled.

8. Good practices

- ★ Record results as the test is done (contemporaneous), never from memory at the end of the day.
- ★ Use "Not Applicable" only with a justification in the observed result.

9. Pitfalls to avoid

- ✗ Editing the acceptance criterion to make a result pass.
- ✗ Closing discrepancies with "retest passed" without root cause.

10 Records and evidence

- Execution with results, evidence and signatures
- Discrepancies with investigation

11 Performance indicators

- Discrepancies per protocol
- Executions rejected
- Time to close discrepancies

12 References

- EU GMP Annex 15 (2015) — Qualification and Validation
- ISPE GAMP 5 Guide, Second Edition (2022) — A Risk-Based Approach to Compliant GxP Computerized Systems
- 21 CFR Part 11 — Electronic Records; Electronic Signatures

13. Demonstration

Case (continued). During OQ-07 one probe reads 124.6 °C for 40 s. The executor sets the verdict *Fail* and creates discrepancy DISC-2027-005 (Major). Investigation: the probe was touching the chamber wall (setup error); impact: none on the equipment; corrective action: probe fixture added to the procedure; retest in round 2 passes. The discrepancy is resolved and closed with signature; round 2 is reviewed and approved.

The screenshot displays the Odoo 19.0 Community software interface for an execution record. The top navigation bar includes 'Validation', 'Planning', 'Execution', and 'Reporting'. The user is logged in as 'My Company (San Francisco)'. The main header shows 'Execution Records' and 'EXE/2026/0002 - Operational Qualification - Ste...'. There are buttons for 'Discrepancies' (1) and 'Signatures' (0). The record is in the 'In Progress' state, with tabs for 'Draft', 'In Progress', 'Completed', 'Reviewed', and 'Approved'. A 'Send message' button is also visible. The record details include:

- SUBJECT:** EXE/2026/0002
- EXECUTION:** Executed by QC Analyst (demo), Start Sep 27, 12:11 PM, End
- Protocol:** PROT/2026/0001 v1 - Operational Qualifi...
- Protocol Type:** OQ - Operational Qualification
- Validation Item:** [EQ-0001] Steam Sterilizer 500 L
- Execution Round:** 1
- Company:** My Company (San Francisco)

The **RESULT SUMMARY** section shows:

Category	Count	Category	Count
Tests	3	Pending	0
Passed	2	Not Applicable	0
Failed	1	Overall Result	Fail
		Open Discrepancies	1

Below the summary is a table with tabs for 'Results', 'Discrepancies', 'Remarks', and 'Electronic Signatures'. The 'Results' tab is active, showing a table with columns: Test Case, Title, Acceptanc..., Observed Result, Verdict, C..., Discrepancy, and Performed... The table contains one row for TC-001, where the observed result is 'Conforms.', the verdict is 'Pass', and a discrepancy is linked (indicated by a checkmark in the 'Discrepancy' column).

On the right, the 'Activity' section shows two events from 'QC Analyst (demo)' at 12:11 PM: 'Execution started.' and 'Validation Execution Record created'.

Figure 108 — Execution in progress with a failed test linked to a discrepancy. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Validation
Planning
Execution
Reporting
7
14
My Company (San Francisco)

New
Discrepancies
DISC/2026/0001 - Failure of test case TC-002
Signatures
0
1 / 1
<
>

Start Investigation
Cancel
Open
Under Investigation
Resolved
Closed
Send message
Log note
Activity
1

DISC/2026/0001

Failure of test case TC-002

CONTEXT		HANDLING	
Execution Record	EXE/2026/0002 - Operational Qualif...	Classification	Major
Protocol	PROT/2026/0001 v1 - Operational Qualific...	Raised by	QC Analyst (demo)
Validation Item	[EQ-0001] Steam Sterilizer 500 L	Raised On	Sep 27, 12:11 PM
Company	My Company (San Francisco)	CAPA Required	☐
		Capa Reference	?

Description
Immediate Action
Investigation
Impacted Test Cases
Electronic Signatures

TC-002 failed: chamber pressure above the acceptance range for 40 s during the plateau.

Today

QC Analyst (demo) 12:11 PM
Validation Execution Discrepancy created

Figure 109 — Validation discrepancy. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Summary report and validation status

Module	ls_validation	Version	_____
Menu	Validation ► Reporting ► Summary Reports	Effective date	_____
Owner (suggested)	Validation Manager	Approved by	_____
Related SOPs	SOP-VAL-003, SOP-VAL-005		

1 Objective

Conclude on the validation of an item on the basis of approved executions, state any restriction, and set the validity that drives the item's validation status.

2 Scope

All validation items after execution of their protocols.

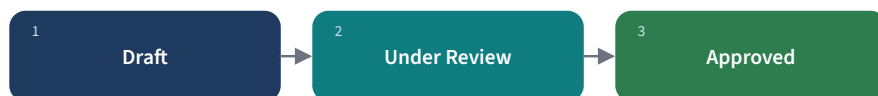
3 Responsibilities

Role (software group)	Responsibility
Author	Writes the report summarising executions and discrepancies.
Reviewer / Approver	Review and approve with signature.

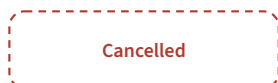
4 Definitions

Term	Definition
Conclusion	Validated, Validated with Restrictions (restrictions mandatory), Not Validated.
Validity	Valid from date and validity in months give <i>Valid Until</i> ; the item becomes Expiring Soon then Expired.

5 Workflow in the software



Other outcomes



Model: ls.validation.report

Figure 110 — Lifecycle of a summary report.

6 Procedure

- Create.** In **Summary Reports** click **New** : title, item, master plan, execution records (approved), report date, summary, conclusion, restrictions, valid from and validity (months).
- Check.** All referenced executions must be approved and concern the item; no linked discrepancy may be open; a failed critical test forbids *Validated*.

- 3 **Review and approve.** Click **Submit for Review** ; the approver clicks **Approve and Sign** .
- 4 **Release for use.** The item's validation status becomes *Validated* for the validity period; inform the owner and production.
- 5 **Restrictions.** If *Validated with Restrictions*, translate each restriction into an operating instruction or label on the equipment.

7. Controls enforced by the software

- ✓ The report must reference at least one approved execution of the same item and have a summary.
- ✓ No linked discrepancy may remain open; failed critical tests forbid the conclusion *Validated*; restrictions are mandatory for *Validated with Restrictions*.
- ✓ Validity cannot start before the report date; approved reports are frozen and cannot be cancelled.

8. Good practices

- ★ Summarise deviations from the protocol and their impact honestly.
- ★ Set validity consistent with the revalidation strategy (SOP-VAL-005).

9. Pitfalls to avoid

- ✗ Using an item before its report is approved.

10 Records and evidence

- Approved summary report with signatures

11 Performance indicators

- Time from last execution to report approval
- Items used while not validated (target 0)

12 References

- EU GMP Annex 15 (2015) — Qualification and Validation
- ISPE GAMP 5 Guide, Second Edition (2022) — A Risk-Based Approach to Compliant GxP Computerized Systems

13. Demonstration

Case. Report for autoclave AC-02: IQ, OQ (round 2) and PQ executions approved, one discrepancy closed, no critical failure. Conclusion *Validated*, valid from 1 April for 36 months. The item status becomes *Validated* until 31 March three years later.

Periodic revalidation

Module	ls_validation	Version	_____
Menu	Validation ► Planning ► Expiring or Expired	Effective date	_____
Owner (suggested)	Validation Manager	Approved by	_____
Related SOPs	SOP-VAL-004, SOP-QA-030, SOP-QA-031		

1 Objective

Keep items in a validated state: review them periodically, revalidate before their validity expires or after significant changes, and retire items that are no longer used.

2 Scope

All validation items with a validity or a periodic review interval.

3 Responsibilities

Role (software group)	Responsibility
Item owner	Provides change, maintenance and deviation history.
Validation Manager	Decides on revalidation scope; plans protocols.

4 Definitions

Term	Definition
Periodic review	Documented evaluation of changes, deviations, maintenance and performance since the last validation, concluding whether revalidation is needed.
Retirement	End of use of an item with a date and a reason.

5 Procedure

- 1 Monitor.** Monthly open **Planning ► Expiring or Expired**.
- 2 Review.** For each item due, review changes (SOP-QA-030), deviations, CAPA, calibration and maintenance history, and performance data; record the conclusion.
- 3 Plan revalidation.** On the item click **Plan Revalidation**: the wizard creates the protocol to write (type, scope reduced to what the review requires).
- 4 Execute and report.** Follow SOP-VAL-002 to SOP-VAL-004; the new report renews the validity.
- 5 Change-triggered.** An approved change with *Requires Revalidation* ticked triggers revalidation before use.
- 6 Retire.** For an item out of use, click **Retire** with the retirement date and reason; **Cancel Retirement** if the decision is reversed.

6. Controls enforced by the software

- ✓ A retirement date is required to retire an item; an item cannot be retired twice.
- ✓ The validation status is derived from the approved reports and their validity.

7. Good practices

- ★ Start revalidation early enough (for example 3 months before expiry).
- ★ Scale revalidation to the review: a full PQ is not always needed.

8. Pitfalls to avoid

- ✗ Items used after *Expired* because nobody watched the list.

9 Records and evidence

- Periodic review conclusions
- Revalidation protocols and reports
- Retirement records

10 Performance indicators

- Items expired while in use (target 0)
- Periodic reviews on time

11 References

- EU GMP Annex 15 (2015) — Qualification and Validation
- ISPE GAMP 5 Guide, Second Edition (2022) — A Risk-Based Approach to Compliant GxP Computerized Systems
- EU GMP Annex 11 — Computerised Systems (check the revision in force)

12. Demonstration

Case. HPLC system HPLC-04 shows *Expiring Soon*. The review finds a pump replaced (change CC-2027-033) and no deviation. The manager clicks **Plan Revalidation** and chooses an OQ limited to pump flow accuracy and gradient composition; the report renews validity for 24 months.

Validation of processes, cleaning, analytical methods and computerised systems (CSV)

Module	ls_validation	Version	_____
Menu	Validation ► Execution ► Protocols (PV, CV, MV, CSV)	Effective date	_____
Owner (suggested)	Validation Manager	Approved by	_____
Related SOPs	SOP-VAL-002, SOP-VAL-003, SOP-DI-001, SOP-QC-010		

1 Objective

Apply the protocol–execution–report cycle to process validation (PV), cleaning validation (CV), analytical method validation (MV) and computerised system validation (CSV), with the specific content each requires.

2 Scope

Manufacturing processes, cleaning procedures, analytical methods and computerised systems registered as validation items — including this Odoo installation and the LS modules used for GxP decisions.

3 Responsibilities

Role (software group)	Responsibility
Process / cleaning / method / system owner	Provides requirements and risk assessment.
Validation Engineer	Writes and executes protocols.
QA	Approves protocols and reports.
IT	Supports CSV (installation, configuration, backup tests).

4 Definitions

Term	Definition
PV	Process validation — demonstration that the process consistently produces product meeting its quality attributes (process design, qualification, continued verification).
CV	Cleaning validation — demonstration that cleaning removes residues to below justified limits.
MV	Analytical method validation — accuracy, precision, specificity, linearity, range, detection/quantitation limits as applicable.
CSV	Computerised system validation — risk-based verification that the configured system meets user requirements and data integrity controls work.

5 Procedure

- 1 Register the item.** Item type Manufacturing Process, Cleaning Procedure, Analytical Method or Computerised System; set GxP impact and criticality.
- 2 Risk assessment.** Identify critical parameters/attributes (PV), worst-case products and locations (CV), method performance characteristics (MV), GxP functions and data flows (CSV); record it in ls_risk_management.

- 3 **Protocol.** Choose protocol type PV, CV, MV or CSV and write test cases with criteria (SOP-VAL-002).
- 4 **CSV specifics for this suite.** Test the workflows and controls you rely on (for example the state gates, segregation of duties, audit trail rules, signature mechanisms, purchase order block), the backup and restore, and the user access matrix. Record the Odoo build and module versions tested. Note: at the time of writing, four OCA modules of the repository failed to install on a fresh Odoo 19.0 database (see the annex *Known limitations*) — include installation checks in the IQ.
- 5 **Execute and report.** SOP-VAL-003 and SOP-VAL-004.
- 6 **Maintain.** Continued process verification for PV; periodic review and revalidation after changes (SOP-VAL-005); for CSV, re-test after each module upgrade under change control.

6. Controls enforced by the software

- ✓ Same controls as SOP-VAL-002 to SOP-VAL-004 (approval before execution, criteria mandatory, discrepancies closed before approval, signatures with re-authentication).

7. Good practices

- ★ For CSV, test what the software claims to enforce: each "control" box of this handbook is a candidate test case.
- ★ For CV, justify limits (e.g. health-based exposure limits) and sampling recovery.
- ★ For MV, follow ICH Q2(R2) characteristics appropriate to the method category.

8. Pitfalls to avoid

- ✗ Validating the software "as delivered" instead of the configuration actually used.
- ✗ Skipping the negative tests (trying the forbidden action and checking it is refused).

9 Records and evidence

- Protocols, executions, discrepancies, reports per item

10 Performance indicators

- GxP systems without current CSV report (target 0)

11 References

- EU GMP Annex 15 (2015) — Qualification and Validation
- FDA Guidance — Process Validation: General Principles and Practices (2011)
- ISPE GAMP 5 Guide, Second Edition (2022) — A Risk-Based Approach to Compliant GxP Computerized Systems
- EU GMP Annex 11 — Computerised Systems (check the revision in force)
- ICH Q2(R2) — Validation of Analytical Procedures
- 21 CFR Part 11 — Electronic Records; Electronic Signatures

12. Demonstration

Case. CSV of Odoo LS — *deviation, CAPA and change control*: 46 test cases, including negative tests such as "attempt to approve a change request assessment as a non-assigned user → refused with the message 'Only ... may complete assessment'", "close a deviation with an open immediate action → refused". Backup/restore test and audit trail rule verification are included. Report *Validated with Restrictions*: the electronic signature policies are not used (no LS model inherits the signature capability) — approvals are confirmed on printed, hand-signed records.

Calibration

ls_calibration

ls_calibration holds the instrument register (criticality, GxP impact, range, tolerance, alert lead time, calibration status Valid / Due Soon / Overdue), calibration plans with test points and interval, calibration records with as-found and as-left readings per point, reference standards, out-of-tolerance impact assessment, review and approval by a Calibration Manager, and internal or external certificates. It depends on the standard Maintenance application (instruments can be linked to equipment).

Menu	Content
Calibration ► Instruments	Instrument register
Calibration ► Calibration Plans / Calibration Records / Certificates	Plans, records, certificates
Calibration ► Calibration Status ► Due and Overdue / Generate Calibration Records	Scheduling
Calibration ► Reporting ► Calibration Analysis	Analysis

Group	Main rights
Calibration / Viewer · Technician	Read; perform calibrations and submit records
Calibration / Manager	Approves records, manages plans and instruments

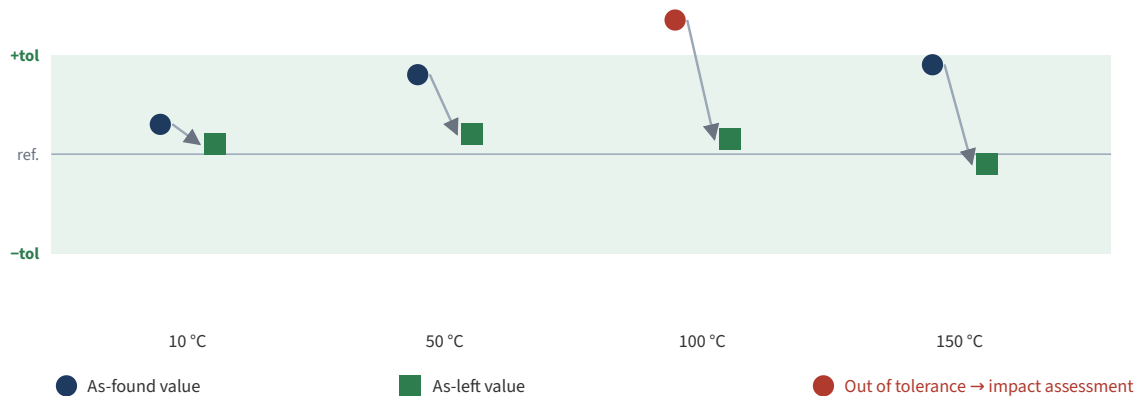


Figure 111 — As-found and as-left readings against the tolerance band (illustrative values).



Model: ls_calibration.record

Figure 112 — Lifecycle of a calibration record.

Calibration

Instruments

Calibration Plans

Calibration Records

Certificates

Calibration Status

Reporting

7

14

My Company (San Francisco)

New

Instruments

[INS/00001] Analytical balance QC laboratory

Plans

Records

Certificates

1 / 1

Set Out of Service

Retire

Draft

In Service

Out of Service

Send message

Log note

Activity

Instrument Name

Analytical balance QC laboratory

Instrument Reference ?

INS/00001

IDENTIFICATION

Instrument Category

Manufacturer

Manufacturer Model

Serial Number

Related Equipment ?

Physical Location ?

Company

CLASSIFICATION

Criticality ?

Gxp Impact ?

Responsible ?

Alert Lead Time (Days) ?

Calibration Status ?

Last Calibration Date ?

Next Calibration Date ?

Metrological Data

Calibration Plans

Calibration Records

Certificates

Notes

MEASURING RANGE

Range Minimum

Range Maximum

0.000000

220.000000

DEFAULT ACCEPTANCE CRITERIA

Tolerance Type ?

Maximum Permissible Error ?

Absolute

0.001000

Today

OdooBot

10:40 AM

Calibration Instrument created

Figure 113 — Instrument form. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Management of the instrument inventory

Module	ls_calibration	Version	_____
Menu	Calibration ► Instruments	Effective date	_____
Owner (suggested)	Calibration Manager (metrology)	Approved by	_____
Related SOPs	SOP-MET-002, SOP-MET-003, SOP-GEN-003		

1 Objective

Keep a complete register of measuring instruments that affect GxP data or product quality, with their criticality, tolerance and status, so that only calibrated instruments are used.

2 Scope

All measuring and test equipment used in production, QC, warehouse, utilities and validation: balances, thermometers, loggers, pressure gauges, pipettes, sensors, reference standards.

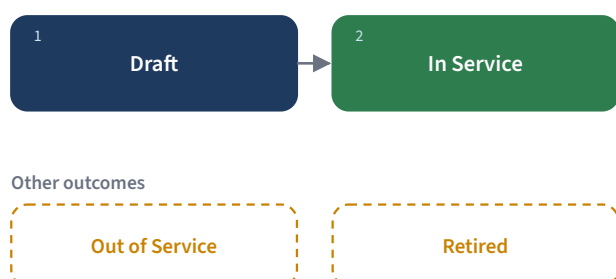
3 Responsibilities

Role (software group)	Responsibility
Calibration Manager	Maintains the register, classifies instruments, sets tolerances.
Instrument owner (Responsible)	Uses the instrument correctly, reports damage, removes it from service when needed.
Technician	Labels instruments and updates locations.

4 Definitions

Term	Definition
Maximum permissible error / tolerance	Accepted deviation, absolute or percentage of reading.
Calibration status	Not Applicable, Not Scheduled, Valid, Due Soon (within the alert lead time), Overdue.

5 Workflow in the software



Model: ls_calibration.instrument

Figure 114 — States of an instrument.

6 Procedure

- 1 **Register.** In **Calibration ► Instruments** click **New** : name, reference (unique tag), related equipment, category, manufacturer, model, serial number, location, responsible, criticality, GxP impact, range, unit, tolerance type and maximum permissible error, alert lead time.
- 2 **Classify.** Set GxP impact (direct, indirect, none) and criticality; instruments with no GxP impact can be recorded for inventory only.
- 3 **Put in service.** After the initial calibration (SOP-MET-002), click **Set In Service** ; label the instrument with its reference and next due date.
- 4 **Monitor status.** Use **Calibration Status ► Due and Overdue** weekly.
- 5 **Out of service.** When damaged, overdue or failed, click **Set Out of Service** and physically segregate it ("do not use" label).
- 6 **Retire.** Click **Retire** when the instrument is disposed of; history is kept.

7. Controls enforced by the software

- ✓ Only draft or out-of-service instruments can be set in service; only in-service instruments can be set out of service; a retired instrument can only be reset to draft.
- ✓ Status is computed from the next calibration date and the alert lead time.

8. Good practices

- ★ One unique physical tag per instrument, matching the reference in the system.
- ★ Include reference standards and data loggers — they are often forgotten.

9. Pitfalls to avoid

- ✗ Instruments moved between rooms without updating the location.
- ✗ Personal instruments (pocket thermometers) used without registration.

10 Records and evidence

- Instrument register with history

11 Performance indicators

- Instruments overdue (target 0)
- Instruments without GxP classification

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 3 — Premises and Equipment
- 21 CFR 211.68 — Automatic, mechanical and electronic equipment
- 21 CFR 211.160 to 211.176 — Laboratory controls
- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 10012:2003 — Measurement management systems

13. Demonstration

Case. A new analytical balance BAL-017 (0.01 mg readability) is registered: criticality *Critical*, GxP *Direct Impact*, range 0–220 g, tolerance absolute 0.1 mg, alert lead time 14 days. After installation calibration it is set in service and labelled.

Calibration

Instruments

Calibration Plans

Calibration Records

Certificates

Calibration Status

Reporting

7

14

My Company (San Francisco)

7

New

Instruments

Q

Search...

1-3 / 3

<

>

☐	Instrument Refere...	Instrument Name	Instrument Categ...	Physical Location	Criticality	Responsible	Last Calibra...	Next Calibr...	Calibration Sta...	State	
☐	INS/00001	Analytical balance QC laboratory		QC laboratory - weighing room	Critical	OdooBot	Mar 13	Mar 13, 2027	Valid	In Service	
☐	INS/00002	Reference thermometer stability cham...		Stability room 1	Major	OdooBot		Oct 17	Due Soon	In Service	
☐	INS/00003	Mass standard set E2		QC laboratory - metrology cabi...	Critical	OdooBot			Not Scheduled	In Service	

Figure 115 — Instrument register with calibration status. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Planning and performing calibrations (as-found and as-left values)

Module	ls_calibration	Version	_____
Menu	Calibration ► Calibration Plans / Calibration Records	Effective date	_____
Owner (suggested)	Calibration Manager	Approved by	_____
Related SOPs	SOP-MET-001, SOP-MET-003, SOP-MET-004		

1 Objective

Calibrate each instrument at a defined interval, against traceable reference standards, recording the value found before any adjustment and the value after adjustment, and have each record approved by someone other than the performer.

2 Scope

All instruments in service with a calibration plan, calibrated internally or by an external provider.

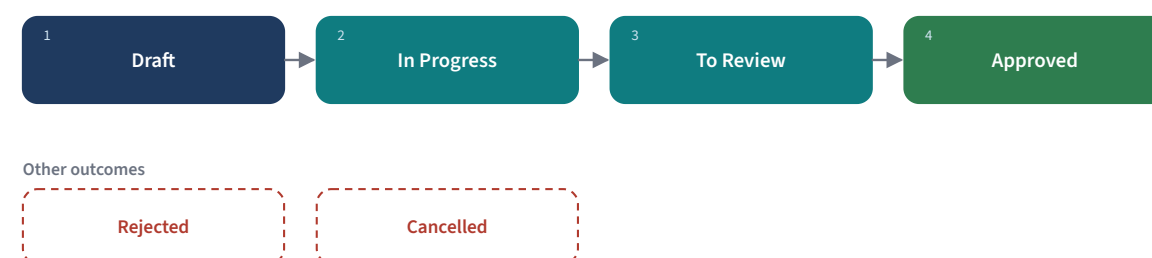
3 Responsibilities

Role (software group)	Responsibility
Calibration Technician	Performs the calibration and records readings; submits for review.
Calibration Manager	Approves or rejects records; cannot approve own calibrations.
External provider	Performs external calibrations and issues certificates.

4 Definitions

Term	Definition
As-found	Reading before any adjustment — shows whether the instrument was within tolerance during its period of use.
As-left	Reading after adjustment — shows the state in which it returns to service.
Reference standard	Calibrated instrument used to calibrate another; it must itself be in calibration.

5 Workflow in the software



Model: ls_calibration.record

Figure 116 — Lifecycle of a calibration record.

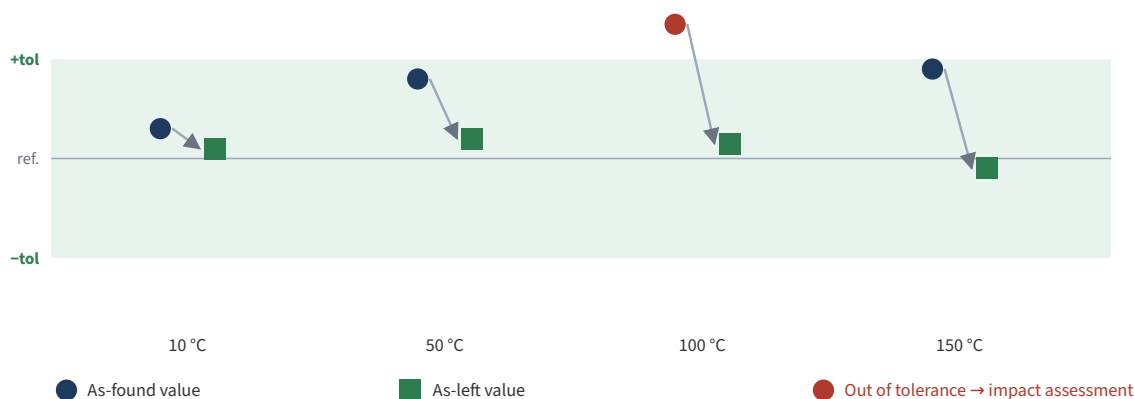


Figure 117 — As-found and as-left per test point (illustrative).

6 Procedure

- 1 **Create the plan.** In **Calibration Plans** click **New** : instrument, description, procedure reference, method, interval and unit, start date, responsible, external provider if any; add test points (nominal value, tolerance type and tolerance). Click **Activate** .
- 2 **Schedule.** Use **Calibration Status ► Generate Calibration Records** or **Create Calibration Record** on the plan for instruments due.
- 3 **Prepare.** Check that the reference standards are in calibration; record ambient temperature and humidity when relevant.
- 4 **Perform.** Click **Start** ; for each test point enter the as-found reading; adjust if needed and enter the as-left reading. The deviations and in-tolerance flags are computed.
- 5 **Conclude.** Record the calibration date, performer, reference standards (or external standard reference), result (Pass, Pass After Adjustment, Fail), adjustment performed, conclusion; if as-found is out of tolerance, complete SOP-MET-003.
- 6 **Submit.** Click **Submit to Review** .
- 7 **Approve.** The Calibration Manager (not the performer) clicks **Approve** or **Reject** with a reason. The next due date is updated.
- 8 **Certificate.** Issue or attach the certificate (SOP-MET-004).

7. Controls enforced by the software

- ✓ A plan needs at least one test point to be activated; records can only be created from active plans; the plan must belong to the instrument.
- ✓ Review requires a calibration date, the performer, at least one test point and at least one reference standard or external standard reference.
- ✓ An instrument cannot be its own reference; an overdue reference standard cannot be used.
- ✓ If as-found readings are out of tolerance, an impact assessment is required.
- ✓ Only a Calibration Manager can approve; the performer cannot approve; rejection requires a reason.
- ✓ Test points can only be edited in draft or in progress; approved or cancelled records are frozen; non-draft records cannot be deleted.

8. Good practices

- ★ Always record as-found before touching the instrument.
- ★ Choose test points covering the range of use, including the points used for release decisions.
- ★ Shorten the interval of instruments often found out of tolerance; extend it only with history.

9. Pitfalls to avoid

- ✗ Adjusting first and recording only the final value.
- ✗ Using an external certificate without checking the points and uncertainties against your tolerance.

10 Records and evidence

- Calibration records with readings, standards and approval
- Certificates

11 Performance indicators

- % calibrations performed before due date
- % as-found out of tolerance per instrument type

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 3 — Premises and Equipment
- 21 CFR 211.68 — Automatic, mechanical and electronic equipment
- 21 CFR 211.160 to 211.176 — Laboratory controls
- ISO/IEC 17025:2017 — Competence of testing and calibration laboratories
- ISO 10012:2003 — Measurement management systems

13. Demonstration

Case. Temperature probe TT-104 (tolerance $\pm 0.5\text{ }^{\circ}\text{C}$) is calibrated at 20, 60 and 121 $^{\circ}\text{C}$ against reference RT-001 (in calibration). As-found at 121 $^{\circ}\text{C}$: 121.7 $^{\circ}\text{C}$ → out of tolerance. The technician adjusts: as-left 121.1 $^{\circ}\text{C}$. Result *Pass After Adjustment*; SOP-MET-003 is triggered for the as-found failure; the manager approves after the impact assessment.

Calibration Instruments Calibration Plans Calibration Records Certificates Calibration Status Reporting 7 14 My Company (San Francisco)

New Calibration Plans CP/00002 - [INS/00002] Reference thermete... Records 0 1 / 1 < >

Create Calibration Record Suspend Set Obsolete Draft Active Obsolete Send message Log note Activity

Plan Reference
CP/00002

SCOPE			SCHEDULE		
Instrument	[INS/00002] Reference thermometer ...		Interval	6	Months
Description ?	Half-yearly calibration of the reference		Start Date ?	Oct 17	
Procedure Reference ?	SOP-MET-004		Last Calibration Date ?		
Responsible	OdooBot		Next Due Date ?	Oct 17	
Company	My Company (San Francisco)		Externally Calibrated ?	☐	

Test Points Method

Point	Nominal Value	Unit of Measurement	Tolerance Type	Tolerance	Lower Limit	Upper Limit
Point 1 - 5 degC	5.000000	degC	Absolute	0.200000	4.800000	5.200000
Point 2 - 25 degC	25.000000	degC	Absolute	0.200000	24.800000	25.200000
Add a line						

Today

OdooBot 10:40 AM
Calibration Plan created

Figure 118 — Calibration plan with test points. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Calibration Instruments Calibration Plans Calibration Records Certificates Calibration Status Reporting 7 14 My Company (San Francisco)

New Calibration Analysis CAL/2026/00001 - [INS/00001] Analytical balan... Certificates 1 1 / 1 < >

Draft In Progress To Review Approved Send message Log note Activity

Record Reference
CAL/2026/00001

IDENTIFICATION		EXECUTION	
Instrument	[INS/00001] Analytical balance QC laborat...	Calibration Date ?	Mar 13, 8:30 AM
Calibration Plan	CP/00001 - [INS/00001] Analytical balance...	Performed By ?	Mitchell Admin
Calibration Type	Periodic	Externally Calibrated	☐
Scheduled Date ?	Mar 11	Ambient Temperature	21.40
Company	My Company (San Francisco)	Relative Humidity	45.00

VERDICT		REVIEW AND APPROVAL	
As Found Status	In Tolerance	Submitted By ?	OdooBot
As Left Status	In Tolerance	Submission Date	Sep 27, 10:40 AM
Result	Pass	Approved By	OdooBot
Adjustment Performed ?	☐	Approval Date	Sep 27, 10:40 AM
Next Due Date ?	Mar 13, 2027		

Test Points Reference Standards Conclusion

Point	Nominal Value	Unit of Me...	Tolerance ...	Tolerance	Lower Limit	Upper Limit	As Found	As Fou
Point 1 - 1...	10.000000	g	Absolute	0.001000	9.999000	10.001000	10.000300	

Today

OdooBot 10:40 AM
Calibration record approved by OdooBot on 2026-09-27 10:40:02. Meaning of the signature: approval of the calibration results and release of the instrument.

OdooBot 10:40 AM
Calibration results submitted for approval by OdooBot.

OdooBot 10:40 AM
Calibration Record created

Figure 119 — Approved calibration record with test points. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Impact assessment of an out-of-tolerance result

Module	ls_calibration	Version	_____
Menu	Calibration ► Calibration Records (Out-of-Tolerance Impact Assessment)	Effective date	_____
Owner (suggested)	Calibration Manager with QA	Approved by	_____
Related SOPs	SOP-MET-002, SOP-QA-020, SOP-QA-021		

1 Objective

When an instrument is found out of tolerance, determine which measurements, batches and decisions made since the last good calibration may be affected, and act on them.

2 Scope

Every calibration record with an as-found reading out of tolerance, and every instrument found damaged or unstable in use.

3 Responsibilities

Role (software group)	Responsibility
Calibration Manager	Documents the metrological impact and the period concerned.
Instrument owner / users	List the uses of the instrument during the period.
QA	Decides on product impact through a deviation.

4 Definitions

Term	Definition
Period at risk	Time between the last calibration found in tolerance and the current as-found failure.
Out-of-tolerance action reference	Reference of the deviation or CAPA opened for the impact.

5 Procedure

- 1 Stop use.** Set the instrument out of service if the as-left is not in tolerance (SOP-MET-001).
- 2 Quantify.** Compare the as-found error with the process or specification tolerance: an instrument error of 0.7 °C may be irrelevant for a 2–8 °C storage alarm set at 7 °C, or critical for a sterilisation cycle.
- 3 Trace uses.** List measurements taken with the instrument during the period at risk (batches, analyses, EM samples, validation tests).
- 4 Assess.** Write the *Out-of-Tolerance Impact Assessment* on the calibration record: error, direction, affected uses, conclusion.
- 5 Escalate.** If product or data may be affected, open a deviation (SOP-QA-020) and record its reference in *Out-of-Tolerance Action Reference*; QA decides on product disposition.
- 6 Close.** Submit the record for approval (SOP-MET-002).

6. Controls enforced by the software

- ✓ A calibration record whose as-found readings are out of tolerance cannot be submitted without an impact assessment.

7. Good practices

- ★ Think about the direction of the error: a thermometer reading low hides over-temperatures.
- ★ Use this assessment to adjust the calibration interval.

8. Pitfalls to avoid

- ✗ Concluding "no impact" without looking at what the instrument was used for.
- ✗ Limiting the period at risk to the last week.

9 Records and evidence

- Impact assessment on the record
- Deviation and product decisions

10 Performance indicators

- Out-of-tolerance events with product impact
- Median time from finding to impact decision

11 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 3 — Premises and Equipment
- 21 CFR 211.68 — Automatic, mechanical and electronic equipment
- 21 CFR 211.192 — Production record review
- ISO 10012:2003 — Measurement management systems

12. Demonstration

Case (continued). TT-104 read 0.7 °C high at 121 °C. It controls autoclave AC-02, whose cycles were therefore actually 0.7 °C cooler than displayed over the last 6 months. The validation data show a minimum F0 margin that covers 0.7 °C; QA opens deviation DEV/2026/00041 to document the evaluation of the 38 cycles; conclusion: sterility assurance maintained, no batch impact.

Calibration certificates

Module	ls_calibration	Version	_____
Menu	Calibration ► Certificates	Effective date	_____
Owner (suggested)	Calibration Manager	Approved by	_____
Related SOPs	SOP-MET-002		

1 Objective

Issue internal certificates and register external certificates so that each calibration is supported by a controlled, retrievable certificate linked to the instrument and the calibration record.

2 Scope

Internal certificates generated after approved calibrations and certificates issued by external laboratories.

3 Responsibilities

Role (software group)	Responsibility
Calibration Manager	Reviews and issues certificates; supersedes wrong certificates.
Technician	Attaches external certificate files.

4 Definitions

Term	Definition
Issuer type	Internal or External (issuing laboratory, accreditation reference).
Superseded	Certificate replaced by a corrected one; the old one remains for history.

5 Workflow in the software



Other outcomes



Model: ls_calibration.certificate

Figure 120 — Lifecycle of a calibration certificate.

6 Procedure

- 1 Create.** From the approved record or in **Certificates** click **New**: certificate number, instrument, calibration record, issuer type, issuing laboratory and accreditation reference for external, issue date, valid until.
- 2 External review.** For an external certificate, check before issuing it in the system: instrument identification, points, results, uncertainty compared with your tolerance, traceability statement, accreditation. Attach the PDF.

3 **Issue.** Click **Issue**. The content is frozen.

4 **Correct.** If an issued certificate is wrong, create the corrected one and click **Supersede** on the old one.

7. Controls enforced by the software

- ✓ A certificate must refer to a calibration record of the same instrument; the validity end cannot precede the issue date.
- ✓ Only a draft certificate can be issued; an external certificate needs its document attached.
- ✓ Issued certificates cannot be modified or deleted; only an issued certificate can be superseded.

8. Good practices

- ★ Prefer accredited laboratories (ISO/IEC 17025) for reference standards.
- ★ Keep the certificate number of the provider, not a new internal number, for external certificates.

9. Pitfalls to avoid

- ✗ Accepting certificates that give results without uncertainty or without as-found values.

10 Records and evidence

- Issued certificates with files

11 Performance indicators

- External certificates rejected at review

12 References

- ISO/IEC 17025:2017 — Competence of testing and calibration laboratories
- EudraLex Vol. 4, EU GMP Part I, Chapter 3 — Premises and Equipment
- ISO 10012:2003 — Measurement management systems

13. Demonstration

Case. The external laboratory returns the certificate of reference standard RT-001 with an expanded uncertainty of 0.05 °C, accredited. The Calibration Manager checks it, attaches the PDF, enters the accreditation reference and clicks **Issue**.

Environmental monitoring

ls_environmental_monitoring

ls_environmental_monitoring manages areas (nested, with cleanroom grade), sampling points (position, selection rationale, critical location), parameters (viable air, settle plates, contact plates, personnel, non-viable particles, temperature, humidity, differential pressure, air velocity, air change rate), methods, approved monitoring plans with frequencies, approved alert/action/specification limits per point, parameter and occupancy state, samples with results evaluated automatically, excursions raised automatically for action and specification breaches (and alert breaches when configured), and trend analyses.

Menu	Content
Environmental Monitoring ► Programme ► Monitoring Plans / Sampling Points / Limits	Programme
Environmental Monitoring ► Operations ► Samples Due / All Samples / Generate Scheduled Samples	Sampling
Environmental Monitoring ► Results ► Results / Threshold Breaches / Excursions	Results
Environmental Monitoring ► Trending ► Run Trend Analysis / Trend Analyses	Trending
Environmental Monitoring ► Configuration ► Areas / Cleanroom Grades / Methods / Parameters	Reference data

Group	Main rights
EM / Viewer · Technician	Read; collect samples, enter results
EM / Manager	Approves plans and limits, reviews and approves samples, closes excursions

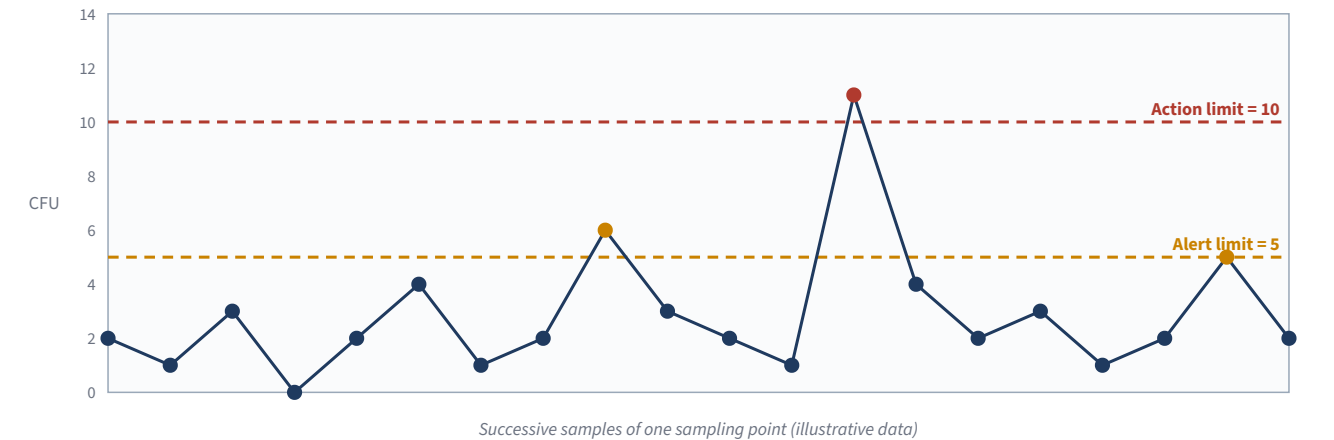
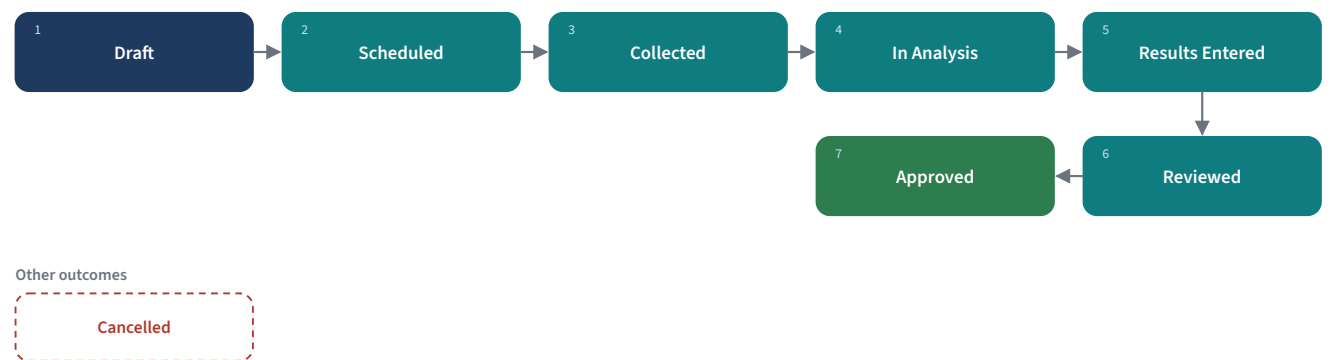


Figure 121 — Alert and action limits on a trend (illustrative values).



Model: ls.env.sample

Figure 122 — Lifecycle of an EM sample.

Monitoring programme: areas, sampling points and rationale

Module	ls_environmental_monitoring	Version	_____
Menu	Environmental Monitoring ► Programme ► Monitoring Plans / Sampling Points	Effective date	_____
Owner (suggested)	EM Manager (QA / microbiology)	Approved by	_____
Related SOPs	SOP-EM-002, SOP-EM-003, SOP-QA-060		

1 Objective

Define, on the basis of a documented risk assessment, which areas and points are monitored, for which parameters, in which occupancy state and how often, and keep this programme approved.

2 Scope

Cleanrooms and controlled areas of all grades, utilities with environmental parameters, and personnel monitoring.

3 Responsibilities

Role (software group)	Responsibility
EM Manager	Designs and approves (with a second person) the programme.
Microbiology / QC	Provides method capabilities.
Production	Provides process knowledge for the choice of critical locations.

4 Definitions

Term	Definition
Critical location	Point where product or sterile surfaces are exposed; flagged on the sampling point.
Occupancy state	At Rest or In Operation (limits and plans distinguish them).

5 Workflow in the software



Other outcomes



Model: ls.env.plan

Figure 123 — Lifecycle of a monitoring plan.

6 Procedure

- 1 Areas.** In **Configuration ► Areas** create the area tree with grades and responsible.

- 2 **Sampling points.** In **Programme ▶ Sampling Points** create each point: code, area, grade, position, *Selection Rationale* (from the risk assessment, airflow studies, process), *Critical Location*.
- 3 **Parameters and methods.** Check parameters (type, quantitative or pass/fail) and methods (which parameter each measures).
- 4 **Plan.** In **Monitoring Plans** click **New**: code, version, primary area, effective from, rationale; add plan lines (point, parameter, method, occupancy state, every N days/weeks/months/years, start date, responsible).
- 5 **Approve.** A user other than the author clicks **Approve**.
- 6 **Revise.** To change an approved plan click **Create Revision**; the new version supersedes the old at approval.
- 7 **Generate samples.** Use **Operations ▶ Generate Scheduled Samples** to create the due samples from approved plans.

7. Controls enforced by the software

- ✓ A plan needs at least one line and must be approved by a user other than its creator; approved plans cannot be cancelled or deleted — they are superseded by revision.
- ✓ The method of a plan line must measure its parameter; point, parameter and plan must belong to the same company; supported frequency units only.

8. Good practices

- ★ Write the selection rationale for every point: an inspector will ask why this point and not another.
- ★ Include worst-case locations (near doors, operator hands, filling needles).

9. Pitfalls to avoid

- ✗ Programmes copied from another site without local risk assessment.
- ✗ Monitoring only At Rest.

10 Records and evidence

- Approved plans with versions
- Sampling point rationale
- Risk assessment

11 Performance indicators

- Points without rationale (target 0)
- Plan review on time

12 References

- EU GMP Annex 1 (2022) — Manufacture of Sterile Medicinal Products
- ISO 14644-1:2015 — Cleanrooms — Classification of air cleanliness by particle concentration
- ICH Q9(R1) — Quality Risk Management

13. Demonstration

Case. Grade A filling zone: 6 points — 2 critical (needle area, stopper bowl) for continuous particle counting and active air each shift In Operation, 4 for settle plates. The rationale refers to the smoke studies of 2026 and to the process FMEA. The plan is approved by the QA microbiologist (not the author).

Environmental Monitoring
Operations
Results
Trending
Programme
Configuration

7
14
My Company (San Francisco)

New
Monitoring Plans
[EMP-STB v1] EM programme - sterile block
1 / 1

Create Revision
Draft
Approved
Superseded

Send message
Log note

1

Plan

EM programme - sterile block

CodeEMP-STB

Effective FromSep 27

Version1

Next Review Due?

Primary Area?Sterile block [STB]

Approved ByQA Reviewer (demo)

CompanyMy Company (San Francisco)

Plan Lines

Rationale

Sampling ...	Area	Parameter	Method	Occupanc...	Every	Frequenc...	Start Date	Next Due	Res	
[C-12] Cor...	Sterile blo...	[SP] Settl...	[M-SP] Se...	In Operation	1	Week(s)	Sep 27	Sep 27		
[A-01] Ne...	Sterile blo...	[AA] Activ...	[M-AA] Ac...	In Operation	1	Day(s)	Sep 27	Sep 27		

QC Analyst (demo) 12:09 PM

Environmental Monitoring Plan created

Figure 124 — Approved monitoring plan with its lines. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Setting and approval of alert and action limits

Module	ls_environmental_monitoring	Version	_____
Menu	Environmental Monitoring ► Programme ► Limits	Effective date	_____
Owner (suggested)	EM Manager	Approved by	_____
Related SOPs	SOP-EM-001, SOP-EM-004, SOP-EM-005		

1 Objective

Set, justify and approve alert, action and, where applicable, specification limits for each sampling point, parameter and occupancy state, and change them only through a revision.

2 Scope

All quantitative and pass/fail parameters of the monitoring plans.

3 Responsibilities

Role (software group)	Responsibility
EM Manager	Proposes limits with justification (regulatory maxima, historical data).
Approver (second EM Manager / QA)	Approves limits; cannot be the creator.

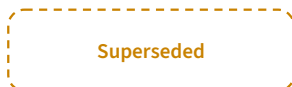
4 Definitions

Term	Definition
Alert limit	Level that signals a potential drift; investigation according to procedure.
Action limit	Level that requires investigation and corrective action; always opens an excursion.
Specification limit	Regulatory or product maximum; always opens an excursion.
Direction	Upper bound (breach above) or lower bound (breach below, e.g. differential pressure).

5 Workflow in the software



Other outcomes



Model: ls.env.limit

Figure 125 — Lifecycle of a limit.

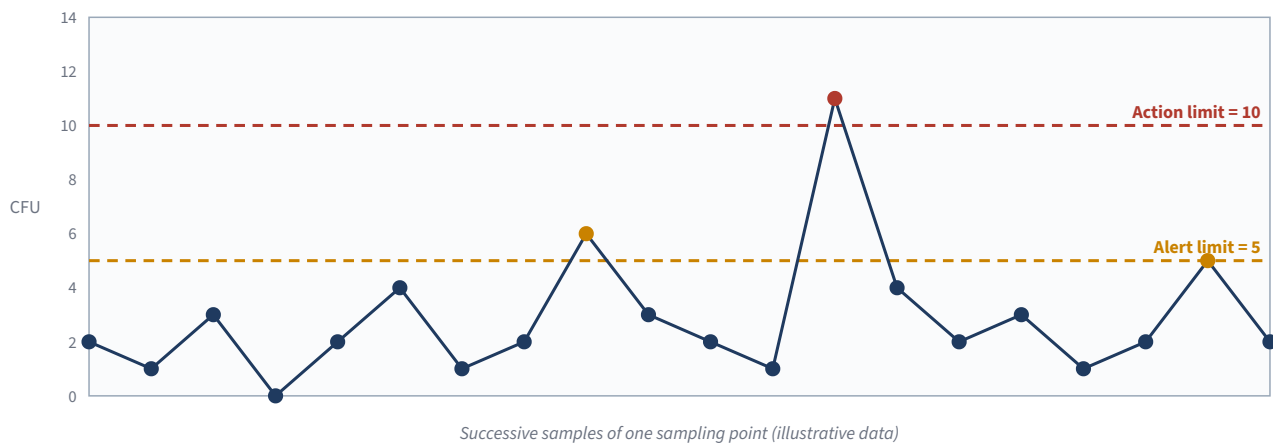


Figure 126 — Alert and action limits (illustrative values).

6 Procedure

- 1 **Collect basis.** Regulatory grade limits (e.g. EU GMP Annex 1 tables) and historical data of the point.
- 2 **Create.** In **Programme ► Limits** click **New** : sampling point, parameter, occupancy state, direction, alert, action and specification limits, *Raise Excursion on Alert* if alerts must be formally investigated, justification, controlling document, effective from.
- 3 **Approve.** A user other than the creator clicks **Approve** .
- 4 **Revise.** To change thresholds click **Create Revision** ; approve the revision (the previous one is superseded).
- 5 **Review.** Review limits yearly with trend data (SOP-EM-005).

7. Controls enforced by the software

- ✓ At least one threshold is required; the order of thresholds must match the direction (for an upper bound alert $\leq$ action $\leq$ specification).
- ✓ Only one approved limit per point, parameter, occupancy state and direction — supersede before approving a replacement.
- ✓ A justification is required; the approver must differ from the creator; approved thresholds are frozen; only draft limits can be deleted.

8. Good practices

- ★ Never set alert limits above regulatory action levels.
- ★ Tighten alert limits on the basis of historical performance (e.g. a statistical percentile).

9. Pitfalls to avoid

- ✗ Samples evaluated as *No Approved Limit* — they never raise an excursion. Check that every plan line has an approved limit.

10 Records and evidence

- Approved limits with justification and versions

11 Performance indicators

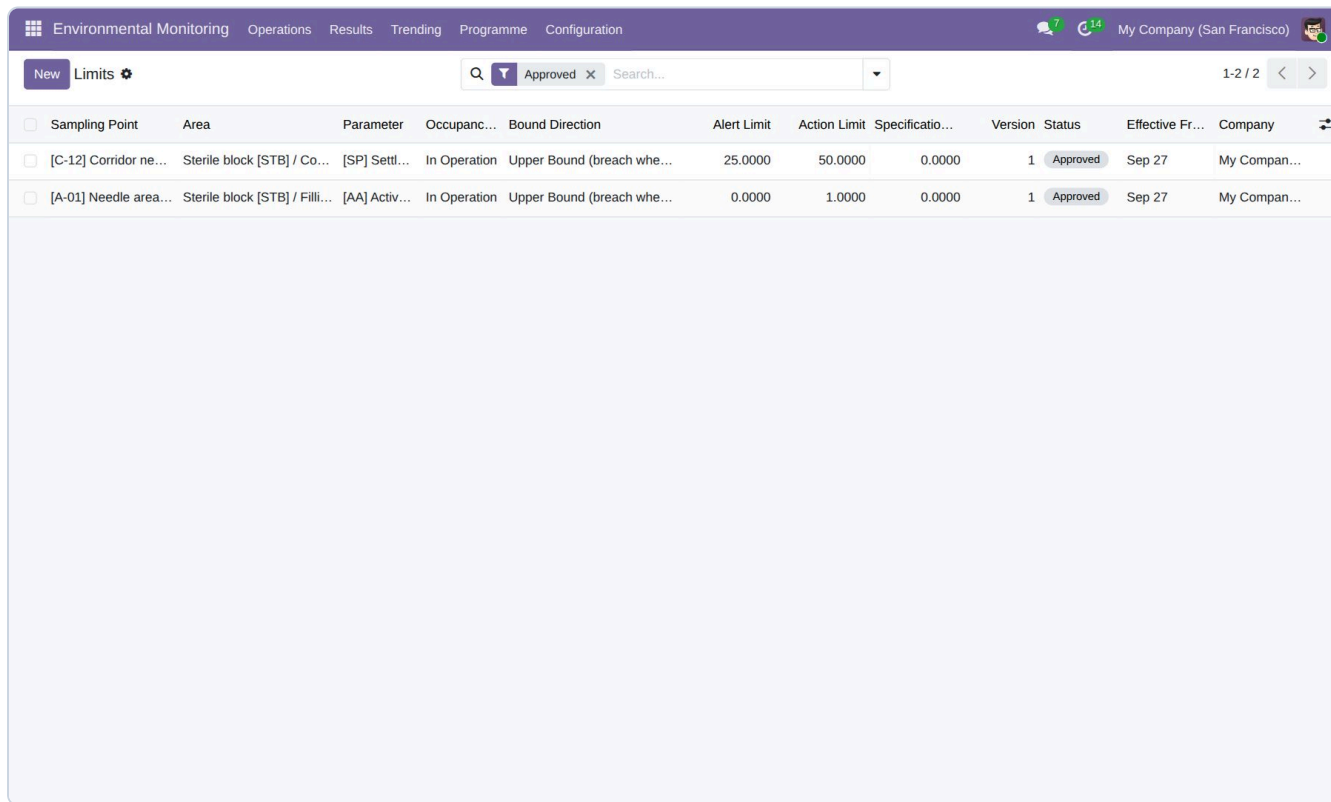
- Plan lines without approved limit (target 0)

12 References

- EU GMP Annex 1 (2022) — Manufacture of Sterile Medicinal Products
- ISO 14644-1:2015 — Cleanrooms — Classification of air cleanliness by particle concentration

13. Demonstration

Case. Point A-01, active air, In Operation, upper bound: action limit 1 CFU/m³ (grade A), no alert limit; specification not used. Point C-12, settle plate, In Operation: alert 25, action 50 CFU/4 h, *Raise Excursion on Alert* ticked because the room had two alerts last quarter. Both approved by the QA microbiologist.



The screenshot displays the 'Limits' configuration page in the Odoo 19.0 Community LS modules interface. The page features a navigation bar at the top with tabs for 'Environmental Monitoring', 'Operations', 'Results', 'Trending', 'Programme', and 'Configuration'. The 'Limits' tab is active, and the page title is 'Limits'. A search bar is located at the top right, with a filter set to 'Approved'. The main content area shows a table of limits with the following columns: Sampling Point, Area, Parameter, Occupancy, Bound Direction, Alert Limit, Action Limit, Specification, Version, Status, Effective From, and Company. Two rows are visible, both with a status of 'Approved'.

Sampling Point	Area	Parameter	Occupancy	Bound Direction	Alert Limit	Action Limit	Specification	Version	Status	Effective From	Company
[C-12] Corridor ne...	Sterile block [STB] / Co...	[SP] Settl...	In Operation	Upper Bound (breach whe...	25.0000	50.0000	0.0000	1	Approved	Sep 27	My Compan...
[A-01] Needle area...	Sterile block [STB] / Filli...	[AA] Activ...	In Operation	Upper Bound (breach whe...	0.0000	1.0000	0.0000	1	Approved	Sep 27	My Compan...

Figure 127 — Approved limits per point, parameter and occupancy state. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Sampling, analysis and review of results

Module	ls_environmental_monitoring	Version	_____
Menu	Environmental Monitoring ► Operations ► Samples	Effective date	_____
Owner (suggested)	EM Manager	Approved by	_____
Related SOPs	SOP-EM-001, SOP-EM-002, SOP-EM-004		

1 Objective

Collect scheduled samples on time, analyse them, record results traceably and have them reviewed and approved by people other than the analyst, so that breaches are detected and escalated.

2 Scope

All scheduled and unscheduled EM samples.

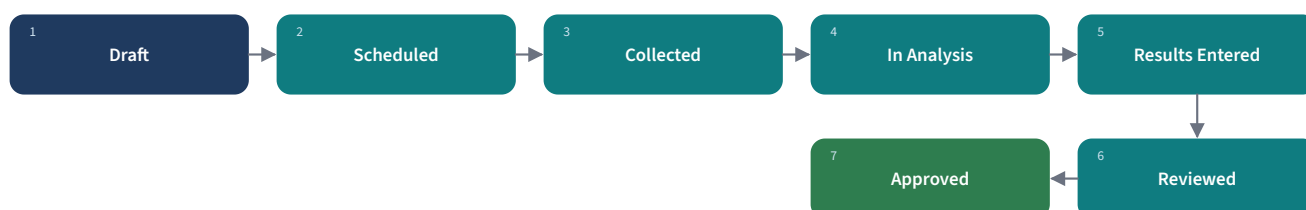
3 Responsibilities

Role (software group)	Responsibility
EM Technician	Collects, analyses, enters results.
EM Manager	Reviews and approves (different from the person who entered the results).

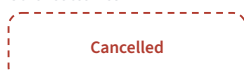
4 Definitions

Term	Definition
Evaluation	Automatic comparison with the approved limit: Within Limits, Alert/Action/Specification Limit Exceeded, No Approved Limit.
Amendment	Traced correction of a result after approval, with a reason, once only.

5 Workflow in the software



Other outcomes



Model: ls.env.sample

Figure 128 — Sample lifecycle.

6 Procedure

1 Prepare. Open **Operations ► Samples Due**; click **Schedule** for the day's samples; print labels.

- 2 **Collect.** Take the sample at the point in the stated occupancy state; click **Record Collection** with the actual date/time and collector; record the batch or campaign reference when relevant.
- 3 **Analyse.** Click **Start Analysis**; incubate/read according to method.
- 4 **Enter results.** Enter the value (or pass/fail) of each result; click **Submit Results**. The evaluation is computed.
- 5 **Review.** An EM Manager other than the result entrant clicks **Review**; if a value is doubtful, **Return to Analysis**.
- 6 **Approve.** An EM Manager other than the result entrant clicks **Approve**; excursions are raised automatically for action and specification breaches (and alerts when configured).
- 7 **Correct after approval.** Use **Amend Result** on the result with a reason; further corrections go through an investigation.
- 8 **Cancel.** A sample not taken is cancelled with a reason (**Cancel**) — never deleted.

7. Controls enforced by the software

- ✓ Collection date/time cannot be in the future; a sample needs at least one parameter before collection; every result needs a value before submission.
- ✓ Review and approval are reserved to the EM Manager role and must be done by users other than the one who entered results.
- ✓ Approved samples are frozen; results on approved samples change only by amendment (once, with reason).
- ✓ Only draft samples can be deleted; cancellation requires a reason; counts cannot be negative.

8. Good practices

- ★ Enter results the day they are read; record colony identification requests in the notes.
- ★ Watch *Threshold Breaches* daily, not only at approval.

9. Pitfalls to avoid

- ✗ Samples collected but recorded days later.
- ✗ Returning a sample to analysis to replace an inconvenient value without justification.

10 Records and evidence

- Samples with results, reviewer and approver
- Amendments

11 Performance indicators

- % samples collected on schedule
- Samples awaiting approval > 3 days

12 References

- EU GMP Annex 1 (2022) — Manufacture of Sterile Medicinal Products
- ISO 14644-1:2015 — Cleanrooms — Classification of air cleanliness by particle concentration
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)

13. Demonstration

Case. Settle plate C-12, 4 h In Operation, read after incubation: 31 CFU. The technician enters 31 and submits; evaluation *Alert Limit Exceeded*. The EM Manager reviews and approves; because *Raise Excursion on Alert* is ticked on the limit, excursion EXC-2026-018 is opened automatically (SOP-EM-004).

Management of excursions

Module	ls_environmental_monitoring	Version	_____
Menu	Environmental Monitoring ► Results ► Excursions	Effective date	_____
Owner (suggested)	EM Manager with QA	Approved by	_____
Related SOPs	SOP-EM-003, SOP-QA-020, SOP-QA-022		

1 Objective

Assess, investigate and close every environmental excursion with a documented product impact decision, and escalate to deviation and CAPA when required.

2 Scope

Excursions raised automatically at sample approval and excursions created manually.

3 Responsibilities

Role (software group)	Responsibility
Excursion owner	Records immediate actions, impact assessment, root cause, corrective actions; proposes closure.
EM Manager / QA (closer)	Closes the excursion; must differ from the owner.
QA	Decides on batch impact through the deviation process.

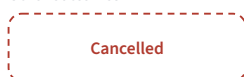
4 Definitions

Term	Definition
Breach type	Alert, Action or Specification Limit Exceeded.
Product impact	No Product Impact Identified, Potential, Confirmed, Not Yet Assessed.

5 Workflow in the software



Other outcomes



Model: ls.env.excursion

Figure 130 — Lifecycle of an excursion.

6 Procedure

- Immediate actions.** Record immediate actions (resampling, cleaning, room check, identification of isolates).
- Assess.** Click **Start Assessment**; write the impact assessment: batches in the room during the period (*Batch / Campaign Reference*), criticality of the point, isolate identity, other results.

- 3 **Investigate.** When required, click **Open Investigation**, determine the root cause and corrective actions. The module cannot create a CAPA or deviation itself: open them in *Is_deviation* / *Is_capa* and write the reference in *External Record Reference*.
- 4 **Propose closure.** Set the product impact and click **Propose Closure**.
- 5 **Close.** A user other than the owner clicks **Close** and writes the closure justification; **Reopen Investigation** if the proposal is insufficient.

7. Controls enforced by the software

- ✓ Excursions cannot be deleted, only cancelled; closed excursions are frozen.
- ✓ An impact assessment is required before investigation and before proposing closure; the product impact must be assessed; a root cause is required when an investigation was required.
- ✓ Closure requires a justification and a user other than the owner.
- ✓ The *create external record* action explains that no corrective action module is linked: the reference must be recorded manually.

8. Good practices

- ★ Identify isolates to species level for grade A/B action breaches.
- ★ Link the excursion to the batch record review: QA must see it before release.

9. Pitfalls to avoid

- ✗ Closing with *No Product Impact Identified* before the batch has been evaluated.
- ✗ Treating repeated alerts individually instead of looking at the trend.

10 Records and evidence

- Excursion with assessment, investigation, closure
- Deviation / CAPA references

11 Performance indicators

- Open excursions > 30 days
- Excursions per area per quarter

12 References

- EU GMP Annex 1 (2022) — Manufacture of Sterile Medicinal Products
- EudraLex Vol. 4, EU GMP Part I, Chapter 1 — Pharmaceutical Quality System
- ICH Q9(R1) — Quality Risk Management

13. Demonstration

Case (continued). EXC-2026-018 (alert, grade C corridor). Immediate action: resample and check door interlock — found faulty. Impact assessment: grade C corridor, no open product; product impact *No Product Impact Identified*. Root cause: door interlock sensor failure; corrective action: sensor replaced (work order WO-2231). The owner proposes closure; the QA microbiologist closes it with justification.

Environmental Monitoring

Operations

Results

Trending

Programme

Configuration

7

14

My Company (San Francisco)

Excursions

EMX/2026/00001

1 / 1

Open Investigation

Propose Closure

Cancel

Open

Impact Assessment

Investigation

Pending Closure

Closed

Send message

Log note

Activity

EMX/2026/00001

DETECTION

Sample

EMS-2026-0918

Sampling Point

[C-12] Corridor near material airlock

Area

Sterile block [STB] / Corridor C-2 [C2]

Detected On

Sep 27, 12:09 PM

Breach Type

Alert Limit Exceeded

Batch / Campaign Reference

HANDLING

Owner

QA Reviewer (demo)

Severity

Minor

Product Impact

No Product Impact Identified

Investigation Required

☐

External Record Reference

Company

My Company (San Francisco)

Breaching Results

Immediate Actions

Impact Assessment

Investigation

Parameter	Measured V...	Outcome	Unit	Alert	Action	Spec	Outcome
[SP] Settle plate 4 h	31.0000		CFU/4h	25.0000	50.0000	0.0000	Alert Limit Exceeded

QA Reviewer (demo)

12:09 PM

None

→

Corridor resampled; airlock door interlock checked - found faulty.

(Immediate Actions)

None

→

Grade C corridor, no open product; filling room pressure cascade maintained (DP records reviewed).

(Impact Assessment)

Not Yet Assessed

→

No Product Impact Identified

(Product Impact)

None

→

Minor

(Severity)

Open

→

Impact Assessment

(Status)

QA Reviewer (demo)

12:09 PM

Environmental Monitoring Excursion created

Figure 131 — Excursion raised automatically at sample approval, in assessment. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Trend analysis

Module	ls_environmental_monitoring	Version	_____
Menu	Environmental Monitoring ► Trending	Effective date	_____
Owner (suggested)	EM Manager	Approved by	_____
Related SOPs	SOP-EM-002, SOP-EM-004, SOP-GEN-001		

1 Objective

Detect adverse trends in environmental data before limits are breached, review them formally, and feed limit reviews and management review.

2 Scope

All EM results, per area, point and parameter.

3 Responsibilities

Role (software group)	Responsibility
EM Manager	Runs and reviews the analyses.
QA	Receives the conclusions for management review.

4 Definitions

Term	Definition
Trend line	Per point and parameter: counts, exceedances, direction (Increasing, Stable, Decreasing, Insufficient Data).
Direction threshold	Change threshold used to call a trend increasing or decreasing.

5 Workflow in the software



Model: ls.env.trend

Figure 132 — Lifecycle of a trend analysis.

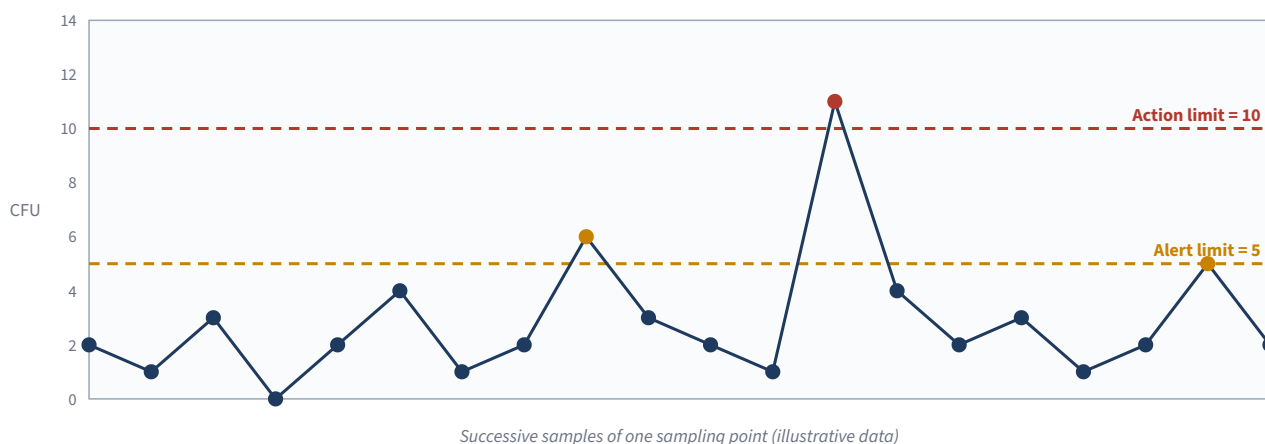


Figure 133 — A trend reveals drifts that single results hide (illustrative).

6 Procedure

- 1 **Run.** Monthly (and quarterly for the report), open **Trending ► Run Trend Analysis**: period, areas, points, parameters, direction threshold; click **Run Analysis** (or create an analysis and click **Compute**).
- 2 **Interpret.** Look at increasing directions, repeated alerts, new isolates, seasonal effects.
- 3 **Conclude.** Write the conclusion and click **Mark as Reviewed**; a reviewed analysis cannot be recomputed — create a new one if needed.
- 4 **Act.** Open a CAPA for adverse trends; revise limits (SOP-EM-002) or the plan (SOP-EM-001) when justified.
- 5 **Report.** Include the quarterly conclusions in management review.

7. Controls enforced by the software

- ✓ The end of the period must not precede its start; only a computed analysis can be reviewed; a conclusion is required; reviewed analyses cannot be recomputed.
- ✓ Counts and thresholds must be non-negative; exceeded count $\leq$ total count.

8. Good practices

- ★ Trend by area as well as by point: a room can drift while each point stays within limits.
- ★ Keep the same period and threshold between analyses for comparability.

9. Pitfalls to avoid

- ✗ Trending only exceedances instead of all results.

10 Records and evidence

- Reviewed trend analyses with conclusions

11 Performance indicators

- Increasing trends without action

12 References

- EU GMP Annex 1 (2022) — Manufacture of Sterile Medicinal Products
- ICH Q10 — Pharmaceutical Quality System

13. Demonstration

Case. Q3 analysis for the grade C area: point C-12 settle plates *Increasing* (mean 8 → 17 CFU), still under the alert limit except once. Conclusion: adverse trend linked to the new material airlock routine; CAPA-2026-044 opened to review the airlock cleaning frequency.

Recalls

ls_recall

ls_recall manages recall plans (coordinator and deputy, decision team, authorities, scope, default effectiveness level, target reconciliation rate, initiation target in hours, mock recall interval), and field actions of five types (recall, market withdrawal, stock recovery, field safety corrective action, mock recall): health hazard classification, depth, public warning, affected lots, distribution traced from stock moves, consignee lines with reconciliation, communications with content checklist, effectiveness checks by level A-E, reports, and gated closure. Lots under an open recall cannot be delivered.

Menu	Content
Recalls ► Configuration ► Recall Plans	Plans
Recalls ► Field Actions ► Initiate Recall / Recalls / All Field Actions	Field actions
Recalls ► Execution ► Consignees / Communications / Effectiveness Checks / Reports	Execution

Group	Main rights
Viewer · Coordinator	Read; run recalls
Manager	Approves plans, closes recalls (including with a justified override)

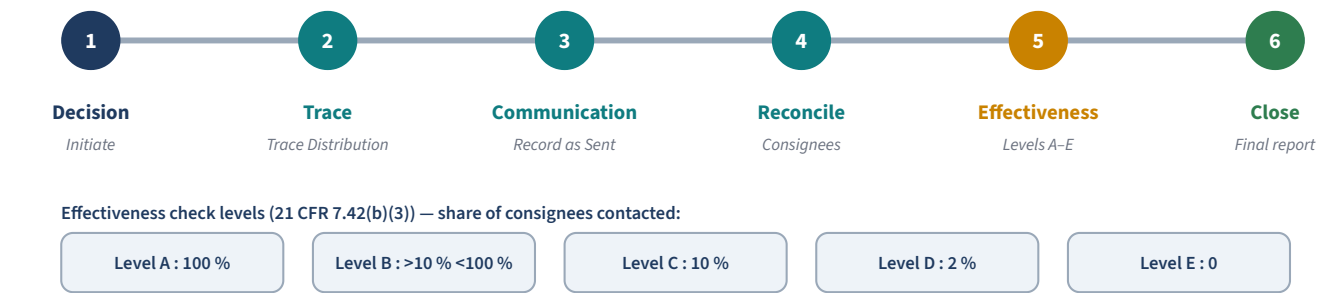
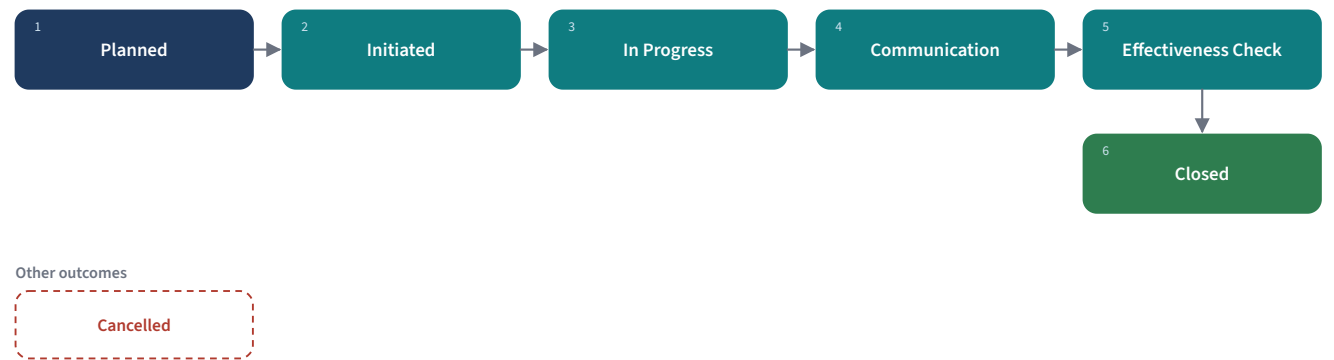


Figure 134 — Recall phases and effectiveness levels.



Model: ls_recall.execution
Figure 135 — Lifecycle of a field action.

Recall plan, with coordinator and deputy

Module	ls_recall	Version	_____
Menu	Recalls ► Configuration ► Recall Plans	Effective date	_____
Owner (suggested)	Head of Quality	Approved by	_____
Related SOPs	SOP-QA-081, SOP-QA-083		

1 Objective

Have an approved plan that allows a recall to be decided and started at any time, including outside office hours, by named people.

2 Scope

All products marketed by the company, or selected products or categories when several plans exist.

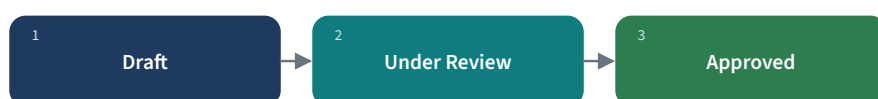
3 Responsibilities

Role (software group)	Responsibility
Head of Quality	Approves the plan.
Recall coordinator and deputy	Initiate and run recalls; available 24/7 by rota.
Decision team	Decides on recalls with the coordinator (QA, regulatory, medical, supply chain, management).

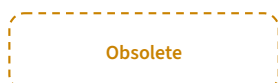
4 Definitions

Term	Definition
Initiation target	Hours between the decision and the first communication.
Target reconciliation rate	Share of distributed quantity to be accounted for before closure.

5 Workflow in the software



Other outcomes



Model: ls_recall.plan

Figure 136 — Lifecycle of a recall plan.

6 Procedure

- Draft.** In **Recall Plans** click **New**: title, coordinator, deputy, decision team, competent authorities, scope (all products, selected products or categories), procedure (the recall procedure text or reference), default effectiveness level, target reconciliation rate, initiation target (hours), mock recall interval (months).

- 2 **Review and approve.** Click **Submit for Review** ; the approver clicks **Approve** (or **Reject**).
- 3 **Publish contacts.** Give the coordinator/deputy contact list to the security desk and switchboard.
- 4 **Revise.** To change an approved plan click **New Revision** ; the old plan is made obsolete when the new one is approved (**Set Obsolete** if withdrawn).
- 5 **Test.** Test the plan through mock recalls (SOP-QA-083).

7. Controls enforced by the software

- ✓ A plan cannot be approved without a documented procedure or without a deputy coordinator.
- ✓ A plan limited to selected products or categories must list them.
- ✓ Approved plans are frozen for key fields — create a new revision; only draft plans can be deleted.

8. Good practices

- ★ Name the deputy from another department than the coordinator's to avoid simultaneous absence.
- ★ Include authority contact details and after-hours numbers in the procedure.

9. Pitfalls to avoid

- ✗ Plans naming people who left the company.

10 Records and evidence

- Approved plan with revisions

11 Performance indicators

- Plan reviewed in the last 12 months
- Coordinator/deputy reachable in mock recall test

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 8 — Complaints and Product Recall
- 21 CFR Part 7, Subpart C — Recalls (7.40 to 7.59)
- MDR Articles 87 to 92 — Vigilance

13. Demonstration

Case. Plan *Recall* — *all products*, coordinator: QA manager; deputy: supply chain manager; initiation target 24 h; default level B at 20 %; target reconciliation 95 %; mock recall every 12 months. Approved by the Head of Quality.

Recalls
Field Actions
Execution
Configuration
7
14
My Company (San Francisco)

New
Recall Plans
[RCP/2026/0001] Recall plan - finished medicin...
Recalls
0
1 / 1

Submit for Review
Draft
Under Review
Approved
Send message
Log note
Activity

Title ?

Recall plan - finished medicinal products

RCP/2026/0001 version 1

RESPONSIBILITIES	SCOPE
Recall Coordinator ? Deputy ? Decision Team ? Competent Authorities ? Company	OdooBot Scope Type All Products National Medicines Agency (demo) X My Company (San Francisco)

Procedure
Default Strategy
Approval

Decision route: the recall coordinator convenes the decision team within four hours of a quality defect being classified as potentially recall-relevant.

Out-of-hours: the deputy coordinator holds the duty telephone and may initiate a class I recall alone.

Disposition: returned product is quarantined in the blocked-stock location pending a disposition decision by the quality unit.

Today

OdooBot
10:40 AM
Recall Plan created

Figure 137 — Recall plan with coordinator and deputy. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Executing a recall or withdrawal: distribution tracing, notification, reconciliation

Module	ls_recall	Version	_____
Menu	Recalls ► Field Actions	Effective date	_____
Owner (suggested)	Recall coordinator	Approved by	_____
Related SOPs	SOP-QA-080, SOP-QA-082, SOP-QA-040, SOP-QA-021		

1 Objective

Remove defective product from the market quickly and completely: decide, trace distribution, notify consignees and authorities, and reconcile quantities.

2 Scope

Recalls, market withdrawals, stock recoveries and field safety corrective actions. Mock recalls follow SOP-QA-083.

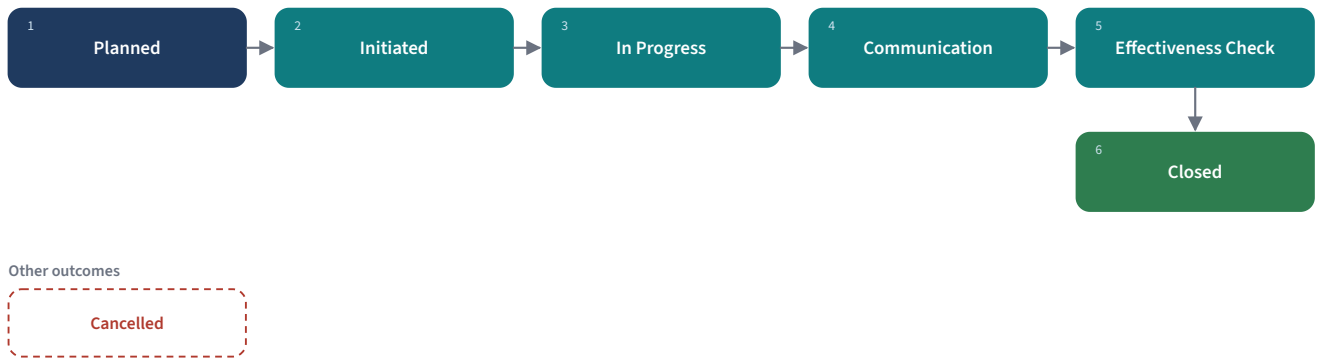
3 Responsibilities

Role (software group)	Responsibility
Recall coordinator (or deputy)	Opens and drives the field action.
Decision team	Decides, classifies the health hazard, approves communications.
Regulatory affairs	Notifies the competent authorities.
Supply chain / customer service	Contacts consignees, receives returns.

4 Definitions

Term	Definition
Classification	Health hazard class I, II or III (structure of 21 CFR 7.3(m)); the authority may assign it.
Depth	Wholesale, Retail or Consumer/User level.
Consignee line	Consignee × lot with quantities shipped, returned, destroyed on site, not recoverable; status Pending Notification → Notified → Response Received → Reconciled, or Discrepancy.

5 Workflow in the software



Model: ls.recall.execution

Figure 138 — Lifecycle of a field action.

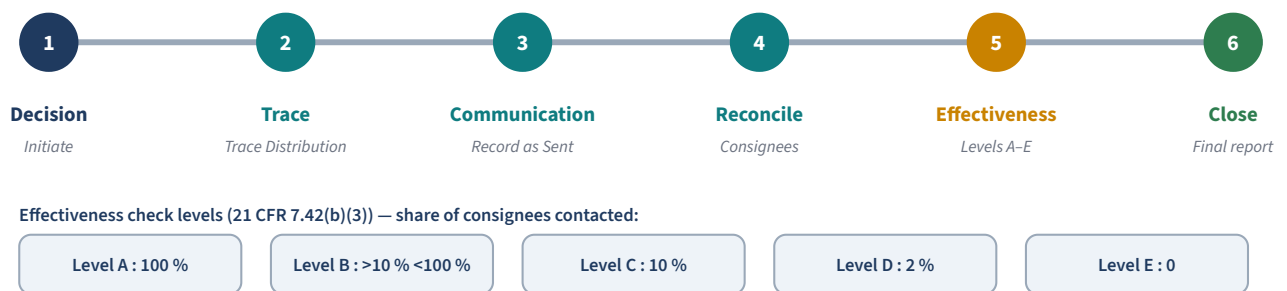


Figure 139 — Phases of a recall.

6 Procedure

- 1 Decide.** The decision team meets; record the decision. Open **Field Actions ▶ Initiate Recall**: action type, plan, product, lots, reason, classification, depth; click **Open Recall**.
- 2 Complete the strategy.** Record the health hazard evaluation (who, when), public warning, effectiveness level (Level B percentage), target completion date, quality defect and CAPA references.
- 3 Initiate.** Click **Initiate** (reason and affected lots required; for a real action, classification, depth and health hazard evaluation required).
- 4 Trace and start.** Click **Trace Distribution and Start**: consignee lines are created from the delivered stock moves of the lots. Add consignees manually for distribution outside the system. Lots under an open recall are blocked for delivery.
- 5 Communicate.** Create communications (Recall Notice, Field Safety Notice, Authority Notification, Public Warning...): recipients, subject, body; confirm the content checklist (product identified, reason and hazard, instructions, response requested, contact point); click **Approve Text**, send, then **Record as Sent**; record acknowledgements.
- 6 Move to communication.** Click **Move to Communication** (at least one communication recorded as sent).
- 7 Reconcile.** Update each consignee line with quantities returned, destroyed, not recoverable and responses; mark notified; investigate discrepancies.
- 8 Notify the authority.** Record authority notification date and reference; issue status reports (**Execution ▶ Reports**) as required.
- 9 Continue.** Go to effectiveness checks (SOP-QA-082).

7. Controls enforced by the software

- ✓ Once initiated, a recall cannot be deleted (cancel it); key fields are under change control.
- ✓ Initiation requires reason and affected lots; a real field action also requires classification, depth and health hazard evaluation.
- ✓ Lots must belong to the product; target completion date $\geq$ decision date; Level B requires a percentage between 10 % and 100 % exclusive.
- ✓ If no distribution is found, the system says so and consignees must be added manually.
- ✓ Communications: content checklist confirmed, recipients and text required, approval before sending; sent communications are frozen.
- ✓ Deliveries of lots under an open recall are refused.
- ✓ A finalised recall cannot be modified or re-traced.

8. Good practices

- ★ Start communication within the plan's initiation target; record the first communication date.
- ★ Call critical consignees (hospitals) in addition to written notices.
- ★ Keep all correspondence attached to the communication records.

9. Pitfalls to avoid

- ✗ Tracing only in the ERP when part of the distribution was handled by a third-party logistician — reconcile with their data.
- ✗ Forgetting samples, returns in transit and internal stock.

10 Records and evidence

- Field action with consignees, communications, reports
- Authority notifications

11 Performance indicators

- Initiation delay (hours) vs target
- Response rate
- Reconciliation rate

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 8 — Complaints and Product Recall
- 21 CFR Part 7, Subpart C — Recalls (7.40 to 7.59)
- MDR Articles 87 to 92 — Vigilance
- Regulation (EC) No 1223/2009, Article 23 — Serious undesirable effects

13. Demonstration

Case. Complaint investigation shows a labelling mix-up on lot 26C044. The decision team decides a Class II recall at pharmacy level. The coordinator opens RCL-2026-002, records the health hazard evaluation, level B 25 %, and initiates. **Trace Distribution and Start** creates 42 consignee lines from deliveries. The recall notice is approved, sent by e-mail and letter within 18 hours; the authority is notified the same day. After 10 days 40 of 42 consignees responded; returned + destroyed = 96 % of shipped quantity.

Recalls
Field Actions
Execution
Configuration

7
14
My Company (San Francisco)

New
Recalls
RCL/2026/0001 - Recall - Paracetamol 500 mg ...

Consignees
0
Messages
0
Checks
0
Reports
0

1 / 1

Initiate
Cancel
Planned
Initiated
In Progress
Communication
Effectiveness Check
Closed

Send message
Log note
Activity

RCL/2026/0001

DECISION

Action Type ?

Recall

Classification ?

Class II

Depth ?

Retail Level

Public Warning ?

No Public Warning

Decision Date ?

Target Completion Date

SCOPE

Company

My Company (San Francisco)

Recall Plan ?

Product

Paracetamol 500 mg tablets (...)

Affected Lots / Serial Numbers ?

LOT-DEMO-2601 X

Recall Coordinator

OdooBot

Reason and Hazard

Strategy

Consignees

Communications

Effectiveness Checks

Reports

Reason ?

Out-of-specification dissolution result obtained during ongoing stability testing at the 12 month time point.

Quality Defect Reference ?

DEV-DEMO-0042

Capa Reference ?

HEALTH HAZARD EVALUATION

Record the assessment of the health hazard, including the population at risk, the seriousness and the likelihood of occurrence.

OdooBot 10:40 AM

Recall record created.

OdooBot 10:40 AM

Recall Execution created

Figure 140 — Recall record. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Recall effectiveness checks

Module	ls_recall	Version	_____
Menu	Recalls ► Execution ► Effectiveness Checks · Field Actions ► Close Recall	Effective date	_____
Owner (suggested)	Recall coordinator	Approved by	_____
Related SOPs	SOP-QA-081, SOP-QA-083		

1 Objective

Verify that consignees received the recall communication and took the requested action, at the level defined in the strategy, and close the recall only when the closure gates are met.

2 Scope

All real field actions; mock recalls use the same mechanism (SOP-QA-083).

3 Responsibilities

Role (software group)	Responsibility
Recall coordinator	Plans and performs checks, prepares closure.
Recall Manager	Closes; may override failed gates only with a written justification.

4 Definitions

Term	Definition
Effectiveness level	A 100 %, B >10 % and <100 %, C 10 %, D 2 %, E no checks (21 CFR 7.42(b)(3)).
Outcome	Contacted – Action Taken, Contacted – No Action Taken, Contacted – Product Not Held, Consignee Not Reachable.

5 Workflow in the software



Other outcomes



Model: ls_recall.effectiveness

Figure 141 — Lifecycle of an effectiveness check.

6 Procedure

- Plan.** Click **Plan Effectiveness Checks**; checks are generated for the number of consignees required by the level.
- Move to the phase.** Click **Move to Effectiveness Check**.

- 3 **Perform.** Contact each selected consignee (telephone, e-mail, letter, visit, portal); record the outcome and date; click **Record as Performed**.
- 4 **Escalate.** For an unreachable consignee or no action taken, click **Escalate and Retry** (a new attempt is planned).
- 5 **Report.** Prepare the final report: situation summary, actions, product disposition, conclusion; **Review**, **Approve** by another person, **Record Submission** to the authority.
- 6 **Close.** Click **Close Recall**; the wizard shows each closure gate as PASS/FAIL (all consignees notified; no quantity discrepancy; required checks performed; reconciliation rate $\geq$ plan target for real actions). Enter the closure justification and attestation and confirm.

7. Controls enforced by the software

- ✓ Level C/D/E set the number of checks from the consignee count; Level B uses the percentage entered.
- ✓ A performed check needs an outcome and a performed date; only planned checks can be performed; finalised recalls freeze their checks.
- ✓ Closure with failed gates is refused unless the override is ticked with a statement, and only a Recall Manager may do it.
- ✓ Reports: reviewed before approval; approver different from reviewer; approved before submission; the recipient must be recorded; one final report per recall.

8. Good practices

- ★ Choose consignees for checks at random, but always include the largest customers.
- ★ Treat "product not held" answers carefully: check the shipment was received.

9. Pitfalls to avoid

- ✗ Closing with an override "to meet the deadline".

10 Records and evidence

- Effectiveness checks with outcomes
- Final report and submission
- Closure justification and attestation

11 Performance indicators

- Effectiveness coverage (%)
- Success rate of checks
- Recalls closed with override (target 0)

12 References

- 21 CFR Part 7, Subpart C — Recalls (7.40 to 7.59)
- EudraLex Vol. 4, EU GMP Part I, Chapter 8 — Complaints and Product Recall

13. Demonstration

Case (continued). Level B 25 % of 42 consignees → 11 checks planned (the number is rounded up by the system). 10 answer *Action Taken*; 1 not reachable → escalated, reached at second attempt. Closure gates: 42/42 notified, no discrepancy, 11/11 checks, reconciliation 96 % $\geq$ 95 %: all PASS. The final report is approved and submitted; the recall is closed.

Mock recall (periodic exercise)

Module	ls_recall	Version	_____
Menu	Recalls ► Field Actions ► Initiate Recall (Mock Recall)	Effective date	_____
Owner (suggested)	Head of Quality	Approved by	_____
Related SOPs	SOP-QA-080, SOP-QA-081, SOP-QA-082		

1 Objective

Test periodically that the recall arrangements work — people reachable, distribution traceable, communication fast — without contacting customers about a real defect.

2 Scope

At the interval set in the recall plan (e.g. yearly), for a representative product and lot.

3 Responsibilities

Role (software group)	Responsibility
Head of Quality	Triggers the exercise unannounced to the coordinator where possible.
Recall coordinator / deputy	Run the exercise as a real recall, up to the agreed limit.
QA	Evaluates and reports the results.

4 Definitions

Term	Definition
Mock recall	Action type <i>Mock Recall</i> ; it is excluded from regulatory KPI reporting and its closure does not check the reconciliation target.

5 Procedure

- 1 Prepare.** Choose a distributed lot and success criteria: time to reach the coordinator, time to trace 100 % of distribution, time to draft and approve the notice.
- 2 Trigger.** Out of hours if possible, call the coordinator; start the timer.
- 3 Run.** Open **Initiate Recall** with action type *Mock Recall*; trace distribution; prepare the communication (not sent to real customers unless agreed); perform effectiveness checks on internal or cooperating consignees.
- 4 Close.** Close the mock recall through the closure wizard.
- 5 Evaluate.** Compare times with the plan targets; record gaps; open CAPA for weaknesses; update the plan's last mock recall date.
- 6 Report.** Report the result in management review.

6. Controls enforced by the software

- ✓ Mock recalls are identified by their action type and excluded from regulatory KPI reporting; the reconciliation gate does not apply to them.

7. Good practices

- ★ Vary the scenario (different product, deputy instead of coordinator, weekend).
- ★ Include the third-party logistician in the exercise.

8. Pitfalls to avoid

- ✗ Announced exercises that test nothing.
- ✗ Sending a mock notice to real customers without warning them.

9 Records and evidence

- Mock recall record
- Evaluation report
- CAPA

10 Performance indicators

- Time to 100 % traceability
- Plan targets met in exercise

11 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 8 — Complaints and Product Recall
- 21 CFR Part 7, Subpart C — Recalls (7.40 to 7.59)

12. Demonstration

Case. Saturday 10:00, the Head of Quality calls the deputy: the coordinator is abroad. The deputy opens a mock recall for lot 26A011 at 10:40; distribution traced to 31 consignees by 11:25; notice drafted and approved by 14:10 (target 24 h: met). Gap: the switchboard list still showed the former deputy's number — CAPA opened.

III

Industry modules

Procedures specific to each industry: pharmaceutical manufacturing, cosmetics, medical devices, medical plastics and the quality control laboratory.

Pharmaceutical — ls_pharma	Medical plastics — ls_medical_plastics
Cosmetics — ls_cosmetics	Laboratory — ls_lab
Medical devices — ls_medical_device	

Pharmaceutical

ls_pharma

ls_pharma covers pharmaceutical manufacturing records and their quality decisions: batches (bulk, packaging, finished product) with yield limits and expiry; batch production and control records with master record check, steps, in-process and laboratory controls, area inspections (line clearance), labelling reconciliation, samples and discrepancies; an append-only batch release decision taken by the quality unit; qualified materials (APIs and excipients); stability studies; GS1 serialisation and aggregation; and CTD dossiers. Its controls cite 21 CFR Part 211; deliveries of lots whose batch is not released are refused.

Menu	Content
Pharmaceutical ► Batches ► Manufacturing Batches / Batch Records / Batch Release / Open Discrepancies	Batches
Pharmaceutical ► Materials ► APIs / Excipients / Pharmaceutical Products	Materials
Pharmaceutical ► Stability ► Stability Studies / Time Points / Stability Samples	Stability
Pharmaceutical ► Serialization ► Serial Numbers / Aggregation / Generate Serial Numbers	Serialisation
Pharmaceutical ► Regulatory ► CTD Dossiers	Regulatory

Group	Main rights
Pharmaceutical / Viewer · Production Operator	Read; execute batch record steps
Pharmaceutical / Production Manager	Manages batches and records
Pharmaceutical / Quality Assurance	Reviews records, decides release
Pharmaceutical / Regulatory Affairs	CTD dossiers
Pharmaceutical / Manager	Configuration

EU Qualified Person certification

The batch release decision of ls_pharma is the quality unit decision described by 21 CFR 211.22 and 211.192. In the EU, certification by a Qualified Person (EU GMP Annex 16) is an additional legal act: state in SOP-QA-090 how and where the QP certifies (register, signature), since the module does not model it.

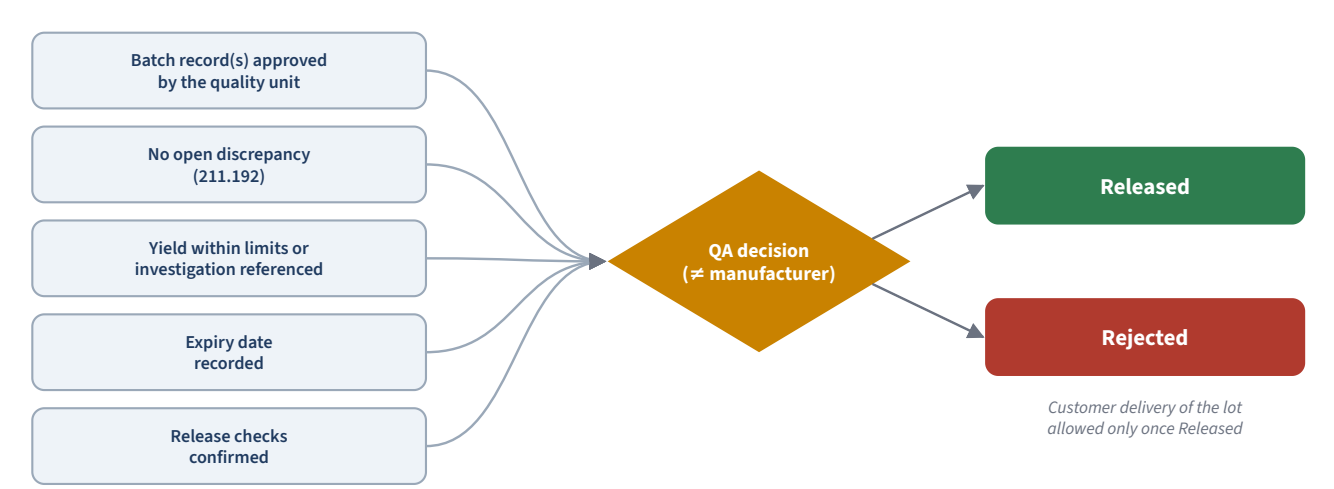
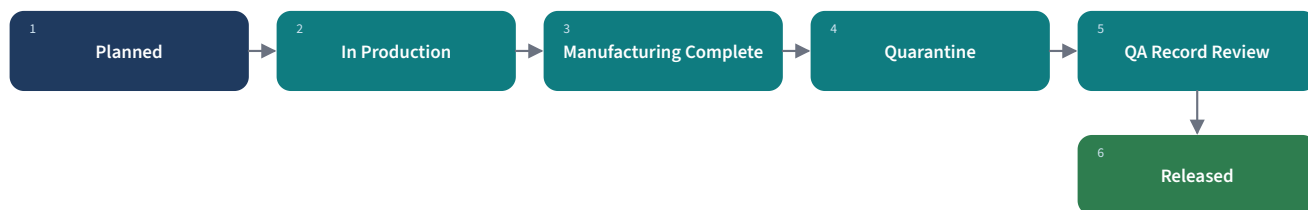


Figure 142 — Release gate of a batch.



Other outcomes



Model: ls.pharma.batch

Figure 143 — Lifecycle of a batch.

Issue and execution of the batch record

Module	ls_pharma	Version	_____
Menu	Pharmaceutical ► Batches ► Manufacturing Batches / Batch Records	Effective date	_____
Owner (suggested)	Production Manager	Approved by	_____
Related SOPs	SOP-PRD-002, SOP-QA-090, SOP-QA-020		

1 Objective

Issue for each batch a production and control record reproduced from the approved master record and checked, and execute it contemporaneously so that every significant step is dated, attributed and, where required, checked by a second person.

2 Scope

Manufacturing and packaging/labelling batch records of all pharmaceutical batches (bulk, packaging, finished product).

3 Responsibilities

Role (software group)	Responsibility
Production Manager	Creates the batch and issues the record.
Checker (second person)	Checks the reproduction of the master record; checks significant steps and component charges.
Production Operators	Execute steps and record values.
QA	Is informed of discrepancies; reviews the record (SOP-QA-090).

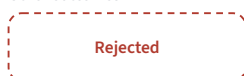
4 Definitions

Term	Definition
Master record	Approved master production and control record; referenced by number and version on each batch record.
Significant step	Step requiring identification of the person performing it and of the person checking it.

5 Workflow in the software



Other outcomes



Model: ls.pharma.batch_record

Figure 144 — Lifecycle of a batch production and control record.

Procedure

- 1 **Create the batch.** In **Manufacturing Batches** click **New** : reference, batch type, product, lot, manufacturing order, BOM, site, planned quantity, theoretical yield, yield limits, shelf life.
- 2 **Issue the record.** Add a batch record: record type (Manufacturing or Packaging and Labelling), master record reference and version; add steps, controls, planned samples and labelling items as defined in the master record.
- 3 **Check the reproduction.** A second person compares the issued record with the master and records *Master Record Checked By / On*.
- 4 **Start.** Click **Start Production** on the batch and **Start Execution** on the record.
- 5 **Execute steps.** For each step line of the record, perform the step, record the value, the date performed and the performer, then click **Done** on the line; for significant or automated steps, record the checker (*Checked or Supervised By*). A step that does not apply is set to *Not Applicable* with a remark.
- 6 **Components.** Record each component charge (lot, quantity, assay) and have it verified by a second person.
- 7 **Controls.** Record in-process and laboratory control results against their limits.
- 8 **Discrepancies.** Record any discrepancy immediately (classification, description) and click **Start Investigation** ; major events also go through SOP-QA-020.
- 9 **Complete.** Click **Complete Execution** when all steps are done or not applicable; record actual yield and click **Complete Manufacturing** on the batch.
- 10 **Submit.** Click **Submit for Review** on the record; place the batch in quarantine (**Place in Quarantine**) pending QA review.

7. Controls enforced by the software

- ✓ Execution cannot start until the reproduction of the master record has been checked, dated and attributed.
- ✓ A step cannot be completed without a date and the person who performed it; automated steps need the checker; *not applicable* needs a remark.
- ✓ A component charged and verified by the same user is refused (21 CFR 211.101(d)).
- ✓ The record cannot complete while steps remain open, and cannot be submitted/approved with open discrepancies; closing a discrepancy needs investigation, conclusion and follow-up.
- ✓ Manufacturing cannot be declared complete without actual yield; a batch without batch record cannot proceed.
- ✓ Expiry after manufacturing date; production end after start; minimum yield $\leq$ maximum; non-draft batches and records cannot be deleted.

8. Good practices

- ★ Record at the time of the operation; never pre-fill or back-date.
- ★ Keep the printed master record page references in the step instructions.
- ★ Record discrepancies even when "corrected immediately".

9. Pitfalls to avoid

- ✗ The same person charging and verifying components to save time.
- ✗ Steps marked done at the end of the shift.

10 Records and evidence

- Batch and batch production and control record with steps, controls, samples, discrepancies

11 Performance indicators

- Right-first-time records (no discrepancy)
- Records submitted within 24 h of completion

12 References

- 21 CFR 211.186 and 211.188 — Master and batch production and control records
- 21 CFR 211.192 — Production record review
- EudraLex Vol. 4, EU GMP Part I, Chapter 4 — Documentation
- EudraLex Vol. 4, EU GMP Part I, Chapter 5 — Production
- ICH Q7 — Good Manufacturing Practice for Active Pharmaceutical Ingredients

13. Demonstration

Case. Batch 26T018 of *Tablets 10 mg*: manufacturing record issued from master MR-TAB10 v7, checked by the shift supervisor. The operator performs step 5 *Dry granulation, 12 min* (value 12 min, done 08:42); the component *API lot A2210, 5.012 kg* is charged by the operator and verified by a second operator. An IPC LOD result is within limits. At the end, actual yield 98.6 % (limits 96–101 %); record submitted for review and batch placed in quarantine.

Step	S.	Recorded ...	Value Unit	Performed...	Checked ...	Date Performed	Status
Dispensin...	done			Productio...	Productio...	Sep 27, 12:08 PM	Completed

Figure 145 — Batch production and control record with steps performed and checked. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Line clearance and label reconciliation

Module	ls_pharma	Version	_____
Menu	Pharmaceutical ► Batches ► Batch Records (Area Inspections, Labelling Control)	Effective date	_____
Owner (suggested)	Packaging Manager	Approved by	_____
Related SOPs	SOP-PRD-001, SOP-QA-090		

1 Objective

Prevent mix-ups by verifying that lines and areas are free of previous materials before and after use, and reconcile the quantities of printed labelling materials issued, used, returned and destroyed.

2 Scope

All packaging and labelling operations, and manufacturing areas at product changeover.

3 Responsibilities

Role (software group)	Responsibility
Operator	Performs the area inspection and label counts.
Verifier	Verifies the inspection (not the same person).
Packaging supervisor	Investigates reconciliation differences.

4 Definitions

Term	Definition
Area inspection (line clearance)	Documented check, before or after use, that no material, document or label of the previous product remains.
Reconciliation	Issued – used – returned – destroyed; the difference must be within the tolerance.

5 Procedure

- Before use.** Add an area inspection: moment *Before Use*, area or line, previous product, performed on/by; record *Pass* or *Fail* with findings; a second person records *Verified By*.
- Fail.** If the inspection fails, describe the findings, clean again, repeat the inspection; do not start.
- Issue labelling.** Add each labelling item (label, leaflet, carton) with its version and code, and the quantity issued.
- Reconcile.** At the end, record used, returned and destroyed; the difference is computed; tick *Reconciled* when within tolerance and record who reconciled.
- Out of tolerance.** Record a discrepancy on the batch record and investigate (missing labels may be on another product).
- After use.** Perform and verify an inspection *After Use*.

6. Controls enforced by the software

- ✓ A failed area inspection requires recorded findings.
- ✓ The inspection cannot be verified by the person who performed it.
- ✓ Labelling quantities cannot be negative; unreconciled labels are counted on the record and shown to QA at review.

7. Good practices

- ★ Use a checklist of places to inspect (under conveyors, reject bins, printers).
- ★ Destroy excess coded labels and record their destruction.

8. Pitfalls to avoid

- ✗ Signing the clearance of a line you did not inspect.
- ✗ Rounding reconciliation differences to zero.

9 Records and evidence

- Area inspections with verification
- Labelling reconciliation

10 Performance indicators

- Failed clearances
- Batches with unreconciled labels at review

11 References

- 21 CFR 211.122 to 211.137 — Packaging and labelling control
- EudraLex Vol. 4, EU GMP Part I, Chapter 5 — Production
- 21 CFR 211.186 and 211.188 — Master and batch production and control records

12. Demonstration

Case. Before packaging batch 26T018, the operator inspects line PK-2 (previous product *Capsules 20 mg*): one leaflet of the previous product found under the feeder → *Fail*, findings recorded, line re-cleaned, second inspection *Pass*, verified by the supervisor. Labels: issued 12 500, used 12 180, returned 300, destroyed 20 → difference 0, reconciled.

Batch record review and batch release by the quality unit

Module	ls_pharma	Version	_____
Menu	Pharmaceutical ► Batches ► Batch Records / Batch Release	Effective date	_____
Owner (suggested)	Head of Quality Assurance	Approved by	_____
Related SOPs	SOP-PRD-001, SOP-PRD-002, SOP-QC-012, SOP-QA-021, SOP-DI-002		

1 Objective

Release a batch for distribution only after an independent quality unit has reviewed and approved its production and control records, confirmed that every discrepancy is investigated and closed, and verified the release checklist.

2 Scope

All pharmaceutical batches made or packaged on site, before distribution.

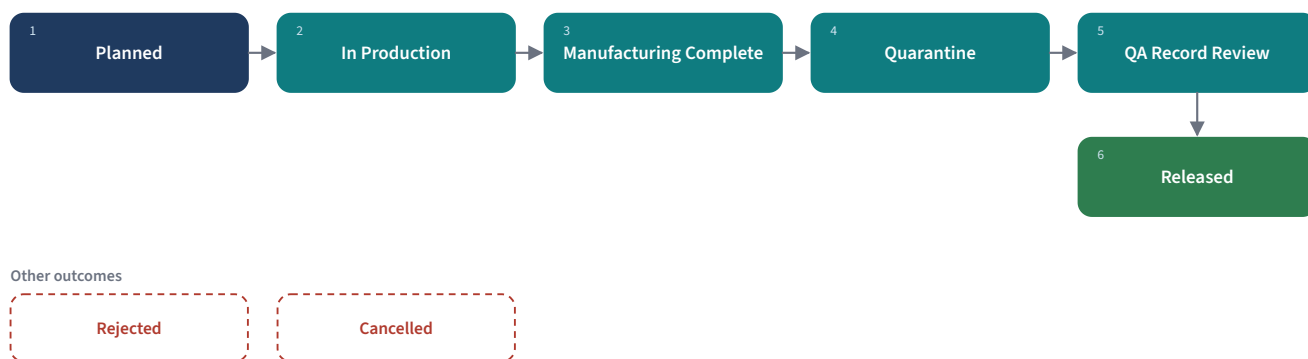
3 Responsibilities

Role (software group)	Responsibility
QA reviewer	Reviews and approves or rejects each batch record.
Release authority (QA; QP in the EU)	Takes and records the release decision; cannot be the manufacturer of the batch.
Production	Answers review questions and completes records returned for completion.

4 Definitions

Term	Definition
Release checklist	Batch records reviewed and approved; discrepancies investigated and closed; yield within limits; component charge-in verified by a second person; laboratory results conform; labelling reconciled; reserve samples retained; covered by the stability programme.
Release decision	Append-only record: Released for Distribution or Rejected, with statement, date, decider and integrity digest.

5 Workflow in the software



Model: ls.pharma.batch

Figure 146 — Batch lifecycle up to release.

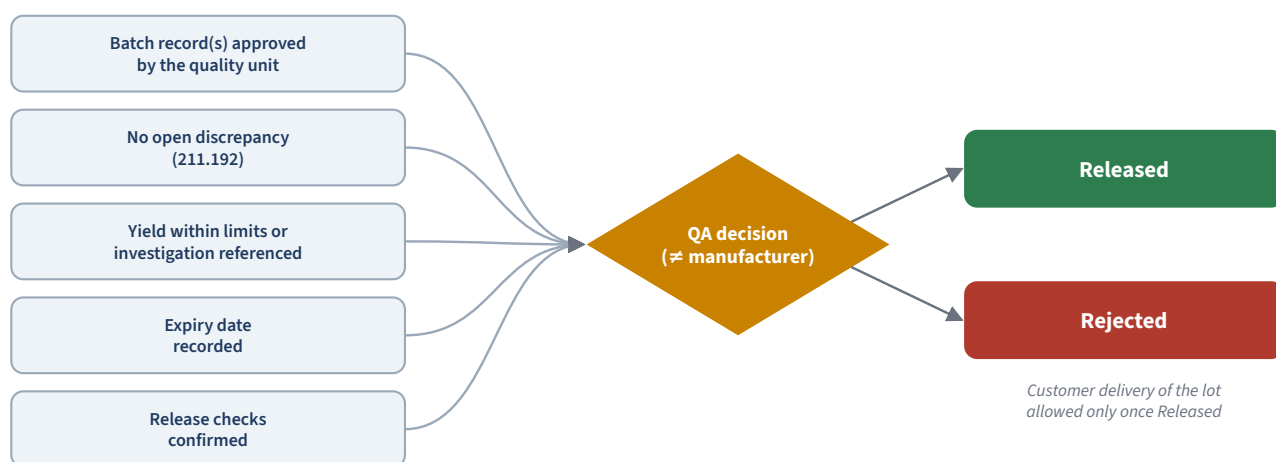


Figure 147 — What the release decision checks.

6 Procedure

- 1 **Review the record.** Open the submitted record: steps, controls (non-conforming controls counter), area inspections, labelling reconciliation (unreconciled counter), samples, discrepancies (open counter), audit trail extract where available (SOP-DI-002).
- 2 **Decide on the record.** Click **Approve** with the review conclusion, **Reject** with a written conclusion, or **Return for Completion** if entries are missing.
- 3 **Check laboratory results.** Confirm that the QC results of the batch are approved (SOP-QC-012) and that no OOS investigation is open.
- 4 **Submit the batch.** Click **Submit for QA Review** on the batch.
- 5 **Decide.** Click **Record Release Decision**: decision, statement, and tick each checklist item after verifying it. The decision is recorded as an append-only release decision with an integrity digest.
- 6 **EU QP certification.** Where applicable, the QP certifies the batch in the QP register according to Annex 16, referencing the release decision.
- 7 **Distribution.** Once released, deliveries of the lot are allowed; a rejected batch is handled as rejected material.
- 8 **Verify integrity.** Use **Verify Integrity** on a release decision during audits.

7. Controls enforced by the software

- ✓ A batch record cannot be reviewed by the user who executed it; rejection needs a written conclusion; approval is refused with open discrepancies; closed records are frozen.
- ✓ A release decision can only be recorded while the batch is under QA review, once per batch, and not by the person who manufactured it.
- ✓ Release is refused when checklist items are not confirmed, when a batch record is not approved, when yield is outside limits without an investigation reference, or without an expiry date.
- ✓ Release decisions are append-only: cannot be modified, duplicated or deleted.
- ✓ Deliveries of lots whose batch is not released are refused (stock move line control).

8. Good practices

- ★ Review by exception using the counters (open discrepancies, non-conforming controls, unreconciled labels) but read the whole record for critical steps.
- ★ Record the reasoning in the decision statement, especially for released batches with closed deviations.

9. Pitfalls to avoid

- ✗ Ticking the checklist without evidence.
- ✗ Releasing before the laboratory results are approved.

10 Records and evidence

- Approved batch records
- Release decision with checklist and digest
- QP register (EU)

11 Performance indicators

- Release lead time
- Batch records returned for completion
- Batches rejected

12 References

- 21 CFR 211.22 — Responsibilities of the quality control unit
- 21 CFR 211.192 — Production record review
- EU GMP Annex 16 — Certification by a Qualified Person and Batch Release
- EudraLex Vol. 4, EU GMP Part I, Chapter 1 — Pharmaceutical Quality System
- ICH Q7 — Good Manufacturing Practice for Active Pharmaceutical Ingredients

13. Demonstration

Case. QA reviews batch 26T018: two records, 0 open discrepancies, 0 non-conforming controls, labels reconciled; yield 98.6 % within 96–101 %; QC results approved. The QA officer (not involved in manufacturing) records *Released for Distribution* with all checklist items ticked. The QP certifies it in the register the same day; the lot can be shipped.

Pharmaceutical Batches Materials Stability Serialization Regulatory Reports Configuration 7 14 My Company (San Francisco)

New Manufacturing Batches 26T018 - Tablets 10 mg Batch Records 1 Stability 0 Serial Numbers 0 1 / 1

Planned In Production Manufacturing Complete Quarantine QA Record Review Released Send message Log note Activity

Batch Reference
26T018

PRODUCT		MANUFACTURING ORDER	
Product	Tablets 10 mg	Manufacturing Order	
Batch Type	Finished Product	Bill of Materials	
Inventory Lot	26T018	Manufacturing Site	
Bulk Batch		Company	My Company (San Francisco)

QUANTITIES AND YIELD		DATES	
Unit of Measure	Units	Production Start	Sep 27, 3:08 AM
Planned Quantity	120,000.00	Production End	Sep 27, 12:08 PM
Theoretical Yield	120,000.00	Manufacturing Date	Sep 27
Actual Yield	118,320.00	Shelf Life (Months)	36
Yield (%)	98.60	Expiry Date	Sep 26, 2029
Yield Limits (%)	96.00 to 101.00	Retest Date	
Yield Investigation Required	☐		

Components Equipment Co-Products Responsibility Notes

Component A.. Active Ing... Excipient Compone... Quantity Ch... Unit of Me... Assay (%) Potency-Co... Charg

Today

Q QA Reviewer (demo) 12:08 PM
Release decision REL/2026/00001 recorded by QA Reviewer (demo): released.

P Production Operator (demo) 12:08 PM
Pharmaceutical Manufacturing Batch created

Figure 148 — Released batch. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Pharmaceutical Batches Materials Stability Serialization Regulatory Reports Configuration 7 14 My Company (San Francisco)

Batch Release REL/2026/00001 - 26T018 1 / 1

Verify Integrity Open Batch Send message Log note

Decision Reference
REL/2026/00001

BATCH		DECISION	
Batch	26T018 - Tablets 10 mg	Decision	Released for Distribution
Product	Tablets 10 mg	Decision Date	Sep 27, 12:08 PM
Company	My Company (San Francisco)	Decided By	QA Reviewer (demo)
		Checklist Complete	☒

CHECKLIST CONFIRMED AT THE TIME OF THE DECISION

Batch Records Reviewed and Approved	☒	Laboratory Results Conform	☒
Discrepancies Investigated and Closed	☒	Labelling Reconciled	☒
Yield Within Limits	☒	Reserve Samples Retained	☒
Component Charge-In Verified by a Second Person	☒	Covered by the Stability Programme	☒

STATEMENT

All checks verified; QC

Today

Q QA Reviewer (demo) 12:08 PM
Batch Release Decision created

Figure 149 — Append-only release decision with its checklist. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Qualification of active substances and excipients

Module	ls_pharma	Version	_____
Menu	Pharmaceutical ► Materials ► APIs / Excipients	Effective date	_____
Owner (suggested)	QC Manager with QA	Approved by	_____
Related SOPs	SOP-QA-070, SOP-QC-010		

1 Objective

Maintain master data of active substances and excipients that are qualified before use: specification basis, approved manufacturers, retest period, storage, and specific requirements (animal origin, controlled substances).

2 Scope

All APIs and excipients used in manufacturing.

3 Responsibilities

Role (software group)	Responsibility
QC / QA	Qualify, restrict or make obsolete materials.
Purchasing	Buys only from approved manufacturers (see SOP-QA-070 / 073).

4 Definitions

Term	Definition
Potency basis	As Is, Anhydrous, Dried or Active Moiety basis; label assay must be positive.
TSE/BSE statement	Required reference for materials of animal origin.

5 Workflow in the software



Other outcomes



Model: ls.pharma.api

Figure 150 — States of a pharmaceutical material.

6 Procedure

- Create.** In **Materials ► APIs** or **Excipients** click **New** : name, internal code, inventory item, pharmacopoeia and monograph, CAS number, approved manufacturers, retest period, storage condition; for APIs: INN, active moiety, potency basis, label assay, controlled substance and schedule, sterile; for excipients: function, novel excipient, maximum daily intake, label declaration.
- Animal origin.** Tick *Of Animal Origin* and record the TSE/BSE statement reference.

3 **Qualify.** When the supplier is approved and the specification is in place, click **Qualify**.

4 **Restrict.** Click **Restrict** to limit use (e.g. pending an investigation); **Declare Obsolete** when no longer used.

7. Controls enforced by the software

- ✓ A controlled substance cannot be qualified until its schedule is recorded; a material of animal origin cannot be qualified without its TSE/BSE statement reference.
- ✓ Label assay > 0; retest period and maximum daily intake non-negative.
- ✓ An obsolete material cannot return to draft; a qualified material cannot be qualified twice.

8. Good practices

- ★ Keep the approved manufacturer list identical to the supplier qualification scope.

9. Pitfalls to avoid

- ✗ Using a new manufacturer's material under an old qualification.

10 Records and evidence

- Material master data with state history

11 Performance indicators

- Materials used while not qualified (target 0)

12 References

- ICH Q7 — Good Manufacturing Practice for Active Pharmaceutical Ingredients
- EudraLex Vol. 4, EU GMP Part I, Chapter 5 — Production
- 21 CFR 211.84 — Testing and approval or rejection of components

13. Demonstration

Case. Gelatin (excipient, function *Coating Agent*) is of bovine origin: the TSE statement reference *EDQM CEP R1-CEP 2000-xxx* is recorded before **Qualify** is accepted.

Stability programme

Module	ls_pharma	Version	_____
Menu	Pharmaceutical ► Stability	Effective date	_____
Owner (suggested)	QC Manager	Approved by	_____
Related SOPs	SOP-QC-014, SOP-QA-090		

1 Objective

Run registration, ongoing, follow-up and post-change stability studies according to approved protocols, pull and test samples on time, and use the results to support shelf life and storage conditions.

2 Scope

Stability studies of finished products and intermediates managed in ls_pharma. Laboratory-side sample testing may be managed in ls_lab (SOP-QC-014).

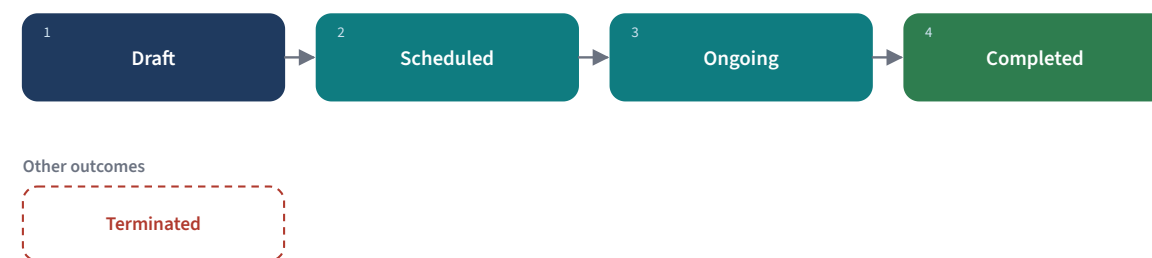
3 Responsibilities

Role (software group)	Responsibility
Stability coordinator	Creates studies, generates schedules, pulls samples.
Analysts	Test and record results.
QC Manager	Concludes studies and proposes shelf life.

4 Definitions

Term	Definition
Storage condition	Long term, intermediate, accelerated or other (temperature, humidity and tolerance).
Time point	Month of a study at a condition; scheduled date derived from study start.

5 Workflow in the software



Model: ls_pharma.stability_study

Figure 151 — Lifecycle of a stability study.

6 Procedure

- 1 Create.** In **Stability Studies** click **New**: title, study type, product, batch, protocol reference, packaging configuration, start, planned duration, storage conditions.
- 2 Schedule.** Click **Generate Schedule** to create time points per condition; review scheduled dates.

- 3 **Start.** Click **Start Study** ; place samples in storage (stability samples).
- 4 **Pull and test.** At each time point click **Pull Samples** , test, record results, then **Mark Tested** and **Complete** . A time point not pulled is recorded with **Mark Missed** and a remark.
- 5 **Out of specification.** Handle non-conforming results through an OOS investigation (SOP-QC-013); significant change is recorded on the study.
- 6 **Conclude.** When all time points are final, write the conclusion (proposed shelf life, storage) and click **Complete Study** ; **Terminate Study** with justification if stopped early.

7. Controls enforced by the software

- ✓ Time points can only be generated in draft or scheduled studies, and require at least one storage condition; a study needs time points to start.
- ✓ A time point cannot be tested before its samples are pulled; tested requires results; missed requires a remark.
- ✓ Completion requires every time point in a final state and a conclusion (21 CFR 211.166); termination requires a justification.
- ✓ Samples cannot be allocated to another study's time point or stored under a different condition; discarding requires a remark.

8. Good practices

- ★ Enrol at least one batch per year per product and strength in the ongoing programme.
- ★ Watch overdue time points weekly.

9. Pitfalls to avoid

- ✗ Pulling samples late without recording the actual date.

10 Records and evidence

- Study, time points, samples, results, conclusion

11 Performance indicators

- Time points pulled on time
- Studies with significant change

12 References

- ICH Q1A(R2) — Stability Testing of New Drug Substances and Products
- 21 CFR 211.166 — Stability testing
- EudraLex Vol. 4, EU GMP Part I, Chapter 6 — Quality Control

13. Demonstration

Case. Ongoing stability of batch 26T018: long term 25 °C/60 % RH and accelerated 40 °C/75 % RH, planned 36 months; time points generated at 0, 3, 6, 9, 12, 18, 24, 36 months. At 6 months accelerated, dissolution is out of specification → OOS investigation opened; the study continues.

GS1 serialisation and aggregation

Module	ls_pharma	Version	_____
Menu	Pharmaceutical ► Serialization	Effective date	_____
Owner (suggested)	Packaging Manager with IT	Approved by	_____
Related SOPs	SOP-PRD-001, SOP-QA-090		

1 Objective

Assign a unique identifier to each saleable unit, commission it only for released packaging, aggregate units into containers with SSCC codes, and keep the status of each identifier accurate until it leaves the company.

2 Scope

Products flagged as requiring serialisation. Connection to national verification systems (e.g. EU hub) is outside the module and must be described separately.

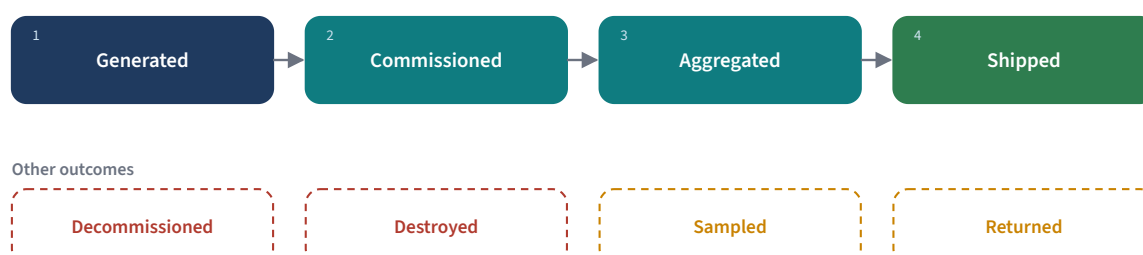
3 Responsibilities

Role (software group)	Responsibility
Packaging operator	Generates, commissions and aggregates.
Packaging Manager	Decommissions with reason; supervises.
IT / serialisation partner	Exchanges data with the repository system.

4 Definitions

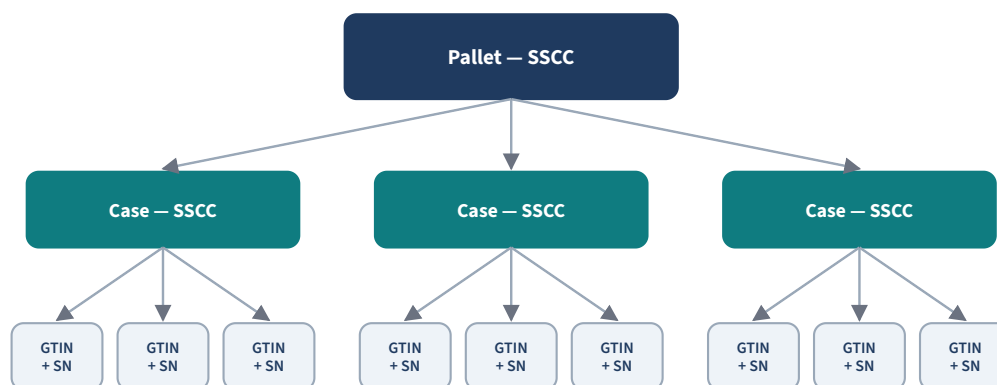
Term	Definition
Unique identifier	GTIN-14 product code + serial number + batch + expiry (+ national reimbursement number when required).
SSCC	18-digit Serial Shipping Container Code with modulo-10 check digit; built from the company GS1 prefix and extension digit set in company settings.
Commissioned / Decommissioned	Active in the supply chain / withdrawn (with reason); other terminal states: destroyed, sampled, returned.

5 Workflow in the software



Model: ls.pharma.serialization

Figure 152 — States of a serialised unit.



Saleable unit: GTIN-14 + serial number (Commissioned before shipping). Container: 18-digit SSCC with modulo-10 check digit.

Figure 153 — Serialisation and aggregation hierarchy.

6 Procedure

- 1 **Configure.** Set the GS1 company prefix, SSCC extension digit and serial length in [Configuration ► Company Settings](#); enter the GTIN-14 on each serialised product.
- 2 **Generate.** Use [Serialization ► Generate Serial Numbers](#) for the batch.
- 3 **Commission.** Click [Commission](#) on units actually printed and verified on the line.
- 4 **Aggregate.** Create containers (bundle, case, pallet) with their SSCC; add units or child containers; click [Pack](#).
- 5 **Ship.** Click [Mark Shipped](#) on containers and units shipped.
- 6 **Correct.** Use [Disaggregate](#) on a packed (not shipped) container to rework; [Decommission](#) units damaged, sampled or destroyed with a reason.

7. Controls enforced by the software

- ✓ GTIN-14 and SSCC formats and check digits are validated; serialised products must carry a product code.
- ✓ Commissioned units cannot be deleted (decommission instead) and their identifier can no longer change; terminal states are final.
- ✓ Only generated units can be commissioned; units must be commissioned before shipping; containers cannot be packed empty or with uncommissioned units, cannot contain themselves, and cannot be disaggregated once shipped.

8. Good practices

- ★ Reconcile generated, commissioned and decommissioned counts per batch at batch record review.
- ★ Keep the rework of aggregated cases under supervision.

9. Pitfalls to avoid

- ✗ Commissioning the full range generated instead of the units actually printed.

10 Records and evidence

- Serial numbers with status history
- Aggregation hierarchy

11 Performance indicators

- Decommissioned units per batch
- Aggregation errors

12 References

- Commission Delegated Regulation (EU) 2016/161 — Safety features on medicinal product packaging
- GS1 General Specifications (GTIN, SSCC, check digit)

13. Demonstration

Case. Batch 26T018 of 12 000 boxes: 12 100 serials generated, 12 000 commissioned; 20 boxes per case (600 cases) and 40 cases per pallet (15 pallets), each with an SSCC. Two damaged boxes are decommissioned with the reason *damaged on line*; their case is disaggregated and repacked.

Preparation of the CTD dossier (Common Technical Document)

Module	ls_pharma	Version	_____
Menu	Pharmaceutical ► Regulatory ► CTD Dossiers	Effective date	_____
Owner (suggested)	Regulatory Affairs Manager	Approved by	_____
Related SOPs	SOP-QA-030, SOP-RA-010		

1 Objective

Plan and track the preparation of marketing authorisation dossiers section by section, record submission, deficiencies and decision, and keep evidence of each section's source document.

2 Scope

New applications, variations, renewals and transfers prepared in CTD format. The module tracks status and references; it does not compile or publish eCTD sequences.

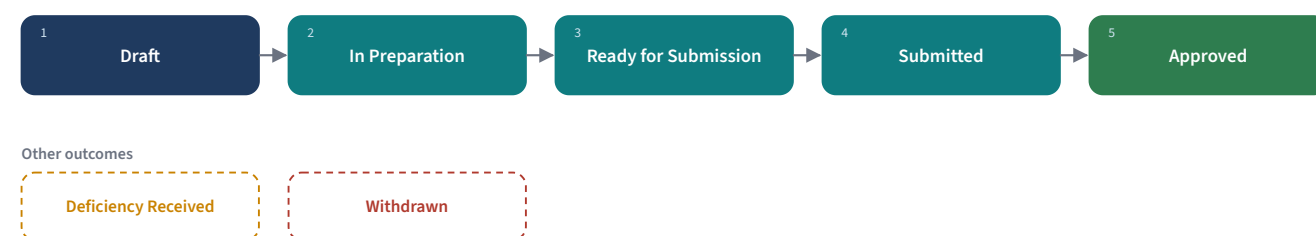
3 Responsibilities

Role (software group)	Responsibility
Regulatory Affairs Manager	Owns the dossier and its planning.
Section owners	Prepare sections and provide document references.
Management	Approves submission.

4 Definitions

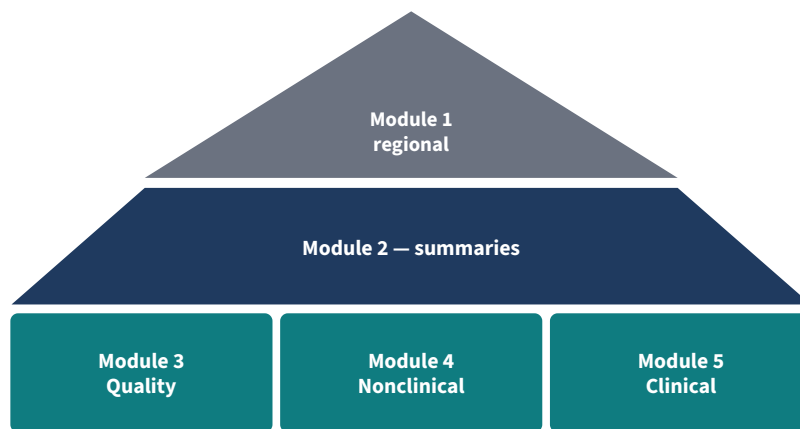
Term	Definition
Section	CTD module (1 to 5), number and title, owner, due date, document reference, status.
Deficiency	Question or objection received from the authority on a submitted dossier.

5 Workflow in the software



Model: ls.pharma.ctd_dossier

Figure 154 — Lifecycle of a CTD dossier.



ICH M4 structure; ls_pharma tracks the status of each section (Load CTD Structure).

Figure 155 — CTD structure.

6 Procedure

- 1 **Create.** In **CTD Dossiers** click **New** : title, product, dossier type, authority, contact, MA holder, version.
- 2 **Load the structure.** Click **Start Preparation** then **Load CTD Structure** to copy the template sections (configured in **Configuration ► CTD Section Template**).
- 3 **Prepare sections.** Assign owners and due dates; each owner clicks **Start Preparation** on the section, then **Mark Ready** with the document reference.
- 4 **Ready.** When all sections are ready click **Mark Ready** on the dossier.
- 5 **Submit.** After internal approval and submission through the authority's channel, click **Record Submission** with the date.
- 6 **Deficiencies.** On receipt of questions click **Record Deficiency** ; set the affected sections to deficiency with a description; answer and mark them ready again.
- 7 **Decision.** Click **Record Approval** with the authorisation number and decision date, or **Withdraw** .

7. Controls enforced by the software

- ✓ A section cannot be declared ready without a document reference; a section in deficiency must describe it; template sections cannot belong to a dossier.
- ✓ The template can only be loaded in preparation; the dossier can only be marked ready when all sections are ready; submission requires *ready*.
- ✓ A deficiency can only be recorded on a submitted dossier; approval requires the authorisation number; decision date $\geq$ submission date; submitted dossiers cannot be deleted.

8. Good practices

- ★ Link each variation dossier to its change request (SOP-QA-030).
- ★ Use the document reference of the controlled document (ls_document_management number and version).

9. Pitfalls to avoid

- ✗ Sections marked ready with drafts as document reference.

10 Records and evidence

- Dossier with sections, submission, deficiencies and decision

11 Performance indicators

- Sections late vs due date
- Deficiency cycles per dossier

12 References

- ICH M4 — Organisation of the Common Technical Document

13. Demonstration

Case. Variation for a new API manufacturer (CC-2026-021 follow-up): dossier type *Variation*; only sections 1.x, 2.3, 3.2.S.2 and 3.2.S.4 apply; owners assigned; all ready in 6 weeks; submission recorded; one deficiency on 3.2.S.4 answered; approval recorded with the authority's decision number.

Cosmetics

ls_cosmetics

ls_cosmetics follows the structure of Regulation (EC) No 1223/2009: an ingredient register (INCI, CAS/EC, regulatory category, nanomaterial, CMR, perfume/aroma), the register of Annex II–VI restrictions, versioned formulations totalling 100 % w/w checked against the annexes, the cosmetic product safety report (CPSR, Part A and Part B, approved by the named safety assessor), the product information file (PIF, Article 11) with its ten-year retention, claims assessed against the six common criteria, labelling particulars of Article 19 with an INCI list generated from the formulation, and the Algerian prior authorisation dossier.

Menu	Content
Cosmetics ► Ingredients ► Ingredient Register / Annex Restrictions	Ingredients and restrictions
Cosmetics ► Products ► Product Formulations / Finished Products	Formulations
Cosmetics ► Safety ► Safety Assessments	CPSR
Cosmetics ► Regulatory ► Product Information Files / Product Claims / Labelling / Algerian Prior Authorisations	Regulatory
Cosmetics ► Reports ► Retention Status / Safety Reports Due for Review	Monitoring

Group	Main rights
Cosmetics: Read Only	Read
Cosmetics: Formulator	Formulations, labels, claims drafts
Cosmetics: Safety Assessor	Safety reports (Part B approval by the named assessor)
Cosmetics: Regulatory Manager	Approvals, PIF, restrictions, Algerian dossiers

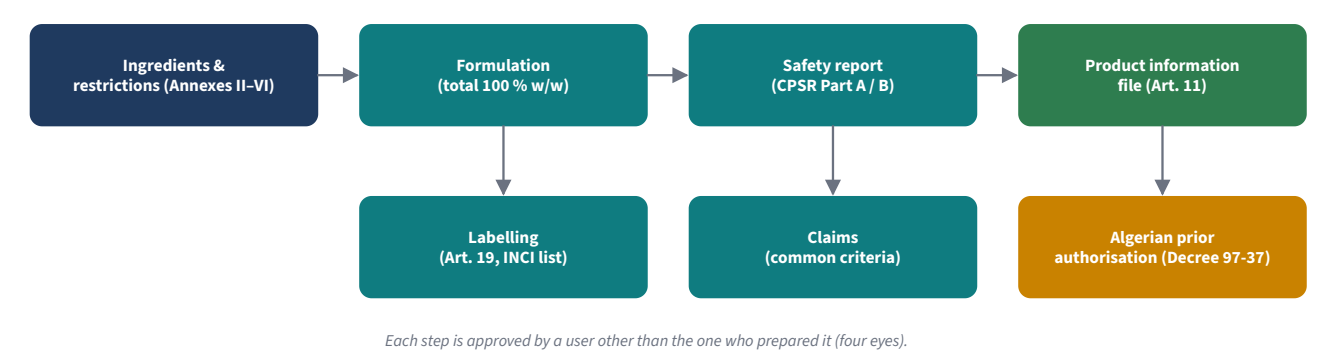
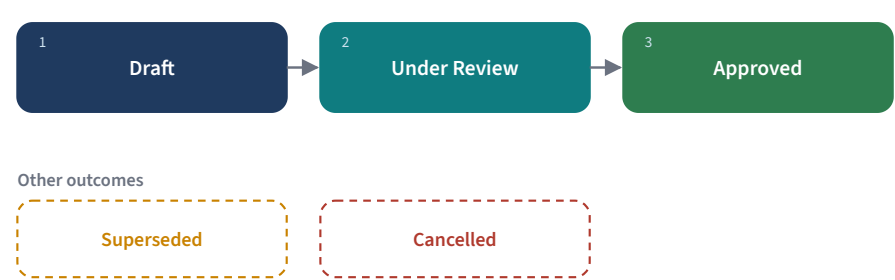


Figure 156 — From ingredients to the PIF and the authorisations.



Model: ls.cosmetic.formulation
Figure 157 — Lifecycle of a formulation.

Development and approval of formulations

Module	ls_cosmetics	Version	_____
Menu	Cosmetics ► Products ► Product Formulations	Effective date	_____
Owner (suggested)	Formulation Manager with the Regulatory Manager	Approved by	_____
Related SOPs	SOP-COS-002, SOP-COS-005, SOP-COS-006, SOP-QA-030		

1 Objective

Record each cosmetic formulation as a controlled, versioned composition in % w/w, check it against the annex restrictions, and approve it by a person other than the one who submitted it.

2 Scope

All cosmetic formulations placed on the market or used for safety assessment.

3 Responsibilities

Role (software group)	Responsibility
Formulator	Creates the formulation and its composition, submits it.
Regulatory Manager	Reviews annex findings and approves (not the submitter).

4 Definitions

Term	Definition
Composition	Lines of ingredients with concentration (% w/w, six decimals); total must equal 100 % within a tolerance of 0.05 %.
Blocking annex finding	Presence of an Annex II substance, or concentration above an Annex III–VI limit; approval is refused.

5 Workflow in the software



Model: ls.cosmetic.formulation

Figure 158 — Lifecycle of a formulation.

6 Procedure

- 1 Check ingredients.** Make sure every ingredient exists in [Ingredient Register](#) with INCI name, category and annex entries (SOP-COS-006).

- 2 **Create.** In **Product Formulations** click **New** : code, name, version, finished product, BOM, normal and reasonably foreseeable use, targeted population, flags (children under three, external intimate hygiene, nanomaterials, CMR).
- 3 **Compose.** Add the lines with concentrations; check *Total (% w/w) = 100* and the counters *Blocking Annex Findings* and *Lines Without Annex Data*.
- 4 **Submit.** Click **Submit for Review** .
- 5 **Approve.** The Regulatory Manager (not the submitter) clicks **Approve** , or **Return to Draft** with comments.
- 6 **Change.** Any change of an approved formulation is a new version: **Create New Version** (under change control, SOP-QA-030); the previous version becomes superseded.

7. Controls enforced by the software

- ✓ A formulation must have a composition to leave draft, and must total 100 % w/w within 0.05 % when submitted.
- ✓ An ingredient linked to an Annex II entry cannot be present in a formulation under review or approved (Article 14(1)(a)); blocking annex findings prevent approval.
- ✓ The submitter cannot approve; approved formulations are frozen; only approved formulations can be revised; non-draft formulations cannot be deleted (retention under Article 11(1)).
- ✓ A concentration rounding to zero at six decimals is refused.

8. Good practices

- ★ Record raw material trade names and their INCI breakdown so the label list is exact.
- ★ Flag early the special populations: they require specific assessment in the CPSR.

9. Pitfalls to avoid

- ✗ Editing the formulation in a spreadsheet and copying only the final list.

10 Records and evidence

- Formulation versions with approval

11 Performance indicators

- Formulations returned to draft
- Lines without annex data at submission

12 References

- Regulation (EC) No 1223/2009 on cosmetic products
- Regulation (EC) No 1223/2009, Article 14 and Annexes II to VI — Restrictions
- ISO 22716:2007 — Cosmetics — Good Manufacturing Practices

13. Demonstration

Case. Face cream *FC-120 v1*: 18 ingredients totalling 100.000 %; phenoxyethanol 0.9 % (Annex V limit 1 %) → within limit; no Annex II substance. Submitted by the formulator, approved by the Regulatory Manager.

Cosmetics

Products

Ingredients

Safety

Regulatory

Reports

7

14

My Company (San Francisco)

New

Product Formulations

FC-120 v1 - Hydrating face cream

Safety Reports

0

1 / 1

Create New Version

Draft

Under Review

Approved

Send message

Log note

Activity

Formulation Code

FC-120

v1

Formulation Name

Hydrating face cream

PRODUCT

Finished Product ?

Bill of Materials ?

Company

My Company (San Francisco)

VERSION HISTORY

Supersedes

Superseded By

COMPOSITION SUMMARY

Ingredients

7

Total (% w/w)

100.000000

REGULATORY FLAGS

Contains Nanomaterials

Contains CMR Substances

Blocking Annex Findings

0

Lines Without Annex Data

7

Some ingredients have no annex entry linked. Compliance with Annexes II to VI has not been evaluated for those lines.

QC Analyst (demo)

12:10 PM

Cosmetic Formulation created

Composition

Use and Population

Approval

Internal Note

Figure 159 — Approved formulation totalling 100 % w/w. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Safety assessment (CPSR, Cosmetic Product Safety Report)

Module	ls_cosmetics	Version	_____
Menu	Cosmetics ► Safety ► Safety Assessments	Effective date	_____
Owner (suggested)	Safety Assessor	Approved by	_____
Related SOPs	SOP-COS-001, SOP-COS-003		

1 Objective

Produce for each product a cosmetic product safety report whose Part A information is complete and whose Part B conclusion is written and approved by the qualified safety assessor named in the report.

2 Scope

All cosmetic products before placing on the market, and each new formulation version.

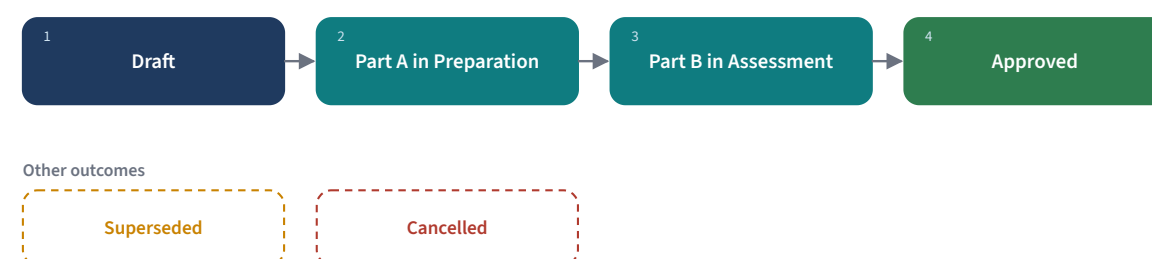
3 Responsibilities

Role (software group)	Responsibility
Formulator / Regulatory	Compile Part A data.
Safety Assessor (named in B4)	Performs the assessment and approves Part B with their own user account.

4 Definitions

Term	Definition
Part A	A1 composition, A2 physico-chemical characteristics and stability, A3 microbiological quality, A4 impurities/packaging, A5 use, A6 exposure to product, A7 exposure to substances, A8 toxicological profiles, A9 undesirable effects, A10 other information.
Part B	B1 conclusion, B2 warnings and instructions, B3 reasoning (including specific assessments), B4 assessor identity, address and qualification.

5 Workflow in the software



Model: ls.cosmetic.safety_assessment

Figure 160 — Lifecycle of a safety report.

6 Procedure

1 Create. In **Safety Assessments** click **New**: reference, title, version, formulation (approved).

- 2 **Part A.** Click **Start Part A** and complete sections A1 to A10; record the lowest margin of safety and its basis.
- 3 **Part B.** Click **Start Part B** once all Part A sections are filled; the assessor completes B1–B4, including the specific assessments for children under three or intimate hygiene products when the formulation is flagged.
- 4 **Approve.** The named assessor, logged in with their own account, clicks **Approve as Safety Assessor**.
- 5 **Review.** Monitor **Reports ▶ Safety Reports Due for Review**; update the report when new data, complaints or formulation changes arise.

7. Controls enforced by the software

- ✓ Part B requires an approved formulation and all Part A sections filled.
- ✓ Specific assessments are required when the formulation is intended for children under three or exclusively for external intimate hygiene.
- ✓ Approval requires every mandatory section, an assessor user account, and must be performed by that assessor.
- ✓ Approved reports are frozen (issue a new version); non-draft reports cannot be deleted.

8. Good practices

- ★ Attach the assessor's qualification evidence (diploma, CV) to the report.
- ★ Use recognised toxicological references and document the source of each profile.

9. Pitfalls to avoid

- ✗ Approving on behalf of an external assessor — the approval must come from the assessor's own account.

10 Records and evidence

- Approved CPSR with version history

11 Performance indicators

- Products marketed without approved CPSR (target 0)
- Reports overdue for review

12 References

- Regulation (EC) No 1223/2009, Article 10 and Annex I — Safety assessment (CPSR)
- Regulation (EC) No 1223/2009 on cosmetic products

13. Demonstration

Case. CPSR for FC-120 v1: Part A completed with stability (3 months accelerated), challenge test, exposure 1.54 g/day; lowest MoS 212 (phenoxyethanol). The external toxicologist named in B4 logs in and clicks **Approve as Safety Assessor**.

Product information file (PIF) and its retention period

Module	ls_cosmetics	Version	_____
Menu	Cosmetics ► Regulatory ► Product Information Files	Effective date	_____
Owner (suggested)	Responsible Person (Regulatory Manager)	Approved by	_____
Related SOPs	SOP-COS-002, SOP-COS-004, SOP-COS-005, SOP-COS-007		

1 Objective

Keep for each product a complete product information file, available at the stated address, and retain it for ten years after the last batch was placed on the market.

2 Scope

All cosmetic products for which the company is the responsible person.

3 Responsibilities

Role (software group)	Responsibility
Responsible Person	Owns the PIF and its availability to authorities.
Regulatory Manager	Compiles, activates, archives.

4 Definitions

Term	Definition
Article 11(2) content	(a) description, (b) safety report, (c) manufacturing method and GMP statement, (d) proof of claimed effects, (e) animal testing data.
Retention period	Ten years from the date the last batch was placed on the market (Article 11(1)).

5 Workflow in the software



Model: ls.cosmetic.pif

Figure 161 — Lifecycle of a product information file.

6 Procedure

- Create.** In **Product Information Files** click **New**: file reference and name, product, responsible person and basis of designation (written mandate reference if applicable), address where the file is kept, languages.
- Complete.** Fill items (a) to (e): description; approved safety report and formulation; approved label; manufacturing method, GMP statement, standard, site and evidence; claims and proof (or justification of absence); animal testing data or the statement that none was performed; notification reference and date.
- Activate.** Click **Activate** when the product is placed on the market (record *First Placed on the Market*).
- Maintain.** Update the file for each formulation, label or claim version; record reviews.

5 **Enter retention.** When the product is discontinued, record the date of the last batch placed on the market and click **Enter Retention Period**; the retention end date is computed.

6 **Archive.** After the retention end date click **Archive**.

7. Controls enforced by the software

- ✓ Activation requires all Article 11(2) items, an approved safety report that concludes the product is safe, and the animal testing statement.
- ✓ The last batch date cannot precede the first placing; retention requires the last batch date; archiving is refused before the ten-year retention end.
- ✓ A file that left draft cannot be deleted; only an active file can return to draft.

8. Good practices

- ★ Keep the PIF in the language(s) accepted by the authorities of each market.
- ★ Check the PIF at each product change, not only at launch.

9. Pitfalls to avoid

- ✗ PIFs pointing to superseded safety reports or labels.

10 Records and evidence

- PIF with contents, activation and retention dates

11 Performance indicators

- Products marketed without active PIF (target 0)
- PIFs not reviewed in 12 months

12 References

- Regulation (EC) No 1223/2009, Article 11 — Product information file
- ISO 22716:2007 — Cosmetics — Good Manufacturing Practices

13. Demonstration

Case. PIF for FC-120: all items completed, CPSR approved (*safe*), label v2 approved, two claims approved, statement *no animal testing performed*. Activated on 1 March 2027. The product is discontinued; last batch placed on the market on 30 June 2031 → retention until 30 June 2041.

Substantiation of claims

Module	ls_cosmetics	Version	_____
Menu	Cosmetics ► Regulatory ► Product Claims	Effective date	_____
Owner (suggested)	Regulatory Manager	Approved by	_____
Related SOPs	SOP-COS-003, SOP-COS-005		

1 Objective

Allow only claims that meet the six common criteria (legal compliance, truthfulness, evidential support, honesty, fairness, informed decision-making) and are supported by adequate evidence.

2 Scope

All claims made on labels, advertising, websites and social media for cosmetic products.

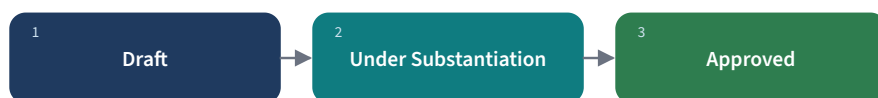
3 Responsibilities

Role (software group)	Responsibility
Marketing / formulator	Proposes claims and evidence.
Assessor	Assesses each criterion with justification.
Regulatory Manager (approver, not the assessor)	Approves or rejects.

4 Definitions

Term	Definition
Common criteria	The six criteria of Regulation (EU) No 655/2013.
Evidence	Studies (with number of subjects), consumer tests, literature, expert opinions — dated, not in the future.

5 Workflow in the software



Other outcomes



Model: ls.cosmetic.claim

Figure 162 — Lifecycle of a claim.

6 Procedure

- Create.** In **Product Claims** click **New**: reference, wording, product, formulation, medium; tick *No Animal Testing Claim* where relevant.

- 2 **Substantiate.** Click **Start Substantiation**; add supporting evidence records; for each of the six criteria tick *Met* and write the justification.
- 3 **Animal testing claims.** Record the Article 20(3) declaration when the claim refers to the absence of animal testing.
- 4 **Decide.** A user other than the assessor clicks **Approve** or **Reject** (with decision reason).
- 5 **Withdraw.** Click **Withdraw** when a claim is no longer used; the substantiation record remains.

7. Controls enforced by the software

- ✓ All six criteria must be marked met with a written justification before approval; an evidentially supported claim must carry evidence.
- ✓ A claim on absence of animal testing needs the Article 20(3) declaration.
- ✓ The assessor cannot approve; only draft claims can be deleted (withdraw instead); state transitions are enforced.
- ✓ Study evidence must record the number of subjects and cannot be dated in the future.

8. Good practices

- ★ Keep the exact wording used on pack and in advertising; a small change is a new claim.
- ★ Prefer studies with adequate design (sample size, controls) for efficacy claims.

9. Pitfalls to avoid

- ✗ Using a claim approved for one formulation on another.

10 Records and evidence

- Claims with evidence, criteria and decision

11 Performance indicators

- Claims used without approval (target 0)
- Rejected claims

12 References

- Regulation (EC) No 1223/2009, Article 20, and Regulation (EU) No 655/2013 — Claims and common criteria

13. Demonstration

Case. Claim "*Hydration +32 % after 4 weeks*" for FC-120: evidence = corneometry study on 30 subjects; all six criteria met with justification; the assessor is the regulatory officer; the Regulatory Manager approves.

Validation of labelling and of the INCI list

Module	ls_cosmetics	Version	_____
Menu	Cosmetics ► Regulatory ► Labelling	Effective date	_____
Owner (suggested)	Regulatory Manager	Approved by	_____
Related SOPs	SOP-COS-001, SOP-COS-003, SOP-COS-004		

1 Objective

Ensure that each label carries all Article 19(1) particulars and an ingredient list generated from the approved formulation, and approve the label before artwork release.

2 Scope

All labels, outer and inner packaging, leaflets and adjacent notices of cosmetic products.

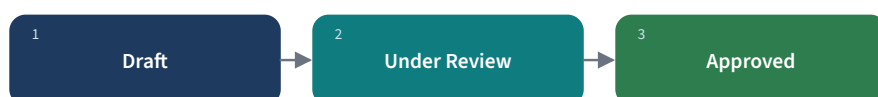
3 Responsibilities

Role (software group)	Responsibility
Regulatory officer	Prepares labelling particulars and generates the ingredient list.
Regulatory Manager	Approves the label.

4 Definitions

Term	Definition
INCI list	Ingredients in descending order of weight at the time of addition; ingredients below 1 % in any order after those above 1 %; nanomaterials followed by (<i>nano</i>); perfume/aroma compositions as <i>parfum/aroma</i> ; colorants may be listed last; +/- marker for shade ranges.
Durability indication	Date of minimum durability, or period after opening when durability exceeds 30 months, per Article 19(1)(c).

5 Workflow in the software



Other outcomes



Model: ls.cosmetic.label

Figure 163 — Lifecycle of labelling particulars.

6 Procedure

- 1 **Create.** In **Labelling** click **New**: label reference, name, version, approved formulation, product, responsible person and address, imported product and country of origin, nominal content (or exemption with reason), durability indication, storage conditions, precautions, professional use, batch number rule, function of the product (or *clear from presentation*), languages, artwork file.
- 2 **Generate the list.** Click **Generate Ingredient List**, choose *List Colorants Last* and the *may contain* marker if applicable, review and **Apply to Label**. Use **Regenerate from Formulation** after a formulation change while the label is draft.
- 3 **Submit.** Click **Submit for Review**.
- 4 **Approve.** The Regulatory Manager checks the artwork against the particulars and clicks **Approve**.
- 5 **Release artwork.** Only approved label versions are sent to the printer (see SOP-PRD-002 for reconciliation of printed labels).

7. Controls enforced by the software

- ✓ Approval requires all Article 19(1) particulars and an approved formulation.
- ✓ An imported product needs its country of origin; the function must be stated or declared clear from presentation; nominal content or a justified exemption is required; period after opening needs a number of months; durability rules of Article 19(1)(c) are checked.
- ✓ Approved labels are frozen (new version) and their ingredient list can no longer be regenerated; only draft labels can be deleted.

8. Good practices

- ★ Compare the printed proof with the approved label word by word, including the INCI list.
- ★ Keep the list prefix ("Ingredients") in the languages of the markets.

9. Pitfalls to avoid

- ✗ Typing the INCI list by hand instead of generating it from the formulation.

10 Records and evidence

- Approved label versions with artwork

11 Performance indicators

- Labels approved first time
- Artwork released without approved label (target 0)

12 References

- Regulation (EC) No 1223/2009, Article 19 — Labelling

13. Demonstration

Case. Label FC-120 v2: 50 ml, PAO 12 months (durability > 30 months), function *face cream*, list generated: *Ingredients: Aqua, Glycerin, ... , Parfum, CI 77891*; approved by the Regulatory Manager before the artwork order.

Maintenance of the register of regulatory restrictions on ingredients

Module	ls_cosmetics	Version	_____
Menu	Cosmetics ► Ingredients ► Annex Restrictions / Ingredient Register	Effective date	_____
Owner (suggested)	Regulatory Manager	Approved by	_____
Related SOPs	SOP-COS-001, SOP-RA-010		

1 Objective

Keep the register of Annex II to VI entries and the ingredient register current with the consolidated regulation, so that formulation checks rely on up-to-date limits.

2 Scope

All annex entries relevant to the ingredients used, and all ingredients of the register.

3 Responsibilities

Role (software group)	Responsibility
Regulatory Manager	Monitors amendments, updates entries, assesses impact on products.

4 Definitions

Term	Definition
Annex entry	Annex (II prohibited, III restricted, IV colorants, V preservatives, VI UV-filters), reference number, substance, CAS/EC, product type/body part, maximum concentration, conditions and warnings, source reference, consolidation date, applicability dates.
CMR substance	Carcinogenic, mutagenic or toxic for reproduction; flag and category must agree.

5 Procedure

- 1 Monitor.** Check monthly for amendments of the annexes (official journal, consolidated version); record them as regulatory watch (SOP-RA-010 principles).
- 2 Update.** Create or update the entries in **Annex Restrictions**: source reference, consolidation date, *Applicable From/Until* for transitional periods.
- 3 Link ingredients.** In **Ingredient Register** link each ingredient to its applicable annex entries; set CMR, nanomaterial and perfume flags.
- 4 Assess impact.** List approved formulations containing the affected ingredients; open a change request for reformulation or relabelling (SOP-QA-030) within the transitional period.
- 5 Record.** Keep the impact assessment with the source reference.

6. Controls enforced by the software

- ✓ *Applicable Until* cannot precede *Applicable From*; a numeric limit must be > 0; Annex II entries cannot carry a permitted concentration.
- ✓ Ingredients: CMR flag and category must agree; an impurity cannot also be a subsidiary technical material; a perfume composition must be labelled *parfum* or *aroma*.
- ✓ Annex II ingredients are refused in formulations under review or approved.

7. Good practices

- ★ Record the consolidation date of the source used, to prove which version was applied.
- ★ Watch the transitional dates: placing on the market and making available often differ.

8. Pitfalls to avoid

- ✗ Entries created from a supplier datasheet instead of the official text.

9 Records and evidence

- Annex entries with sources
- Impact assessments

10 Performance indicators

- Amendments not assessed within 30 days

11 References

- Regulation (EC) No 1223/2009, Article 14 and Annexes II to VI — Restrictions
- Regulation (EC) No 1223/2009 on cosmetic products

12. Demonstration

Case. An amendment restricts a fragrance allergen to 0.01 % in leave-on products from a future date. The Regulatory Manager updates the Annex III entry with *Applicable From*, finds 3 formulations above the limit and opens CC-2027-008 for reformulation before the date.

Cosmetics

Products

Ingredients

Safety

Regulatory

Reports

7

14

My Company (San Francisco)

New

Annex Restrictions

Substance Name

Substance name as published

IDENTIFICATION

Annex

Reference Number ?

Inci Name

Cas Number

Ec Number

Company ?

My Company (San Francisco)

LIMIT

Numeric Limit Applies ?

☐

Product Type / Body Part ?

PROVENANCE

Source Reference ?

Consolidation Date ?

Applicable From

Applicable Until

Conditions of Use

Label Wording

Other Column

Internal Note

Conditions as published in the annex

Figure 164 — Annex restriction record. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Prior authorisation dossier in Algeria

Module	ls_cosmetics	Version	_____
Menu	Cosmetics ► Regulatory ► Algerian Prior Authorisations	Effective date	_____
Owner (suggested)	Regulatory Manager (Algeria)	Approved by	_____
Related SOPs	SOP-COS-003, SOP-RA-010, SOP-RA-011		

1 Objective

Prepare, submit and follow the prior authorisation dossier required in Algeria for cosmetic products, track the decision deadline and formal notices, and keep the authorisation reference.

2 Scope

Cosmetic products manufactured or imported for the Algerian market. The legal texts are those cited by the module (Executive Decree 97-37 as amended by 10-114); their current wording must be verified by Regulatory Affairs before use of this SOP.

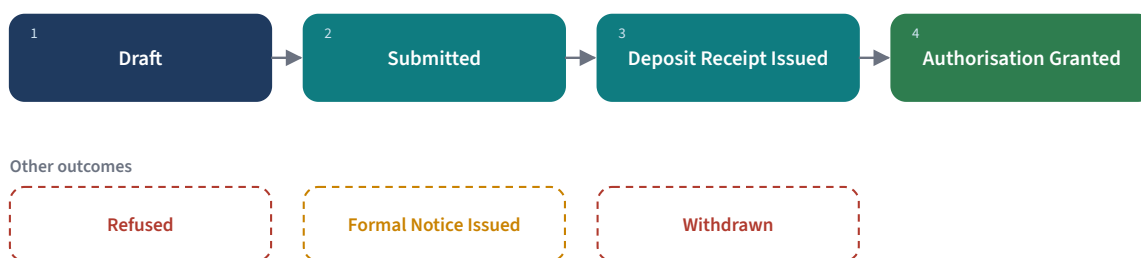
3 Responsibilities

Role (software group)	Responsibility
Regulatory Manager	Compiles the dossier, submits it to the Direction de Wilaya du Commerce, records decisions.
Operator's legal/finance	Provide the administrative documents.

4 Definitions

Term	Definition
Deposit receipt	Receipt issued at submission; the 45-day decision period coded in the module runs from its date.
Formal notice	Notice that an element of the authorisation fails; the module computes a one-month (30-day) compliance deadline.
Dossier items	16 items: commercial register extract, tax identification number, statutes, CNRC filing certificate, tax roll extract, CNAS/CASNOS certificate, denomination, use and directions, qualitative composition, analyses, toxicity testing, batch identification, precautions, label model, responsible persons, trademark.

5 Workflow in the software



Model: ls.cosmetic.dz_authorization

Figure 165 — Lifecycle of an Algerian prior authorisation dossier.

6 Procedure

- 1 **Create.** In **Algerian Prior Authorisations** click **New** : denomination, authorisation type, product, PIF, formulation, operator, wilaya direction, submission mode.
- 2 **Complete.** Tick each of the 16 items when provided and attach documents; *Dossier Complete* becomes true.
- 3 **Submit.** Click **Submit Dossier** with the submission date.
- 4 **Receipt.** Click **Register Deposit Receipt** with its reference and date; the decision due date is computed (45 days).
- 5 **Decision.** Click **Record Authorisation** with the authorisation reference, or **Record Refusal** with the reasoned refusal.
- 6 **Formal notice.** If notified, click **Register Formal Notice** with date and subject; act within the compliance deadline; click **Resolve Formal Notice**.
- 7 **Withdrawal.** Record the withdrawal of an authorisation with **Record Withdrawal**.

7. Controls enforced by the software

- ✓ Submission requires a complete dossier (missing items are listed).
- ✓ Receipt date $\geq$ submission; decision date $\geq$ receipt; a granted dossier needs its authorisation reference; a refusal needs its reason.
- ✓ Formal notices need a date and a subject; state transitions are enforced; submitted dossiers cannot be deleted.

8. Good practices

- ★ Keep scans of the stamped deposit receipt and of the decision in the dossier.
- ★ Record open questions about the procedure as regulatory questions (SOP-RA-011) rather than guessing.

9. Pitfalls to avoid

- ✗ Relying on the coded deadlines without checking the text in force.

10 Records and evidence

- Dossier with items, receipt, decision and notices

11 Performance indicators

- Decisions overdue vs computed due date
- Dossiers refused

12 References

- Algeria — Executive Decree No. 97-37 of 14 January 1997, as amended by Executive Decree No. 10-114 of 18 April 2010 (as cited in ls_cosmetics; to be verified by Regulatory Affairs)

13. Demonstration

Case. Dossier DZ-2027-004 for FC-120: 16/16 items provided; submitted 3 February; deposit receipt 5 February → decision due 22 March; authorisation recorded on 15 March with its reference, linked to the PIF.

Medical devices

ls_medical_device

ls_medical_device follows the structure of Regulation (EU) 2017/745 (MDR): device register with risk class and classification rationale, Basic UDI-DI and UDI-DI assignments, risk management files (ISO 14971) with risks, controls and residual acceptability, clinical evaluations and PMCF evaluation reports, technical documentation by section (Annex II/III, or Annex XIII for custom-made devices), CE marking and certificate records with notified bodies, post-market surveillance plans and periodic reports. A device cannot be placed on the market without approved technical documentation and, when its class requires it, a valid certificate.

Menu	Content
Medical Devices ▶ Devices ▶ Device Register / UDI Assignments	Devices and UDI
Medical Devices ▶ Risk Management ▶ Risk Management Files / Risks	ISO 14971
Medical Devices ▶ Clinical ▶ Clinical Evaluations / PMCF Evaluation Reports	Clinical
Medical Devices ▶ Regulatory ▶ Technical Documentation / Documentation Sections / CE Marking	Regulatory
Medical Devices ▶ Post-Market Surveillance ▶ Surveillance Plans / Periodic Reports / Open Reporting Period	PMS
Medical Devices ▶ Monitoring ▶ Expired Certificates / Overdue Post-Market Reports	Monitoring

Group	Main rights
Viewer · User	Read; prepare records
Regulatory Affairs · Manager	Approve files, change market status, manage certificates

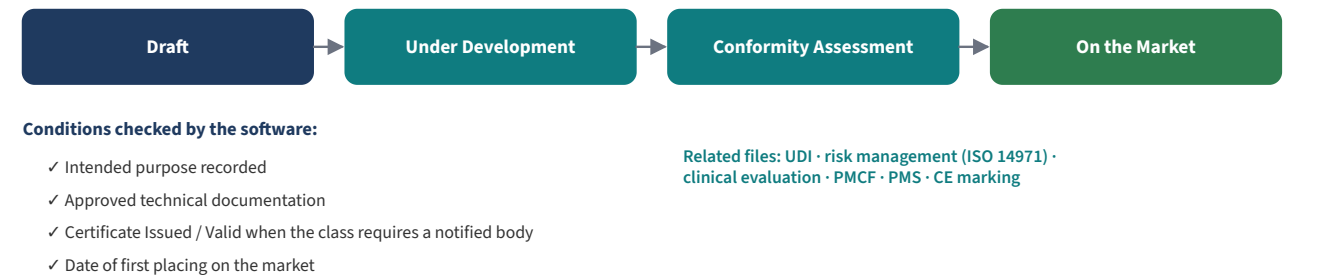


Figure 166 — Device lifecycle and the gates checked by the software.

Device registration and justification of its risk class

Module	ls_medical_device	Version	_____
Menu	Medical Devices ► Devices ► Device Register	Effective date	_____
Owner (suggested)	Regulatory Affairs Manager	Approved by	_____
Related SOPs	SOP-MD-002, SOP-MD-005, SOP-MD-006		

1 Objective

Register every medical device with its intended purpose and characteristics, justify its risk class by the applicable classification rule, and control its market status through documented gates.

2 Scope

All medical devices designed, manufactured or placed on the market by the company, including custom-made devices.

3 Responsibilities

Role (software group)	Responsibility
Regulatory Affairs	Registers devices, classifies them, changes market status.
R&D / quality	Provide intended purpose, characteristics and design documentation.

4 Definitions

Term	Definition
Risk class	Class I, IIa, IIb, III (with sub-cases) according to the classification rules of MDR Annex VIII; configured in Configuration ► Risk Classes .
Market status	Draft → Under Development → Conformity Assessment → On the Market; Suspended; Withdrawn.

5 Workflow in the software



Other outcomes



Model: ls.md.device

Figure 167 — Lifecycle of a device.



Conditions checked by the software:

- ✓ Intended purpose recorded
- ✓ Approved technical documentation
- ✓ Certificate Issued / Valid when the class requires a notified body
- ✓ Date of first placing on the market

Related files: UDI · risk management (ISO 14971) · clinical evaluation · PMCF · PMS · CE marking

Figure 168 — Gates before placing on the market.

6 Procedure

- 1 **Register.** In **Device Register** click **New** : reference, name, model/catalogue reference, nomenclature code, product, legal manufacturer, authorised representative, risk class, classification rule and rationale, intended purpose, characteristics (implantable, sterile, measuring function, reusable surgical instrument, software, custom-made, direct marking), documentation retention years.
- 2 **Develop.** Click **Start Development** .
- 3 **Conformity assessment.** Click **Start Conformity Assessment** once the intended purpose is recorded; prepare the technical documentation, risk file, clinical evaluation and CE marking (SOP-MD-003 to 006).
- 4 **Place on the market.** Click **Place on the Market** with the date of first placing on the market.
- 5 **Suspend / resume / withdraw.** Use **Suspend** , **Resume** , **Withdraw** , or **Change Status with Justification** to record the reason.

7. Controls enforced by the software

- ✓ Changing the market status requires the Regulatory Affairs or Manager access level.
- ✓ Conformity assessment requires the intended purpose; placing on the market requires the date, completed conformity assessment, approved technical documentation and, when the class requires a notified body, an issued or valid certificate.
- ✓ A custom-made implantable device requires a classification rationale; withdrawal date $\geq$ first placing date.

8. Good practices

- ★ Write the classification rationale citing the rule and the reasoning for borderline cases.
- ★ Keep the intended purpose identical in the IFU, technical documentation and clinical evaluation.

9. Pitfalls to avoid

- ✗ Changing the intended purpose without reclassification.

10 Records and evidence

- Device record with status history and reasons

11 Performance indicators

- Devices on market without valid certificate (target 0)

12 References

- Regulation (EU) 2017/745 on medical devices (MDR)
- ISO 13485:2016 — Medical devices — Quality management systems

13. Demonstration

Case. Sterile single-use catheter CAT-14: class IIa by rule 7 (surgically invasive, short term), rationale recorded; development started; after approved technical documentation and a certificate issued by the notified body, placed on the market on 1 September.

The screenshot displays the Odoo 19.0 Medical Devices application interface. The top navigation bar includes modules like Medical Devices, Devices, Risk Management, Clinical, Regulatory, Post-Market Surveillance, Monitoring, and Configuration. The user is logged in as 'My Company (San Francisco)'. The main header shows the 'Device Register' form for 'Demo Implantable Pulse Generator' with a reference 'MD/2026/00002'. The form is divided into several sections: IDENTIFICATION, ECONOMIC OPERATORS, CLASSIFICATION, and CHARACTERISTICS. The 'Under Development' status is highlighted in the top navigation bar. The right sidebar shows a chat window with 'OdooBot' and a message 'Medical Device created'.

Reference ?
MD/2026/00002

Device Name ?
Demo Implantable Pulse Generator

IDENTIFICATION

Model or Catalogue Reference ? DEMO-IPG-200

Basic UDI-DI ?

Nomenclature Code ?

Product ?

Company ? My Company (San Francisco)

ECONOMIC OPERATORS

Legal Manufacturer ?

Authorised Representative ?

CLASSIFICATION

Risk Class ? Class III

Classification Rule ? Annex VIII, rule to be confirmed by the

Notified Body Involvement Required ? ☒

CHARACTERISTICS

Implantable ? ☒

Placed on the Market Sterile ? ☒

Has a Measuring Function ? ☐

Reusable Surgical Instrument ? ☐

Send message Log note Activity

Today

OdooBot 10:40 AM
Medical Device created

Figure 169 — Device register form. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Assignment of the UDI (Unique Device Identification)

Module	ls_medical_device	Version	_____
Menu	Medical Devices ► Devices ► UDI Assignments	Effective date	_____
Owner (suggested)	Regulatory Affairs Manager	Approved by	_____
Related SOPs	SOP-MD-001		

1 Objective

Assign one Basic UDI-DI per device and UDI-DIs per packaging level from an accredited issuing entity, define the production identifier, register them in the UDI database and retire them in a controlled way.

2 Scope

All devices subject to UDI obligations.

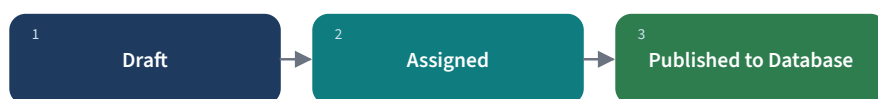
3 Responsibilities

Role (software group)	Responsibility
Regulatory Affairs	Assigns and publishes identifiers.
Packaging / labelling	Applies the UDI carrier and direct marking.

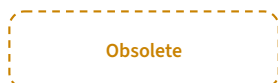
4 Definitions

Term	Definition
Basic UDI-DI	Primary identifier of a device model, key in the UDI database; one per device.
UDI-DI / UDI-PI	Device identifier per packaging level / production identifier (lot, serial, manufacturing date, expiry, software version).

5 Workflow in the software



Other outcomes



Model: ls.md.udi

Figure 170 — Lifecycle of a UDI assignment.

6 Procedure

- 1 Create.** In **UDI Assignments** click **New** ; device, identifier kind (Basic UDI-DI or UDI-DI), identifier, issuing entity (note when *Other*), packaging level and quantity, carrier type, direct marking, production identifier components.
- 2 Assign.** Record the assignment date and click **Assign** .

- 3 **Publish.** After registration in the UDI database, record the database name, submission date and reference, and click **Publish to Database**.
- 4 **Retire.** Click **Mark Obsolete** with the reason when the identifier is no longer used; a change requiring a new UDI-DI is a new assignment.

7. Controls enforced by the software

- ✓ A device carries a single Basic UDI-DI.
- ✓ Leaving draft requires the assignment date; publication requires the database name and submission date; retirement requires a reason.
- ✓ Assigned identifiers cannot be deleted; published identifiers cannot be modified (create a new assignment).

8. Good practices

- ★ Define in this SOP which changes trigger a new UDI-DI (e.g. change of packaging quantity, sterilisation state).

9. Pitfalls to avoid

- ✗ Reusing a UDI-DI after a change that requires a new one.

10 Records and evidence

- UDI assignments with publication data

11 Performance indicators

- Devices on market with unpublished UDI (target 0)

12 References

- MDR Article 27 and Annex VI — Unique Device Identification
- GS1 General Specifications (GTIN, SSCC, check digit)

13. Demonstration

Case. CAT-14: Basic UDI-DI assigned (GS1), UDI-DI for the unit (box of 1) and the shelf box (10), PI = lot + expiry; published in the database with its reference.

Risk management according to ISO 14971

Module	ls_medical_device	Version	_____
Menu	Medical Devices ► Risk Management	Effective date	_____
Owner (suggested)	Risk management lead (Regulatory / R&D)	Approved by	_____
Related SOPs	SOP-QA-060, SOP-MD-004, SOP-MD-007		

1 Objective

Maintain for each device a risk management file identifying hazards, hazardous situations and harms, estimating and controlling risks, evaluating residual risks and the overall benefit-risk, and updating it with production and post-production information.

2 Scope

All devices of the register, throughout their lifecycle.

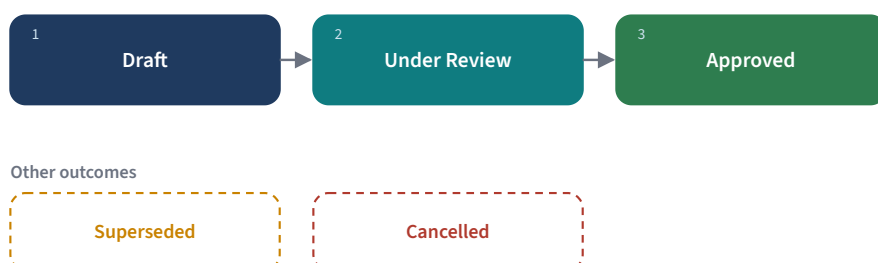
3 Responsibilities

Role (software group)	Responsibility
Responsible person of the file	Leads the analysis; cannot approve the file.
Cross-functional team	R&D, clinical, quality, manufacturing, post-market.
Regulatory Affairs / Manager	Approves the file.

4 Definitions

Term	Definition
Risk index	Combination of severity and probability; the residual index cannot exceed the initial index.
Residual acceptability	Acceptability after control; a difference from the value proposed by the matrix needs a justification.

5 Workflow in the software



Model: ls.md.technical_file

Figure 171 — Lifecycle of the regulated files (risk file, clinical, technical documentation, PMS).

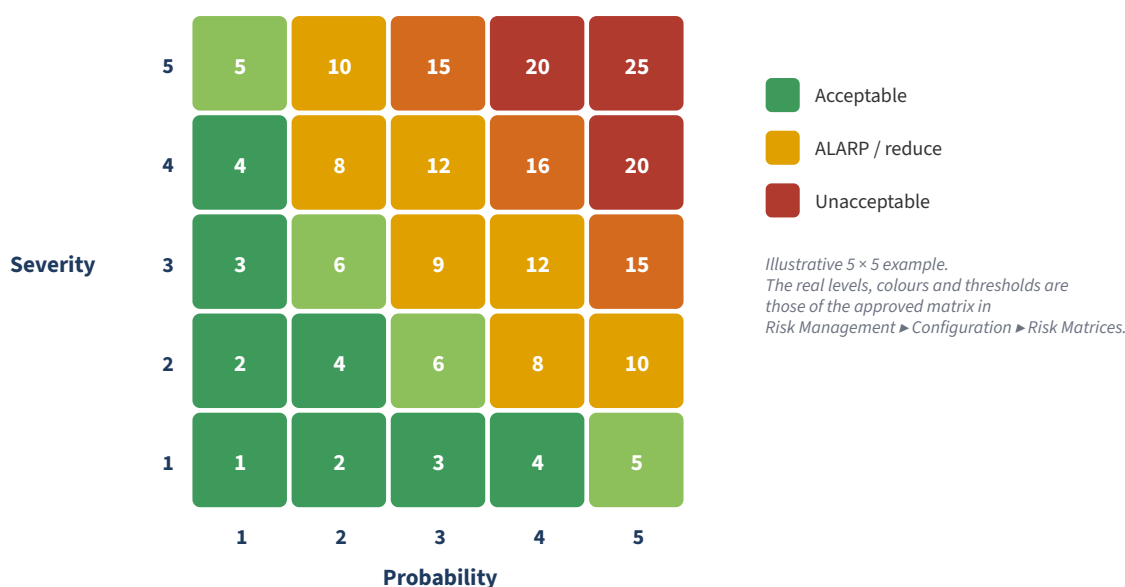


Figure 172 — Illustrative risk matrix.

6 Procedure

- 1 **Create the file.** In **Risk Management Files** click **New**; title, device, version, scope, risk acceptability policy, standard reference, responsible.
- 2 **Identify risks.** Add risks: identifier, hazard and category, sequence of events, hazardous situation, harm, affected party, initial severity and probability.
- 3 **Control.** For each risk choose the control option, describe the measure, record implementation and effectiveness evidence, residual severity and probability, residual acceptability (justify deviations from the proposed value), disclosure of residual risk in the IFU.
- 4 **New hazards.** If a control introduces a new hazard, tick it, describe it and add it as a new risk.
- 5 **Conclude.** Write the overall benefit-risk conclusion and tick overall residual risk acceptable when justified.
- 6 **Review and approve.** Click **Submit for Review**; the approver (Regulatory Affairs/Manager, not the responsible person) clicks **Approve**.
- 7 **Update.** Feed production and post-production information (complaints, PMS, PMCF); create a new version when risks change (**Supersede** the old one).

7. Controls enforced by the software

- ✓ A file cannot be submitted without at least one risk; approval requires the overall benefit-risk conclusion, no unacceptable residual risk and no risk without a control measure.
- ✓ Residual index $\leq$ initial index; a reduced residual risk needs a control measure; a new hazard needs a description; an acceptability different from the matrix proposal needs a justification.
- ✓ Approved files are closed for editing — create a new version.
- ✓ Approval requires the Regulatory Affairs or Manager access level; the author/responsible person cannot approve (segregation of duties).
- ✓ Only a draft can be submitted; only a record under review can be approved or returned to draft; approved records are frozen — create a new version; non-draft records cannot be deleted (cancel instead).

8. Good practices

- ★ Link device risks to the general register (ls_risk_management) only if you need its workflow; keep one reference file per device.
- ★ Review the file at each PMS report.

9. Pitfalls to avoid

- ✗ Using "information for safety" as the main control for high risks.

10 Records and evidence

- Approved risk management file with versions

11 Performance indicators

- Files not updated after a PMS report
- Risks without control measure (target 0)

12 References

- ISO 14971:2019 — Medical devices — Application of risk management
- Regulation (EU) 2017/745 on medical devices (MDR)

13. Demonstration

Case. CAT-14 risk *R-07 tip detachment* → *vessel embolism*: initial S5/P3; controls: bonding process validated (protective), pull test on each lot (protective), IFU warning (information). Residual S5/P1 acceptable with justification. The file is approved by Regulatory Affairs.

Clinical evaluation and PMCF (Post-Market Clinical Follow-up)

Module	ls_medical_device	Version	_____
Menu	Medical Devices ► Clinical	Effective date	_____
Owner (suggested)	Clinical evaluation lead	Approved by	_____
Related SOPs	SOP-MD-003, SOP-MD-007		

1 Objective

Demonstrate conformity with the relevant safety and performance requirements through a planned clinical evaluation, and keep it updated with post-market clinical follow-up.

2 Scope

All devices; the depth depends on class, novelty and available evidence.

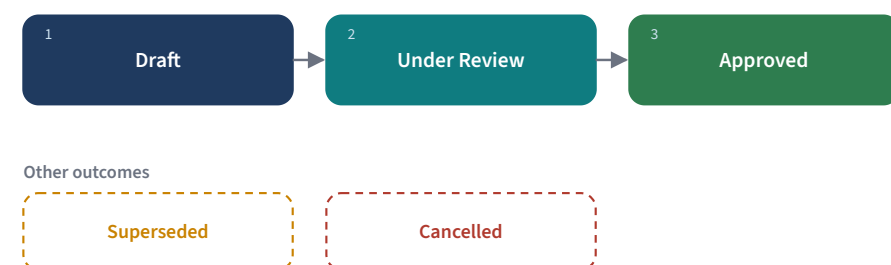
3 Responsibilities

Role (software group)	Responsibility
Evaluator (qualified)	Plans and writes the clinical evaluation; cannot approve it.
Regulatory Affairs / Manager	Approves.
PMCF author	Writes PMCF evaluation reports.

4 Definitions

Term	Definition
Equivalence	Claim that clinical data of another device apply; requires a demonstration.
PMCF	Continuous process to collect clinical data after placing on the market; a justification is required when not applied.

5 Workflow in the software



Model: ls.md.technical_file

Figure 173 — Lifecycle of a clinical evaluation / PMCF report.

6 Procedure

- Plan.** In **Clinical Evaluations** click **New**: device, version, evaluator and qualification, plan scope, requirements supported by clinical data, intended benefits, target population, acceptance criteria, literature search protocol.

- 2 **Collect and appraise.** Record evidence sources, equivalence (and its demonstration), clinical investigation (reference) or the justification for none; appraisal and analysis of data.
- 3 **Conclude.** Write the benefit-risk conclusion, confirm conformity, record residual gaps; state whether PMCF applies (plan summary) or not (justification).
- 4 **Approve.** Submit for review; Regulatory Affairs/Manager (not the evaluator) approves.
- 5 **PMCF reports.** Periodically create PMCF evaluation reports: period, activities, data, findings, new or emerging risks, off-label use, benefit-risk conclusion, actions; approve (not the author).
- 6 **Update.** When a PMCF report requires it, update the clinical evaluation (new version) and the risk file.

7. Controls enforced by the software

- ✓ Equivalence claimed requires a demonstration; a declared clinical investigation needs its reference, otherwise a justification is required; PMCF applicable needs a plan summary, not applicable needs a justification.
- ✓ Approval requires the benefit-risk conclusion; the evaluator/author cannot approve.
- ✓ PMCF report: new risks and required actions must be described; the linked clinical evaluation must belong to the same device.
- ✓ Approval requires the Regulatory Affairs or Manager access level; the author/responsible person cannot approve (segregation of duties).
- ✓ Only a draft can be submitted; only a record under review can be approved or returned to draft; approved records are frozen — create a new version; non-draft records cannot be deleted (cancel instead).

8. Good practices

- ★ Keep the literature search reproducible (databases, dates, search strings).
- ★ Align acceptance criteria with the state of the art.

9. Pitfalls to avoid

- ✗ Equivalence claimed on superficial similarity.

10 Records and evidence

- Clinical evaluations and PMCF reports with versions

11 Performance indicators

- Clinical evaluations past their review date
- PMCF reports overdue

12 References

- MDR Article 61 and Annex XIV — Clinical evaluation and PMCF
- Regulation (EU) 2017/745 on medical devices (MDR)

13. Demonstration

Case. Clinical evaluation of CAT-14 v2: literature search on 3 databases, 41 relevant publications, no equivalence claimed, own clinical data from a 60-patient study; PMCF applicable (registry follow-up). The annual PMCF report identifies no new risk; benefit-risk remains acceptable.

Technical documentation

Module	ls_medical_device	Version	_____
Menu	Medical Devices ► Regulatory ► Technical Documentation	Effective date	_____
Owner (suggested)	Regulatory Affairs Manager	Approved by	_____
Related SOPs	SOP-MD-001, SOP-MD-006		

1 Objective

Compile, for each device, technical documentation structured by section according to the applicable annex, with evidence for each section, approved and retained for the required period.

2 Scope

Technical documentation (Annex II and III) of all devices; Annex XIII documentation for custom-made devices.

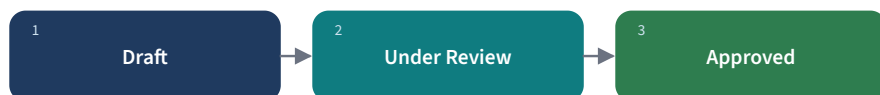
3 Responsibilities

Role (software group)	Responsibility
Section owners	Provide the evidence references.
Responsible person	Compiles; cannot approve.
Regulatory Affairs / Manager	Approves.

4 Definitions

Term	Definition
Section	Part of the documentation from a template (Configuration ► Documentation Section Templates), mandatory or not; marked complete with an evidence reference, or not applicable with a justification.

5 Workflow in the software



Other outcomes



Model: ls.md.technical_file

Figure 174 — Lifecycle of technical documentation.

6 Procedure

- Create.** In **Technical Documentation** click **New**: title, device, annex (Annex II/III, or Annex XIII for custom-made), version, responsible, storage location; load the sections from the templates.

- 2 **Complete sections.** For each section record the evidence reference (controlled document number and version) and mark it complete; for a non-applicable section record the justification (Annex II point 4(a) for GSPR).
- 3 **Check completeness.** *Missing Mandatory Sections* must be zero.
- 4 **Approve.** Submit for review; Regulatory Affairs/Manager (not the responsible person) approves. This is a gate for placing on the market (SOP-MD-001).
- 5 **Maintain.** Changes to the device create a new version of the documentation; retain until the date computed from the device retention years.

7. Controls enforced by the software

- ✓ Custom-made devices use Annex XIII and other devices cannot; the file must contain sections; approval requires all mandatory sections complete.
- ✓ A section complete requires an evidence reference; a non-applicable section requires a justification; approved documentation is closed for editing.
- ✓ Approval requires the Regulatory Affairs or Manager access level; the author/responsible person cannot approve (segregation of duties).
- ✓ Only a draft can be submitted; only a record under review can be approved or returned to draft; approved records are frozen — create a new version; non-draft records cannot be deleted (cancel instead).

8. Good practices

- ★ Use `ls_document_management` references so that each evidence is a controlled version.

9. Pitfalls to avoid

- ✗ Sections marked complete with a folder name instead of a document reference.

10 Records and evidence

- Approved technical documentation with sections and versions

11 Performance indicators

- Devices with incomplete documentation

12 References

- MDR Annexes II and III — Technical documentation
- ISO 13485:2016 — Medical devices — Quality management systems

13. Demonstration

Case. TD of CAT-14 (Annex II/III): 26 sections; section *Biocompatibility* complete with evidence *REP-BIO-2027-03 v1*; section *Software* not applicable with justification; 0 mandatory section missing; approved.

CE marking and relationship with the notified body

Module	ls_medical_device	Version	_____
Menu	Medical Devices ► Regulatory ► CE Marking · Configuration ► Notified Bodies	Effective date	_____
Owner (suggested)	Regulatory Affairs Manager	Approved by	_____
Related SOPs	SOP-MD-001, SOP-MD-005, SOP-MD-007		

1 Objective

Record the conformity assessment route, the notified body and its certificates, the EU declaration of conformity and the affixing of CE marking, and keep certificate status (valid, suspended, withdrawn, expired) under control.

2 Scope

All devices placed on the EU market.

3 Responsibilities

Role (software group)	Responsibility
Regulatory Affairs	Manages certificates and the relationship with the notified body.
Management	Signs the declaration of conformity.

4 Definitions

Term	Definition
Conformity assessment route	Annex(es) applied; routes involving a notified body require one to be selected.
Notified body	Registered with its designation verification date (not in the future).

5 Workflow in the software



Other outcomes



Model: ls.md.ce_marking

Figure 175 — Lifecycle of a CE marking / certificate record.

6 Procedure

- 1 Register the notified body.** In **Configuration ► Notified Bodies** record name, number and designation verification date.
- 2 Create the record.** In **CE Marking** click **New**: device, route, notified body, scope, conditions and restrictions, declaration of conformity number and date, Basic UDI-DI.

- 3 **Submit.** Click **Submit to Notified Body** when the application is filed.
- 4 **Certificate issued.** Record the certificate number, issue and expiry dates, and click **Record Certificate Issued** ; then **Activate** (status Valid).
- 5 **Affix CE.** Record that CE marking is affixed and its date.
- 6 **Surveillance.** Record the next surveillance audit due; monitor **Monitoring ► Expired Certificates**.
- 7 **Suspension / withdrawal.** Use **Suspend** or **Withdraw** with reason and date; assess the device market status (SOP-MD-001).

7. Controls enforced by the software

- ✓ Managing certificate status requires the Regulatory Affairs or Manager access level.
- ✓ A route involving a notified body requires one; CE affixed requires its date; issued requires certificate number and issue date; an expired certificate cannot be activated.
- ✓ Only valid certificates can be suspended; suspension and withdrawal need a reason; issued records are frozen for their identifying data and cannot be deleted.

8. Good practices

- ★ Plan recertification at least 12 months before expiry.
- ★ Keep the notified body correspondence attached to the record.

9. Pitfalls to avoid

- ✗ Placing a new variant on the market outside the certificate scope.

10 Records and evidence

- CE marking and certificate records
- Declarations of conformity

11 Performance indicators

- Certificates expiring within 12 months
- Devices on market with expired certificate (target 0)

12 References

- MDR Articles 20 and 52, Annexes IX to XI — CE marking and conformity assessment
- Regulation (EU) 2017/745 on medical devices (MDR)

13. Demonstration

Case. CAT-14: route Annex IX (QMS + technical documentation assessment) with notified body NB 0000 (placeholder number); certificate issued 15 August, expiry 5 years later; activated; CE affixed 1 September.

Post-market surveillance and periodic reports

Module	ls_medical_device	Version	_____
Menu	Medical Devices ► Post-Market Surveillance	Effective date	_____
Owner (suggested)	PMS lead (Regulatory Affairs)	Approved by	_____
Related SOPs	SOP-QA-040, SOP-QA-041, SOP-MD-003, SOP-MD-004		

1 Objective

Collect and analyse post-market data according to an approved plan, report periodically (PMS report or periodic safety update report depending on class), and feed the risk file, clinical evaluation and labelling.

2 Scope

All devices on the market.

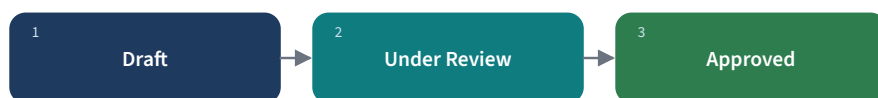
3 Responsibilities

Role (software group)	Responsibility
PMS lead / responsible person	Writes the plan and reports; cannot approve own documents.
Regulatory Affairs / Manager	Approves; records notified body submissions.
Complaint and vigilance functions	Provide data (SOP-QA-040, SOP-QA-041).

4 Definitions

Term	Definition
PMS plan elements	Information sources, proactive collection, indicators and thresholds, analysis methods, communication methods, referenced procedures, corrective action process, traceability tools, PMCF plan or justification (Annex III section 1.1).
Periodic report	Type and interval set on the device/plan; created per period with <i>Open Reporting Period</i> .

5 Workflow in the software



Other outcomes



Model: ls.md.technical_file

Figure 176 — Lifecycle of a PMS plan / periodic report.

6 Procedure

- 1 Plan.** In **Surveillance Plans** create the plan with all Annex III elements, PMCF plan reference or justification, report type and interval, plan review frequency; submit and approve.

- 2 **Open the period.** Use **Open Reporting Period** to create the periodic reports due.
- 3 **Report.** Fill the report: sales volume, user population, complaints, serious and non-serious incidents, FSCA, trend signal (with description), analysis summary, actions, updates required (risk file, clinical evaluation, labelling/IFU), benefit-risk determination and conclusion.
- 4 **Approve.** Submit for review (analysis summary required); approve (not the author).
- 5 **Submit.** When required, send it to the notified body and click **Record Notified Body Submission**.
- 6 **Act.** Implement the updates flagged in the report (SOP-MD-003, SOP-MD-004, SOP-QA-030); monitor **Monitoring**
► **Overdue Post-Market Reports.**

7. Controls enforced by the software

- ✓ Approving a plan or report requires the Regulatory Affairs or Manager access level; the responsible person/author cannot approve.
- ✓ The plan must contain every Annex III section 1.1 element, and a PMCF plan reference or a justification for its absence.
- ✓ Report periods of the same device cannot overlap; submission requires the analysis summary; approval requires the benefit-risk determination and conclusion; a trend signal requires its description; only approved reports can be recorded as submitted to the notified body.
- ✓ Approval requires the Regulatory Affairs or Manager access level; the author/responsible person cannot approve (segregation of duties).
- ✓ Only a draft can be submitted; only a record under review can be approved or returned to draft; approved records are frozen — create a new version; non-draft records cannot be deleted (cancel instead).

8. Good practices

- ★ Define trend thresholds numerically in the plan (e.g. complaint rate per 10 000 units).
- ★ Reconcile incident counts with the vigilance records before approval.

9. Pitfalls to avoid

- ✗ Reports written from memory rather than from complaint and incident data.

10 Records and evidence

- Approved PMS plan and periodic reports
- Notified body submission references

11 Performance indicators

- Overdue periodic reports (target 0)
- Trend signals and their follow-up

12 References

- MDR Articles 83 to 86 and Annex III — Post-market surveillance, PMS report and PSUR
- MDR Articles 87 to 92 — Vigilance
- ISO 13485:2016 — Medical devices — Quality management systems

13. Demonstration

Case. CAT-14 (class IIa): PSUR every 2 years. The 2027–2028 report: 1.2 million units, 38 complaints, 2 serious incidents reported, no FSCA, no trend signal; risk file update not required; benefit-risk favourable. Approved and made available to the notified body.

Medical plastics

ls_medical_plastics

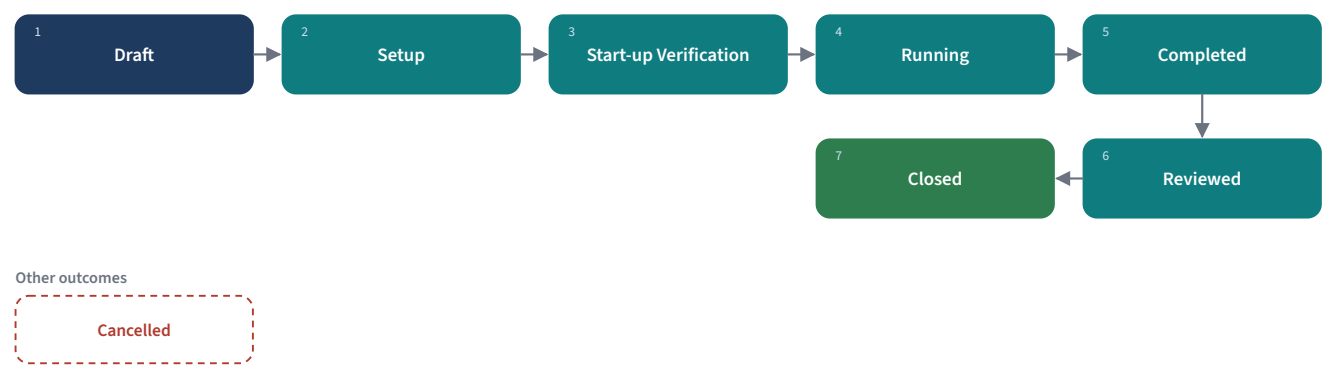
ls_medical_plastics supports injection moulding of primary packaging and device components: components with approved material grades, qualified material grades, moulds (tools) with a cavity register, shot counters and preventive maintenance by shots and by time, versioned moulding parameter specifications (process windows) approved by people other than the author, moulding runs with setup, start-up verification, immutable parameter readings, material consumption, scrap by cavity and independent review, and a traceability enquiry by lot, material lot, component or tool.

Menu	Content
Medical Plastics ▶ Products ▶ Components / Packaging	Components
Medical Plastics ▶ Tools ▶ Tool Register / Maintenance Records / Due For Maintenance	Tools
Medical Plastics ▶ Production ▶ Injection Moulding / Moulding Parameters / Parameter Readings	Production
Medical Plastics ▶ Reports ▶ Moulding Analysis / Traceability Enquiry	Reports
Medical Plastics ▶ Configuration ▶ Material Grades / Scrap Reasons	Configuration

Group	Main rights
Medical Plastics / Viewer · Operator	Read; run production, record readings
Medical Plastics / Tool Technician	Tool maintenance
Medical Plastics / Process Engineer	Parameter specifications, run review
Medical Plastics / Manager	Approvals and configuration



Figure 177 — Maintenance triggered by shots or time (illustrative).



Model: ls.mp.injection_molding

Figure 178 — Lifecycle of a moulding run.

Qualification and commissioning of moulds

Module	ls_medical_plastics	Version	_____
Menu	Medical Plastics ► Tools ► Tool Register	Effective date	_____
Owner (suggested)	Tooling / Process Engineering Manager	Approved by	_____
Related SOPs	SOP-MP-002, SOP-MP-003, SOP-VAL-002		

1 Objective

Register each mould with its cavities and the components it produces, qualify it before first production, and place it in service only when qualified with at least one active cavity.

2 Scope

Injection moulds, family moulds, inserts, hot runners, core pin sets and fixtures used for GMP components.

3 Responsibilities

Role (software group)	Responsibility
Tool Technician	Registers the tool and cavity register, records maintenance.
Process Engineer	Qualifies the tool (IQ/OQ/PQ as defined by validation).
Manager	Places in service, quarantines, decommissions.

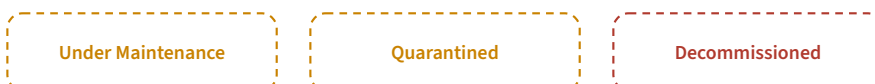
4 Definitions

Term	Definition
Cavity register	One record per cavity number with marking and status (Active, Blocked, Removed).
Requalification interval	Months after which the tool must be requalified.

5 Workflow in the software



Other outcomes



Model: ls.mp.tool

Figure 179 — Lifecycle of a mould.

6 Procedure

- 1 Register.** In **Tool Register** click **New**: name, code, type, tool maker, serial number, acquisition date, steel, storage location, default work centre, components produced, cavity count, opening shot count, maintenance intervals (shots, months), requalification interval.
- 2 Cavities.** Check the cavity register (number and marking of each cavity).

- 3 **Qualify.** Perform the tool qualification (dimensional study per cavity, process capability) under SOP-VAL-002; record the qualification date and click **Qualify**.
- 4 **Place in service.** Click **Place In Service** (at least one active cavity required).
- 5 **Quarantine / decommission.** Use **Quarantine** when the tool is suspect and **Decommission** at end of life (no run in progress).
- 6 **Requalify.** Monitor *Requalification Due*; requalify after major repair or modification.

7. Controls enforced by the software

- ✓ Qualification and placing in service require a qualification date; transitions are allowed only from the proper states; a tool without active cavity cannot be placed in service.
- ✓ The cavity count cannot be reduced below a registered cavity number (parts remain traceable to that cavity).
- ✓ A tool with runs in progress cannot be sent to maintenance, quarantined or decommissioned.

8. Good practices

- ★ Engrave cavity numbers on the parts so that scrap and complaints can be traced to the cavity.
- ★ Store tools with their last-shot samples.

9. Pitfalls to avoid

- ✗ Moving a tool to another press without checking the specification work centre restriction.

10 Records and evidence

- Tool record, cavity register, qualification evidence

11 Performance indicators

- Tools in service past requalification (target 0)

12 References

- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 15378:2017 — Primary packaging materials for medicinal products
- GHTF/SG3/N99-10:2004 — Quality Management Systems — Process Validation Guidance
- EU GMP Annex 15 (2015) — Qualification and Validation

13. Demonstration

Case. Mould T-032 (8 cavities) for syringe plunger SP-5: registered with maintenance interval 250 000 shots or 6 months; OQ/PQ done with a dimensional study of all 8 cavities; qualified on 12 May; placed in service.

Medical Plastics
Production
Products
Tools
Reports
Configuration
7
14
My Company (San Francisco)

New
Tool Register
[T-032] Mould T-032 (8 cavities)
Runs 1
Maintenance 0
1 / 1

Send To Maintenance
Quarantine
Decommission
Draft
Qualified
In Service
Send message
Log note
Activity

Tool Name
Mould T-032 (8 cavities)

Tool Code
T-032

IDENTIFICATION		PRODUCTION	
Tool Type	Injection Mould	Default Work Centre ?	
Tool Maker		Cavity Count ?	8
Manufacturer Serial Number	TM-2025-0457	Active Cavities	8
Acquisition Date		Blocked Cavities	0
Steel / Material of Construction	1.2083 stainless		
Storage Location ?			
Company	My Company (San Francisco)		
Active	☒		

SHOT COUNTING		PREVENTIVE MAINTENANCE	
Opening Shot Count ?	0	Maintenance Interval (shots) ?	250,000
Recorded Shots ?	0	Maintenance Interval	6

Today

OdooBot 12:10 PM
Medical Plastics Moulding Tool created

Figure 180 — Mould in service with its cavity register and shot counters. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Preventive maintenance of moulds (by shot count and by time) and cavity blocking

Module	ls_medical_plastics	Version	_____
Menu	Medical Plastics ► Tools ► Due For Maintenance / Maintenance Records	Effective date	_____
Owner (suggested)	Tooling Manager	Approved by	_____
Related SOPs	SOP-MP-001, SOP-GEN-003		

1 Objective

Maintain moulds before wear affects parts, by whichever of the shot-count and time intervals comes first, and block faulty cavities with a recorded reason so that production continues on good cavities only.

2 Scope

All moulds in service.

3 Responsibilities

Role (software group)	Responsibility
Tool Technician	Performs maintenance and records findings and actions.
Process Engineer	Decides on cavity blocking and requalification.
Operator	Reports defects by cavity.

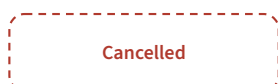
4 Definitions

Term	Definition
Maintenance status	OK, Due Soon, Overdue, from shots since last maintenance and months since last maintenance.
Resets maintenance baseline	Flag on a maintenance event that resets the shot counter baseline.

5 Workflow in the software



Other outcomes



Model: ls.mp.tool.maintenance

Figure 181 — Lifecycle of a maintenance event.



Preventive maintenance is triggered by shot count or elapsed time, whichever comes first.

Figure 182 — Shots since last maintenance against the interval.

6 Procedure

- 1 **Monitor.** Daily, open **Tools ► Due For Maintenance**.
- 2 **Plan.** Create a maintenance record: tool, type (preventive, corrective, cleaning, repair, modification, requalification), planned date.
- 3 **Stop the tool.** When runs are finished, click **Send To Maintenance** on the tool.
- 4 **Perform.** Click **Start** on the maintenance; record shot count at intervention, performer or provider, findings, actions taken, cavities affected, *Requalification Required*, *Resets Maintenance Baseline*; click **Complete**.
- 5 **Return to service.** Click **Return To Service** on the tool (wizard); requalify first when required.
- 6 **Block a cavity.** When a cavity produces defects, click **Block** on the cavity with a reason; **Unblock** after repair and verification.

7. Controls enforced by the software

- ✓ A maintenance cannot be completed without recording the actions taken; only draft maintenance can start; completed maintenance cannot be cancelled or deleted (tool history).
- ✓ A cavity cannot be blocked or removed without a recorded reason; only active cavities can be blocked.
- ✓ A tool with runs in progress cannot be sent to maintenance.

8. Good practices

- ★ Keep the last parts of the run before maintenance and the first parts after it for comparison.
- ★ Trend corrective maintenance per tool to adjust the preventive interval.

9. Pitfalls to avoid

- ✗ Blocking a cavity physically without recording it — parts counts and traceability become wrong.

10 Records and evidence

- Maintenance records
- Cavity status history

11 Performance indicators

- Tools overdue for maintenance (target 0)
- Corrective / preventive ratio

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 3 — Premises and Equipment
- ISO 13485:2016 — Medical devices — Quality management systems

13. Demonstration

Case. T-032 reaches 236 000 shots (Due Soon). After the current run the technician sends it to maintenance, cleans vents and replaces ejector pins, records the shot count, ticks *Resets Maintenance Baseline*, completes. Cavity 6 showed flash: the engineer blocks it with reason *flash on parting line*; production continues on 7 cavities until repair.

Process windows: drafting, review and approval

Module	ls_medical_plastics	Version	_____
Menu	Medical Plastics ► Production ► Moulding Parameters	Effective date	_____
Owner (suggested)	Process Engineering Manager	Approved by	_____
Related SOPs	SOP-MP-004, SOP-QA-030, SOP-VAL-006		

1 Objective

Define for each component/tool combination the validated process window (parameters with target, minimum and maximum, critical parameters and monitoring frequency), approved through author–reviewer–approver separation.

2 Scope

All moulding parameter specifications used in production.

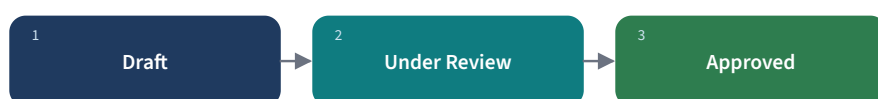
3 Responsibilities

Role (software group)	Responsibility
Author (Process Engineer)	Writes the specification from process validation data.
Reviewer	Reviews (not the author).
Approver	Approves (neither author nor reviewer).

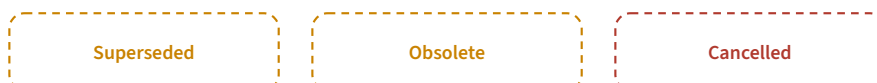
4 Definitions

Term	Definition
Critical parameter	Parameter whose deviation may affect quality; recording is mandatory and out-of-tolerance at start-up blocks the run.
Monitoring frequency	At setup only, at start-up verification, hourly, once per shift, once per lot, continuous.

5 Workflow in the software



Other outcomes



Model: ls.mp.molding_parameter

Figure 183 — Lifecycle of a parameter specification.

6 Procedure

- Create.** In **Moulding Parameters** click **New**: component, tool, work centre (optional restriction), version, effective from, periodic review interval; for versions after the first, the reason for change.

- 2 **Add parameters.** For each parameter: name, code, unit, value type (numeric or qualitative), target, minimum, maximum (or expected value), critical, monitoring frequency, recording required.
- 3 **Submit.** Click **Submit For Review**.
- 4 **Review.** The reviewer clicks **Record Review** (or **Return To Draft**).
- 5 **Approve.** The approver clicks **Approve**; the previous approved version is superseded.
- 6 **Change.** Use **New Version** under change control; **Set Obsolete** when the combination is no longer produced (no run in progress).

7. Controls enforced by the software

- ✓ The tool must be declared as producing the component; one approved specification per component and tool — supersede before approving another.
- ✓ A specification needs at least one parameter; later versions need a reason for change.
- ✓ Author ≠ reviewer ≠ approver; approval requires review.
- ✓ $\text{Min} \leq \text{target} \leq \text{max}$; qualitative parameters need an expected value; critical parameters must have recording enabled.
- ✓ Approved specifications cannot be cancelled (make obsolete); a specification used by runs in progress cannot be made obsolete; non-draft or referenced specifications cannot be deleted.

8. Good practices

- ★ Derive windows from the process validation (DoE, OQ/PQ), not from habits.
- ★ Keep the number of critical parameters small and meaningful.

9. Pitfalls to avoid

- ✗ Widening a window after a failed run without change control.

10 Records and evidence

- Approved specifications with versions

11 Performance indicators

- Specifications past periodic review

12 References

- GHTF/SG3/N99-10:2004 — Quality Management Systems — Process Validation Guidance
- ISO 13485:2016 — Medical devices — Quality management systems
- EU GMP Annex 15 (2015) — Qualification and Validation

13. Demonstration

Case. SP-5 on T-032, v2: melt temperature 230 °C (220–240, critical, hourly), injection pressure 850 bar (800–900, critical, continuous), cooling time 8 s (7–9), visual aspect *no flash* (qualitative, once per lot). Author and reviewer from process engineering, approver from QA; v1 superseded.

Medical Plastics
Production
Products
Tools
Reports
Configuration

7
14
My Company (San Francisco)

New
Moulding Parameters
SP-5 on T-032 - SP-5/T-032 v1
Runs
1
1 / 1

New Version
Set Obsolete

Draft
Under Review
Approved

Send message
Log note
Activity

SP-5 on T-032

Version 1

PRODUCTION CONTEXT

EFFECTIVITY

Component
[SP-5] Syringe plunger SP-5
Tool
[T-032] Mould T-032 (8 cavities)
Work Centre
My Company (San Francisco)
Company
My Company (San Francisco)
Active

Effective From
Sep 27
Periodic Review Interval (months)
24
Next Review Due
Sep 27, 2028
Review Overdue

APPROVAL RECORD

Author
QC Analyst (demo)
Reviewed By
QA Reviewer (demo)
Reviewed On
Sep 27, 12:10 PM

Approved By
QA Manager (demo)
Approved On
Sep 27, 12:10 PM
Parameter Count
4
Critical Parameters
2

Parameters
Reason For Change
Notes

Parameter...	Parameter	Unit	Value Type	Minimum	Target	Maximum	Expected ...	C..	Monitr
MT	Melt temp...	°C	Numeric	220.0000	230.0000	240.0000			Hourly
IP	Injection p...	bar	Numeric	800.0000	850.0000	900.0000			Continuou.

QC Analyst (demo)
12:10 PM

Medical Plastics Moulding Parameter Specification created

Figure 184 — Approved process window (author, reviewer and approver are different users). (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Start-up, operation and closure of a moulding run

Module	ls_medical_plastics	Version	_____
Menu	Medical Plastics ► Production ► Injection Moulding	Effective date	_____
Owner (suggested)	Moulding Production Manager	Approved by	_____
Related SOPs	SOP-MP-003, SOP-MP-005, SOP-MP-006, SOP-QA-020		

1 Objective

Run production only with a released component, a tool in service and an approved specification; verify start-up parameters before running; record readings, material and scrap; have the run reviewed by someone other than its operator or setter before closure.

2 Scope

All injection moulding runs of GMP components.

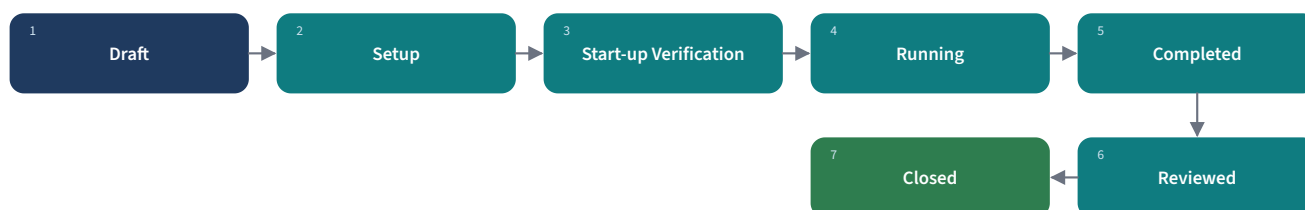
3 Responsibilities

Role (software group)	Responsibility
Setter	Sets up the tool and press.
Operator	Runs production, records readings and scrap.
Reviewer (Process Engineer / QA)	Reviews the run; cannot be the operator or setter.

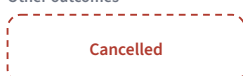
4 Definitions

Term	Definition
Start-up verification	Readings of every parameter required at start-up, within tolerance for critical ones, before the run can start.
Shot counter	Tool counter at start and end; shots = end – start.

5 Workflow in the software



Other outcomes



Model: ls.mp.injection_molding

Figure 185 — Lifecycle of a moulding run.

6 Procedure

- 1 **Create the run.** In **Injection Moulding** click **New** : component, tool, work centre, manufacturing order, produced lot, operator, shift, shot counter at start. The approved specification is proposed.
- 2 **Setup.** Click **Start Setup** ; the setter records setup readings (**Record Readings**); click **Setup Complete** .
- 3 **Start-up verification.** Record start-up verification readings for the required parameters and the material consumption (grade, lot, quantity); click **Confirm Start-up and Run** .
- 4 **Run.** Record in-process readings at the specified frequency; record scrap by reason and cavity; block cavities if needed (SOP-MP-002).
- 5 **Complete.** Record the end shot counter, quantities produced and rejected; click **Complete Run** .
- 6 **Deviations.** If readings were out of tolerance, open a deviation (SOP-QA-020) and record its reference.
- 7 **Review.** The reviewer (not operator nor setter) clicks **Review** , or **Return For Correction** with a comment.
- 8 **Close.** Click **Close Run** .

7. Controls enforced by the software

- ✓ Production requires a released component, a tool in service declared for the component, and an approved specification for this component/tool (and work centre if restricted).
- ✓ Start-up verification must be complete; critical parameters out of tolerance at start-up block the run; material consumption must be recorded before running.
- ✓ Completion requires the end shot counter ($\geq$ start); rejected $\leq$ produced.
- ✓ Review requires completion, a reviewer who is neither operator nor setter, and a deviation reference if readings were out of tolerance; closure requires review.
- ✓ Readings cannot be captured outside setup/start-up/running; closed runs are frozen; runs past draft cannot be deleted; cancellation requires a reason.

8. Good practices

- ★ Record readings at the time they are taken — the timestamp is part of the evidence.
- ★ Quarantine parts produced between the last good reading and an out-of-tolerance one.

9. Pitfalls to avoid

- ✗ Starting production "while waiting" for the start-up verification.

10 Records and evidence

- Run with readings, material, scrap, review

11 Performance indicators

- Reject rate per run
- Runs with out-of-tolerance readings
- Runs returned for correction

12 References

- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 15378:2017 — Primary packaging materials for medicinal products
- GHTF/SG3/N99-10:2004 — Quality Management Systems — Process Validation Guidance

13. Demonstration

Case. Run MR-2027-118: SP-5 on T-032 (7 active cavities), lot SP5-2711. Setup and start-up readings within limits; material PP grade lot R-5531 recorded; 7 in-process readings; injection pressure once at 905 bar (max 900) for 10 min → parts of that period quarantined, deviation DEV/2027/00012 recorded. Completed: 96 600 shots, 676 200 parts, 1 900 rejects. Reviewed by the process engineer and closed.

The screenshot displays the Odoo 19.0 Community LS modules demonstration software interface. The top navigation bar includes links for Medical Plastics, Production, Products, Tools, Reports, and Configuration. The user is logged in as 'My Company (San Francisco)'. The main content area shows a 'New' button and a 'Readings' button. The status bar indicates the run is 'Completed'. The main content area is divided into two sections: 'MR-2027-118' and 'Production Operator (demo)'. The 'MR-2027-118' section contains a warning message: 'Out of tolerance readings recorded. A deviation reference must be recorded before this run can be reviewed.' Below this, there are two tables: 'PRODUCTION CONTEXT' and 'SPECIFICATION APPLIED'. The 'PRODUCTION CONTEXT' table lists details such as Component, Tool, Work Centre, Manufacturing Order, Produced Lot, and Company. The 'SPECIFICATION APPLIED' table lists details such as Parameter, Specification, Specification Version, and Cavities in Production. The 'TIMING AND PERSONNEL' table lists details such as Start, End, Shift, Operator, and Setter. The 'COUNTERS AND OUTPUT' table lists details such as Shot Counter at Start, Shot Counter at End, Shots, Quantity Produced, Start-up Scrap, Quantity Rejected, and Quantity Accepted.

PRODUCTION CONTEXT		SPECIFICATION APPLIED	
Component	[SP-5] Syringe plunger SP-5	Parameter	SP-5 on T-032 - SP-5/T-032 v1
Tool	[T-032] Mould T-032 (8 cavities)	Specification	
Work Centre		Specification Version	1
Manufacturing Order		Cavities in Production	8
Produced Lot	SP5-2711		
Company	My Company (San Francisco)		

TIMING AND PERSONNEL		COUNTERS AND OUTPUT	
Start	Sep 27, 12:10 PM	Shot Counter at Start	0
End	Sep 27, 12:10 PM	Shot Counter at End	96,600
Shift	Shift 1	Shots	96,600
Operator	Production Operator (demo)	Quantity Produced	676,200.000
Setter	Production Operator 2 (demo)	Start-up Scrap	0.000
		Quantity Rejected	0.000
		Quantity Accepted	676,200.000

Figure 186 — Completed run awaiting review, with an out-of-tolerance reading and its deviation reference. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Correction of parameter readings (without altering the original value)

Module	ls_medical_plastics	Version	_____
Menu	Medical Plastics ► Production ► Parameter Readings	Effective date	_____
Owner (suggested)	Moulding Production Manager with QA	Approved by	_____
Related SOPs	SOP-MP-004, SOP-DI-002		

1 Objective

Correct an erroneous parameter reading by a new, justified reading that supersedes the original, so that the original value, the correction and its reason all remain visible.

2 Scope

All parameter readings of moulding runs.

3 Responsibilities

Role (software group)	Responsibility
Operator / setter	Records the correcting reading with its reason.
Reviewer	Checks corrections at run review.

4 Definitions

Term	Definition
Superseding reading	New reading linked to the reading it corrects (same run, same parameter), with a mandatory reason; the original is marked superseded.

5 Procedure

- 1 Detect.** An erroneous reading is noticed (typing error, wrong instrument, wrong parameter).
- 2 Record the correction.** Use **Record Readings** on the run, select the parameter, enter the correct value, select the reading it supersedes and write the reason for correction.
- 3 Never edit.** Do not try to modify or delete the original reading — the system refuses it.
- 4 Review.** At run review, check each correction: reason plausible, time close to the original, same person or supervisor.

6. Controls enforced by the software

- ✓ Readings cannot be modified or deleted.
- ✓ A correcting reading must record a reason, belong to the same run and address the same parameter as the reading it supersedes.
- ✓ Qualitative parameters require an observed value; no reading on a run that is not open for readings.

7. Good practices

- ★ Explain the correction factually ("typed 285 instead of 258, see press display photo").

8. Pitfalls to avoid

- ✗ Using corrections to bring out-of-tolerance values back into range without evidence — a data integrity breach.

9 Records and evidence

- Original and superseding readings with reasons

10 Performance indicators

- Corrections per 100 readings
- Corrections without evidence found at review

11 References

- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)
- MHRA — GxP Data Integrity Guidance and Definitions (2018)
- EU GMP Annex 11 — Computerised Systems (check the revision in force)

12. Demonstration

Case. The operator types a cooling time of 80 s instead of 8.0 s. Two minutes later she records a new reading 8.0 s superseding the first, reason "*decimal point omitted — press display shows 8.0 s*". At review the engineer sees both readings and accepts the correction.

Material traceability

Module	ls_medical_plastics	Version	_____
Menu	Medical Plastics ► Configuration ► Material Grades · Reports ► Traceability Enquiry	Effective date	_____
Owner (suggested)	Moulding Quality Manager	Approved by	_____
Related SOPs	SOP-MP-004, SOP-QA-070, SOP-QA-081		

1 Objective

Use only qualified material grades approved for each component, record the material lots consumed by each run, and be able to trace forward (material lot → parts) and backward (part lot → material, tool, cavity, run) within minutes.

2 Scope

All resins, masterbatches and colorants used for GMP components.

3 Responsibilities

Role (software group)	Responsibility
Quality / Process Engineer	Qualify material grades and approve them for components.
Operators	Record material lots at start-up and at each change.
Quality	Performs traceability enquiries for complaints and recalls.

4 Definitions

Term	Definition
Qualification status	Draft, Under Evaluation, Qualified, Conditionally Qualified (conditions recorded), Rejected, Obsolete.
Direct drug contact	Grade used for parts in contact with the drug; a lot reference is mandatory at consumption.

5 Procedure

- 1 Qualify the grade.** In **Material Grades**: polymer type, resin manufacturer, approved suppliers, compendial references, master file, change notification agreement, properties; **Start Evaluation**, then **Qualify** (or **Qualify Conditionally** with conditions) with the qualification date and requalification due date.
- 2 Approve for components.** On the component list the approved material grades and the primary grade; release the component (**Release**).
- 3 Record consumption.** On each run, record material grade, lot and quantity before start-up and at every lot change.
- 4 Trace.** Use **Reports ► Traceability Enquiry** by produced lot, material lot, component and period, or tool and period; **Open Results** or **Print Records**.
- 5 Change of material.** A resin change (grade, supplier, site) goes through change control and requalification.

6. Controls enforced by the software

- ✓ A grade cannot be qualified without a qualification date; conditional qualification requires recorded conditions; requalification date after qualification.
- ✓ Only qualified grades approved for the component can be consumed; a direct-drug-contact grade requires a lot reference; the lot must belong to the product.
- ✓ A component cannot be released without at least one approved grade; the primary grade must be among the approved grades; sterile components need a sterilisation method.

7. Good practices

- ★ Obtain a change notification agreement from resin suppliers.
- ★ Test the traceability enquiry during mock recalls (SOP-QA-083).

8. Pitfalls to avoid

- ✗ Mixing two resin lots in the dryer without recording both.

9 Records and evidence

- Material grades with qualification
- Material consumption per run
- Traceability enquiry outputs

10 Performance indicators

- Time to answer a traceability enquiry
- Runs without material record (target 0)

11 References

- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 15378:2017 — Primary packaging materials for medicinal products
- ISO 10993-1 — Biological evaluation of medical devices

12. Demonstration

Case. A resin supplier notifies a contamination risk on lot R-5531. The quality manager runs the enquiry *By Material Lot*: 3 runs, lots SP5-2711 to SP5-2713, all on T-032. The lots are blocked and a deviation opened within 1 hour.

Medical Plastics
Production
Products
Tools
Reports
Configuration

7
14
My Company (San Francisco)

New
Material Grades
[PP-MG-01] PP homopolymer medical grade
1 / 1

Set Obsolete
Draft
Under Evaluation
Qualified
Send message
Log note
Activity

Grade Name ?
PP homopolymer medical grade

Internal Code ?
PP-MG-01

IDENTIFICATION

Polymer Type
PP - Polypropylene
Resin Manufacturer ?
Inventory Product ?
Consumption Unit ?
kg
Company
My Company (San Francisco)
Active

INTENDED USE
Direct Drug Contact ?
Food Contact
Master File Reference ?
DMF 12345 (example)
Change Notification Agreement ?

Qualification
Technical Data
Supply
Components
Notes

Qualified On
Mar 11
Requalification Due ?
Mar 10, 2028

QUALIFICATION NOTES
Basis of qualification and any conditions attached to it

Today

OdooBot 12:10 PM
Medical Plastics Material Grade created

Figure 187 — Qualified material grade. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

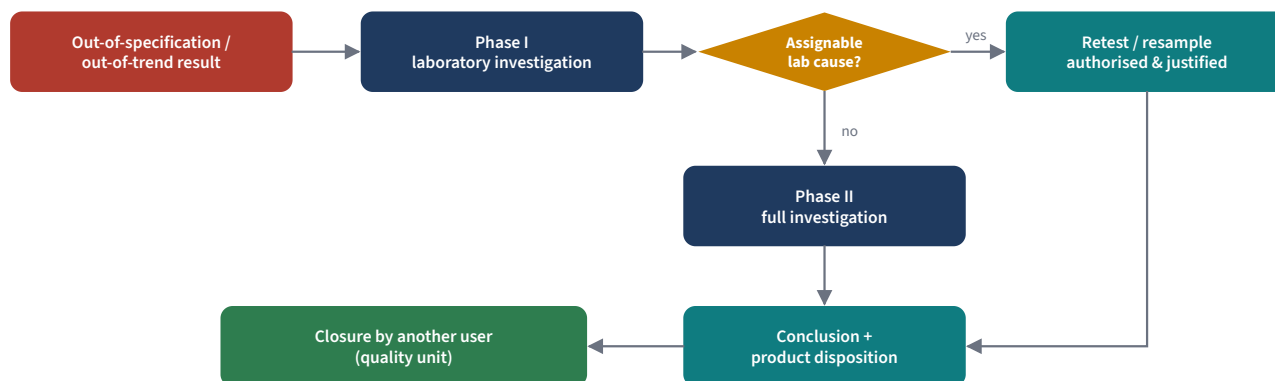
Laboratory

ls_lab

ls_lab supports the quality control laboratory: controlled test methods and product specifications (draft, review, approval, versions), samples from receipt to reporting, results with automatic evaluation and second-person review, OOS/OOT investigations in two phases with retest and resample authorisation, stability studies with time points and pull samples, and certificates of analysis. Approvals and certificate issue record a *signature intent* (user, time, meaning), which the module states is **not** a 21 CFR Part 11 electronic signature. Lots held by an open investigation cannot be delivered.

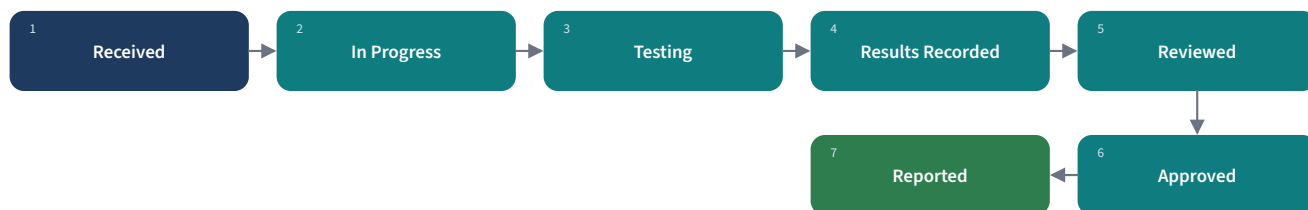
Menu	Content
Laboratory ► Tests ► Test Methods / Test Results	Methods and results
Laboratory ► Specifications ► Product Specifications / OOS / OOT	Specifications and investigations
Laboratory ► Samples ► Sample Receipt / Sample Register	Samples
Laboratory ► Stability ► Stability Studies / Time Points / Stability Samples	Stability
Laboratory ► Reports ► Certificates of Analysis / Test Result Analysis	Reporting

Group	Main rights
Laboratory Viewer · Analyst	Read; perform tests and record results
Laboratory Reviewer	Review results and samples (not their own)
Laboratory Manager	Approve methods, specifications, samples; close investigations

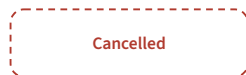


The lot stays blocked for delivery while the investigation is open (control on stock.move.line).

Figure 188 — OOS investigation phases.



Other outcomes



Model: *ls.lab.sample*

Figure 189 — Lifecycle of a laboratory sample.

Approval of test methods and specifications

Module	ls_lab	Version	_____
Menu	Laboratory ► Tests ► Test Methods · Specifications ► Product Specifications	Effective date	_____
Owner (suggested)	QC Manager	Approved by	_____
Related SOPs	SOP-VAL-006, SOP-QC-012, SOP-QA-030		

1 Objective

Use only approved, validated test methods and approved specifications with explicit acceptance criteria, and control their versions.

2 Scope

Test methods and specifications for raw materials, packaging, in-process, finished product, stability and water.

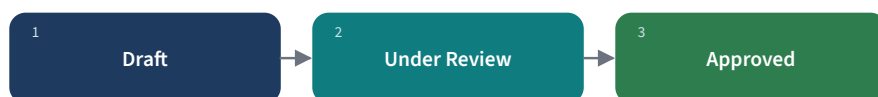
3 Responsibilities

Role (software group)	Responsibility
Analyst / QC specialist	Drafts methods and specifications.
Laboratory Manager	Approves; makes obsolete with a reason.

4 Definitions

Term	Definition
Method validation status	Not Required, Planned, In Progress, Validated, Revalidation Due (with validation reference).
Criterion type	Between Min and Max, Not Less Than, Not More Than, Text Match, Pass/Fail, Informative Only; <i>Mandatory</i> and <i>Show on CoA</i> flags.

5 Workflow in the software



Other outcomes



Model: ls.lab.test_method

Figure 190 — Lifecycle of a test method and of a specification.

6 Procedure

- Method.** In **Test Methods** click **New**: name, code, version, technique, result type, unit, decimals, reference document, procedure summary, instrument category, validation status and reference. **Submit for Review** then **Approve**.

- 2 **Specification.** In **Product Specifications** click **New** : name, code, version, product, type, effective date; add criteria lines (approved method, criterion type and limits, unit, mandatory, show on CoA). **Submit for Review** then **Approve**.
- 3 **Revise.** Use **Create Revision** for any change; the new version supersedes the old when approved.
- 4 **Obsolete.** Click **Set Obsolete** with the obsolescence reason; only one approved specification per product and type.

7. Controls enforced by the software

- ✓ A specification without criteria cannot be approved; only approved methods can be used in a specification; $\text{min} \leq \text{max}$; a text criterion needs the expected text.
- ✓ Only one approved specification per product and type — make the old one obsolete first.
- ✓ Approved records are frozen (blocked fields listed); criteria of approved specifications cannot change; obsolescence requires a reason; only draft records can be deleted.

8. Good practices

- ★ Link each method to its validation report (SOP-VAL-006) before approval.
- ★ Align criteria with the registered specification (marketing authorisation, pharmacopoeia).

9. Pitfalls to avoid

- ✗ Approving a method whose validation is still *In Progress* without a documented justification.

10 Records and evidence

- Approved methods and specifications with versions

11 Performance indicators

- Methods with revalidation due
- Specifications revised per year

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 6 — Quality Control
- 21 CFR 211.160 to 211.176 — Laboratory controls
- ICH Q2(R2) — Validation of Analytical Procedures

13. Demonstration

Case. Specification *Tablets 10 mg — finished product v4*: assay 95.0–105.0 % (HPLC method AM-012 v3, validated), dissolution NLT 80 % (Q) at 30 min, related substances NMT 0.5 %, appearance text match. Approved; v3 made obsolete with reason "*new impurity limit per variation 2027-02*".

Receipt and registration of samples

Module	ls_lab	Version	_____
Menu	Laboratory ► Samples ► Sample Receipt	Effective date	_____
Owner (suggested)	QC Manager	Approved by	_____
Related SOPs	SOP-QC-012, SOP-QC-010		

1 Objective

Register every sample received by the laboratory with its identity, specification and storage, so that it can be tested against the right criteria and traced.

2 Scope

Raw material, packaging, in-process, finished product, stability, retain, water and cleaning verification samples.

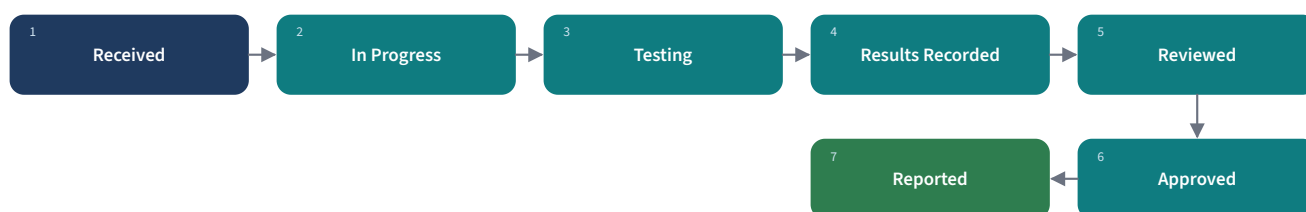
3 Responsibilities

Role (software group)	Responsibility
Sample receiver	Checks labelling and condition, registers the sample.
Laboratory Manager	Sets priorities and due dates.

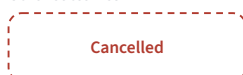
4 Definitions

Term	Definition
Sample type	Raw Material, Packaging Material, In Process, Finished Product, Stability, Retain, Water, Cleaning Verification.

5 Workflow in the software



Other outcomes



Model: ls.lab.sample

Figure 192 — Sample lifecycle.

6 Procedure

- Check.** Check label, container integrity, quantity and storage condition on arrival; refuse or annotate non-conforming samples.
- Register.** In **Sample Receipt** click **New**: sample type, product, lot, batch reference, approved specification, quantity, sampling date, receipt date, due date, priority, sampled by, sampling point, storage condition.

- 3 **Start.** Click **Start** to accept the sample into the laboratory; test result lines are prepared from the specification.
- 4 **Store.** Store the sample in the recorded condition until testing.
- 5 **Cancel.** If the sample cannot be tested, use **Cancel** with a reason — never delete.

7. Controls enforced by the software

- ✓ The specification must be approved; the lot must belong to the product.
- ✓ State transitions are enforced; cancelled samples keep their reason; records that left draft cannot be deleted.

8. Good practices

- ★ Label the container with the system sample reference.
- ★ Record the actual receipt time for time-sensitive samples (microbiology, water).

9. Pitfalls to avoid

- ✗ Testing a sample against the specification of another product version.

10 Records and evidence

- Sample register

11 Performance indicators

- Samples received vs tested on time
- Samples cancelled

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 6 — Quality Control
- 21 CFR 211.160 to 211.176 — Laboratory controls
- ISO/IEC 17025:2017 — Competence of testing and calibration laboratories

13. Demonstration

Case. Finished product sample of batch 26T018 received: 60 tablets, specification *Tablets 10 mg — FP v4* (approved), priority normal, due in 5 working days; registered and started.

Entry, review and approval of results

Module	ls_lab	Version	_____
Menu	Laboratory ► Tests ► Test Results · Samples ► Sample Register	Effective date	_____
Owner (suggested)	QC Manager	Approved by	_____
Related SOPs	SOP-QC-011, SOP-QC-013, SOP-QC-015, SOP-DI-002		

1 Objective

Record each result contemporaneously with its analyst, date and instrument, have it reviewed by a second analyst, and approve the sample only when all mandatory results are reviewed and no investigation is open.

2 Scope

All results of laboratory samples.

3 Responsibilities

Role (software group)	Responsibility
Analyst	Performs the test and records the result.
Laboratory Reviewer	Reviews results (not their own) and the sample.
Laboratory Manager (approver)	Approves the sample (not the reviewer).

4 Definitions

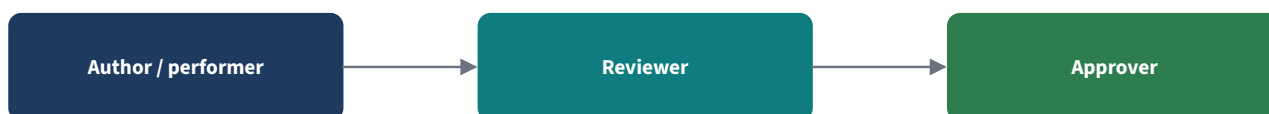
Term	Definition
Evaluation	Pending, Conforms, Out Of Specification, Informative — computed against the specification line.
Out of trend	Flag set by the analyst/reviewer with a mandatory justification.

5 Workflow in the software



Model: ls.lab.test_result

Figure 193 — Lifecycle of a test result.



The software refuses the same person in two roles on one record wherever the rule exists
(e.g. result review, SOP approval, batch release, excursion closure, protocol approval).

Figure 194 — Analyst, reviewer and approver are different people.

6 Procedure

- 1 **Test.** Click **Start Testing** on the sample.
- 2 **Record.** For each result line enter the numeric, text or pass/fail result, analyst, test date, instrument reference, remarks; click **Record Result**. Attach raw data or reference it.
- 3 **OOS / OOT.** If the evaluation is *Out Of Specification* or the result is out of trend, open an investigation (SOP-QC-013) — do not retest on your own.
- 4 **Confirm.** When all mandatory results are recorded, click **Confirm Results Recorded**.
- 5 **Review.** A reviewer who did not enter results reviews each result (**Review** on the result) and then the sample (**Review**).
- 6 **Approve.** The approver (not the reviewer) clicks **Approve** and confirms the signature intent (meaning *Approved*).
- 7 **Report.** Click **Mark Reported** once the results are communicated or the CoA issued (SOP-QC-015).

7. Controls enforced by the software

- ✓ A result cannot be recorded on a sample not open for entry, nor entered twice; a criterion must belong to the sample's specification.
- ✓ The analyst cannot review their own result; the result entrant cannot review the sample; the reviewer cannot approve it.
- ✓ Approval requires every mandatory test recorded and reviewed and no open investigation.
- ✓ Reviewed results cannot be modified (record a retest under an investigation); results with data cannot be deleted; results linked to an investigation cannot be reset.
- ✓ An out-of-trend flag requires a justification; a retest requires a documented retest authorisation.

8. Good practices

- ★ Review raw data and the audit trail, not only the value typed (SOP-DI-002).
- ★ Record instrument references to link results with calibration status.

9. Pitfalls to avoid

- ✗ Resetting a result to draft to change an inconvenient value.
- ✗ Averaging an OOS result with retest results.

10 Records and evidence

- Results with analyst, reviewer, dates
- Sample approval with signature intent

11 Performance indicators

- Right-first-time results
- Review turnaround
- OOS rate

12 References

- 21 CFR 211.160 to 211.176 — Laboratory controls
- EudraLex Vol. 4, EU GMP Part I, Chapter 6 — Quality Control
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)
- ISO/IEC 17025:2017 — Competence of testing and calibration laboratories

13. Demonstration

Case. Sample of 26T018: 6 results entered by analyst A; assay 99.2 % (Conforms). Reviewer B reviews each result and the sample; manager C approves (signature intent *Approved*, timestamp recorded); sample marked reported after the CoA is issued.

The screenshot displays the Odoo 19.0 Community LS modules interface for a sample record. The top navigation bar includes links for Laboratory, Samples, Tests, Specifications, Stability, Reports, and Configuration. The breadcrumb trail shows the path: New > Sample Register > LS-2026-0412 - Tablets 10 mg. The record is currently in the 'Approved' state, with a status bar showing 'Mark Reported', 'Received', 'In Progress', 'Testing', 'Results Recorded', 'Reviewed', 'Approved', and 'Reported'. The 'Approved' state is highlighted. The record details are as follows:

MATERIAL		SPECIFICATION	
Product	Tablets 10 mg	Specification ?	[SPEC-TAB10-FP v4] Tablets 10 mg - finis...
Lot / Serial Number	26T018	Overall Result	Conforms
Batch Reference ?	26T018	Storage Condition	
Sample Type	Finished Product	Priority	☆
Quantity	60.0000	Company	My Company (San Francisco)
Unit of Measure			

SAMPLING AND RECEIPT		REVIEW AND APPROVAL	
Sampling Date		Reviewed By	QA Reviewer (demo)
Sampled By	QC Analyst (demo)	Review Date	Sep 27, 12:08 PM
Sampling Point	Packaging line PK-2	Approved By	QA Manager (demo)
Receipt Date	Sep 27, 12:08 PM	Approval Date	Sep 27, 12:08 PM
Received By	QC Analyst (demo)		
Due Date			

The right sidebar shows the activity log for the sample, including a message from QA Manager (demo) at 12:08 PM, indicating the sample was reviewed and approved by QA Manager (demo). The activity log also shows a message from QC Analyst (demo) at 12:08 PM, indicating the laboratory sample was created.

Figure 195 — Approved sample: results entered, reviewed and approved by three different users. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Investigation of OOS (out-of-specification) and OOT (out-of-trend) results

Module	ls_lab	Version	_____
Menu	Laboratory ► Specifications ► OOS / OOT	Effective date	_____
Owner (suggested)	QC Manager with QA	Approved by	_____
Related SOPs	SOP-QC-012, SOP-QA-020, SOP-QA-021, SOP-QA-090		

1 Objective

Investigate every OOS or OOT result in a structured way — laboratory phase first, full investigation when no laboratory cause is found — authorise retests or resamples only with written scientific justification, and decide on the product with QA.

2 Scope

All OOS and OOT results of release, stability, raw material and in-process testing.

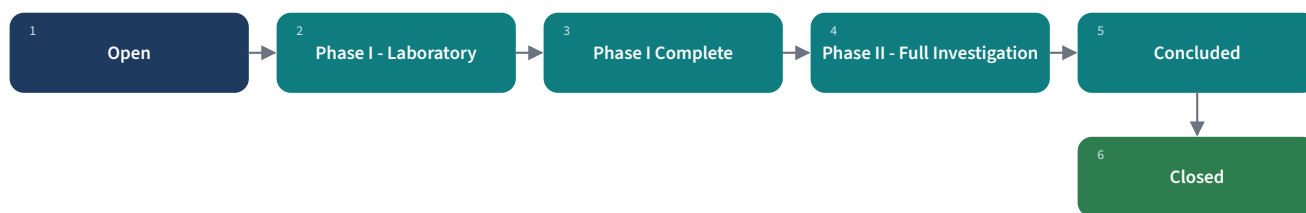
3 Responsibilities

Role (software group)	Responsibility
Investigator (analyst's supervisor)	Conducts Phase I and Phase II.
QA / quality unit approver	Approves closure; must not be the investigator.
Production	Contributes to the manufacturing review in Phase II.

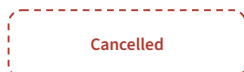
4 Definitions

Term	Definition
Phase I	Laboratory investigation: analyst interview, calculations, instrument performance, standards and reagents, sample integrity, method adherence.
Phase II	Full investigation including manufacturing review, when Phase I finds no assignable laboratory cause.
Final conclusion	Assignable Laboratory Cause, Assignable Manufacturing Cause, No Assignable Cause Identified; disposition Reject, Release, Remain In Quarantine, Further Investigation Required.

5 Workflow in the software

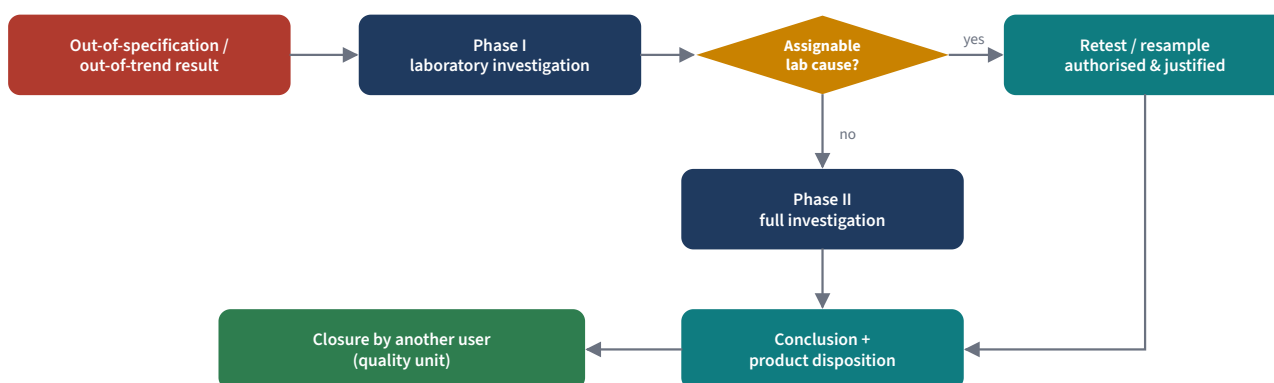


Other outcomes



Model: ls.lab.oos

Figure 196 — Lifecycle of an OOS/OOT investigation.



The lot stays blocked for delivery while the investigation is open (control on stock.move.line).

Figure 197 — Phase I, Phase II, conclusion and closure.

6 Procedure

- 1 **Open.** In **Specifications ► OOS / OOT** click **New**: type (OOS or OOT), originating result, investigator, description. The lot is held: deliveries are refused while it is open.
- 2 **Phase I.** Click **Start Phase I**; tick each check performed (analyst interviewed, calculation verified, instrument performance, standards and reagents, sample integrity, method adherence); record Phase I findings and conclusion; click **Complete Phase I**.
- 3 **Laboratory cause.** If an assignable laboratory cause is identified, invalidate the result with justification and, when justified in writing, click **Authorise Retest**; the retest is recorded as a retest linked to the investigation.
- 4 **Phase II.** If no laboratory cause, click **Start Phase II**: review manufacturing (batch record, deviations, equipment), determine root cause; resample only with **Authorise Resample** and justification; record Phase II findings; **Complete Phase II**.
- 5 **Conclude.** Click **Conclude** with the final conclusion, product disposition and justification, CAPA reference if any.
- 6 **Close.** A quality unit member who did not conduct the investigation clicks **Close**.

7. Controls enforced by the software

- ✓ Investigations cannot be deleted — cancel with a reason.
- ✓ Phase I needs findings and a conclusion; Phase II applies only when Phase I found no laboratory cause, and is mandatory in that case before concluding.
- ✓ Retest and resample require a written scientific justification; a retest result cannot be recorded without authorisation.
- ✓ Closure requires a final conclusion and a product disposition, by a user other than the investigator.
- ✓ Lots held by an open investigation cannot be delivered; a sample cannot be approved while an investigation is open.

8. Good practices

- ★ Never test into compliance: the number of retests and the decision rule are defined before retesting.
- ★ Report OOS trends in management review.

9. Pitfalls to avoid

- ✗ Invalidating an OOS result without evidence of a laboratory error.
- ✗ Closing the OOS with *Release* while a manufacturing cause is under investigation.

10 Records and evidence

- Investigation with phases, checks, authorisations, conclusion, disposition

11 Performance indicators

- OOS invalidated in Phase I (%)
- Median time to closure
- OOS per product

12 References

- FDA Guidance — Investigating Out-of-Specification (OOS) Test Results for Pharmaceutical Production (Revision 1, 2022)
- 21 CFR 211.192 — Production record review
- EudraLex Vol. 4, EU GMP Part I, Chapter 6 — Quality Control

13. Demonstration

Case. Dissolution of 26T018 at 30 min: 76 % (NLT 80 %). Phase I: vessel 4 paddle height found out of tolerance at post-test check → assignable laboratory cause; the manager authorises a retest with justification. Retest on 6 new units: 88–93 %. Conclusion *Assignable Laboratory Cause*, disposition *Release*; closed by QA (not the investigator); CAPA opened for apparatus verification frequency.

Figure 198 — OOS investigation in Phase I with laboratory checks. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Stability studies in the laboratory

Module	ls_lab	Version	_____
Menu	Laboratory ► Stability	Effective date	_____
Owner (suggested)	QC Manager	Approved by	_____
Related SOPs	SOP-QC-002, SOP-QC-012, SOP-QC-013		

1 Objective

Run stability studies with an approved protocol, generate pull samples at each time point, test them against the stability specification and record missed time points with reasons.

2 Scope

Stability studies managed on the laboratory side (long term, intermediate, accelerated, in-use, photostability, ongoing/annual). Programme-level stability of pharmaceutical batches may be managed in ls_pharma (SOP-QC-002); choose one module per study to avoid double records.

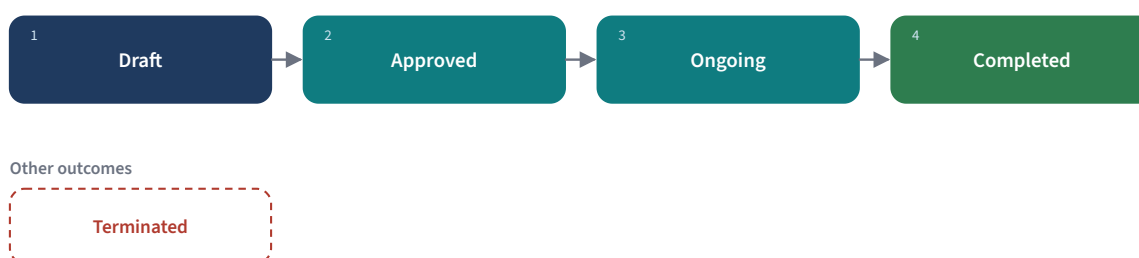
3 Responsibilities

Role (software group)	Responsibility
Stability analyst	Pulls and tests samples.
Laboratory Manager	Approves protocol, starts, completes or terminates studies.

4 Definitions

Term	Definition
Time point	Planned, Sampled, Tested, Completed, Missed, Cancelled.

5 Workflow in the software



Model: ls.lab.stability_study

Figure 199 — Lifecycle of a laboratory stability study.

6 Procedure

- 1 Create.** In **Stability Studies** click **New** : product, lot, batch, study type, storage condition, container closure, specification, protocol reference, planned end; add time points.
- 2 Approve and start.** Click **Approve Protocol** ; set the start date; click **Start Study** .
- 3 Pull.** At each time point click **Create Pull Sample** on the time point: a laboratory sample is created and tested (SOP-QC-012).

4 **Progress.** Click **Mark Tested** then **Complete** on the time point; if not pulled, **Mark Missed** with notes.

5 **Close.** Click **Complete** on the study when all time points are final, or **Terminate** with a reason.

7. Controls enforced by the software

- ✓ A study cannot be approved without time points; a start date is required to begin (due dates are derived from it).
- ✓ Pull samples only for ongoing studies and planned time points; only sampled time points can be marked tested, only tested ones completed, only planned ones missed (with notes).
- ✓ Completion requires every time point completed, missed or cancelled; termination requires a reason; the lot must belong to the product.

8. Good practices

- ★ Pull within the time window defined in the protocol and record the actual date.

9. Pitfalls to avoid

- ✗ Managing the same study in both ls_pharma and ls_lab.

10 Records and evidence

- Study, time points, samples, results

11 Performance indicators

- Time points missed
- Studies completed on time

12 References

- ICH Q1A(R2) — Stability Testing of New Drug Substances and Products
- 21 CFR 211.166 — Stability testing

13. Demonstration

Case. Photostability study on 26T018 per ICH Q1B conditions: 2 time points (exposed, dark control); pull samples created and tested; both completed; study completed with conclusion *photostable in the proposed pack*.

Issue of certificates of analysis

Module	ls_lab	Version	_____
Menu	Laboratory ► Reports ► Certificates of Analysis	Effective date	_____
Owner (suggested)	QC Manager	Approved by	_____
Related SOPs	SOP-QC-012, SOP-QA-090		

1 Objective

Issue certificates of analysis only from approved samples, showing the results and conclusion against the specification, and revise them in a traceable way.

2 Scope

Certificates of analysis for customers and for internal release.

3 Responsibilities

Role (software group)	Responsibility
QC officer	Prepares the certificate.
Laboratory Manager	Issues it (signature intent <i>Issued</i>).

4 Definitions

Term	Definition
Conclusion	Conforms To Specification / Does Not Conform To Specification (Pending until issued).
Revision	New version superseding an issued certificate, with a revision reason.

5 Workflow in the software



Other outcomes



Model: ls.lab.coa

Figure 200 — Lifecycle of a certificate of analysis.

6 Procedure

- 1 Create.** In **Reports ► Certificates of Analysis** click **New** and select the approved sample; enter version, customer, remarks; results flagged *Show on CoA* in the specification are included.
- 2 Issue.** Click **Issue Certificate** and confirm the signature intent.
- 3 Send.** Send the PDF to the customer or attach it to the batch file.

4 **Revise.** If a correction is needed, click **Create Revision** with the revision reason; the previous certificate becomes superseded.

5 **Cancel.** Only a draft certificate can be cancelled, with a reason.

7. Controls enforced by the software

- ✓ A certificate can only be issued from an approved sample; issued certificates cannot be modified or deleted.
- ✓ Only an issued certificate can be revised, with a revision reason; only drafts can be cancelled, with a reason.

8. Good practices

- ★ Include the specification version and the method references on the certificate.
- ★ Check the customer-specific requirements (units, additional tests) before issue.

9. Pitfalls to avoid

- ✗ Editing the PDF manually after issue.

10 Records and evidence

- Issued and superseded certificates

11 Performance indicators

- Certificates revised after issue

12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 6 — Quality Control
- 21 CFR 211.160 to 211.176 — Laboratory controls
- ICH Q7 — Good Manufacturing Practice for Active Pharmaceutical Ingredients

13. Demonstration

Case. CoA for batch 26T018: 6 results, conclusion *Conforms To Specification*; issued by the Laboratory Manager. A customer asks for the dissolution profile: a revision v2 is created with reason "*customer request — add dissolution at 15/20/30 min*"; v1 is superseded.

IV

Regulatory affairs

Regulatory intelligence for import and export. Only phase 1 of ls_import_export (the cited provision registry and regulatory questions) exists; import-export operations (authorisations, customs, banking) are later phases and are not covered.

Import-export (phase 1) — ls_import_export

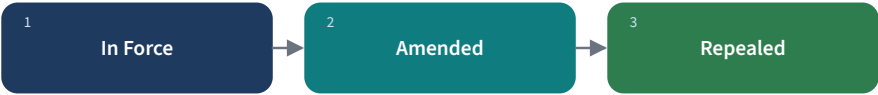
Import-export compliance (phase 1)

ls_import_export

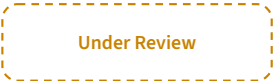
ls_import_export is built on a *Truth Protocol*: the module asserts no regulatory requirement that is not first recorded as a cited provision with a source, an effective date and an in-force status. The registry ships **empty**; every provision is entered by a person who read the source. Expected but unverified rules are recorded as **regulatory questions** for a qualified adviser. A compliance requirement cannot exist without at least one cited provision. Phase 1 is non-behavioural: it does not yet block purchases, sales or shipments.

Menu	Content
Import & Export Compliance ► Registry ► Provisions / Compliance Requirements / Regulatory Questions	Registry

Group	Main rights
Viewer · User · Officer · Manager	From read-only to registry maintenance and approval



Other outcomes



Model: *ls.import_export.provision*
Figure 202 — *Statuses of a provision.*

Regulatory watch: register of cited texts

Module	ls_import_export	Version	_____
Menu	Import & Export Compliance ► Registry ► Provisions / Compliance Requirements	Effective date	_____
Owner (suggested)	Regulatory Affairs Manager (import-export)	Approved by	_____
Related SOPs	SOP-RA-011, SOP-COS-006, SOP-COS-007		

1 Objective

Maintain a registry of the legal and regulatory provisions that govern regulated imports and exports, each cited from its official source with effective date and status, and derive compliance requirements only from cited provisions.

2 Scope

Algerian and international texts applicable to the import and export of medicinal products, medical devices and cosmetics (e.g. pharmaceutical agency, ministry, customs, central bank, foreign trade rules), as selected by Regulatory Affairs.

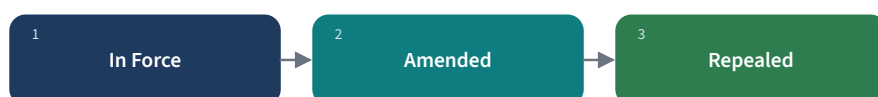
3 Responsibilities

Role (software group)	Responsibility
Regulatory Officer	Reads the source, records provisions and citations.
Regulatory Affairs Manager	Validates entries and requirements; decides on status changes.

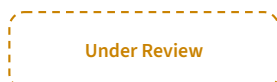
4 Definitions

Term	Definition
Provision	Legal text or article recorded with source, effective date and status: In Force, Amended, Repealed, Under Review.
Compliance requirement	Operational obligation derived from one or more cited provisions.

5 Workflow in the software



Other outcomes



Model: ls.import_export.provision

Figure 203 — Statuses of a provision.

6 Procedure

- 1 Monitor sources.** Define the official sources to monitor (official journal, agency websites, customs notes) and the frequency (e.g. weekly).

- 2 **Record a provision.** In **Registry ► Provisions** create the provision with its official reference, title, source, effective date and status; add citations (article, exact wording or summary with reference).
- 3 **Derive requirements.** In **Compliance Requirements** create the requirement and cite at least one provision.
- 4 **Update.** When a text is amended or repealed, change the status and create the new provision; review the requirements citing it.
- 5 **Open questions.** When a rule is expected but not verified, record a regulatory question (SOP-RA-011) instead of a requirement.

7. Controls enforced by the software

- ✓ A compliance requirement must cite at least one provision; the last cited provision cannot be removed.
- ✓ The registry contains nothing that has not been entered by a person — there is no pre-loaded law.

8. Good practices

- ★ Record the date on which the source was consulted.
- ★ Never paraphrase a legal obligation without the citation next to it.

9. Pitfalls to avoid

- ✗ Entering requirements from memory or from a consultant's slide without the text.

10 Records and evidence

- Provisions with citations
- Requirements with cited provisions

11 Performance indicators

- Provisions under review for more than 30 days
- Requirements whose provisions were amended

12. Demonstration

Case. The officer reads a newly published executive decree on the import of medical devices, records it as a provision *In Force* from its effective date with citations of the articles on prior import authorisation, and creates the requirement "*Import programme approval before first import of the year*" citing those articles.

Handling open regulatory questions

Module	ls_import_export	Version	_____
Menu	Import & Export Compliance ► Registry ► Regulatory Questions	Effective date	_____
Owner (suggested)	Regulatory Affairs Manager	Approved by	_____
Related SOPs	SOP-RA-010, SOP-COS-007		

1 Objective

Record, assign and answer regulatory questions whose answer is not verified, so that no rule is assumed or guessed and each answer is traceable to a source or a qualified adviser.

2 Scope

Questions about import-export rules, and more generally any regulatory point where the applicable requirement is uncertain.

3 Responsibilities

Role (software group)	Responsibility
Any user	Raises a question with context.
Regulatory Affairs Manager	Assigns, obtains the answer (authority, lawyer, consultant), records it.
Management	Decides how to operate while the question is open.

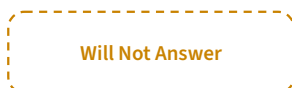
4 Definitions

Term	Definition
Status	Open, Answered, Will Not Answer.

5 Workflow in the software



Other outcomes



Model: ls.import_export.regulatory.question

Figure 204 — Statuses of a regulatory question.

6 Procedure

- 1 Raise.** In **Regulatory Questions** create the question: precise wording, context, products concerned, urgency.
- 2 Assign.** Assign it to a responsible person and a due date.

- 3 **Answer.** Obtain the answer from an authoritative source; record the answer, its source (letter, meeting minutes, legal opinion) and date; set *Answered*. If a provision results, record it (SOP-RA-010).
- 4 **Decline.** If the question will not be pursued, set *Will Not Answer* with the reason.
- 5 **Interim decision.** While open, record the interim operating decision taken by management (e.g. the most conservative interpretation).

7. Controls enforced by the software

- ✓ Statuses are limited to Open, Answered and Will Not Answer.

8. Good practices

- ★ Prefer written answers from authorities; keep minutes of oral answers.

9. Pitfalls to avoid

- ✗ Treating an informal answer as a legal requirement without recording its source.

10 Records and evidence

- Questions with answers and sources

11 Performance indicators

- Open questions older than 60 days

12. Demonstration

Case. Question: *"Does a change of the importer's storage site require a new import authorisation for cosmetics?"* — assigned to the regulatory manager; answer received by letter from the competent directorate after 3 weeks; recorded with the letter reference; the provision cited in the letter is added to the registry.



Procedures supported by OCA modules

Procedures supported by community (OCA) modules present in the repository. Before relying on them, note the installation findings reported in the annex *Known limitations*: on a fresh Odoo 19.0 database, auditlog, base_user_role, mgmtsystem_kpi and mgmtsystem_review (through its dependency base_external_dbsource) failed to install because of view references renamed from *tree* to *list*. Screenshots are therefore given only for maintenance_plan, maintenance_equipment_certification and quality_control_oca, which installed.

Management review — mgmtsystem_review	Roles and access review — base_user_role
Quality indicators — mgmtsystem_kpi	Technical audit log — auditlog
Preventive maintenance — maintenance_plan	Generic inspections — quality_control_oca
Equipment certificates — maintenance_equipment_certification	

Management review

Module	mgmtsystem_review	Version	_____
Menu	Management Systems ► Management System ► Reviews	Effective date	_____
Owner (suggested)	General Manager with the Head of Quality	Approved by	_____
Related SOPs	SOP-QA-005, SOP-GEN-002, SOP-QA-062, SOP-EM-005		

1 Objective

Review at planned intervals the suitability, adequacy and effectiveness of the quality system and decide on improvements and resources.

2 Scope

The whole quality management system, at least once a year (more often in the first years or after major changes).

3 Responsibilities

Role (software group)	Responsibility
Top management	Chairs the review, takes decisions.
Head of Quality	Prepares inputs, records the review.
Process owners	Present their indicators and actions.

4 Definitions

Term	Definition
Inputs	Status of previous actions; changes; customer feedback and complaints; process performance and product conformity; nonconformities and CAPA; audit results; monitoring results; supplier performance; risks and opportunities; adequacy of resources; new regulatory requirements.
Outputs	Decisions on improvement, changes to the QMS, and resource needs.

5 Workflow in the software

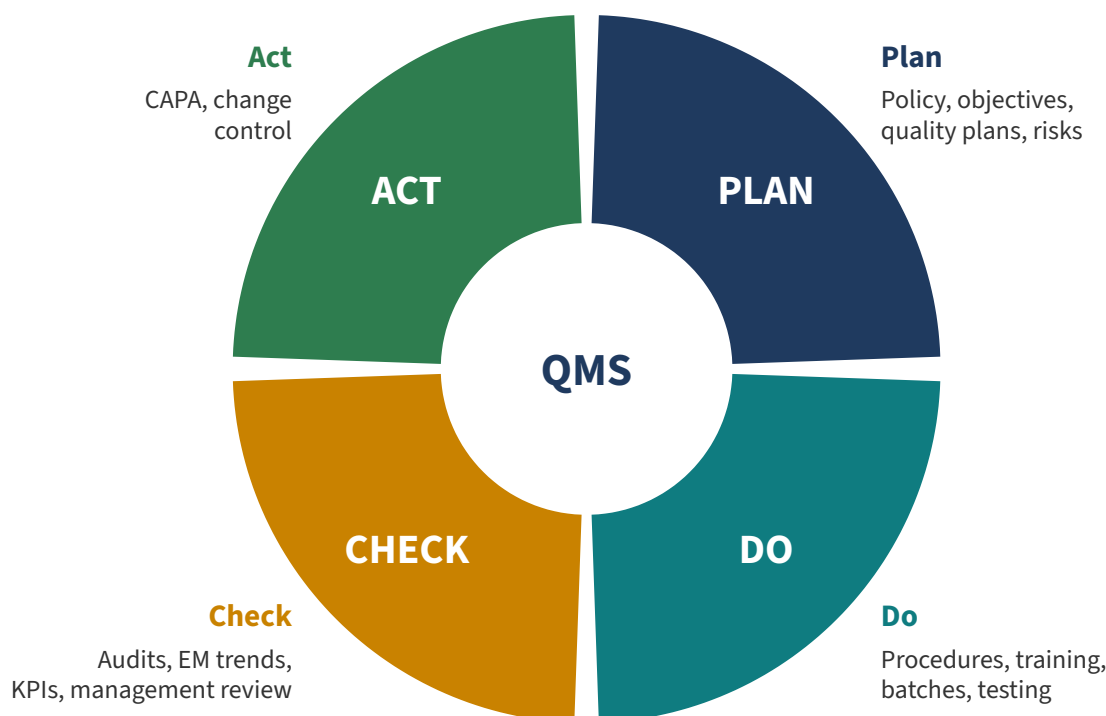


Figure 205 — Management review closes the quality cycle.

6 Procedure

- 1 **Prepare inputs.** Collect from the suite: objectives (SOP-QA-005), CAPA status (ls_capa), deviations and complaints trends, audit results (ls_audit), supplier ratings, EM trends, risk profile, training compliance, recalls, validation status.
- 2 **Create the review.** In the management review record (mgmtsystem_review): name, date, participants, policy, changes; update the KPI history with the dedicated button (KPIs from mgmtsystem_kpi).
- 3 **Hold the meeting.** Discuss each input; record decisions and actions as review lines.
- 4 **Conclude and close.** Write the conclusion; close the review.
- 5 **Follow up.** Turn decisions into CAPA, changes or objectives with owners and dates; report their status at the next review.
- 6 **Alternative.** If mgmtsystem_review is not installed (see limitations), record the review as a quality record of type *Management Review* in ls_qms (SOP-QA-004), with the minutes attached.

7. Controls enforced by the software

- ✓ The review record has states Open and Closed; closed reviews keep their KPI history snapshot.

8. Good practices

- ★ Send the input pack one week before the meeting.
- ★ Record decisions as verifiable actions, not as intentions.

9. Pitfalls to avoid

- ✗ A review reduced to a slide presentation without decisions.

10 Records and evidence

- Management review record or quality record with minutes
- Actions arising

11 Performance indicators

- Actions from the last review completed on time

12 References

- ISO 9001:2015 — Quality management systems — Requirements
- ISO 13485:2016 — Medical devices — Quality management systems
- ICH Q10 — Pharmaceutical Quality System
- EudraLex Vol. 4, EU GMP Part I, Chapter 1 — Pharmaceutical Quality System

13. Demonstration

Case. Annual review 2027: inputs from 14 audits, 62 deviations, 11 CAPA overdue, EM trends stable except corridor C. Decisions: add one QA reviewer (resource), revise the deviation classification SOP (change CC-2028-002), new objective on CAPA timeliness. Recorded, closed, actions assigned.

Quality indicators (KPI)

Module	mgmtsystem_kpi	Version	_____
Menu	Management Systems ► KPI	Effective date	_____
Owner (suggested)	Head of Quality	Approved by	_____
Related SOPs	SOP-QA-005, SOP-GEN-001		

1 Objective

Define a small set of quality indicators with clear formulas, sources, frequencies and thresholds, and review them regularly.

2 Scope

Indicators of the quality system and GMP processes (e.g. deviation closure time, CAPA on time, complaint rate, right-first-time batches, OOS rate, EM excursions, training compliance, supplier rating).

3 Responsibilities

Role (software group)	Responsibility
Head of Quality	Owns the KPI list.
Process owners	Provide values and comment trends.

4 Definitions

Term	Definition
KPI	Indicator computed periodically from a defined source, with thresholds (green/amber/red).
KPI history	Stored values over time, snapshotted into management reviews.

5 Procedure

- 1 Define.** For each indicator write: name, formula, data source (menu or report of the suite), frequency, owner, thresholds, direction.
- 2 Configure.** Create the KPI in mgmtsystem_kpi (based on the OCA *kpi* module) with its computation and thresholds; or, if the module is not available, follow it as a quality objective with measurements in ls_qms (SOP-QA-005).
- 3 Collect.** Compute or enter values at the defined frequency; check the source data.
- 4 Review.** Monthly in the quality meeting; quarterly and annually in management review (SOP-GEN-001).

6. Controls enforced by the software

- ✓ Indicator computation and history are managed by the kpi / mgmtsystem_kpi modules; check the installation limits before relying on them.

7. Good practices

- ★ Prefer indicators computed from system data rather than typed by hand.
- ★ Pair each lagging indicator (e.g. complaints) with a leading one (e.g. CAPA on time).

8. Pitfalls to avoid

- ✗ Too many indicators, none of them acted upon.

9 Records and evidence

- KPI definitions and history

10 Performance indicators

- KPIs out of threshold without action

11 References

- ISO 9001:2015 — Quality management systems — Requirements
- ICH Q10 — Pharmaceutical Quality System

12. Demonstration

Case. KPI *CAPA closed on time (%)* = $\text{CAPA closed} \leq \text{due date} / \text{CAPA closed}$, monthly, source [Quality ► CAPA ► CAPA Records](#), green $\geq 90\%$, red $< 75\%$. In March: 71 % → red; discussed in the quality meeting; action: weekly CAPA review.

Preventive maintenance of equipment

Module	maintenance_plan	Version	_____
Menu	Maintenance ► Maintenance ► Maintenance Plans	Effective date	_____
Owner (suggested)	Maintenance Manager	Approved by	_____
Related SOPs	SOP-MET-001, SOP-MP-002, SOP-VAL-005		

1 Objective

Maintain GMP equipment at planned intervals so that it remains in its qualified state, and record every intervention.

2 Scope

Production, utility and laboratory equipment registered in Odoo Maintenance. Moulds follow SOP-MP-002; measuring instruments follow SOP-MET-002.

3 Responsibilities

Role (software group)	Responsibility
Maintenance Manager	Defines plans and kinds of maintenance.
Technicians	Perform maintenance requests and record work done.
QA	Is informed of interventions that may affect the qualified state.

4 Definitions

Term	Definition
Maintenance plan	Per equipment: kind, interval and interval step, duration, start date, planning horizon, instructions (PDF, slide or text); generates maintenance requests; next maintenance date computed.

5 Procedure

- Create plans.** On each equipment add maintenance plans: kind, interval (e.g. 3 months), duration, start date, horizon, instructions.
- Execute.** Perform the generated maintenance requests; record the work done and parts replaced; close the request.
- Assess impact.** When an intervention may affect the qualified state (e.g. replacement of a critical part), raise a change request (SOP-QA-030) or a revalidation (SOP-VAL-005) before release to production.
- Review.** Review overdue requests weekly; adjust intervals with failure history.

6. Controls enforced by the software

- ✓ Next maintenance dates are computed from the plan interval; requests are generated within the planning horizon.

7. Good practices

- ★ Attach the GMP-relevant instructions (cleaning after maintenance, line clearance) to the plan.

8. Pitfalls to avoid

- ✗ Equipment returned to production after intervention without QA being informed.

9 Records and evidence

- Maintenance plans and requests with work done

10 Performance indicators

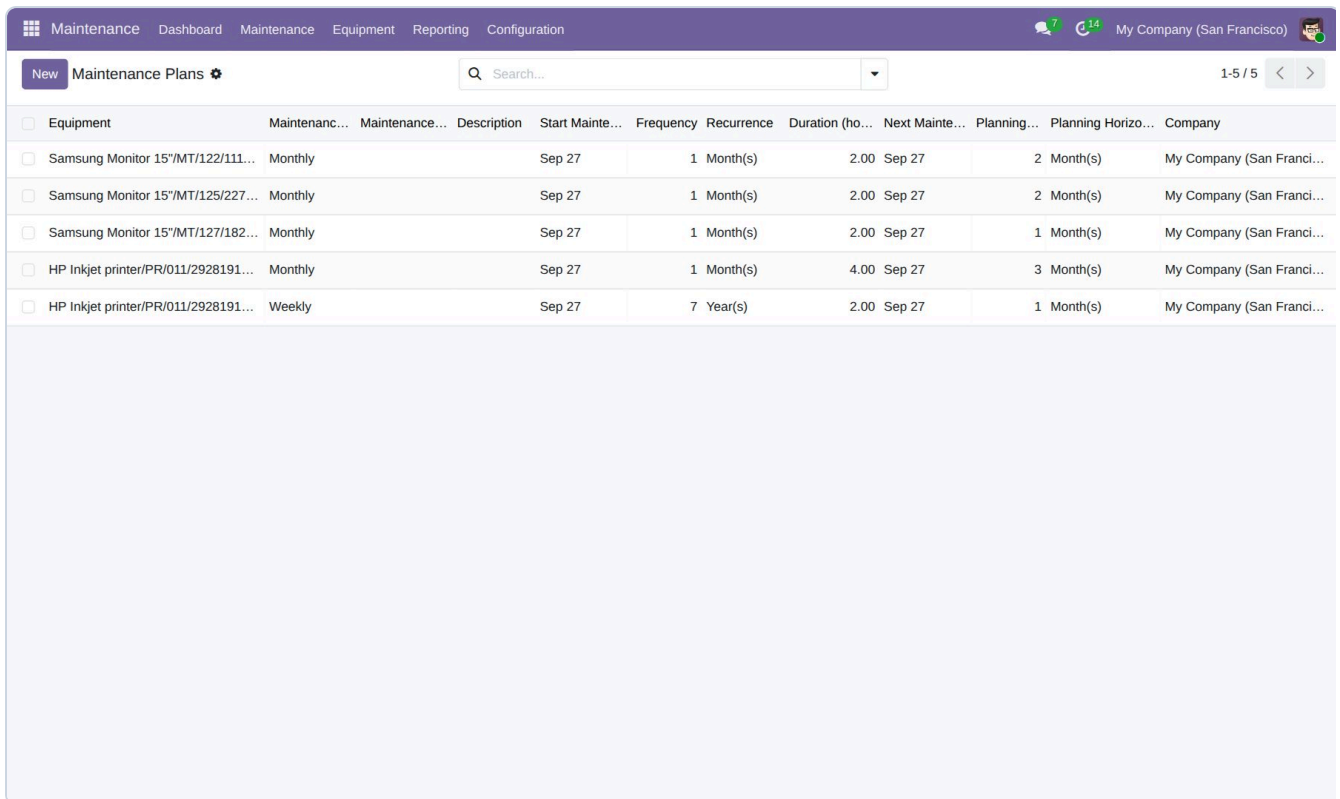
- Preventive maintenance done on time (%)
- Corrective interventions per equipment

11 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 3 — Premises and Equipment
- 21 CFR 211.68 — Automatic, mechanical and electronic equipment
- ISO 13485:2016 — Medical devices — Quality management systems

12. Demonstration

Case. Tablet press TP-02: plan *Preventive* — 3 months, 4 h, instructions PDF; the request of June is done: turret cleaned, punches inspected; no part replaced → no revalidation needed; request closed.



☐ Equipment	Maintenanc...	Maintenance...	Description	Start Mainte...	Frequency	Recurrence	Duration (ho...	Next Mainte...	Planning...	Planning Horizo...	Company
☐ Samsung Monitor 15"/MT/122/111...	Monthly			Sep 27		1 Month(s)	2.00	Sep 27		2 Month(s)	My Company (San Franci...
☐ Samsung Monitor 15"/MT/125/227...	Monthly			Sep 27		1 Month(s)	2.00	Sep 27		2 Month(s)	My Company (San Franci...
☐ Samsung Monitor 15"/MT/127/182...	Monthly			Sep 27		1 Month(s)	2.00	Sep 27		1 Month(s)	My Company (San Franci...
☐ HP Inkjet printer/PR/011/2928191...	Monthly			Sep 27		1 Month(s)	4.00	Sep 27		3 Month(s)	My Company (San Franci...
☐ HP Inkjet printer/PR/011/2928191...	Weekly			Sep 27		7 Year(s)	2.00	Sep 27		1 Month(s)	My Company (San Franci...

Figure 206 — Preventive maintenance plans (maintenance_plan). (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Regulatory certificates of equipment

Module	maintenance_equipment_certification	Version	_____
Menu	Maintenance ► Configuration ► Certificates / Certificate Reminder Rules	Effective date	_____
Owner (suggested)	Maintenance Manager / EHS	Approved by	_____
Related SOPs	SOP-GEN-003, SOP-MET-004		

1 Objective

Keep the regulatory certificates of equipment (pressure vessels, lifting equipment, electrical installations, etc.) current, with reminders before renewal dates.

2 Scope

Equipment subject to statutory inspections or certifications.

3 Responsibilities

Role (software group)	Responsibility
Maintenance Manager	Records certificates and renewals.
EHS	Defines which equipment needs which certificate.

4 Definitions

Term	Definition
Certificate	Name, certificate number, date, renewal date, file, notes, linked equipment.
Reminder rule	Number of days before renewal at which a reminder is sent (activated in settings).

5 Procedure

- 1 Configure reminders.** Enable certificate expiration notifications in the settings and create reminder rules (e.g. 60 and 15 days before).
- 2 Record.** On the equipment add each certificate with number, date, renewal date and file.
- 3 Renew.** Schedule the inspection before the renewal date; record the new certificate; archive the old one.
- 4 Stop use.** If a certificate expires, stop using the equipment until renewed.

6. Controls enforced by the software

- ✓ Reminders are sent according to the rules when notifications are enabled.

7. Good practices

- ★ Keep inspection reports with their certificate.

8. Pitfalls to avoid

- ✗ Relying on the inspection body to remind you.

9 Records and evidence

- Equipment certificates with files

10 Performance indicators

- Expired certificates of equipment in use (target 0)

11 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 3 — Premises and Equipment

12. Demonstration

Case. Autoclave AC-02 pressure vessel certificate renewed every 24 months; reminders at 60 and 15 days; the inspection is performed 3 weeks before the renewal date and the new certificate attached.

Role-based access management and periodic review of access rights

Module	base_user_role	Version	_____
Menu	Settings ► Users & Companies ► Roles	Effective date	_____
Owner (suggested)	System Owner with IT	Approved by	_____
Related SOPs	SOP-DI-011, SOP-QA-013		

1 Objective

Grant access rights through documented roles matching job functions and the segregation of duties of the SOPs, and review them periodically.

2 Scope

All users of the Odoo installation, all LS module groups and standard Odoo groups.

3 Responsibilities

Role (software group)	Responsibility
System Owner	Defines roles, approves access requests, performs the periodic review.
IT administrator	Implements approved changes; never grants rights without approval.
Line managers	Request access and removal for their staff.

4 Definitions

Term	Definition
Role	Named set of groups (base_user_role) assigned to users, optionally with <i>From</i> and <i>To</i> dates.
Access matrix	Document listing, per job function, the roles and the LS groups granted.

5 Procedure

- Design roles.** Build roles from the responsibilities tables of the SOPs (e.g. *QA officer* = Deviation Investigator + CAPA Coordinator + Document Approver + Laboratory Reviewer...). Check that no role combines incompatible duties (e.g. batch manufacturer and release authority).
- Request and approve.** Access requests are made by the line manager and approved by the System Owner; record them (ticket or quality record).
- Assign.** Assign roles to users with dates when temporary (holidays, projects). Without base_user_role (see limitations), assign groups directly and keep the access matrix as the reference.
- Link to training.** Grant critical roles only to users with valid qualifications (SOP-QA-013).
- Remove.** Remove access the day a person leaves or changes job.
- Review.** Every 6 months (critical systems) export users and groups, compare with the matrix and HR list, correct differences, record the review.

6. Controls enforced by the software

- ✓ Role lines can be limited in time (From/To); each LS module enforces its own group checks and segregation-of-duties rules regardless of the role mechanism.

7. Good practices

- ★ Give administrators separate personal accounts for administration and for business use.
- ★ Keep at least two people per critical approval role.

8. Pitfalls to avoid

- ✗ Granting *Manager* groups to everyone to avoid access errors — this defeats segregation of duties.

9 Records and evidence

- Access matrix
- Access requests and approvals
- Periodic review records

10 Performance indicators

- Differences found at review
- Leavers with active accounts (target 0)

11 References

- EU GMP Annex 11 — Computerised Systems (check the revision in force)
- 21 CFR Part 11 — Electronic Records; Electronic Signatures
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)

12. Demonstration

Case. Semi-annual review: 142 active users; 3 leavers still active (disabled the same day), 2 users with Deviation Manager who changed jobs (removed), 1 missing Laboratory Reviewer backup (granted after training). Review recorded as a quality record.

Technical audit log

Module	auditlog	Version	_____
Menu	Settings ► Technical ► Audit ► Rules / Logs (menu defined by the module)	Effective date	_____
Owner (suggested)	IT administrator with the System Owner	Approved by	_____
Related SOPs	SOP-DI-001, SOP-DI-002		

1 Objective

Use the OCA auditlog module, where installed, to log technical operations (including reads, sessions and HTTP requests) useful for security investigations, and keep it distinct from the GxP audit trail.

2 Scope

Technical and security logging of selected models. GxP record changes are covered by ls_audit_trail (SOP-DI-001 to 004).

3 Responsibilities

Role (software group)	Responsibility
IT administrator	Configures rules and investigates security events.
System Owner / QA	Approves the scope and reviews findings.

4 Definitions

Term	Definition
auditlog rule	Model with log_read, log_write, log_create, log_unlink options, log type (full or fast), capture of the record.
Difference with ls_audit_trail	auditlog entries are not hash-chained; ls_audit_trail provides tamper-evidence, verification and evidence packs.

5 Procedure

- 1 Decide the need.** Define which technical events must be logged (e.g. reads of personal data, access to configuration) and why ls_audit_trail does not cover them.
- 2 Configure.** Create rules for the selected models and subscribe them; keep reads logging limited (volume).
- 3 Review.** Review the logs during security investigations and periodically for administrator activity.
- 4 Retain and purge.** Define retention and purge the logs by the module's scheduled action according to the IT policy.

6. Controls enforced by the software

- ✓ At the time of writing, auditlog failed to install on a fresh Odoo 19.0 database (view reference `view_auditlog_http_request_tree` not found). Fix and validate the module before relying on this SOP.

7. Good practices

- ★ Do not duplicate GxP auditing in two tools with different rules — use ls_audit_trail for GxP.

8. Pitfalls to avoid

- ✗ Logging reads on large models, degrading performance.

9 Records and evidence

- auditlog rules
- Investigation extracts

10 Performance indicators

- Security investigations using the log

11 References

- EU GMP Annex 11 — Computerised Systems (check the revision in force)
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)

12. Demonstration

Case. Reads of the HR employee model are logged to support investigations of personal data access; changes to GxP records remain in ls_audit_trail.

Generic inspection controls, for example at reception

Module	quality_control_oqa	Version	_____
Menu	Quality Control ► Inspections ► Inspections · Tests ► Tests	Effective date	_____
Owner (suggested)	QC Manager	Approved by	_____
Related SOPs	SOP-QC-011, SOP-QA-070		

1 Objective

Perform simple, questionnaire-based inspections (e.g. visual and documentary checks at reception) with recorded answers and a pass/fail outcome, where a full laboratory sample is not needed.

2 Scope

Incoming visual/identity checks, packaging inspections and other attribute inspections. Analytical testing is managed in *ls_lab*.

3 Responsibilities

Role (software group)	Responsibility
Inspector	Performs the inspection and records answers.
Supervisor	Approves inspections waiting for approval.

4 Definitions

Term	Definition
Test	Set of questions (qualitative with allowed values, or quantitative with min/max).
Inspection states	Plan, Draft, Ready, Waiting supervisor approval, Quality success, Quality failed, Canceled.

5 Procedure

- 1 Define tests.** In [Quality Control ► Tests ► Tests](#) create the test and its questions with acceptable values or ranges; group tests by category.
- 2 Create the inspection.** Create an inspection for the object (e.g. the product/lot received) and select the test. Note: automatic creation on receipts requires a stock integration module that is not present in the repository — inspections are created manually.
- 3 Record answers.** Answer each question; the outcome is computed; confirm the inspection.
- 4 Approve.** Inspections requiring approval go to *Waiting supervisor approval*; the supervisor approves.
- 5 Act on failure.** A failed inspection leads to quarantine and a deviation or supplier complaint.

6. Controls enforced by the software

- ✓ The inspection outcome (success) is computed from the answers against the defined values or ranges.

7. Good practices

- ★ Keep tests short and specific to the material category.

8. Pitfalls to avoid

- ✗ Using generic inspections for analytical tests that need method, specification and review (use ls_lab).

9 Records and evidence

- Inspections with answers and outcome

10 Performance indicators

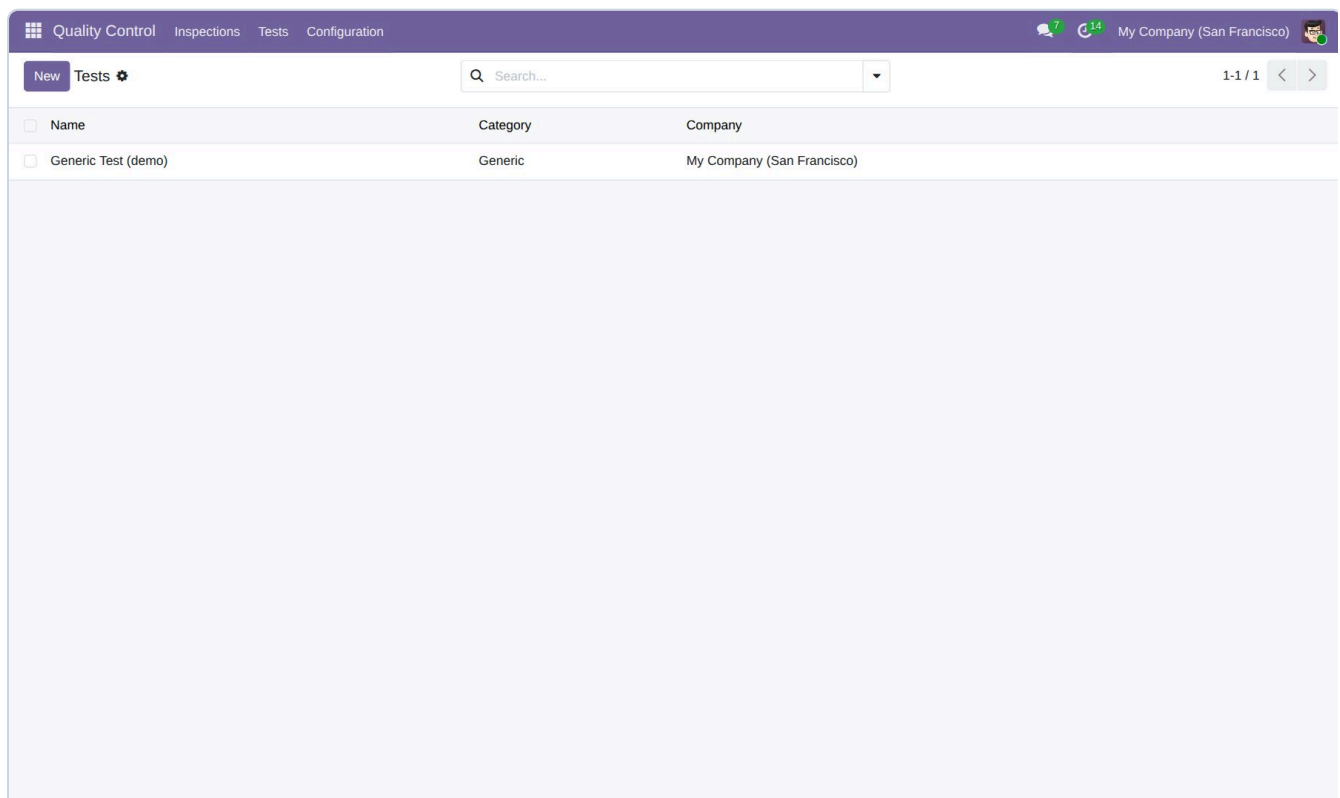
- Failed inspections per supplier

11 References

- ISO 13485:2016 — Medical devices — Quality management systems
- 21 CFR 211.84 — Testing and approval or rejection of components
- EudraLex Vol. 4, EU GMP Part I, Chapter 5 — Production

12. Demonstration

Case. Reception of printed cartons: test *Carton — reception* with 6 questions (supplier label present, quantity matches, artwork version, colour vs standard, no damage, CoC present). One carton lot has artwork version 3 instead of 4 → *Quality failed*, lot quarantined, supplier complaint opened.



The screenshot displays the 'Quality Control' module interface in Odoo 19.0. The top navigation bar includes 'Quality Control', 'Inspections', 'Tests', and 'Configuration'. On the right, there are notification icons for 7 messages and 14 alerts, along with the company name 'My Company (San Francisco)'. Below the navigation bar, a 'New Tests' button and a search field are visible. The main content area shows a table with the following data:

☐	Name	Category	Company
☐	Generic Test (demo)	Generic	My Company (San Francisco)

Figure 207 — Inspection test definitions (*quality_control_oca*). (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Part VI — Procedures needed but not covered by the suite

These procedures depend on infrastructure and organisation, not on the modules. They are nevertheless required by GMP and data integrity guidance (EU GMP Annex 11, PIC/S PI 041-1) for the suite to be used in a regulated way. Each outline below lists what the SOP must contain; the content must be written with IT and the hosting provider.

Proposed code	Procedure	What it must contain
SOP-IT-010	Backup and restore	Scope (database, filestore, configuration), frequency and retention, encryption, off-site copy, restore test at defined intervals with evidence (restore to a test environment and verification of records, audit trail chain and signature chain — SOP-DI-002, SOP-DI-012), responsibilities.
SOP-IT-011	Business continuity and disaster recovery	Recovery time and recovery point objectives per process, fallback on paper (forms, later transcription and verification), crisis roles, periodic exercise, return to normal.
SOP-IT-012	IT incident management	Detection, classification (including data integrity impact), escalation to QA, investigation, link to deviation/CAPA, communication, closure.
SOP-IT-013	Physical security of servers	Location, access control and logs, environment (power, temperature, fire), visitor rules, cloud provider controls when hosted.
SOP-IT-014	Qualification of the hosting	Supplier assessment of the hosting/cloud provider (SOP-QA-070), service agreement, infrastructure qualification (IQ), security certifications, change notification.
SOP-VAL-010	Validation of the suite itself (IQ, OQ, PQ of the system)	User requirements, supplier assessment of the modules, risk-based test strategy (SOP-VAL-006), installation qualification of Odoo and modules (versions, installation log without errors), operational qualification of the controls listed in this handbook, performance qualification with real data and users, traceability matrix, report, periodic review and change control of upgrades.

Why these matter here

Several controls of the suite depend on the database (append-only triggers, hash chains, scheduled integrity checks). A restore, a migration or a server change can therefore affect regulated evidence: backup/restore and validation procedures must include the verification of the audit trail and signature chains after any restore.

Annexes

Annex A — SOP register

Register of the SOPs of this handbook, to steer their drafting. Copy it into your document system and complete owners, dates and status.

Code	Title	Module	Owner (suggested)	Drafting status	Target date
SOP-QA-001	Drafting, review, approval and publication of quality documents	ls_qms · ls_document_management	Head of Quality Assurance	To draft / adapt	
SOP-QA-002	Revision, withdrawal and archiving of documents	ls_qms · ls_document_management	QMS Manager	To draft / adapt	
SOP-QA-003	Periodic review of documents	ls_qms	QMS Manager	To draft / adapt	
SOP-QA-004	Management and retention of quality records	ls_qms · ls_document_management	QMS Manager	To draft / adapt	
SOP-QA-005	Quality objectives and measurement of their achievement	ls_qms	Head of Quality / Management	To draft / adapt	
SOP-QA-010	Creation and approval of training courses	ls_training	Training Manager	To draft / adapt	
SOP-QA-011	Session planning and issue of certifications	ls_training	Training Manager	To draft / adapt	
SOP-QA-012	Training linked to the publication of a new document	ls_qms · ls_training	QMS Manager with the Training Manager	To draft / adapt	
SOP-QA-013	Follow-up of qualifications and requalification	ls_training	Training Manager	To draft / adapt	
SOP-QA-020	Reporting, assessment and classification of deviations	ls_deviation	Head of Quality Assurance	To draft / adapt	
SOP-QA-021	Investigation, extension to other batches and product disposition	ls_deviation	Head of Quality Assurance	To draft / adapt	
SOP-QA-022	CAPA management from identification to closure	ls_capa	CAPA Manager (QA)	To draft / adapt	
SOP-QA-023	Verification of CAPA effectiveness	ls_capa	CAPA Manager (QA)	To draft / adapt	
SOP-QA-030	Change request and impact assessment	ls_change_control	Change Control Manager	To draft / adapt	
SOP-QA-031	Approval, implementation and verification of changes	ls_change_control	Change Control Manager	To draft / adapt	
SOP-QA-040	Reception, assessment and investigation of complaints	ls_complaint	Complaint Manager (QA)	To draft / adapt	
SOP-QA-041	Adverse events: reportability assessment and reporting deadlines	ls_complaint	Vigilance officer / Complaint Manager	To draft / adapt	
SOP-QA-050	Internal audit programme	ls_audit	Audit Manager	To draft / adapt	
SOP-QA-051	Auditor qualification and impartiality	ls_audit	Audit Manager	To draft / adapt	
SOP-QA-052	Conducting an audit: preparation, fieldwork, findings	ls_audit	Lead auditor	To draft / adapt	

Code	Title	Module	Owner (suggested)	Drafting status	Target date
SOP-QA-053	Audit report and follow-up of findings	ls_audit	Audit Manager	To draft / adapt	
SOP-QA-060	Quality risk management: matrix, assessment, control and acceptance of residual risk	ls_risk_management	Risk Manager	To draft / adapt	
SOP-QA-061	FMEA (failure modes and effects analysis)	ls_risk_management	Risk Manager	To draft / adapt	
SOP-QA-062	Periodic review of risks	ls_risk_management	Risk Manager	To draft / adapt	
SOP-QA-070	Qualification and approval of suppliers	ls_supplier_qualification	Supplier Manager (QA)	To draft / adapt	
SOP-QA-071	Supplier audits	ls_supplier_qualification	Supplier Manager (QA)	To draft / adapt	
SOP-QA-072	Supplier performance evaluation and requalification	ls_supplier_qualification	Supplier Manager (QA)	To draft / adapt	
SOP-QA-073	Control of purchases from non-approved suppliers	ls_supplier_qualification · purchase	Supplier Manager with the Purchasing Manager	To draft / adapt	
SOP-DI-001	Definition of the audit trail scope (rules per model)	ls_audit_trail	Data Integrity lead / System owner	To draft / adapt	
SOP-DI-002	Audit trail review and integrity verification	ls_audit_trail	QA / Data Integrity lead	To draft / adapt	
SOP-DI-003	Evidence packs for inspections	ls_audit_trail	Audit Trail Auditor (QA)	To draft / adapt	
SOP-DI-004	Retention and controlled purge of audit data	ls_audit_trail	QA with the Audit Trail Administrator	To draft / adapt	
SOP-DI-010	Electronic signature policy: meanings, policies, signer commitment	ls_electronic_signature	QA with the System Owner	To draft / adapt	
SOP-DI-011	Management of credentials, account lockout and security alerts	ls_electronic_signature	IT security with QA	To draft / adapt	
SOP-DI-012	Verification of the signature chain	ls_electronic_signature	Signature Auditor (QA)	To draft / adapt	
SOP-VAL-001	Validation master plan (VMP)	ls_validation	Validation Manager	To draft / adapt	
SOP-VAL-002	Drafting and approval of qualification protocols: DQ, IQ, OQ, PQ	ls_validation	Validation Manager	To draft / adapt	
SOP-VAL-003	Protocol execution and management of discrepancies	ls_validation	Validation Manager	To draft / adapt	
SOP-VAL-004	Summary report and validation status	ls_validation	Validation Manager	To draft / adapt	
SOP-VAL-005	Periodic revalidation	ls_validation	Validation Manager	To draft / adapt	

Code	Title	Module	Owner (suggested)	Drafting status	Target date
SOP-VAL-006	Validation of processes, cleaning, analytical methods and computerised systems (CSV)	ls_validation	Validation Manager	To draft / adapt	
SOP-MET-001	Management of the instrument inventory	ls_calibration	Calibration Manager (metrology)	To draft / adapt	
SOP-MET-002	Planning and performing calibrations (as-found and as-left values)	ls_calibration	Calibration Manager	To draft / adapt	
SOP-MET-003	Impact assessment of an out-of-tolerance result	ls_calibration	Calibration Manager with QA	To draft / adapt	
SOP-MET-004	Calibration certificates	ls_calibration	Calibration Manager	To draft / adapt	
SOP-EM-001	Monitoring programme: areas, sampling points and rationale	ls_environmental_monitoring	EM Manager (QA / microbiology)	To draft / adapt	
SOP-EM-002	Setting and approval of alert and action limits	ls_environmental_monitoring	EM Manager	To draft / adapt	
SOP-EM-003	Sampling, analysis and review of results	ls_environmental_monitoring	EM Manager	To draft / adapt	
SOP-EM-004	Management of excursions	ls_environmental_monitoring	EM Manager with QA	To draft / adapt	
SOP-EM-005	Trend analysis	ls_environmental_monitoring	EM Manager	To draft / adapt	
SOP-QA-080	Recall plan, with coordinator and deputy	ls_recall	Head of Quality	To draft / adapt	
SOP-QA-081	Executing a recall or withdrawal: distribution tracing, notification, reconciliation	ls_recall	Recall coordinator	To draft / adapt	
SOP-QA-082	Recall effectiveness checks	ls_recall	Recall coordinator	To draft / adapt	
SOP-QA-083	Mock recall (periodic exercise)	ls_recall	Head of Quality	To draft / adapt	
SOP-PRD-001	Issue and execution of the batch record	ls_pharma	Production Manager	To draft / adapt	
SOP-PRD-002	Line clearance and label reconciliation	ls_pharma	Packaging Manager	To draft / adapt	
SOP-QA-090	Batch record review and batch release by the quality unit	ls_pharma	Head of Quality Assurance	To draft / adapt	
SOP-QC-001	Qualification of active substances and excipients	ls_pharma	QC Manager with QA	To draft / adapt	
SOP-QC-002	Stability programme	ls_pharma	QC Manager	To draft / adapt	

Code	Title	Module	Owner (suggested)	Drafting status	Target date
SOP-PRD-003	GS1 serialisation and aggregation	ls_pharma	Packaging Manager with IT	To draft / adapt	
SOP-RA-001	Preparation of the CTD dossier (Common Technical Document)	ls_pharma	Regulatory Affairs Manager	To draft / adapt	
SOP-COS-001	Development and approval of formulations	ls_cosmetics	Formulation Manager with the Regulatory Manager	To draft / adapt	
SOP-COS-002	Safety assessment (CPSR, Cosmetic Product Safety Report)	ls_cosmetics	Safety Assessor	To draft / adapt	
SOP-COS-003	Product information file (PIF) and its retention period	ls_cosmetics	Responsible Person (Regulatory Manager)	To draft / adapt	
SOP-COS-004	Substantiation of claims	ls_cosmetics	Regulatory Manager	To draft / adapt	
SOP-COS-005	Validation of labelling and of the INCI list	ls_cosmetics	Regulatory Manager	To draft / adapt	
SOP-COS-006	Maintenance of the register of regulatory restrictions on ingredients	ls_cosmetics	Regulatory Manager	To draft / adapt	
SOP-COS-007	Prior authorisation dossier in Algeria	ls_cosmetics	Regulatory Manager (Algeria)	To draft / adapt	
SOP-MD-001	Device registration and justification of its risk class	ls_medical_device	Regulatory Affairs Manager	To draft / adapt	
SOP-MD-002	Assignment of the UDI (Unique Device Identification)	ls_medical_device	Regulatory Affairs Manager	To draft / adapt	
SOP-MD-003	Risk management according to ISO 14971	ls_medical_device	Risk management lead (Regulatory / R&D)	To draft / adapt	
SOP-MD-004	Clinical evaluation and PMCF (Post-Market Clinical Follow-up)	ls_medical_device	Clinical evaluation lead	To draft / adapt	
SOP-MD-005	Technical documentation	ls_medical_device	Regulatory Affairs Manager	To draft / adapt	
SOP-MD-006	CE marking and relationship with the notified body	ls_medical_device	Regulatory Affairs Manager	To draft / adapt	
SOP-MD-007	Post-market surveillance and periodic reports	ls_medical_device	PMS lead (Regulatory Affairs)	To draft / adapt	
SOP-MP-001	Qualification and commissioning of moulds	ls_medical_plastics	Tooling / Process Engineering Manager	To draft / adapt	
SOP-MP-002	Preventive maintenance of moulds (by shot count and by time) and cavity blocking	ls_medical_plastics	Tooling Manager	To draft / adapt	
SOP-MP-003	Process windows: drafting, review and approval	ls_medical_plastics	Process Engineering Manager	To draft / adapt	

Code	Title	Module	Owner (suggested)	Drafting status	Target date
SOP-MP-004	Start-up, operation and closure of a moulding run	ls_medical_plastics	Moulding Production Manager	To draft / adapt	
SOP-MP-005	Correction of parameter readings (without altering the original value)	ls_medical_plastics	Moulding Production Manager with QA	To draft / adapt	
SOP-MP-006	Material traceability	ls_medical_plastics	Moulding Quality Manager	To draft / adapt	
SOP-QC-010	Approval of test methods and specifications	ls_lab	QC Manager	To draft / adapt	
SOP-QC-011	Receipt and registration of samples	ls_lab	QC Manager	To draft / adapt	
SOP-QC-012	Entry, review and approval of results	ls_lab	QC Manager	To draft / adapt	
SOP-QC-013	Investigation of OOS (out-of-specification) and OOT (out-of-trend) results	ls_lab	QC Manager with QA	To draft / adapt	
SOP-QC-014	Stability studies in the laboratory	ls_lab	QC Manager	To draft / adapt	
SOP-QC-015	Issue of certificates of analysis	ls_lab	QC Manager	To draft / adapt	
SOP-RA-010	Regulatory watch: register of cited texts	ls_import_export	Regulatory Affairs Manager (import-export)	To draft / adapt	
SOP-RA-011	Handling open regulatory questions	ls_import_export	Regulatory Affairs Manager	To draft / adapt	
SOP-GEN-001	Management review	mgmtsystem_review	General Manager with the Head of Quality	To draft / adapt	
SOP-GEN-002	Quality indicators (KPI)	mgmtsystem_kpi	Head of Quality	To draft / adapt	
SOP-GEN-003	Preventive maintenance of equipment	maintenance_plan	Maintenance Manager	To draft / adapt	
SOP-GEN-004	Regulatory certificates of equipment	maintenance_equipment_certification	Maintenance Manager / EHS	To draft / adapt	
SOP-IT-001	Role-based access management and periodic review of access rights	base_user_role	System Owner with IT	To draft / adapt	
SOP-IT-002	Technical audit log	auditlog	IT administrator with the System Owner	To draft / adapt	
SOP-QC-020	Generic inspection controls, for example at reception	quality_control_oca	QC Manager	To draft / adapt	
SOP-IT-010	Backup and restore	— (not covered)	IT / QA	To draft / adapt	
SOP-IT-011	Business continuity and disaster recovery	— (not covered)	IT / QA	To draft / adapt	
SOP-IT-012	IT incident management	— (not covered)	IT / QA	To draft / adapt	

Code	Title	Module	Owner (suggested)	Drafting status	Target date
SOP-IT-013	Physical security of servers	— (not covered)	IT / QA	To draft / adapt	
SOP-IT-014	Qualification of the hosting	— (not covered)	IT / QA	To draft / adapt	
SOP-VAL-010	Validation of the suite (IQ, OQ, PQ)	— (not covered)	IT / QA	To draft / adapt	

Annex B — Glossary

Term	Meaning
ALCOA+	Attributable, Legible, Contemporaneous, Original, Accurate, plus Complete, Consistent, Enduring, Available — expected properties of GxP data.
Audit trail	Secure, computer-generated, time-stamped record of creation, modification and deletion of records.
CAPA	Corrective and Preventive Action.
CoA	Certificate of Analysis.
CPSR	Cosmetic Product Safety Report (Regulation (EC) No 1223/2009, Annex I).
CSV	Computerised System Validation.
CTD / eCTD	Common Technical Document (ICH M4) / its electronic format.
DQ / IQ / OQ / PQ	Design / Installation / Operational / Performance Qualification.
EM	Environmental Monitoring.
FMEA / RPN	Failure Modes and Effects Analysis / Risk Priority Number (S × O × D).
FSCA	Field Safety Corrective Action (medical devices).
GTIN / SSCC	Global Trade Item Number / Serial Shipping Container Code (GS1).
GxP	Good practices (manufacturing, laboratory, distribution, clinical...).
INCI	International Nomenclature of Cosmetic Ingredients.
MDR	Regulation (EU) 2017/745 on medical devices.
OOS / OOT	Out of specification / out of trend result.
PIF	Product Information File (cosmetics, Article 11).
PMCF / PMS / PSUR	Post-Market Clinical Follow-up / Post-Market Surveillance / Periodic Safety Update Report.
QP	Qualified Person (EU GMP Annex 16).
Segregation of duties	Rule that the same person cannot perform and approve (or review and approve) the same record.
SOP	Standard Operating Procedure.
UDI (Basic UDI-DI, UDI-DI, UDI-PI)	Unique Device Identification and its components.
VMP	Validation Master Plan.

Annex C — Known limitations observed while preparing this handbook

- OCA modules that failed to install** on a fresh Odoo 19.0 Community database (source build of the 19.0 branch dated 26 September 2026) during the preparation of this handbook: *auditlog* (external ID `auditlog.view_auditlog_http_request_tree` not found), *base_user_role* (`base_user_role.view_res_users_role_tree` not found), and *mgmtsystem_kpi* and *mgmtsystem_review* through their dependency *base_external_dbsource* (`base_external_dbsource.view_dbsource_tree` not found). In each case the view was renamed from *tree* to *list* but a `ref=` still uses the old identifier. The project's own

container image (19.0-20260723) and database were not available for comparison; the production database may already contain these modules from an earlier installation. SOP-GEN-001, SOP-GEN-002, SOP-IT-001 and SOP-IT-002 describe fall-back procedures.

- **Electronic signatures:** no production model of the 22 LS modules inherits *ls.signature.mixin*; signature policies can therefore not yet be applied to LS records (see the Electronic signatures chapter). *ls_validation* re-authenticates signers; *ls_lab* records a signature intent that it states is not a Part 11 signature; *ls_supplier_qualification* keeps a chained decision log without re-authentication.
- **Hand-offs between modules** (deviation → CAPA, complaint → CAPA, audit finding → CAPA, excursion → deviation...) are made by typed references, not by relational links. The procedures must require the references to be written and checked.
- **Translations:** the LS modules ship no French translation; the interface and the screenshots are in English.
- **Training:** *ls_training* states that it does not implement 21 CFR Part 11 (no electronic signature, no field-level audit trail of its own). There is no automatic link between a published document and a training course (SOP-QA-012).
- **Supplier purchase control** is installed at level *warn* by default; SOP-QA-073 requires *block* for GMP companies.
- **quality_control_oca:** no stock integration module is present in the repository; inspections are created manually.
- **Demonstration data:** many LS models ship no demo records. Additional demonstration records were created by the scripts in `docs/sop_handbook/scenarios`, which drive the real workflows with separate demonstration users (so that segregation-of-duties rules apply). Models still without data are shown as the empty form of a new record, with its real fields and status bar.

Annex D — How this handbook was produced

Workflows, buttons, menus, groups and controls were extracted from the module source code (repository `youneskh/LSS`, branch `master`) and from a database where the 22 LS modules were installed with demonstration data on Odoo 19.0 Community. State lists come from that database (file `data/states_from_db.txt`). Screenshots were captured from that running instance. The book is generated by `docs/sop_handbook/build.py`; regenerate it after module changes so that the procedures stay aligned with the software.